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(54) COMPOSITIONS AND METHODS FOR THE TREATMENT OF CONGENITAL ICHTHYOSES

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(60) Provisional application No. 62/895,045, filed on Sep. 3, 2019.

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(52) **U.S. Cl.**

CPC ***C12N 15/86*** (2013.01); ***A61P 17/00*** (2018.01); ***C07K 14/47*** (2013.01); ***C12N 15/8695*** (2013.01); *C12N 2710/16643* (2013.01); *C12N 2710/16671* (2013.01)

(57) **ABSTRACT**

The present disclosure provides recombinant nucleic acids comprising one or more polynucleotides encoding an ichthyosis-associated polypeptide; viruses comprising the recombinant nucleic acids; compositions comprising the recombinant nucleic acids and/or viruses; methods of their use; and articles of manufacture or kits thereof.

Specification includes a Sequence Listing.

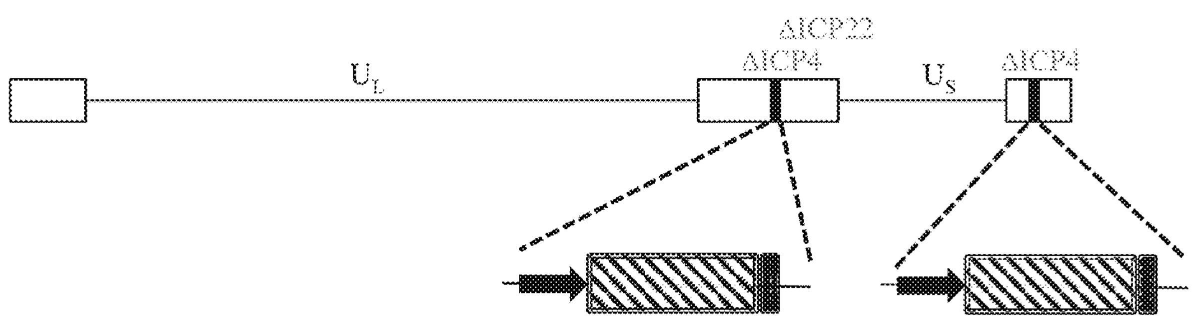


FIG. 1A

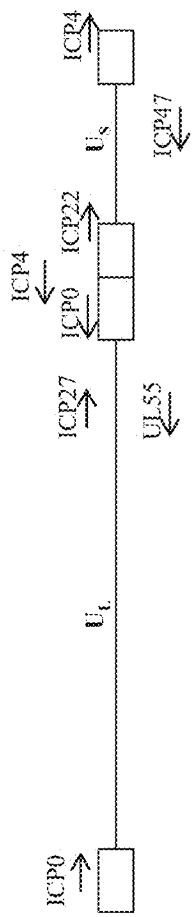

ICP0
U_L
ICP27
UL55
ICP0
ICP4
ICP22
U_S
ICP47
ICP4

FIG. 1B

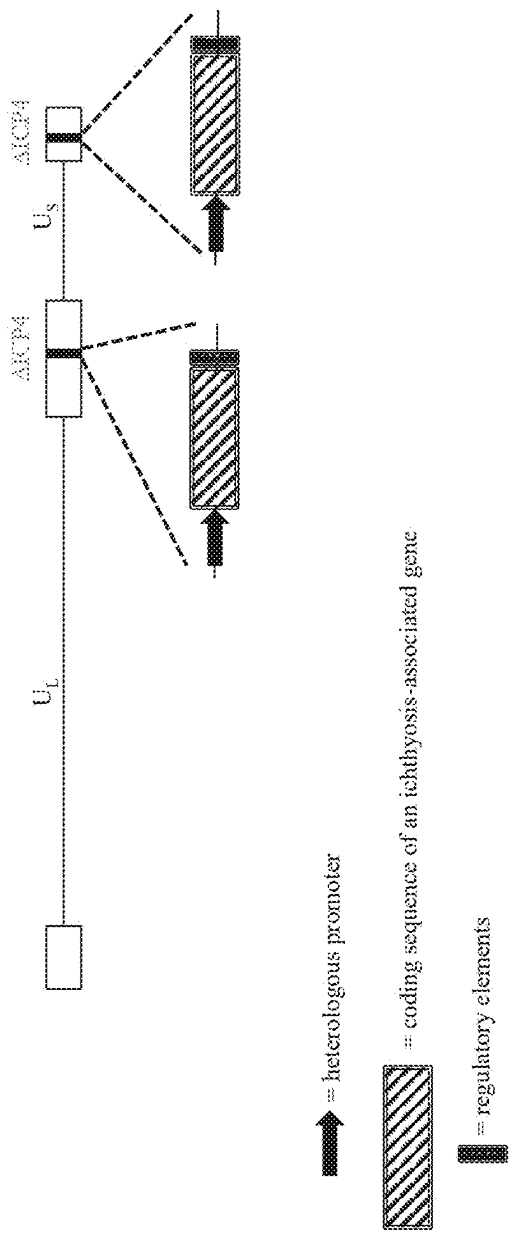

U_L
ΔICP4
U_S
ΔICP4
= heterologous promoter
= coding sequence of an ichthyosis-associated gene
= regulatory elements

FIG. 1C

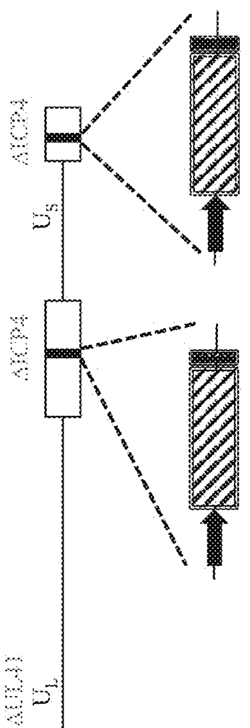
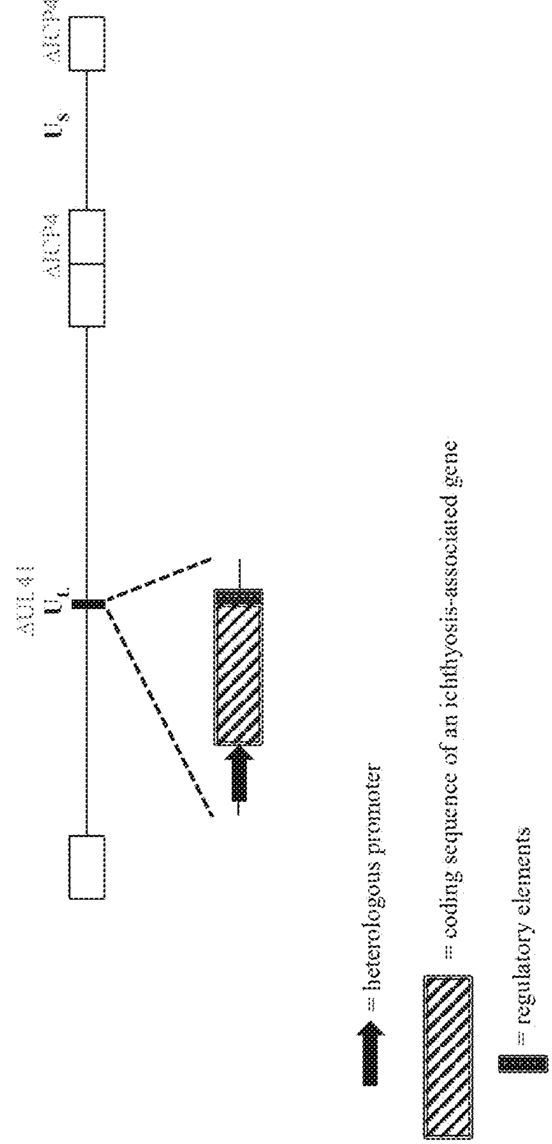

ΔUL41
U_L
ΔICP4
U_S
ΔICP4

FIG. 1D


ΔUL41
ΔICP4
ΔICP4
= heterologous promoter
= coding sequence of an ichthyosis-associated gene
= regulatory elements

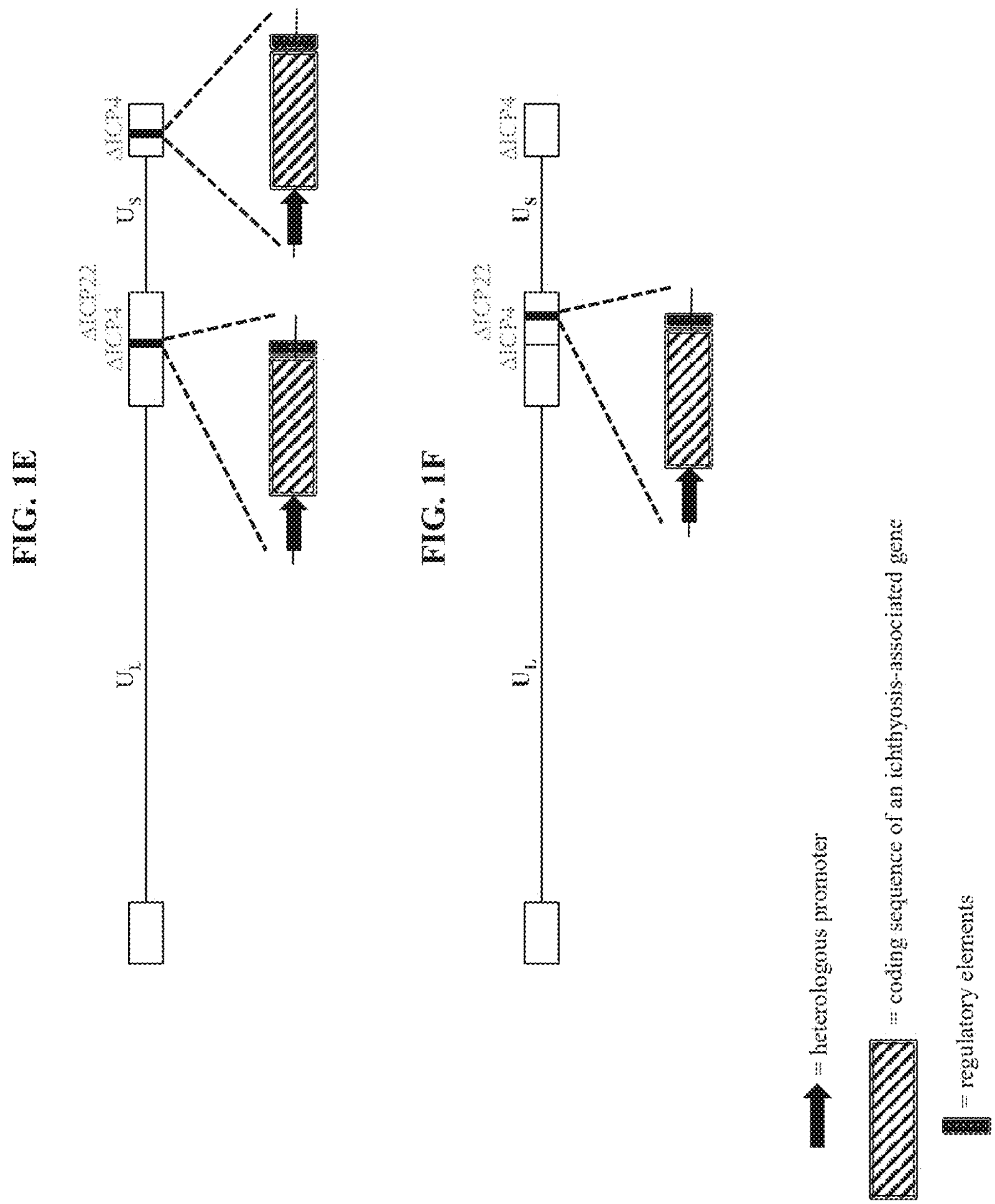
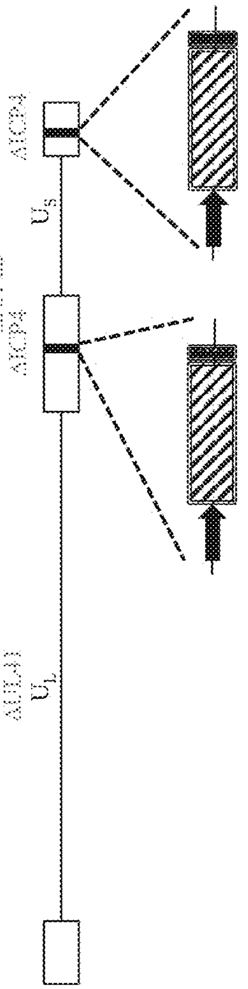

ΔICP4
ΔICP22
ΔICP4
U_S
U_L
ΔICP4
ΔICP22
ΔICP4
U_S
U_L
= heterologous promoter
= coding sequence of an ichthyosis-associated gene
= regulatory elements

FIG. 1E

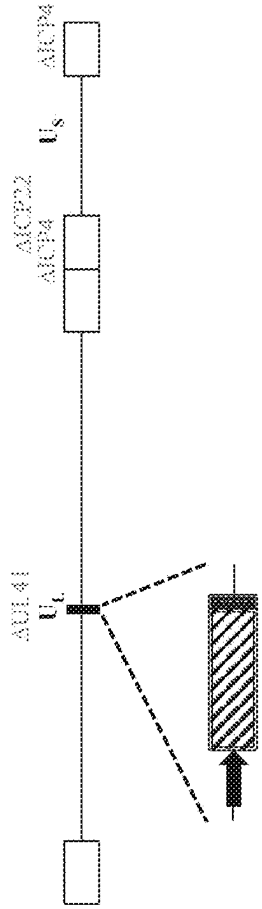
FIG. 1F

FIG. 1G

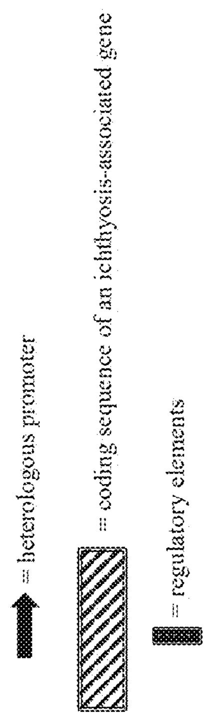


ΔUL43
U_L
ΔICP4
ΔICP22
U_S
ΔICP4

FIG. 1H



ΔUL41
U_L
ΔICP4
ΔICP22
U_S
ΔICP4



= heterologous promoter
= coding sequence of an ichthyosis-associated gene
= regulatory elements

FIG. 1I

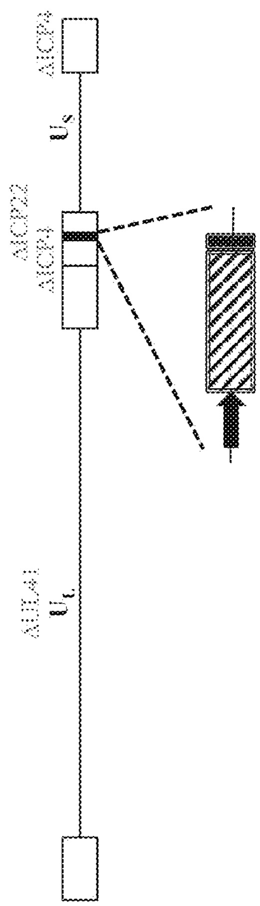


FIG. 2A

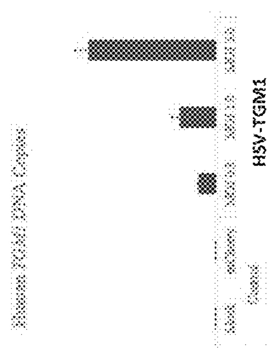


FIG. 2B

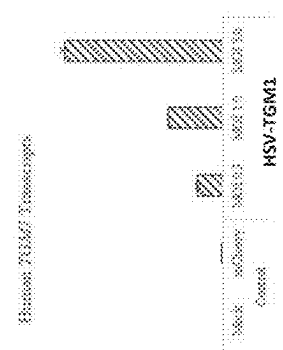


FIG. 2C

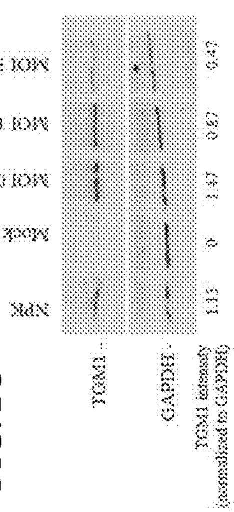


FIG. 2D

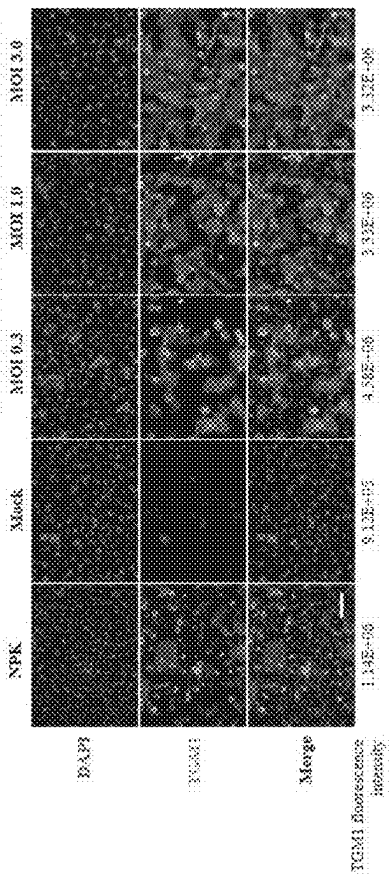


FIG. 2E

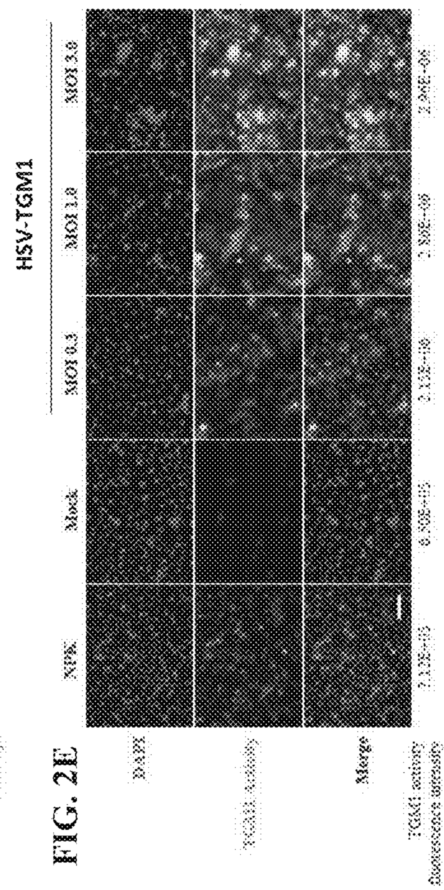


FIG. 2G

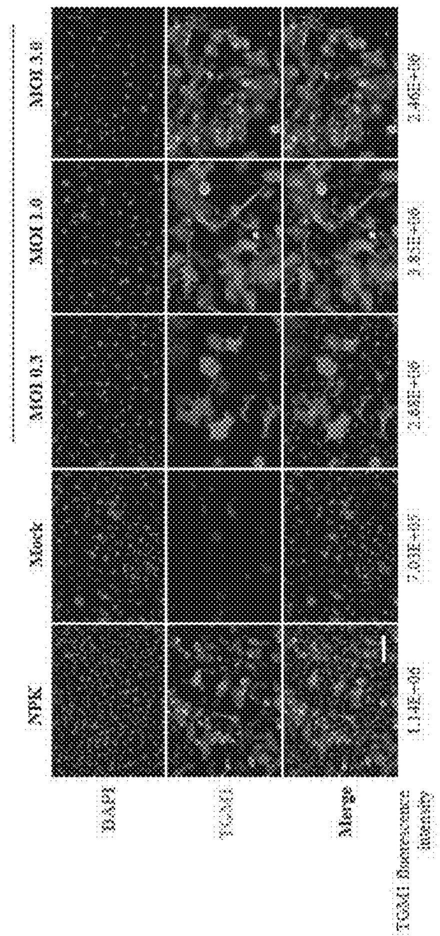

HSV-TGM1
NPK
Mock
MOI 0.3
MOI 1.0
MOI 3.0
DAPI
TGM1
Merge
TGM1 fluorescence intensity

FIG. 2F

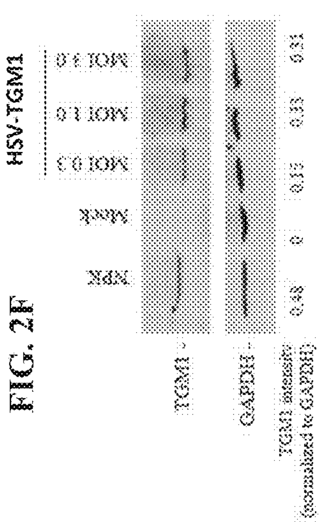

HSV-TGM1
NPK
Mock
MOI 0.3
MOI 1.0
MOI 3.0
TGM1
GAPDH
TGM1 intensity (normalized to GAPDH)
0.48
0
0.15
0.33
0.31

FIG. 3A

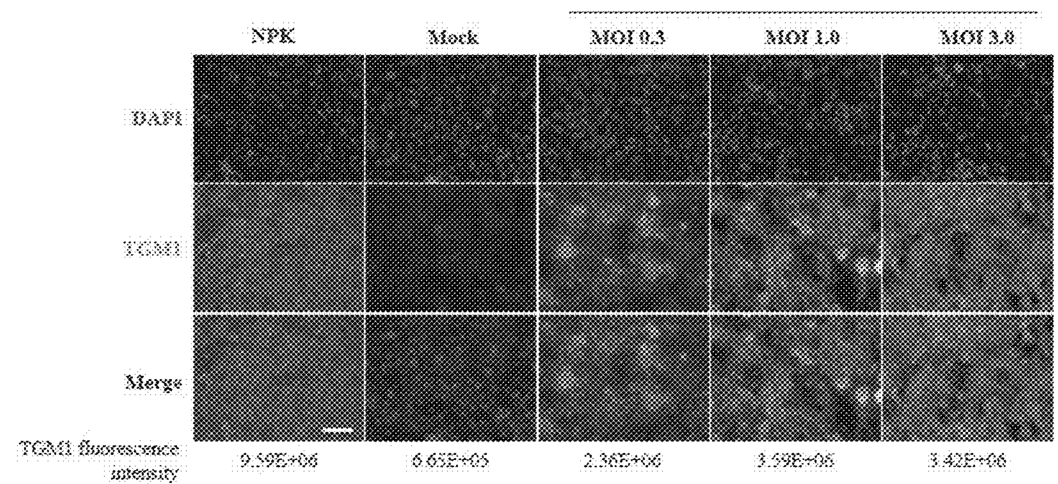

HSV-TGM1
NPK
Mock
MOI 0.3
MOI 1.0
MOI 3.0
DAPI
TGM1
Merge
TGM1 fluorescence intensity
9.59E+06
6.65E+05
2.36E+06
3.59E+06
3.42E+06

FIG. 3B

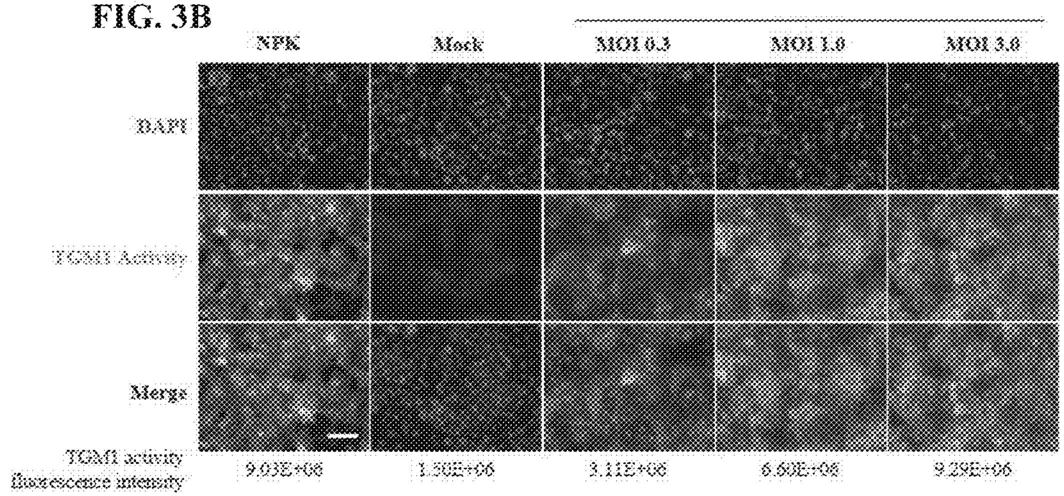

HSV-TGM1
NPK
Mock
MOI 0.3
MOI 1.0
MOI 3.0
DAPI
TGM1 Activity
Merge
TGM1 activity fluorescence intensity
9.03E+06
1.50E+06
3.11E+06
6.60E+06
9.29E+06

FIG. 4

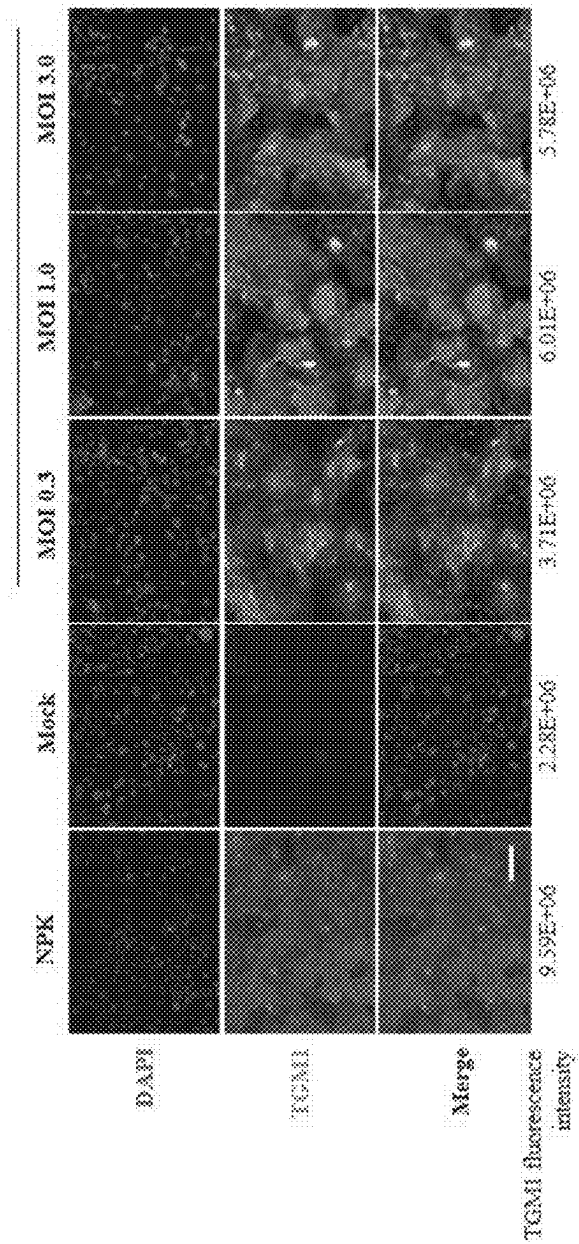

HSV-TGM1
NPK
Mock
MOI 0.3
MOI 1.0
MOI 3.0
DAPI
TGM1
Merge
TGM1 fluorescence intensity
9.59E+06
2.28E+06
3.71E+06
6.01E+06
5.78E+06

FIG. 5A

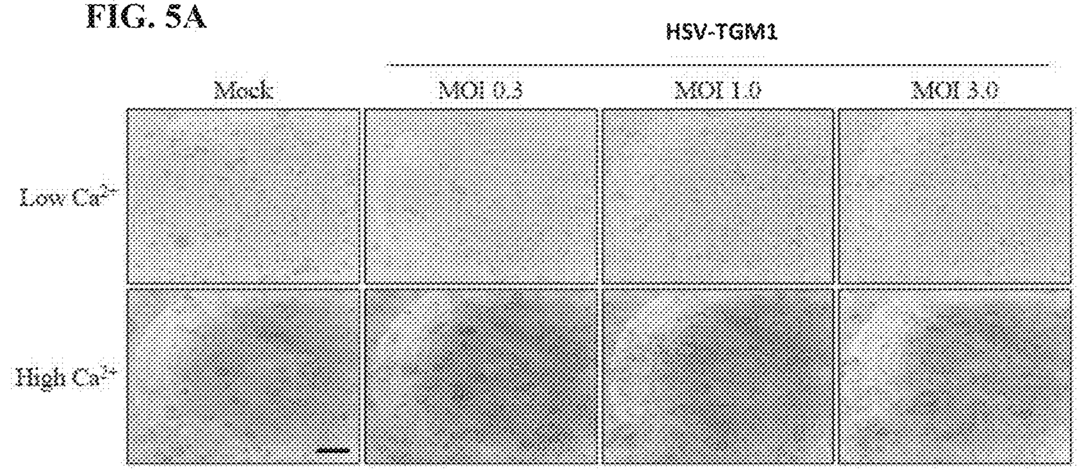

HSV-TGM1
Mock
MOI 0.3
MOI 1.0
MOI 3.0
Low Ca2+
High Ca2+

FIG. 5B

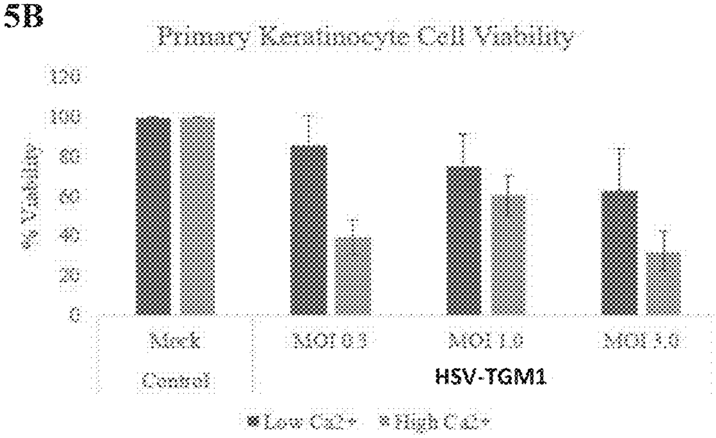

Primary Keratinocyte Cell Viability
% Viability
120
100
80
60
40
20
0
Mock
Control
MOI 0.3
MOI 1.0
MOI 3.0
HSV-TGM1
Low Ca2+
High Ca2+

FIG. 6A

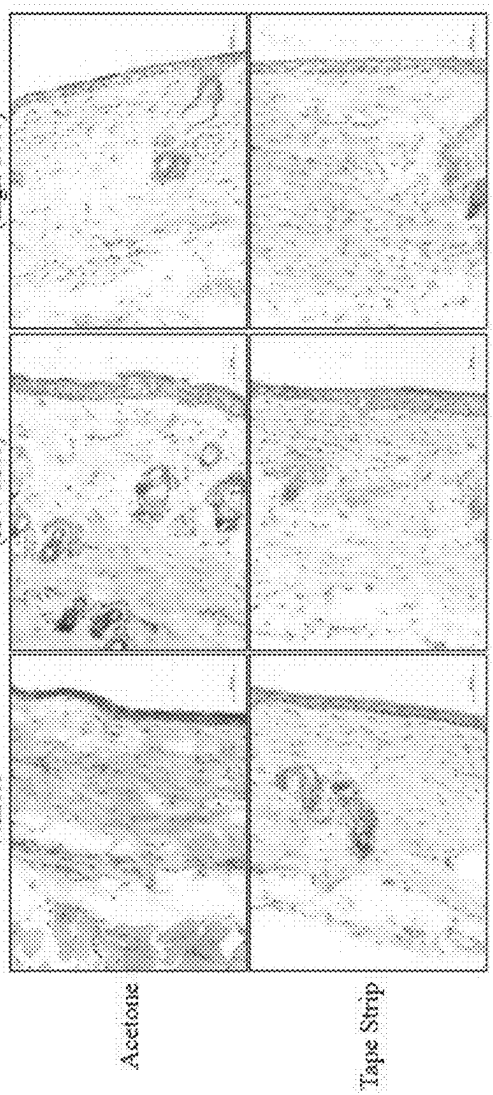


FIG. 6B

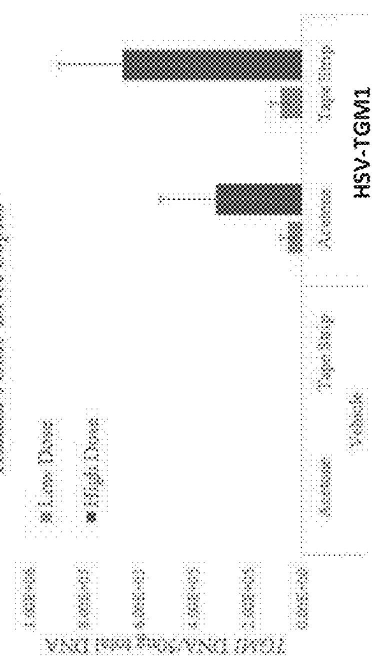


FIG. 6C

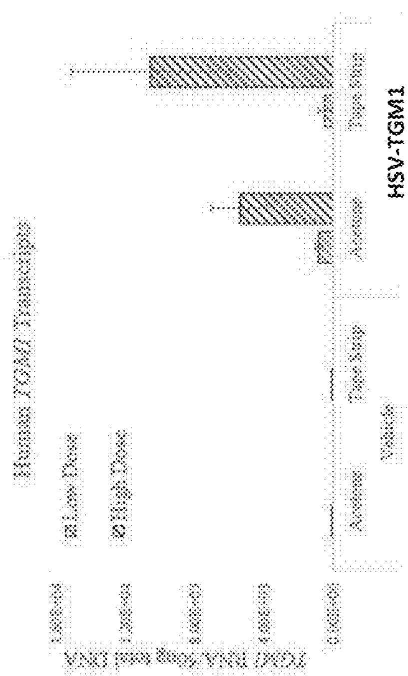


FIG. 6D

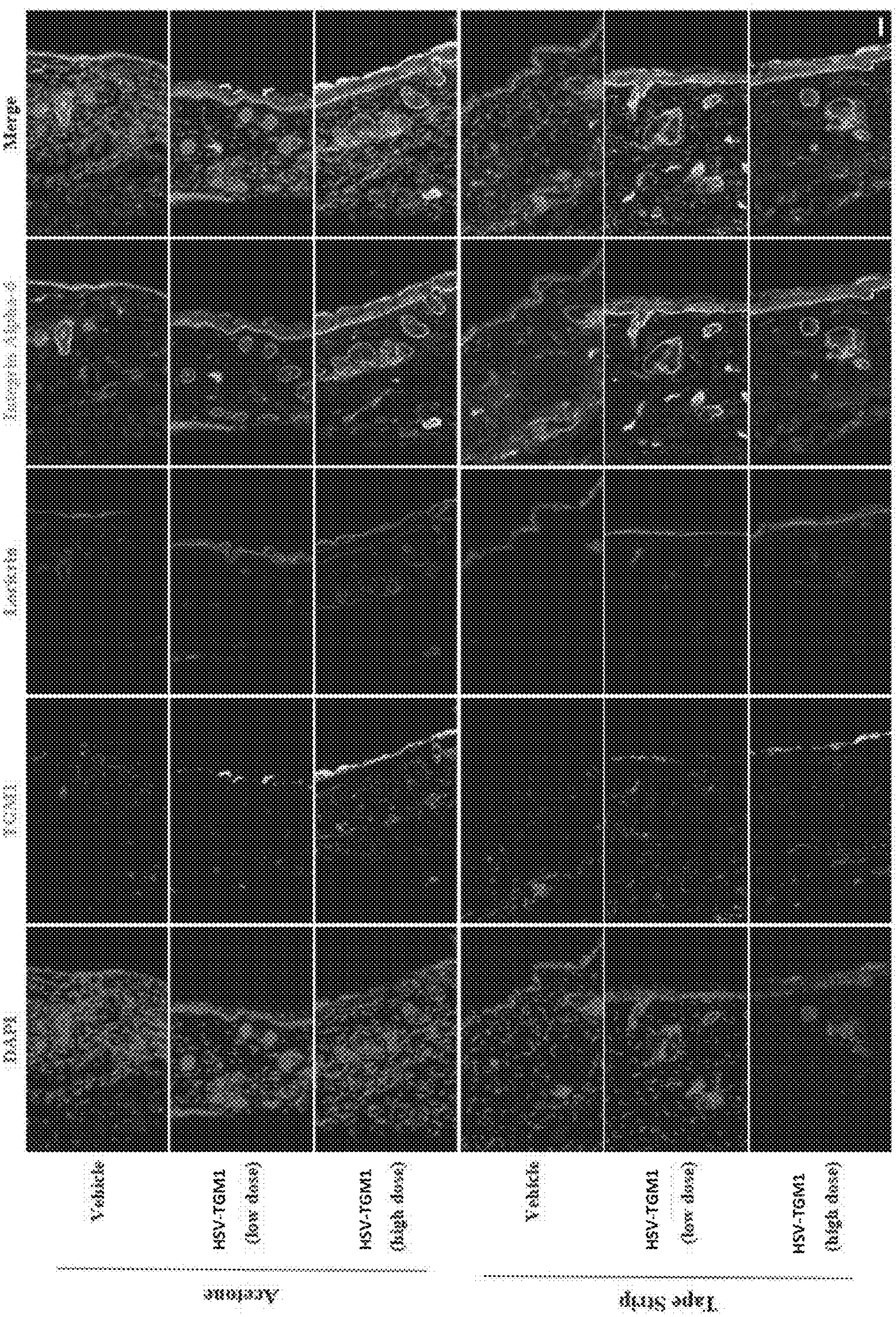

DAPI
TGM1
Loricrin
Integrin Alpha-6
Merge
Vehicle
HSV-TGM1 (low dose)
HSV-TGM1 (high dose)
Vehicle
HSV-TGM1 (low dose)
HSV-TGM1 (high dose)
Acetone
Tape Strip

FIG. 7

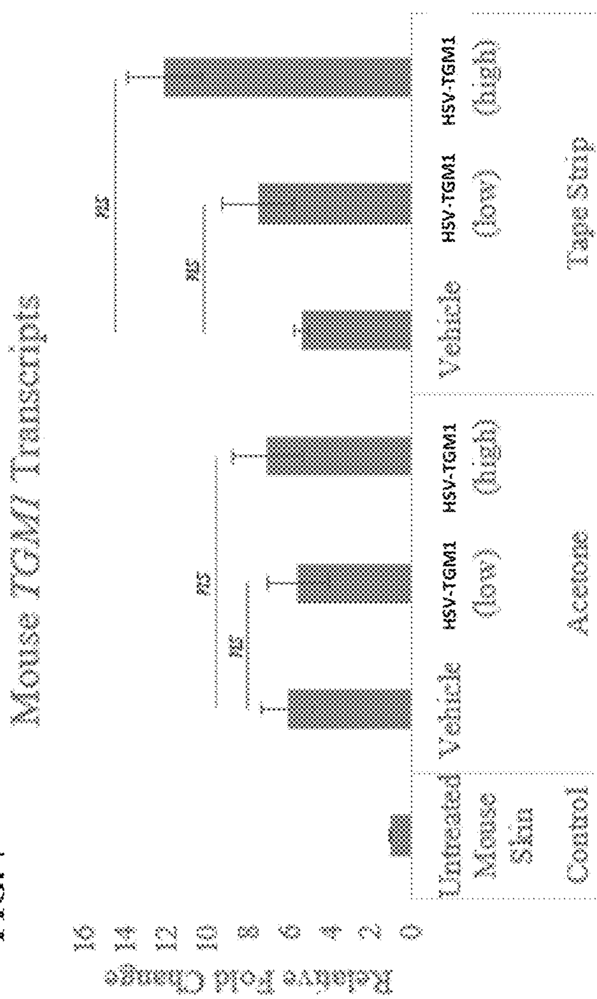

Mouse TGM1 Transcripts
Relative Fold Change
16
14
12
10
8
6
4
2
0
ns
ns
ns
ns
Untreated Mouse Skin
Control
Vehicle
HSV-TGM1 (low)
HSV-TGM1 (high)
Acetone
Vehicle
HSV-TGM1 (low)
HSV-TGM1 (high)
Tape Strip

FIG. 8A

FIG. 8B

FIG. 8C

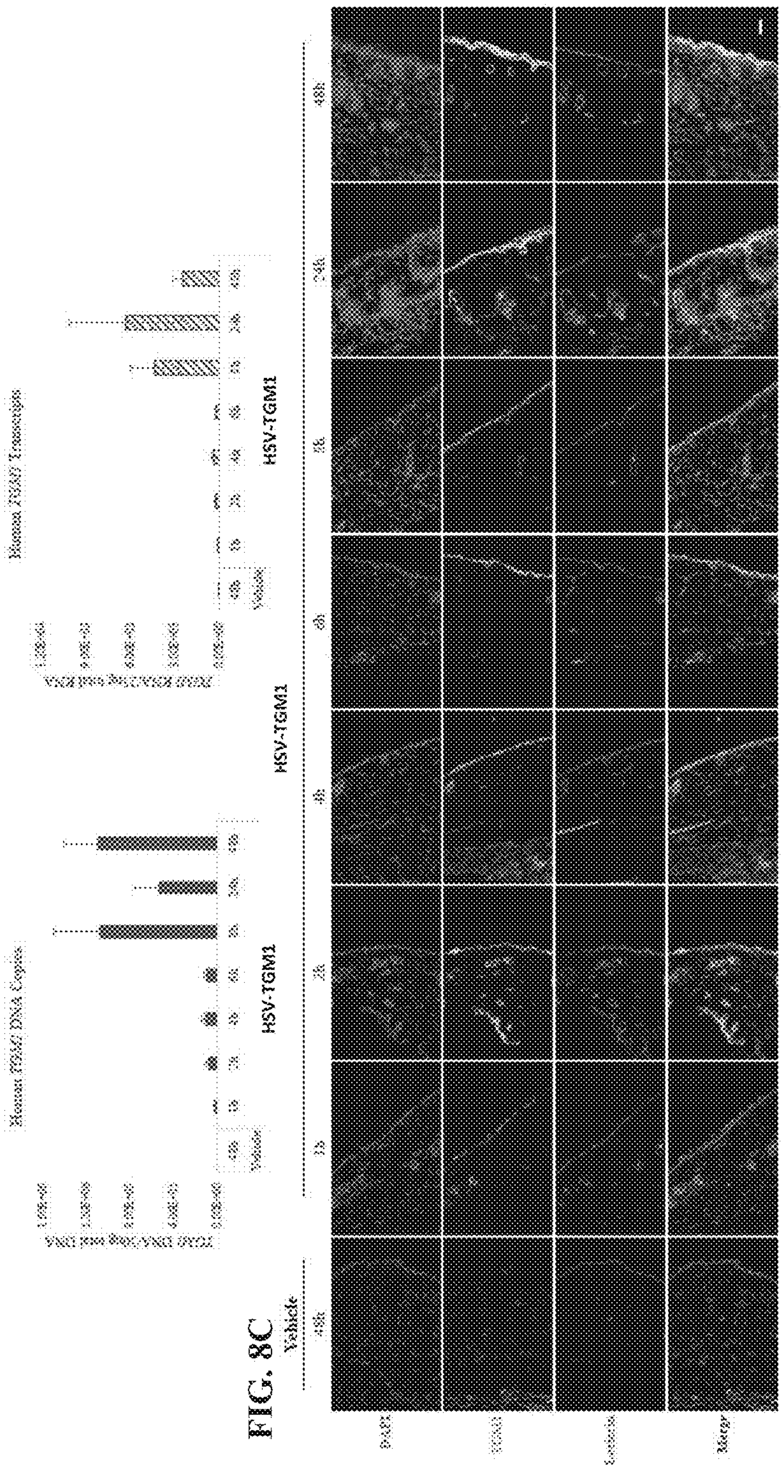


FIG. 9A

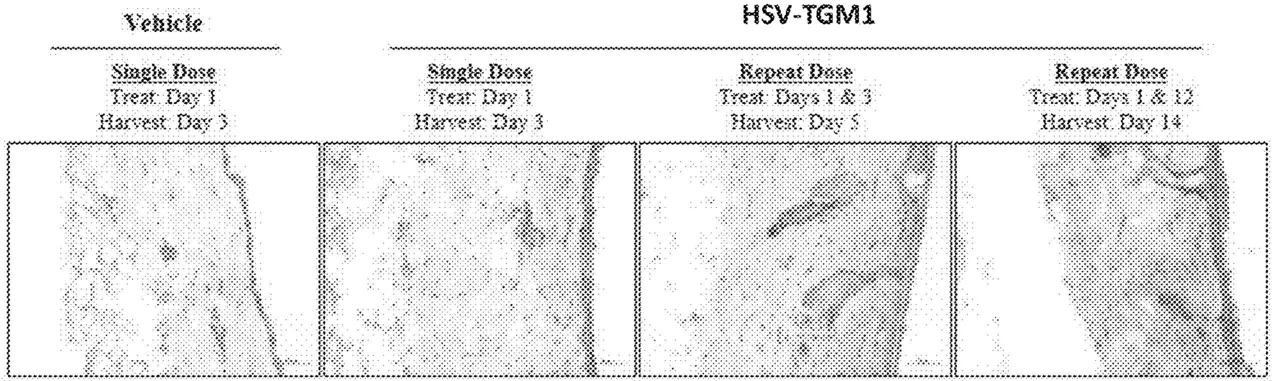


FIG. 9B

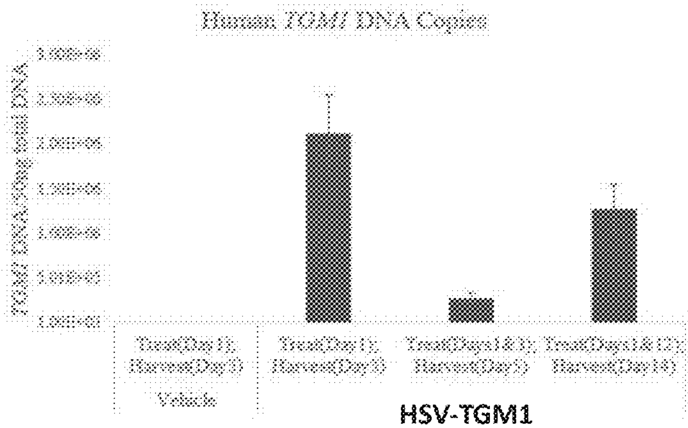


FIG. 9C

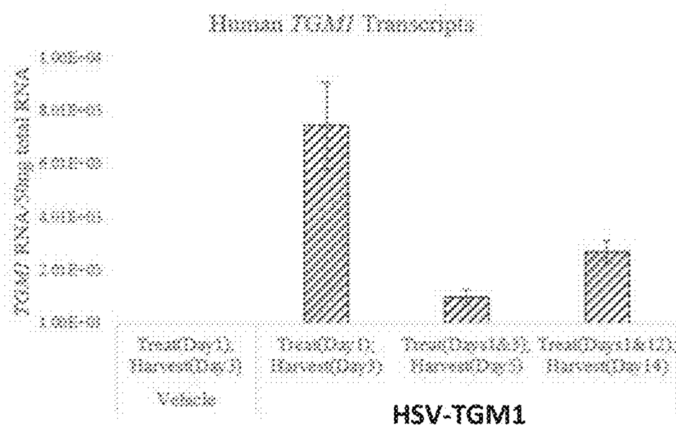
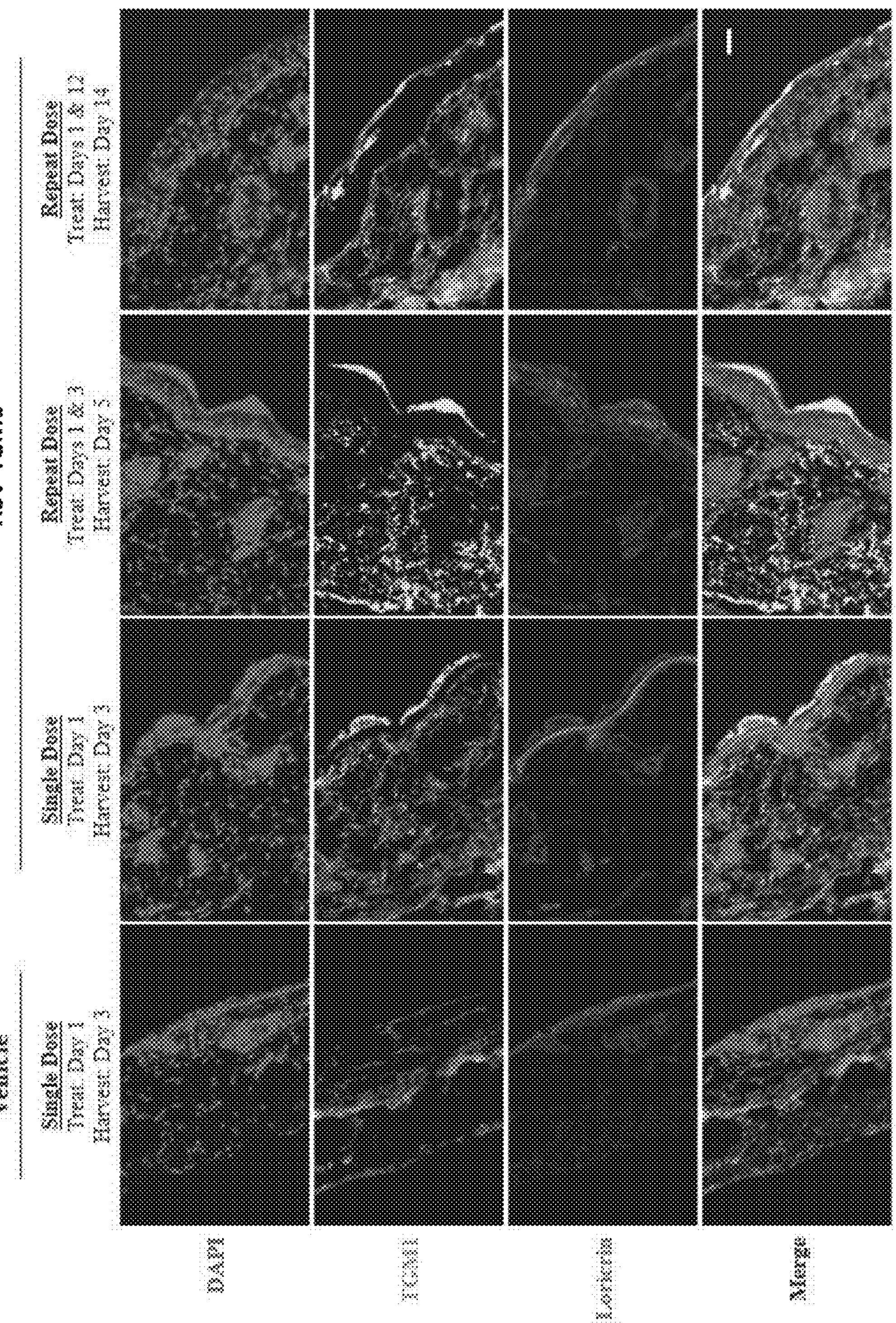


FIG. 9D


Vehicle
Single Dose
Treat: Day 1
Harvest: Day 3
HSV-TGM1
Single Dose
Treat: Day 1
Harvest: Day 3
Repeat Dose
Treat: Days 1 & 3
Harvest: Day 5
Repeat Dose
Treat: Days 1 & 12
Harvest: Day 14
DAPI
TGM1
Loricrin
Merge

COMPOSITIONS AND METHODS FOR THE TREATMENT OF CONGENITAL ICHTHYOSES

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/639,878, filed Sep. 2, 2020, which is the U.S. National Stage of PCT Application No. PCT/US2020/049070, filed Sep. 2, 2020, which claims the priority benefit of U.S. Provisional Application Ser. No. 62/895,045, filed Sep. 3, 2019, each of which is incorporated herein by reference in its entirety.

REFERENCE TO A SEQUENCE LISTING

[0002] The Sequence Listing associated with this application is filed in electronic format via Patent Center and is hereby incorporated by reference into the specification in its entirety. The name of the file containing the Sequence Listing is 2501905.xml. The size of the file is 371,065 bytes, and the file was created on Apr. 29, 2025.

FIELD OF THE INVENTION

[0003] The present disclosure relates, in part, to recombinant nucleic acids, viruses, medicaments, pharmaceutical compositions, and methods of their use for treating subjects harboring loss-of-function mutations in, and/or pathogenic variants of, one or more ichthyosis-associated genes and/or for providing prophylactic, palliative, or therapeutic relief of one or more signs or symptoms of congenital ichthyosis, e.g., X-linked ichthyosis (XLI), epidermolytic ichthyoses (EI), ichthyosis vulgaris (IV), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), harlequin ichthyosis (HI), etc.

BACKGROUND

[0004] Congenital ichthyoses are a heterogenous group of disorders manifesting at birth or infancy with visible scaling and/or thickening of the skin, which may be accompanied by variable degrees of redness (erythema), skin fragility, and/or blistering, as well as abnormalities of the hair, nails, and/or mucus membranes. Scaling and/or thickening of the outermost layer of the skin (hyperkeratosis) may be generalized or localized, may involve other organ systems (syndromic ichthyosis), or may be limited to skin and skin appendages (non-syndromic ichthyosis). The management of congenital ichthyoses is a life-long endeavor, which remains largely symptomatic. At present, disease treatment or management is generally supportive, and commonly focuses on skin lubrication and/or reducing scaling. Thus, there exists a clear need for novel treatment options for all forms of congenital ichthyoses. The present disclosure addresses this and other needs.

[0005] All references cited herein, including patent applications, patent publications, non-patent literature, and UniProtKB/Swiss-Prot Accession numbers are herein incorporated by reference in their entirety, as if each individual reference were specifically and individually indicated to be incorporated by reference.

BRIEF SUMMARY

[0006] In some embodiments, provided herein are recombinant nucleic acids (e.g., recombinant herpes viral genomes) comprising the coding sequence of one or more ichthyosis-associated genes (e.g., a polynucleotide encoding a wild-type and/or functional ichthyosis-associated polypeptide) for use in viruses (e.g., herpes viruses), compositions, pharmaceutical formulations, medicaments, and/or methods useful for supplementing or treating ichthyosis-associated gene deficiencies in a subject in need thereof (e.g., a subject naturally harboring pathogenic variants of such gene(s)) and/or for providing prophylactic, palliative, or therapeutic relief of a wound, disorder, or disease of the skin in a subject having, or at risk of developing, one or more signs or symptoms of congenital ichthyosis (e.g., X-linked ichthyosis, lamellar ichthyosis, harlequin ichthyosis, etc.).

[0007] Certain aspects of the present disclosure relate to a recombinant herpes virus genome comprising one or more polynucleotides encoding an ichthyosis-associated polypeptide. In some embodiments, the recombinant herpes virus genome is replication competent. In some embodiments, the recombinant herpes virus genome is replication defective. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes virus genome is selected from a recombinant herpes simplex virus genome, a recombinant varicella zoster virus genome, a recombinant human cytomegalovirus genome, a recombinant herpesvirus 6A genome, a recombinant herpesvirus 6B genome, a recombinant herpesvirus 7 genome, a recombinant Kaposi's sarcoma-associated herpesvirus genome, and any derivatives thereof. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes virus genome is a recombinant herpes simplex virus genome. In some embodiments, the recombinant herpes simplex virus genome is a recombinant type 1 herpes simplex virus (HSV-1) genome, a recombinant type 2 herpes simplex virus (HSV-2) genome, or any derivatives thereof. In some embodiments, the recombinant herpes simplex virus genome is a recombinant type 1 herpes simplex virus (HSV-1) genome.

[0008] In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome has been engineered to reduce or eliminate expression of one or more toxic herpes simplex virus genes. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation. In some embodiments, the inactivating mutation is in a herpes simplex virus gene. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the herpes simplex virus gene. In some embodiments, the herpes simplex virus gene is selected from Infected Cell Protein (ICP) 0, ICP4, ICP22, ICP27, ICP47, thymidine kinase (tk), Long Unique Region (UL) 41, and UL55. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in one or both copies of the ICP4 gene. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP22 gene. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL41 gene. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in one or both copies of the ICP0 gene. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP27 gene. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL55 gene. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the Joint region. In some embodiments, the recombinant herpes simplex virus genome comprises a deletion of the Joint region. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within one or both copies of the ICP4 viral gene loci. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the ICP22 viral gene locus. In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the UL41 viral gene locus.

[0009] In some embodiments that may be combined with any of the preceding embodiments, the ichthyosis-associated polypeptide is not a transglutaminase (TGM) polypeptide. In some embodiments, the ichthyosis-associated polypeptide is not a transglutaminase 1 (TGM1) polypeptide or a transglutaminase 5 (TGM5) polypeptide. In some embodiments that may be combined with any of the preceding embodiments, the ichthyosis-associated polypeptide is selected from an ATP-binding cassette sub-family A member 12 polypeptide (ABCA12), a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide (ABHD5), an Aldehyde dehydrogenase family 3 member A2 polypeptide (ALDH3A2), an Arachidonate 12-lipoxygenase 12R-type polypeptide (ALOX12B), a Hydroperoxide isomerase ALOXE3 polypeptide (ALOXE3), an AP-1 complex subunit sigma-1A polypeptide (AP1S1), an Arylsulfatase E polypeptide (ARSE), a Caspase-14 polypeptide (CASP14), a Corneodesmosin polypeptide (CDSN), a Ceramide synthase 3 polypeptide (CERS3), a Carbohydrate sulfotransferase 8 polypeptide (CHST8), a Claudin-1 polypeptide (CLDN1), a Cystatin-A polypeptide (CSTA), a Cytochrome P450 4F22 polypeptide (CYP4F22), a 3-beta-hydroxysteroid-Delta(8), Delta(7)-isomerase polypeptide (EBP), an Elongation of very long chain fatty acids protein 4 polypeptide (ELOVL4), a Filaggrin polypeptide (FLG), a Filaggrin 2 polypeptide (FLG2), a Gap junction beta-2 polypeptide (GJB2), a Gap junction beta-3 polypeptide (GJB3), a Gap junction beta-4 polypeptide (GJB4), a Gap junction beta-6 polypeptide (GJB6), a 3-ketodihydrosphingosine reductase polypeptide (KDSR), a Keratin, type II cytoskeletal 1 polypeptide (KRT1), a Keratin, type II cytoskeletal 2 epidermal polypeptide (KRT2), a Keratin, type I cytoskeletal 9 polypeptide (KRT9), a Keratin, type I cytoskeletal 10 polypeptide (KRT10), a Lipase member N polypeptide (LIPN), a Loricrin polypeptide (LOR), a Membrane-bound transcription factor site-2 protease polypeptide (MBTPS2), a Magnesium transporter NIPA4 polypeptide (NIPAL4), a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide (NSDHL), a Peroxisomal targeting signal 2 receptor polypeptide (PEX7), a D-3-phosphoglycerate dehydrogenase polypeptide (PHGDH), a Phytanoyl-CoA dioxygenase, peroxisomal polypeptide (PHYH), Patatin-like phospholipase domain-containing protein 1 polypeptide (PNPLA1), a Proteasome maturation protein polypeptide (POMP), a Phosphoserine aminotransferase polypeptide (PSAT1), a Short-chain dehydrogenase/reductase family 9C member 7 polypeptide (SDR9C7), a Serpin B8 polypeptide (SERPINB8), a Long-chain fatty acid transport protein 4 polypeptide (SLC27A4), a Synaptosomal-associated protein 29 polypeptide (SNAP29), a Suppressor of tumorigenicity 14 protein polypeptide (ST14), a Steryl-sulfatase polypeptide (STS), a Sulfotransferase 2B1 polypeptide (SULT2B1), a Vacuolar protein sorting-associated protein 33B polypeptide (VPS33B), and a CAAX prenyl protease 1 homolog polypeptide (ZMPSTE24). In some embodiments that may be combined with any of the preceding embodiments, the ichthyosis-associated polypeptide is a human ichthyosis-associated polypeptide. In some embodiments that may be combined with any of the preceding embodiments, the ichthyosis-associated polypeptide comprises a sequence having at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to an amino acid sequence selected from SEQ ID NOS: 102-152 or 155. In some embodiments that may be combined with any of the preceding embodiments, the ichthyosis-associated polypeptide is selected from ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, ARSE, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, ELOVL4, KDSR, LIPN, MBTPS2, NIPAL4, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4, SNAP29, ST14, STS, VPS33B, and ZMPSTE24. In some embodiments, the ichthyosis-associated polypeptide is selected from ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, ELOVL4, KDSR, LIPN, NIPAL4, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4, SNAP29, ST14, VPS33B, and ZMPSTE24. In some embodiments, the ichthyosis-associated polypeptide is selected from ARSE, MBTPS2, and STS. In some embodiments, the ichthyosis-associated polypeptide is STS.

[0010] In some embodiments that may be combined with any of the preceding embodiments, the recombinant herpes virus genome has reduced cytotoxicity when introduced into a target cell as compared to a corresponding wild-type herpes virus genome. In some embodiments, the target cell is a cell of the epidermis and/or dermis. In some embodiments, the target cell is a human cell.

[0011] Other aspects of the present disclosure relate to a herpes virus comprising any of the recombinant herpes virus genomes described herein. In some embodiments, the herpes virus is replication competent. In some embodiments, the herpes virus is replication defective. In some embodiments that may be combined with any of the preceding embodiments, the herpes virus has reduced cytotoxicity as compared to a corresponding wild-type herpes virus. In some embodiments that may be combined with any of the preceding embodiments, the herpes virus is selected from a herpes simplex virus, a varicella zoster virus, a human cytomegalovirus, a herpesvirus 6A, a herpesvirus 6B, a herpesvirus 7, and a Kaposi's sarcoma-associated herpesvirus. In some embodiments that may be combined with any of the preceding embodiments, the herpes virus is a herpes simplex virus. In some embodiments, the herpes simplex virus is a type 1 herpes simplex virus (HSV-1), a type 2 herpes simplex virus (HSV-2), or any derivatives thereof. In some embodiments, the herpes simplex virus is a type 1 herpes simplex virus (HSV-1).

[0012] Other aspects of the present disclosure related to a pharmaceutical composition comprising any of the recombinant herpes virus genomes and/or any of the herpes viruses described herein and a pharmaceutically acceptable excipient. In some embodiments, the pharmaceutical composition is suitable for topical, transdermal, subcutaneous, intradermal, oral, intranasal, intratracheal, sublingual, buccal, rectal, vaginal, inhaled, intravenous, intraarterial, intramuscular, intracardiac, intraosseous, intraperitoneal, transmucosal, intravitreal, subretinal, intraarticular, peri-articular, local, or epicutaneous administration. In some embodiments, the pharmaceutical composition is suitable for topical, transdermal, subcutaneous, intradermal, or transmucosal administration. In some embodiments, the pharmaceutical composition is suitable for topical, transdermal, or intradermal administration. In some embodiments, the pharmaceutical composition is suitable for topical administration.

[0013] Other aspects of the present disclosure relate to the use of any of the recombinant herpes virus genomes and/or herpes viruses described herein as a medicament.

[0014] Other aspects of the present disclosure relate to the use of any of the recombinant herpes virus genomes and/or herpes viruses described herein in a therapy.

[0015] Other aspects of the present disclosure relate to the use of any of the herpes viruses and/or pharmaceutical compositions described herein in the manufacture of a medicament for treating one or more forms of congenital ichthyosis.

[0016] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of congenital ichthyosis in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the congenital ichthyosis is selected from harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and restrictive dermopathy.

[0017] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of harlequin ichthyosis (HI) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding an ABCA12 polypeptide.

[0018] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Chanarin-Dorfman syndrome (CDS) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding an ABHD5 polypeptide.

[0019] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Sjogren-Larsson syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding an ALDH3A2 polypeptide.

[0020] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of autosomal recessive congenital ichthyosis (ARCI) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from ALOX12B, ALOXE3, CASP14, CERS3, CYP4F22, LIPN, NIPAL4, PNPLA1, SDR9C7, SLC27A4, ST14, and SULT2B1.

[0021] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding an AP1S1 polypeptide.

[0022] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of chondrodysplasia *punctata* 1 (CDPX1) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding an ARSE polypeptide.

[0023] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of chondrodysplasia *punctata* 2 (CDPX2) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding an EBP polypeptide.

[0024] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of peeling skin syndrome (PSS) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from CDSN, CHST8, CSTA, FLG2, and SERPINB8.

[0025] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a CLDN1 polypeptide.

[0026] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis vulgaris in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a FLG polypeptide.

[0027] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of keratitis-ichthyosis-deafness (KID) syndrome, Clouston syndrome, and/or palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from GJB2 and GJB6.

[0028] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of erythrokeratodermia *variabilis* (EKV) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from GJB3 and GJB4.

[0029] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of progressive symmetric erythrokeratodermia in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a KDSR polypeptide.

[0030] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of epidermolytic ichthyosis (EI) and/or superficial epidermolytic ichthyosis (SEI) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from KRT1, KRT2, and KRT10.

[0031] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of epidermolytic palmoplantar keratoderma (EPPK) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a KRT9 polypeptide.

[0032] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of loricrin keratoderma in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a LOR polypeptide.

[0033] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a MBTPS2 polypeptide.

[0034] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a NSDHL polypeptide.

[0035] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Refsum disease in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from PEX7 and PHYH.

[0036] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Neu-Laxova in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from PHGDH and PSAT1.

[0037] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a POMP polypeptide.

[0038] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis prematurity syndrome (IPS) in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a SLC27A4 polypeptide.

[0039] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a SNAP29 polypeptide.

[0040] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of X-linked ichthyosis in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a STS polypeptide.

[0041] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a VPS33B polypeptide.

[0042] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of restrictive dermopathy in a subject in need thereof comprising administering to the subject an effective amount of any of the herpes viruses and/or pharmaceutical compositions described herein. In some embodiments, the recombinant herpes virus genome comprises one or more polynucleotides encoding a ZMPSTE24 polypeptide.

[0043] In some embodiments that may be combined with any of the preceding embodiments, the subject is a human. In some embodiments that may be combined with any of the preceding embodiments, the subject's genome comprises a pathogenic variant of an ichthyosis-associated gene. In some embodiments that may be combined with any of the preceding embodiments, the subject's genome comprises a loss-of-function mutation in an ichthyosis-associated gene.

[0044] In some embodiments that may be combined with any of the preceding embodiments, the herpes virus or pharmaceutical composition is administered topically, transdermally, subcutaneously, epicutaneously, intradermally, orally, sublingually, buccally, rectally, vaginally, intravenously, intraarterially, intramuscularly, intraosseously, intracardially, intraperitoneally, transmucosally, intravitreally, subretinally, intraarticularly, peri-articularly, locally, or via inhalation to the subject. In some embodiments, the herpes virus or pharmaceutical composition is administered topically, transdermally, subcutaneously, intradermally, or transmucosally to the subject. In some embodiments, the herpes virus or pharmaceutical composition is administered topically, transdermally, or intradermally to the subject. In some embodiments, the herpes virus or pharmaceutical composition is administered topically to the subject. In some embodiments that may be combined with any of the preceding embodiments, the skin of the subject is abraded or made more permeable prior to administration.

BRIEF DESCRIPTION OF THE DRAWINGS

[0045] FIGS. **1**A-**1**I show schematics of wild-type and modified herpes simplex virus genomes. FIG. **1**A shows a wild-type herpes simplex virus genome. FIG. **1**B shows a modified herpes simplex virus genome comprising deletions of the coding sequence of ICP4 (both copies), with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at each of the ICP4 loci. FIG. **1**C shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies) and UL41, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at each of the ICP4 loci. FIG. **1**D shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies) and UL41, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at the UL41 locus. FIG. **1**E shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies) and ICP22, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at each of the ICP4 loci. FIG. **1**F shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies) and ICP22, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at the ICP22 locus. FIG. **1**G shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies), UL41, and ICP22, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at each of the ICP4 loci. FIG. **1**H shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies), UL41, and

ICP22, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at the UL41 locus. FIG. 1I shows a modified herpes simplex virus genome comprising deletions of the coding sequences of ICP4 (both copies), UL41, and ICP22, with an expression cassette containing a polynucleotide encoding an ichthyosis-associated polypeptide integrated at the ICP22 locus.

[0046] FIGS. 2A-2G show in vitro assessments of KB105 in immortalized and primary TGM1-deficient human ARCI keratinocytes grown in low calcium cell culture medium. FIG. 2A shows dose-dependent detection of human TGM1 DNA copies at increasing multiplicities of infection (MOIs) of KB105 in immortalized keratinocytes, as assessed by qPCR. Data is presented as the average of two replicates ±SEM. FIG. 2B shows dose-dependent expression of human TGM1 transcripts at increasing MOIs of KB105 in immortalized keratinocytes, as assessed by qRT-PCR. Data is presented as the average of two replicates ±SEM. FIG. 2C shows KB105-mediated TGM1 protein expression in infected immortalized keratinocytes by western blot. FIG. 2D shows representative immunofluorescence images of human TGM1 protein expression upon KB105 infection of immortalized keratinocytes. FIG. 2E shows representative immunofluorescence images of KB105-dependent TGM1 enzymatic activity in immortalized keratinocytes. FIG. 2F shows KB105-mediated TGM1 protein expression in infected primary cells by western blot analysis. FIG. 2G shows representative immunofluorescence images of human TGM1 protein expression upon KB105 infection of primary cells. For these experiments, uninfected (mock) and HSV-mCherry infected (mCherry) cells were used as negative controls; normal primary keratinocytes (NPK) were used as a positive control. DAPI staining was used to visualize nuclei. GAPDH was used as a loading control. For western blot and immunofluorescence analyses, quantification of protein levels and fluorescence intensities are provided for each condition. Bar: 130 μm.

[0047] FIGS. 3A-3B show in vitro assessments of KB105 in immortalized TGM1-deficient human ARCI keratinocytes grown in high calcium cell culture medium. FIG. 3A shows representative immunofluorescence images of human TGM1 protein expression upon KB105 infection of immortalized keratinocytes. FIG. 3B shows representative immunofluorescence images of KB105-dependent TGM1 enzymatic activity in immortalized keratinocytes. Uninfected (mock) cells were used as negative controls; normal primary keratinocytes (NPK) were used as a positive control. DAPI staining was used to visualize nuclei. Quantification of fluorescence intensities are provided for each condition. Bar: 130 μm.

[0048] FIG. 4 shows in vitro assessment of KB105 in primary TGM1-deficient human ARCI keratinocytes grown in high calcium cell culture medium. Representative immunofluorescence images of human TGM1 protein expression upon KB105 infection of primary keratinocytes. Uninfected (mock) cells were used as negative controls; normal primary keratinocytes (NPK) were used as a positive control. DAPI staining was used to visualize nuclei. Quantification of fluorescence intensities are provided for each condition. Bar: 130 μm.

[0049] FIGS. 5A-5B show viability assessments after KB105 infection of primary TGM1-deficient human ARCI keratinocytes grown in low and high calcium cell culture medium. FIG. 5A shows representative brightfield images of primary keratinocytes, grown in low or high calcium cell culture medium, 48 hours after infection with KB105 at the indicated MOIs. Uninfected (mock) cells were used as a negative control. FIG. 5B shows viability assessment of KB105-infected primary LI patient keratinocytes at 48h post-infection as determined by MTS Assay. For each condition, data is presented as the average of three separate experiments (with triplicate wells) ±SEM. Bar: 370 μm.

[0050] FIGS. 6A-6D show in vivo evaluation of KB105 via multiple routes of topical delivery to BALB/c mice. FIG. 6A shows representative hematoxylin and eosin (H&E)-stained samples harvested from tape stripped or acetone permeabilized BALB/c mouse skin treated topically with either KB105 (low or high dose) or negative control (vehicle). FIG. 6B shows dose-dependent detection of human TGM1 DNA copies in mouse skin biopsies harvested 48 hours after permeabilization by tape stripping or acetone treatment and application of KB105 (low or high dose) or negative control (vehicle), as assessed by qPCR. FIG. 6C shows dose-dependent expression of human TGM1 transcripts in mouse skin biopsies harvested 48 hours after permeabilization by tape stripping or acetone treatment and application of KB105 (low or high dose) or negative control (vehicle), as assessed by qRT-PCR. For each vehicle control condition in the qPCR and qRT-PCR analysis, data is presented as the average of two tissue samples (two replicates tested/tissue sample) ±SEM; for each KB105 condition, data is presented as the average of four tissue samples (two replicates tested/tissue sample) ±SEM. FIG. 6D shows representative immunofluorescence images of human TGM1, mouse loricrin, and mouse integrin alpha-6 protein localization in mouse skin biopsies harvested 48 hours after skin barrier disruption by acetone treatment or tape stripping and application of KB105 (low or high dose) or negative control (vehicle). DAPI staining was used to visualize nuclei. Bar: 50 μm.

[0051] FIG. 7 shows in vivo evaluation of mouse TGM1 transcription upon KB105 infection via multiple routes of topical delivery to BALB/c mice. Fold change of mouse TGM1 RNA copies in skin biopsies harvested 48 hours after permeabilization by tape stripping or acetone treatment and application of KB105 (low or high dose) or vehicle alone relative to untreated control skin, as assessed by qPCR. For each vehicle condition, data is presented as the average of two tissue samples (two replicates tested/tissue sample) ±SEM; for each KB105 condition, data is presented as the average of four tissue samples (two replicates tested/tissue sample) ±SEM. ns: not significant ($p>0.05$), as determined by two-tailed Student's T-test. Bar: 50 μm.

[0052] FIGS. 8A-8C show in vivo short-term pharmacokinetics of KB105 upon topical delivery to BALB/c mice. FIG. 8A shows detection of human TGM1 DNA copies in skin biopsies harvested at the indicated timepoints from BALB/c mice treated topically with either KB105 or negative control (vehicle). FIG. 8B shows detection of human TGM1 transcripts in skin biopsies harvested at the indicated timepoints from BALB/c mice treated topically with either KB105 or negative control (vehicle). For each vehicle control condition in the qPCR and qRT-PCR analysis, data is presented as the average of two tissue samples (two replicates tested/tissue sample) ±SEM; for each KB105 condition, data is presented as the average of four tissue samples (two replicates tested/tissue sample) ±SEM. FIG.

8C shows representative immunofluorescence images of human TGM1 and mouse loricrin protein localization in mouse skin biopsies harvested at the indicated timepoints from BALB/c mice treated topically with either KB105 or negative control (vehicle). DAPI staining was used to visualize nuclei. Bar: 50 μm.

[0053] FIGS. **9**A-**9**D show in vivo pharmacokinetics of KB105 upon single and repeat topical delivery to BALB/c mice. FIG. **9**A shows H&E-stained skin biopsies taken from BALB/c mice treated and harvested at the indicated timepoints after a single or repeat topical dose of either KB105 or negative control (vehicle). FIG. **9**B shows detection of human TGM1 DNA copies in skin biopsies taken from BALB/c mice treated and harvested at the indicated timepoints after a single or repeat topical dose of either KB105 or negative control (vehicle). FIG. **9**C shows detection of human TGM1 transcripts in skin biopsies taken from BALB/c mice treated and harvested at the indicated timepoints after a single or repeat topical dose of either KB105 or negative control (vehicle). For each vehicle control condition in the qPCR and qRT-PCR analysis, data is presented as the average of two tissue samples (two replicates tested/tissue sample) ±SEM; for each KB105 condition, data is presented as the average of four or six tissue samples (two replicates tested/tissue sample) ±SEM. FIG. **9**D shows representative immunofluorescence images of human TGM1 and mouse loricrin protein colocalization in skin biopsies taken from BALB/c mice treated and harvested at the indicated timepoints after a single or repeat topical dose of either KB105 or negative control (vehicle). DAPI staining was used to visualize nuclei. Bar: 50 μm.

DETAILED DESCRIPTION

[0054] In some embodiments, the present disclosure relates to recombinant nucleic acids (e.g., recombinant herpes viral genomes) comprising one or more polynucleotides encoding one or more ichthyosis-associated polypeptides (e.g., encoding wild-type and/or functional ichthyosis-associated polypeptides), and/or use of these recombinant nucleic acids in viruses (e.g., herpes viruses), compositions, formulations, medicaments, and/or methods in order to supplement or treat endogenous ichthyosis-associated gene deficiencies (e.g., in a subject whose genome naturally harbors a pathogenic variant of the ichthyosis-associated gene(s)). Without wishing to be bound by theory, it is believed that the recombinant nucleic acids, viruses, compositions, formulations, medicaments, and methods described herein will help to treat the existing skin abnormalities in individuals suffering from congenital ichthyosis (such as individuals suffering from X-linked ichthyosis), as well as prevent or delay reformation of wounds or skin abnormalities in treated subjects.

[0055] The following description sets forth exemplary methods, parameters, and the like. It should be recognized, however, that such a description is not intended as a limitation on the scope of the present disclosure but is instead provided as a description of exemplary embodiments.

I. General Techniques

[0056] The techniques and procedures described or referenced herein are generally well understood and commonly employed using conventional methodology by those skilled in the art, such as, for example, the widely utilized methodologies described in Sambrook et al., *Molecular Cloning: A Laboratory Manual* 3d edition (2001) Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y.; *Current Protocols in Molecular Biology* (F. M. Ausubel, et al. eds., (2003)); the series *Methods in Enzymology* (Academic Press, Inc.): *PCR 2: A Practical Approach* (M. J. MacPherson, B. D. Hames and G. R. Taylor eds. (1995)), Harlow and Lane, eds. (1988); *Oligonucleotide Synthesis* (M. J. Gait, ed., 1984); *Methods in Molecular Biology*, Humana Press; *Cell Biology: A Laboratory Notebook* (J. E. Cellis, ed., 1998) Academic Press; *Animal Cell Culture* (R. I. Freshney), ed., 1987); *Introduction to Cell and Tissue Culture* (J. P. Mather and P. E. Roberts, 1998) Plenum Press; *Cell and Tissue Culture: Laboratory Procedures* (A. Doyle, J. B. Griffiths, and D. G. Newell, eds., 1993-8) J. Wiley and Sons; *Gene Transfer Vectors for Mammalian Cells* (J. M. Miller and M. P. Calos, eds., 1987); *PCR: The Polymerase Chain Reaction*, (Mullis et al., eds., 1994); *Short Protocols in Molecular Biology* (Wiley and Sons, 1999).

II. Definitions

[0057] Before describing the present disclosure in detail, it is to be understood that the present disclosure is not limited to particular compositions or biological systems, which can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

[0058] As used herein, the singular forms "a", "an" and "the" include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to "a molecule" optionally includes a combination of two or more such molecules, and the like.

[0059] As used herein, the term "and/or" may include any and all combinations of one or more of the associated listed items. For example, the term "a and/or b" may refer to "a alone", "b alone", "a or b", or "a and b"; the term "a, b, and/or c" may refer to "a alone", "b alone", "c alone", "a or b", "a or c", "b or c", "a, b, or c", "a and b", "a and c", "b and c", or "a, b, and c"; etc.

[0060] As used herein, the term "about" refers to the usual error range for the respective value readily known to the skilled person in this technical field. Reference to "about" a value or parameter herein includes (and describes) embodiments that are directed to that value or parameter per se.

[0061] It is understood that aspects and embodiments of the present disclosure include "comprising", "consisting", and "consisting essentially of" aspects and embodiments.

[0062] As used herein, the terms "polynucleotide", "nucleic acid sequence", "nucleic acid", and variations thereof shall be generic to polydeoxyribonucleotides (containing 2-deoxy-D-ribose), to polyribonucleotides (containing D-ribose), to any other type of polynucleotide that is an N-glycoside of a purine or pyrimidine base, and to other polymers containing non-nucleotidic backbones, provided that the polymers contain nucleobases in a configuration that allows for base pairing and base stacking, as found in DNA and RNA. Thus, these terms include known types of nucleic acid sequence modifications, for example, substitution of one or more of the naturally occurring nucleotides with an analog, and inter-nucleotide modifications.

[0063] As used herein, a nucleic acid is "operatively linked" or "operably linked" when it is placed into a functional relationship with another nucleic acid sequence. For example, a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence; or a ribosome binding site is operably linked to a coding sequence if it is positioned so as to facilitate translation. Generally, "operatively linked" or "operably linked" means that the DNA or RNA sequences being linked are contiguous.

[0064] As used herein, the term "vector" refers to discrete elements that are used to introduce heterologous nucleic acids into cells for either expression or replication thereof. An expression vector includes vectors capable of expressing nucleic acids that are operatively linked with regulatory sequences, such as promoter regions, that are capable of effecting expression of such nucleic acids. Thus, an expression vector may refer to a DNA or RNA construct, such as a plasmid, a phage, recombinant virus, or other vector that, upon introduction into an appropriate host cell, results in expression of the nucleic acids. Appropriate expression vectors are well known to those of skill in the art and include those that are replicable in eukaryotic cells and those that remain episomal or those which integrate into the host cell genome.

[0065] As used herein, an "open reading frame" or "ORF" refers to a continuous stretch of nucleic acids, either DNA or RNA, that encode a protein or polypeptide. Typically, the nucleic acids comprise a translation start signal or initiation codon, such as ATG or AUG, and a termination codon.

[0066] As used herein, an "untranslated region" or "UTR" refers to untranslated nucleic acids at the 5' and/or 3' ends of an open reading frame. The inclusion of one or more UTRs in a polynucleotide may affect post-transcriptional regulation, mRNA stability, and/or translation of the polynucleotide.

[0067] As used herein, the term "transgene" refers to a polynucleotide that is capable of being transcribed into RNA and translated and/or expressed under appropriate conditions, after being introduced into a cell. In some embodiments, it confers a desired property to a cell into which it was introduced, or otherwise leads to a desired therapeutic or diagnostic outcome.

[0068] As used herein, the terms "polypeptide," "protein," and "peptide" are used interchangeably and may refer to a polymer of two or more amino acids.

[0069] As used herein, a "subject", "host", or an "individual" refers to any animal classified as a mammal, including humans, domestic and farm animals, and zoo, sports, or pet animals, such as dogs, horses, cats, cows, as well as animals used in research, such as mice, rats, hamsters, rabbits, and non-human primates, etc. In some embodiments, the mammal is human.

[0070] As used herein, the terms "pharmaceutical formulation" or "pharmaceutical composition" refer to a preparation which is in such a form as to permit the biological activity of the active ingredient(s) to be effective, and which contains no additional components which are unacceptably toxic to a subject to which the composition or formulation would be administered. "Pharmaceutically acceptable" excipients (e.g., vehicles, additives) are those which can reasonably be administered to a subject mammal to provide an effective dose of the active ingredient(s) employed.

[0071] As used herein, "cutaneous administration" or "cutaneously administering" refers to the delivery of a composition to a subject by contacting, directly or otherwise, a formulation comprising the composition to all ("systemic") or a portion ("topical") of the skin of a subject. The term encompasses several routes of administration including, but not limited to, topical and transdermal. Topical administration may be used as a means to deliver a composition to the epidermis or dermis of a subject, or to specific strata thereof.

[0072] As used herein, an "effective amount" is at least the minimum amount required to affect a measurable improvement or prevention of one or more symptoms of a particular disorder. An "effective amount" may vary according to factors such as the disease state, age, sex, and weight of the patient. An effective amount is also one in which any toxic or detrimental effects of the treatment are outweighed by the therapeutically beneficial effects. For prophylactic use, beneficial or desired results include results such as eliminating or reducing the risk, lessening the severity, or delaying the onset of the disease, its complications and intermediate pathological phenotypes presenting during development of the disease. For therapeutic use, beneficial or desired results include clinical results such as decreasing one or more symptoms resulting from the disease, increasing the quality of life of those suffering from the disease, decreasing the dose of other medications used to treat symptoms of the disease, delaying the progression of the disease, and/or prolonging survival. An effective amount can be administered in one or more administrations. For purposes of the present disclosure, an effective amount of a recombinant nucleic acid, virus, and/or pharmaceutical composition is an amount sufficient to accomplish prophylactic or therapeutic treatment either directly or indirectly. As is understood in the clinical context, an effective amount of a recombinant nucleic acid, virus, and/or pharmaceutical composition may or may not be achieved in conjunction with another drug, compound, or pharmaceutical composition. Thus, an "effective amount" may be considered in the context of administering one or more therapeutic agents, and a single agent may be considered to be given in an effective amount if, in conjunction with one or more other agents, a desirable result may be or is achieved.

[0073] As used herein, "treatment" refers to clinical intervention designed to alter the natural course of the individual or cell being treated during the course of clinical pathology. Desirable effects of treatment include decreasing the rate of disease/disorder/defect progression, ameliorating, or palliating the disease/disorder/defect state, and remission or improved prognosis. For example, an individual is successfully "treated" if one or more symptoms associated with congenital ichthyosis (e.g., X-linked ichthyosis, LI, CIE, HI, etc.) are mitigated or eliminated.

[0074] As used herein, the term "delaying progression of" a disease/disorder/defect refers to deferring, hindering, slowing, retarding, stabilizing, and/or postponing development of the disease/disorder/defect. This delay can be of varying lengths or time, depending on the history of the disease/disorder/defect and/or the individual being treated. As is evident to one of ordinary skill in the art, a sufficient or significant delay can, in effect, encompass prevention, in that the individual does not develop the disease.

III. Recombinant Nucleic Acids

[0075] Certain aspects of the present disclosure relate to recombinant nucleic acids (e.g., isolated recombinant nucleic acids) comprising one or more (e.g., one or more, two or more, three or more, four or more, five or more, ten or more, etc.) polynucleotides encoding an ichthyosis-associated polypeptide (e.g., a human ichthyosis-associated polypeptide such as a Steryl-sulfatase polypeptide). In some embodiments, the present disclosure relates to recombinant nucleic acids (e.g., isolated recombinant nucleic acids) comprising one or more (e.g., one or more, two or more, three or more, four or more, five or more, ten or more, etc.) polynucleotides encoding two or more (e.g., two or more, three or more, four or more, five or more, etc.) ichthyosis-associated polypeptides. In some embodiments, the recombinant nucleic acid comprises one or more polynucleotides encoding two or more identical ichthyosis-associated polypeptides. In some embodiments, the recombinant nucleic acid comprises one or more polynucleotides encoding two or more different ichthyosis-associated polypeptides.

[0076] In some embodiments, the recombinant nucleic acid is a vector. In some embodiments, the recombinant nucleic acid is a viral vector. In some embodiments, the recombinant nucleic acid is a herpes viral vector. In some embodiments, the recombinant nucleic acid is a herpes simplex virus amplicon. In some embodiments, the recombinant nucleic acid is a recombinant herpes virus genome. In some embodiments, the recombinant nucleic acid is a recombinant herpes simplex virus genome. In some embodiments, the recombinant herpes simplex virus genome is a recombinant type 1 herpes simplex virus (HSV-1) genome.

Polynucleotides Encoding Ichthyosis-Associated Polypeptides

[0077] In some embodiments, the present disclosure relates to a recombinant nucleic acid comprising one or more polynucleotides comprising the coding sequence of an ichthyosis-associated gene. In some embodiments, an ichthyosis-associated gene is a wild-type and/or functional version of a gene that has been identified as comprising a pathogenic variant and/or loss-of-function mutation that is correlated with, causative of, or contributes to one or more forms of congenital ichthyosis (e.g., a pathogenic variant and/or loss-of function mutation in gene identified in a patient suffering from one or more of harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and/or restrictive dermopathy). Genes harboring pathogenic variants and/or loss-of-function mutations that are correlated with, causative of, or contribute to one or more forms of congenital ichthyosis include, e.g., ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, ARSE, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, EBP, ELOVL4, FLG, FLG2, GJB2, GJB3, GJB4, GJB6, KDSR, KRT1, KRT2, KRT9, KRT10, LIPN, LOR, MBTPS2, NIPAL4, NSDHL, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4, SNAP29, ST14, STS, SULT2B1, VPS33B, and ZMPSTE24.

[0078] The coding sequence of any suitable ichthyosis-associated gene (including any isoform thereof) known in the art may be encoded by a polynucleotide of the present disclosure, including, for example, an ABCA12 gene (such as a human ABCA12 gene, e.g., as disclosed by NCBI Gene ID: 26154), an ABHD5 gene (such as a human ABHD5 gene, e.g., as disclosed by NCBI Gene ID: 51099), an ALDH3A2 gene (such as a human ALDH3A2 gene, e.g., as disclosed by NCBI Gene ID: 224), an ALOX12B gene (such as a human ALOX12B gene, e.g., as disclosed by NCBI Gene ID: 242), an ALOXE3 gene (such as a human ALOXE3 gene, e.g., as disclosed by NCBI Gene ID: 59344), an AP1S1 gene (such as a human AP1S1 gene, e.g., as disclosed by NCBI Gene ID: 1174), an ARSE gene (such as a human ARSE gene, e.g., as disclosed by NCBI Gene ID: 415), a CASP14 gene (such as a human CASP14 gene, e.g., as disclosed by NCBI Gene ID: 23581), a CDSN gene (such as a human CDSN gene, e.g., as disclosed by NCBI Gene ID: 1041), a CERS3 gene (such as a human CERS3 gene, e.g., as disclosed by NCBI Gene ID: 204219), a CHST8 gene (such as a human CHST8 gene, e.g., as disclosed by NCBI Gene ID: 64377), a CLDN1 gene (such as a human CLDN1 gene, e.g., as disclosed by NCBI Gene ID: 9076), a CSTA gene (such as a human CSTA gene, e.g., as disclosed by NCBI Gene ID: 1475), a CYP4F22 gene (such as a human CYP4F22 gene, e.g., as disclosed by NCBI Gene ID: 126410), an EBP gene (such as a human EBP gene, e.g., as disclosed by NCBI Gene ID: 10682), an ELOVL4 gene (such as a human ELOVL4 gene, e.g., as disclosed by NCBI Gene ID: 6785), a FLG gene (such as a human FLG gene, e.g., as disclosed by NCBI Gene ID: 2312), a FLG2 gene (such as a human FLG2 gene, e.g., as disclosed by NCBI Gene ID: 388698), a GJB2 gene (such as a human GJB2 gene, e.g., as disclosed by NCBI Gene ID: 2706), a GJB3 gene (such as a human GJB3 gene, e.g., as disclosed by NCBI Gene ID: 2707), a GJB4 gene (such as a human GJB4 gene, e.g., as disclosed by NCBI Gene ID: 127534), a GJB6 gene (such as a human GJB6 gene, e.g., as disclosed by NCBI Gene ID: 10804), a KDSR gene (such as a human KDSR gene, e.g., as disclosed by NCBI Gene ID: 2531), a KRT1 gene (such as a human KRT1 gene, e.g., as disclosed by NCBI Gene ID: 3848), a KRT2 gene (such as a human KRT2 gene, e.g., as disclosed by NCBI Gene ID: 3849), a KRT9 gene (such as a human KRT9 gene, e.g., as disclosed by NCBI Gene ID: 3857), a KRT10 gene (such as a human KRT10 gene, e.g., as disclosed by NCBI Gene ID: 3858), an LIPN gene (such as a human LIPN gene, e.g., as disclosed by NCBI Gene ID: 643418), an LOR gene (such as a human LOR gene, e.g., as disclosed by NCBI Gene ID: 4014), an MBTPS2 gene (such as a human MBTPS2 gene, e.g., as disclosed by NCBI Gene ID: 51360), an NIPAL4 gene (such as a human NIPAL4 gene, e.g., as disclosed by NCBI Gene ID: 348938), an NSDHL gene (such as a human NSDHL gene, e.g., as disclosed by NCBI Gene ID: 50814), a PEX7 gene (such as a human PEX7 gene, e.g., as disclosed by NCBI Gene ID: 5191), a PHGDH gene (such as a human PHGDH gene, e.g., as disclosed by NCBI Gene ID: 26227), a PHYH gene (such as a human PHYH gene, e.g., as disclosed by NCBI Gene ID: 5264), a PNPLA1 gene (such as a human PNPLA1 gene, e.g., as disclosed by NCBI Gene ID: 285848), a POMP gene (such as a human POMP gene, e.g., as disclosed by NCBI Gene ID: 51371), a PSAT1 gene (such as a human PSAT1 gene, e.g., as disclosed by NCBI Gene ID: 29968), an SDR9C7 gene (such as a human SDR9C7 gene, e.g., as disclosed by NCBI Gene ID: 121214), a SERPINB8 gene (such as a human SERPINB8 gene, e.g., as disclosed by NCBI Gene ID: 5271), an SLC27A4 gene (such as a human SLC27A4 gene, e.g., as disclosed by NCBI Gene ID: 10999), a SNAP29 gene (such as a human SNAP29 gene, e.g., as disclosed by NCBI Gene ID: 9342), an ST14 gene (such as a human ST14 gene, e.g., as disclosed by NCBI Gene ID: 6768), an STS gene (such as a human STS gene, e.g., as disclosed by NCBI Gene ID: 412), a SULT2B1 gene (such as a human SULT2B1 gene, e.g., as disclosed by NCBI Gene ID: 6820), a VPS33B gene (such as a human VPS33B gene, e.g., as disclosed by NCBI Gene ID: 26276), and a ZMPSTE24 gene (such as a human ZMPSTE24 gene, e.g., as disclosed by NCBI Gene ID: 10269). Methods of identifying ichthyosis-associated gene homologs/orthologs from additional species are known to one of ordinary skill in the art, including, for example, using a nucleic acid sequence alignment program such as the BLAST® blastn suite. In some embodiments, a polynucleotide of the preset disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of any of the ichthyosis-associated genes (and/or coding sequences thereof) described herein. In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ichthyosis-associated gene.

[0079] In some embodiments, a polynucleotide of the present disclosure comprises a codon-optimized variant of the coding sequence of any of the ichthyosis-associated genes described herein or known in the art. In some embodiments, use a of a codon-optimized variant of the coding sequence of an ichthyosis-associated gene increases stability and/or yield of heterologous expression (RNA and/or protein) of the encoded polypeptide in a target cell, as compared to the stability and/or yield of heterologous expression of a corresponding, non-codon-optimized, wild-type sequence. Any suitable method known in the art for performing codon optimization of a sequence for expression in one or more target cells (e.g., one or more human cells) may be used, including, for example, by the methods described by Fath et al. (PLOS One. 2011 Mar. 3; 6 (3): e17596).

[0080] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ABCA12 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to a sequence selected from SEQ ID NOS: 1-4. In some embodiments, a polynucleotide of the present disclosure comprises a sequence selected from SEQ ID NOS: 1-4. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 1 or SEQ ID NO: 2.

[0081] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 1 or SEQ ID NO: 2 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 2000, at least 3000, at least 4000, at least 5000, at least 6000, at least 7000, but fewer than 7788 consecutive nucleotides of SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-7785 of SEQ ID NO: 1 or SEQ ID NO: 2. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-7785 of SEQ ID NO: 1 or SEQ ID NO: 2.

[0082] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 3 or SEQ ID NO: 4. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 3 or SEQ ID NO: 4 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 2000, at least 3000, at least 4000, at least 5000, at least 6000, but fewer than 6834 consecutive nucleotides of SEQ ID NO: 3 or SEQ ID NO: 4. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-6831 of SEQ ID NO: 3 or SEQ ID NO: 4. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-6831 of SEQ ID NO: 3 or SEQ ID NO: 4.

[0083] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ABHD5 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 5 or SEQ ID NO: 6. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 5 or SEQ ID NO: 6.

[0084] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 5 or SEQ ID NO: 6. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 5 or SEQ ID NO: 6 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1050 consecutive nucleotides of SEQ ID NO: 5 or SEQ ID NO: 6. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1047 of SEQ ID NO: 5 or SEQ ID NO: 6. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1047 of SEQ ID NO: 5 or SEQ ID NO: 6.

[0085] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ALDH3A2 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 7 or SEQ ID NO: 8. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 7 or SEQ ID NO: 8.

[0086] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 7 or SEQ ID NO: 8. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 7 or SEQ ID NO: 8 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1527 consecutive nucleotides of SEQ ID NO: 7 or SEQ ID NO: 8. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1524 of SEQ ID NO: 7 or SEQ ID NO: 8. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1524 of SEQ ID NO: 7 or SEQ ID NO: 8.

[0087] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ALOX12B gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 9 or SEQ ID NO: 10. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 9 or SEQ ID NO: 10.

[0088] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 9 or SEQ ID NO: 10. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 9 or SEQ ID NO: 10 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, but fewer than 2106 consecutive nucleotides of SEQ ID NO: 9 or SEQ ID NO: 10. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-2103 of SEQ ID NO: 9 or SEQ ID NO: 10. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-2106 of SEQ ID NO: 9 or SEQ ID NO: 10.

[0089] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ALOXE3 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to a sequence selected from SEQ ID NOS: 11-14. In some embodiments, a polynucleotide of the present disclosure comprises a sequence selected from SEQ ID NOS: 11-14.

[0090] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 11 or SEQ ID NO: 12. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 11 or SEQ ID NO: 12 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, but fewer than 2136 consecutive nucleotides of SEQ ID NO: 11 or SEQ ID NO: 12. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-2133 of SEQ ID NO: 11 or SEQ ID NO: 12. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-2133 of SEQ ID NO: 11 or SEQ ID NO: 12.

[0091] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 13 or SEQ ID

NO: 14. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 13 or SEQ ID NO: 14 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, at least 2250, at least 2500, but fewer than 2532 consecutive nucleotides of SEQ ID NO: 13 or SEQ ID NO: 14. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-2529 of SEQ ID NO: 13 or SEQ ID NO: 14. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-2529 of SEQ ID NO: 13 or SEQ ID NO: 14.

[0092] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human AP1S1 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 15 or SEQ ID NO: 16. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 15 or SEQ ID NO: 16.

[0093] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 15 or SEQ ID NO: 16. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 15 or SEQ ID NO: 16 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, but fewer than 477 consecutive nucleotides of SEQ ID NO: 15 or SEQ ID NO: 16. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-474 of SEQ ID NO: 15 or SEQ ID NO: 16. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-474 of SEQ ID NO: 15 or SEQ ID NO: 16.

[0094] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ARSE gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 17 or SEQ ID NO: 18. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 17 or SEQ ID NO: 18.

[0095] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 17 or SEQ ID NO: 18. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 17 or SEQ ID NO: 18 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1770 consecutive nucleotides of SEQ ID NO: 17 or SEQ ID NO: 18. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1767 of SEQ ID NO: 17 or SEQ ID NO: 18. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1767 of SEQ ID NO: 17 or SEQ ID NO: 18.

[0096] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CASP14 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 19 or SEQ ID NO: 20. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 19 or SEQ ID NO: 20.

[0097] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 19 or SEQ ID NO: 20. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 19 or SEQ ID NO: 20 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, but fewer than 729 consecutive nucleotides of SEQ ID NO: 19 or SEQ ID NO: 20. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-726 of SEQ ID NO: 19 or SEQ ID NO: 20. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-726 of SEQ ID NO: 19 or SEQ ID NO: 20.

[0098] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CDSN gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 21 or SEQ ID NO: 22. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 21 or SEQ ID NO: 22.

[0099] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 21 or SEQ ID NO: 22. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 21 or SEQ ID NO: 22 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1590 consecutive nucleotides of SEQ ID NO: 21 or SEQ ID NO: 22. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1587 of SEQ ID NO: 21 or SEQ ID NO: 22. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1587 of SEQ ID NO: 21 or SEQ ID NO: 22.

[0100] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CERS3 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 23 or SEQ ID NO: 24. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 23 or SEQ ID NO: 24.

[0101] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 23 or SEQ ID NO: 24. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 23 or SEQ ID NO: 24 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1152 consecutive nucleotides of SEQ ID NO: 23 or SEQ ID NO: 24. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1149 of SEQ ID NO: 23 or SEQ ID NO: 24. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1149 of SEQ ID NO: 23 or SEQ ID NO: 24.

[0102] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CHST8 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 25 or SEQ ID NO: 26. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 25 or SEQ ID NO: 26.

[0103] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 25 or SEQ ID NO: 26. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 25 or SEQ ID NO: 26 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, but fewer than 1275 consecutive nucleotides of SEQ ID NO: 25 or SEQ ID NO: 26. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1272 of SEQ ID NO: 25 or SEQ ID NO: 26. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-272 of SEQ ID NO: 25 or SEQ ID NO: 26.

[0104] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CLDN1 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 27 or SEQ ID NO: 28. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 27 or SEQ ID NO: 28.

[0105] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 27 or SEQ ID NO: 28. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 27 or SEQ ID NO: 28 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, but fewer than 636 consecutive nucleotides of SEQ ID NO: 27 or SEQ ID NO: 28. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-633 of SEQ ID NO: 27 or SEQ ID NO: 28. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-633 of SEQ ID NO: 27 or SEQ ID NO: 28.

[0106] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CSTA gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 29 or SEQ ID NO: 30. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 29 or SEQ ID NO: 30.

[0107] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 29 or SEQ ID NO: 30. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 29 or SEQ ID NO: 30 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, but fewer than 297 consecutive nucleotides of SEQ ID NO: 29 or SEQ ID NO: 30. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-294 of SEQ ID NO: 29 or SEQ ID NO: 30. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-294 of SEQ ID NO: 29 or SEQ ID NO: 30.

[0108] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human CYP4F22 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 31 or SEQ ID NO: 32. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 31 or SEQ ID NO: 32.

[0109] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 31 or SEQ ID NO: 32. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 31 or SEQ ID NO: 32 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1596 consecutive nucleotides of SEQ ID NO: 31 or SEQ ID NO: 32. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1593 of SEQ ID NO: 31 or SEQ ID NO: 32. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1593 of SEQ ID NO: 31 or SEQ ID NO: 32.

[0110] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human EBP gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 33 or SEQ ID NO: 34. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 33 or SEQ ID NO: 34.

[0111] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 33 or SEQ ID NO: 34. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 33 or SEQ ID NO: 34 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, but fewer than 693 consecutive nucleotides of SEQ ID NO: 33 or SEQ ID NO: 34. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-690 of SEQ ID NO: 33 or SEQ ID NO: 34. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-690 of SEQ ID NO: 33 or SEQ ID NO: 34.

[0112] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ELOVL4 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 35 or SEQ ID NO: 36. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 35 or SEQ ID NO: 36.

[0113] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 35 or SEQ ID NO: 36. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 35 or SEQ ID NO: 36 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 945 consecutive nucleotides of SEQ ID NO: 35 or SEQ ID NO: 36. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least

92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-942 of SEQ ID NO: 35 or SEQ ID NO: 36. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-942 of SEQ ID NO: 35 or SEQ ID NO: 36.

[0114] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human FLG gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 37 or SEQ ID NO: 38. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 37 or SEQ ID NO: 38.

[0115] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 37 or SEQ ID NO: 38. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 37 or SEQ ID NO: 38 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, at least 3000, at least 4000, at least 5000, at least 6000, at least 7000, at least 8000, at least 9000, at least 10000, at least 11000, at least 12000, but fewer than 12186 consecutive nucleotides of SEQ ID NO: 37 or SEQ ID NO: 38. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-12183 of SEQ ID NO: 37 or SEQ ID NO: 38. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-12183 of SEQ ID NO: 37 or SEQ ID NO: 38.

[0116] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human FLG2 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 39. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 39.

[0117] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 39. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 39 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, at least 3000, at least 4000, at least 5000, at least 6000, at least 7000, but fewer than 7176 consecutive nucleotides of SEQ ID NO: 39. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-7173 of SEQ ID NO: 39. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-7173 of SEQ ID NO: 39.

[0118] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human GJB2 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 40 or SEQ ID NO: 41. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 40 or SEQ ID NO: 41.

[0119] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 40 or SEQ ID NO: 41. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 40 or SEQ ID NO: 41 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, but fewer than 681 consecutive nucleotides of SEQ ID NO: 40 or SEQ ID NO: 41. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-678 of SEQ ID NO: 40 or SEQ ID NO: 41. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-678 of SEQ ID NO: 40 or SEQ ID NO: 41.

[0120] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human GJB3 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 42 or SEQ ID NO: 43. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 42 or SEQ ID NO: 43.

[0121] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 42 or SEQ ID NO: 43. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 42 or SEQ

ID NO: 43 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 813 consecutive nucleotides of SEQ ID NO: 42 or SEQ ID NO: 43. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-810 of SEQ ID NO: 42 or SEQ ID NO: 43 In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-810 of SEQ ID NO: 42 or SEQ ID NO: 43.

[0122] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human GJB4 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 44 or SEQ ID NO: 45. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 44 or SEQ ID NO: 45.

[0123] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 44 or SEQ ID NO: 45. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 44 or SEQ ID NO: 45 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 801 consecutive nucleotides of SEQ ID NO: 44 or SEQ ID NO: 45. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-798 of SEQ ID NO: 44 or SEQ ID NO: 45. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-798 of SEQ ID NO: 44 or SEQ ID NO: 45.

[0124] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human GJB6 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 46 or SEQ ID NO: 47. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 46 or SEQ ID NO: 47.

[0125] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 46 or SEQ ID NO: 47. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 46 or SEQ ID NO: 47 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 786 consecutive nucleotides of SEQ ID NO: 46 or SEQ ID NO: 47. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-783 of SEQ ID NO: 46 or SEQ ID NO: 47. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-783 of SEQ ID NO: 46 or SEQ ID NO: 47.

[0126] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human KDSR gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 48 or SEQ ID NO: 49. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 48 or SEQ ID NO: 49.

[0127] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 48 or SEQ ID NO: 49. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 48 or SEQ ID NO: 49 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 999 consecutive nucleotides of SEQ ID NO: 48 or SEQ ID NO: 49. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-996 of SEQ ID NO: 48 or SEQ ID NO: 49. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-996 of SEQ ID NO: 48 or SEQ ID NO: 49.

[0128] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human KRT1 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 50 or SEQ ID NO: 51. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 50 or SEQ ID NO: 51.

[0129] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 50 or SEQ ID NO: 51. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 50 or SEQ ID NO: 51 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1935 consecutive nucleotides of SEQ ID NO: 50 or SEQ ID NO: 51. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1932 of SEQ ID NO: 50 or SEQ ID NO: 51. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1932 of SEQ ID NO: 50 or SEQ ID NO: 51.

[0130] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human KRT2 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 52 or SEQ ID NO: 53. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 52 or SEQ ID NO: 53.

[0131] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 52 or SEQ ID NO: 53. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 52 or SEQ ID NO: 53 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1920 consecutive nucleotides of SEQ ID NO: 52 or SEQ ID NO: 53. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1917 of SEQ ID NO: 52 or SEQ ID NO: 53. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1917 of SEQ ID NO: 52 or SEQ ID NO: 53.

[0132] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human KRT9 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 54 or SEQ ID NO: 55. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 54 or SEQ ID NO: 55.

[0133] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 54 or SEQ ID NO: 55. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 54 or SEQ ID NO: 55 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1872 consecutive nucleotides of SEQ ID NO: 54 or SEQ ID NO: 55. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1869 of SEQ ID NO: 54 or SEQ ID NO: 55. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1869 of SEQ ID NO: 54 or SEQ ID NO: 55.

[0134] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human KRT10 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 56 or SEQ ID NO: 57. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 56 or SEQ ID NO: 57.

[0135] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 56 or SEQ ID NO: 57. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 56 or SEQ ID NO: 57 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1755 consecutive nucleotides of SEQ ID NO: 56 or SEQ ID NO: 57. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1752 of SEQ ID NO: 56 or SEQ ID NO: 57. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1752 of SEQ ID NO: 56 or SEQ ID NO: 57.

[0136] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human LIPN gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 58 or SEQ ID NO: 59. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 58 or SEQ ID NO: 59.

[0137] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 58 or SEQ ID NO: 59. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 58 or SEQ ID NO: 59 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1197 consecutive nucleotides of SEQ ID NO: 58 or SEQ ID NO: 59. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1194 of SEQ ID NO: 58 or SEQ ID NO: 59. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1194 of SEQ ID NO: 58 or SEQ ID NO: 59.

[0138] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human LOR gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 60 or SEQ ID NO: 61. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 60 or SEQ ID NO: 61.

[0139] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 60 or SEQ ID NO: 61. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 60 or SEQ ID NO: 61 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 939 consecutive nucleotides of SEQ ID NO: 60 or SEQ ID NO: 61. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-936 of SEQ ID NO: 60 or SEQ ID NO: 61. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-936 of SEQ ID NO: 60 or SEQ ID NO: 61.

[0140] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human MBTPS2 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 62 or SEQ ID NO: 63. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 62 or SEQ ID NO: 63.

[0141] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 62 or SEQ ID NO: 63. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 62 or SEQ ID NO: 63 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1560 consecutive nucleotides of SEQ ID NO: 62 or SEQ ID NO: 63. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1557 of SEQ ID NO: 62 or SEQ ID NO: 63. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1557 of SEQ ID NO: 62 or SEQ ID NO: 63.

[0142] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human NIPAL4 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to a sequence selected from SEQ ID NOS: 64-67. In some embodiments, a polynucleotide of the present disclosure comprises a sequence selected from SEQ ID NOS: 64-67. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 64 or SEQ ID NO: 65.

[0143] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 64 or SEQ ID NO: 65. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 64 or SEQ ID NO: 65 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, but fewer than 1401 consecutive nucleotides of SEQ ID NO: 64 or SEQ ID NO: 65. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1398 of SEQ ID NO: 64 or SEQ ID NO: 65. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1398 of SEQ ID NO: 64 or SEQ ID NO: 65.

[0144] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 66 or SEQ ID NO: 67. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 66 or SEQ ID NO: 67 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, but fewer than 1344 consecutive nucleotides of SEQ ID NO: 66 or SEQ ID NO: 67. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1341 of SEQ ID NO: 66 or SEQ ID NO: 67. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1341 of SEQ ID NO: 66 or SEQ ID NO: 67.

[0145] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human NSDHL gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 68 or SEQ ID NO: 69. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 68 or SEQ ID NO: 69.

[0146] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 68 or SEQ ID NO: 69. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 68 or SEQ ID NO: 69 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1122 consecutive nucleotides of SEQ ID NO: 68 or SEQ ID NO: 69. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1119 of SEQ ID NO: 68 or SEQ ID NO: 69. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1119 of SEQ ID NO: 68 or SEQ ID NO: 69.

[0147] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human PEX7 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 70 or SEQ ID NO: 71. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 70 or SEQ ID NO: 71.

[0148] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 70 or SEQ ID NO: 71. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 70 or SEQ ID NO: 71 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 972 consecutive nucleotides of SEQ ID NO: 70 or SEQ ID NO: 71. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-969 of SEQ ID NO: 70 or SEQ ID NO: 71. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-969 of SEQ ID NO: 70 or SEQ ID NO: 71.

[0149] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human PHGDH gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 72 or SEQ ID NO: 73. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 72 or SEQ ID NO: 73.

[0150] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 72 or SEQ ID NO: 73. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 72 or SEQ ID NO: 73 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1602 consecutive nucleotides of SEQ ID NO: 72 or SEQ ID NO: 73. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least

80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1599 of SEQ ID NO: 72 or SEQ ID NO: 73. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1599 of SEQ ID NO: 72 or SEQ ID NO: 73.

[0151] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human PHYH gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 74 or SEQ ID NO: 75. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 74 or SEQ ID NO: 75.

[0152] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 74 or SEQ ID NO: 75. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 74 or SEQ ID NO: 75 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1017 consecutive nucleotides of SEQ ID NO: 74 or SEQ ID NO: 75. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1014 of SEQ ID NO: 74 or SEQ ID NO: 75. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1014 of SEQ ID NO: 74 or SEQ ID NO: 75.

[0153] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human PNPLA1 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to a sequence selected from SEQ ID NOS: 76-81. In some embodiments, a polynucleotide of the present disclosure comprises a sequence selected from SEQ ID NOS: 76-81.

[0154] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 76 or SEQ ID NO: 77. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 76 or SEQ ID NO: 77 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1599 consecutive nucleotides of SEQ ID NO: 76 or SEQ ID NO: 77. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1596 of SEQ ID NO: 76 or SEQ ID NO: 77. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1596 of SEQ ID NO: 76 or SEQ ID NO: 77.

[0155] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 78 or SEQ ID NO: 79. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 78 or SEQ ID NO: 79 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, but fewer than 1314 consecutive nucleotides of SEQ ID NO: 78 or SEQ ID NO: 79. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1311 of SEQ ID NO: 78 or SEQ ID NO: 79. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1311 of SEQ ID NO: 78 or SEQ ID NO: 79.

[0156] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 80 or SEQ ID NO: 81. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 80 or SEQ ID NO: 81 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, but fewer than 1341 consecutive nucleotides of SEQ ID NO: 80 or SEQ ID NO: 81. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1338 of SEQ ID NO: 80 or SEQ ID NO: 81. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1338 of SEQ ID NO: 80 or SEQ ID NO: 81.

[0157] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human POMP gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 82 or SEQ ID NO: 83. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 82 or SEQ ID NO: 83.

[0158] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 82 or SEQ ID NO: 83. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 82 or SEQ ID NO: 83 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, but fewer than 426 consecutive nucleotides of SEQ ID NO: 82 or SEQ ID NO: 83. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-423 of SEQ ID NO: 82 or SEQ ID NO: 83. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-423 of SEQ ID NO: 82 or SEQ ID NO: 83.

[0159] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human PSAT1 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 84 or SEQ ID NO: 85. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 84 or SEQ ID NO: 85.

[0160] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 84 or SEQ ID NO: 85. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 84 or SEQ ID NO: 85 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1113 consecutive nucleotides of SEQ ID NO: 84 or SEQ ID NO: 85. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1110 of SEQ ID NO: 84 or SEQ ID NO: 85. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1110 of SEQ ID NO: 84 or SEQ ID NO: 85.

[0161] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human SDR9C7 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 86 or SEQ ID NO: 87. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 86 or SEQ ID NO: 87.

[0162] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 86 or SEQ ID NO: 87. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 86 or SEQ ID NO: 87 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 942 consecutive nucleotides of SEQ ID NO: 86 or SEQ ID NO: 87. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-939 of SEQ ID NO: 86 or SEQ ID NO: 87. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-939 of SEQ ID NO: 86 or SEQ ID NO: 87.

[0163] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human SERPINB8 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 88 or SEQ ID NO: 89. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 88 or SEQ ID NO: 89.

[0164] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 88 or SEQ ID NO: 89. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 88 or SEQ ID NO: 89 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, but fewer than 1125 consecutive nucleotides of SEQ ID NO: 88 or SEQ ID NO: 89. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1122 of SEQ ID NO: 88 or SEQ ID NO: 89. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1125 of SEQ ID NO: 88 or SEQ ID NO: 89.

[0165] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human SLC27A4 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 90 or SEQ ID NO: 91. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 90 or SEQ ID NO: 91.

[0166] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 90 or SEQ ID NO: 91. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 90 or SEQ ID NO: 91 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1932 consecutive nucleotides of SEQ ID NO: 90 or SEQ ID NO: 91. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1929 of SEQ ID NO: 90 or SEQ ID NO: 91. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1929 of SEQ ID NO: 90 or SEQ ID NO: 91.

[0167] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human SNAP29 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 92 or SEQ ID NO: 93. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 92 or SEQ ID NO: 93.

[0168] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 92 or SEQ ID NO: 93. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 92 or SEQ ID NO: 93 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, but fewer than 777 consecutive nucleotides of SEQ ID NO: 92 or SEQ ID NO: 93. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-774 of SEQ ID NO: 92 or SEQ ID NO: 93. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-774 of SEQ ID NO: 92 or SEQ ID NO: 93.

[0169] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ST14 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 94 or SEQ ID NO: 95. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 94 or SEQ ID NO: 95.

[0170] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 94 or SEQ ID NO: 95. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 94 or SEQ ID NO: 95 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, at least 2500, but fewer than 2568 consecutive nucleotides of SEQ ID NO: 94 or SEQ ID NO: 95. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-2565 of SEQ ID NO: 94 or SEQ ID NO: 95. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-2565 of SEQ ID NO: 94 or SEQ ID NO: 95.

[0171] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human STS gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 96 or SEQ ID NO: 97. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 96 or SEQ ID NO: 97.

[0172] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 96 or SEQ ID NO: 97. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 96 or SEQ ID NO: 97 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, but fewer than 1752 consecutive nucleotides of SEQ ID NO: 96 or SEQ ID NO: 97. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least

80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1749 of SEQ ID NO: 96 or SEQ ID NO: 97. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1749 of SEQ ID NO: 96 or SEQ ID NO: 97.

[0173] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human SULT2B1 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 153 or SEQ ID NO: 154. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 153 or SEQ ID NO: 154.

[0174] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 153 or SEQ ID NO: 154. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 153 or SEQ ID NO: 154 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 600, at least 700, at least 800, at least 900, at least 1000, at least 1050, but fewer than 1098 consecutive nucleotides of SEQ ID NO: 153 or SEQ ID NO: 154. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1095 of SEQ ID NO: 153 or SEQ ID NO: 154. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1095 of SEQ ID NO: 153 or SEQ ID NO: 154.

[0175] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human VPS33B gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 98 or SEQ ID NO: 99. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 98 or SEQ ID NO: 99.

[0176] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 98 or SEQ ID NO: 99. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 98 or SEQ ID NO: 99 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, but fewer than 1854 consecutive nucleotides of SEQ ID NO: 98 or SEQ ID NO: 99. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1851 of SEQ ID NO: 98 or SEQ ID NO: 99. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1851 of SEQ ID NO: 98 or SEQ ID NO: 99.

[0177] In some embodiments, a polynucleotide of the present disclosure comprises the coding sequence of a human ZMPSTE24 gene (or a codon-optimized variant thereof). In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of SEQ ID NO: 100 or SEQ ID NO: 101. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of SEQ ID NO: 100 or SEQ ID NO: 101.

[0178] In some embodiments, a polynucleotide of the present disclosure comprises a 5' truncation, a 3' truncation, or a fragment of the sequence of SEQ ID NO: 100 or SEQ ID NO: 101. In some embodiments, the 5' truncation, 3' truncation, or fragment of the sequence of SEQ ID NO: 100 or SEQ ID NO: 101 is a polynucleotide that has at least 25, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 250, at least 300, or at least 350, at least 400, at least 450, at least 500, at least 750, at least 1000, at least 1250, but fewer than 1428 consecutive nucleotides of SEQ ID NO: 100 or SEQ ID NO: 101. In some embodiments, a polynucleotide of the present disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of nucleic acids 1-1425 of SEQ ID NO: 100 or SEQ ID NO: 101. In some embodiments, a polynucleotide of the present disclosure comprises the sequence of nucleic acids 1-1425 of SEQ ID NO: 100 or SEQ ID NO: 101.

[0179] In some embodiments, expression of an ichthyosis-associated gene (e.g., as described above) in one or more cells of a subject in need thereof (e.g., a subject harboring one or more pathogenic variants and/or loss-of-function mutations in one or both copies of the corresponding endogenous gene) is beneficial for providing prophylactic, palliative, and/or therapeutic relief of one or more signs or symptoms of an autosomal dominant, autosomal semi-dominant, autosomal recessive, X-linked dominant, and/or X-linked recessive form of congenital ichthyosis.

[0180] In some embodiments, expression of an ichthyosis-associated gene (e.g., ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, ARSE, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, ELOVL4, KDSR, LIPN, MBTPS2, NIPAL4, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4,

SNAP29, ST14, STS, SULT2B1, VPS33B, ZMPSTE24) in one or more cells of a subject in need thereof (e.g., a subject harboring one or more pathogenic variants and/or loss-of-function mutations in one or both copies of the corresponding endogenous gene) is beneficial for providing prophylactic, palliative, and/or therapeutic relief of one or more signs or symptoms of an autosomal recessive and/or X-linked recessive form of congenital ichthyosis.

[0181] In some embodiments, expression of an ichthyosis-associated gene (e.g., ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, ELOVL4, KDSR, LIPN, NIPAL4, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4, SNAP29, ST14, SULT2B1, VPS33B, ZMPSTE24) in one or more cells of a subject in need thereof (e.g., a subject harboring one or more pathogenic variants and/or loss-of-function mutations in one or both copies of the corresponding endogenous gene) is beneficial for providing prophylactic, palliative, and/or therapeutic relief of one or more signs or symptoms of an autosomal recessive form of congenital ichthyosis.

[0182] In some embodiments, expression of an ichthyosis-associated gene (e.g., ARSE, MBTPS2, STS) in one or more cells of a subject in need thereof (e.g., a subject harboring one or more pathogenic variants and/or loss-of-function mutations in one or both copies of the corresponding endogenous gene) is beneficial for providing prophylactic, palliative, and/or therapeutic relief of one or more signs or symptoms of an X-linked recessive form of congenital ichthyosis.

[0183] A polynucleotide of the present disclosure (e.g., comprising the coding sequence of an ichthyosis-associated gene (i.e., encoding an ichthyosis-associated polypeptide)) may further encode additional coding and non-coding sequences. Examples of additional coding and non-coding sequences may include, but are not limited to, sequences encoding additional polypeptide tags (e.g., encoded in-frame with the ichthyosis-associated polypeptide in order to produce a fusion protein), introns (e.g., native, modified, or heterologous introns), 5' and/or 3' UTRs (e.g., native, modified, or heterologous 5' and/or 3' UTRs), and the like. Examples of suitable polypeptide tags may include, but are not limited, to any combination of purification tags, such as his-tags, flag-tags, maltose binding protein and glutathione-S-transferase tags, detection tags, such as tags that may be detected photometrically (e.g., green fluorescent protein, red fluorescent protein, etc.) and tags that have a detectable enzymatic activity (e.g., alkaline phosphatase, etc.), tags containing secretory sequences, signal sequences, leader sequences, and/or stabilizing sequences, protease cleavage sites (e.g., furin cleavage sites, TEV cleavage sites, Thrombin cleavage sites, etc.), and the like. In some embodiments, the 5' and/or 3'UTRs increase the stability, localization, and/or translational efficiency of the polynucleotides. In some embodiments, the 5' and/or 3'UTRs improve the level and/or duration of protein expression. In some embodiments, the 5' and/or 3'UTRs include elements (e.g., one or more miRNA binding sites, etc.) that may block or reduce off-target expression (e.g., inhibiting expression in specific cell types (e.g., neuronal cells), at specific times in the cell cycle, at specific developmental stages, etc.). In some embodiments, the 5' and/or 3'UTRs include elements (e.g., one or more miRNA binding sites, etc.) that may enhance effector protein expression in specific cell types (such as human keratinocytes and/or fibroblasts).

[0184] In some embodiments, a polynucleotide of the present disclosure is operably linked to one or more (e.g., one or more, two or more, three or more, four or more, five or more, ten or more, etc.) regulatory sequences. The term "regulatory sequence" may include enhancers, insulators, promoters, and other expression control elements (e.g., polyadenylation signals). Any suitable enhancer(s) known in the art may be used, including, for example, enhancer sequences from mammalian genes (such as globin, elastase, albumin, a-fetoprotein, insulin and the like), enhancer sequences from a eukaryotic cell virus (such as SV40 enhancer on the late side of the replication origin (bp 100-270), the cytomegalovirus early promoter enhancer, the polyoma enhancer on the late side of the replication origin, adenovirus enhancers, and the like), and any combinations thereof. Any suitable insulator(s) known in the art may be used, including, for example, herpes simplex virus (HSV) chromatin boundary (CTRL/CTCF-binding/insulator) elements CTRL1 and/or CTRL2, chicken hypersensitive site 4 insulator (cHS4), human HNRPA2B1-CBX3 ubiquitous chromatin opening element (UCOE), the scaffold/matrix attachment region (S/MAR) from the human interferon beta gene (IFNB1), and any combinations thereof. Any suitable promoter (e.g., suitable for transcription in mammalian host cells) known in the art may be used, including, for example, promoters obtained from the genomes of viruses (such as polyoma virus, fowlpox virus, adenovirus (such as Adenovirus 2), bovine papilloma virus, avian sarcoma virus, cytomegalovirus, a retrovirus, hepatitis-B virus, Simian Virus 40 (SV40), and the like), promoters from heterologous mammalian genes (such as the actin promoter (e.g., the β-actin promoter), a ubiquitin promoter (e.g., a ubiquitin C (UbC) promoter), a phosphoglycerate kinase (PGK) promoter, an immunoglobulin promoter, from heat-shock protein promoters, and the like), promoters from native and/or homologous mammalian genes, synthetic promoters (such as the CAGG promoter), and any combinations thereof, provided such promoters are compatible with the host cells. Regulatory sequences may include those which direct constitutive expression of a nucleic acid, as well as tissue-specific regulatory and/or inducible or repressible sequences.

[0185] In some embodiments, a polynucleotide of the present disclosure is operably linked to one or more heterologous promoters. In some embodiments, the one or more heterologous promoters are one or more of constitutive promoters, tissue-specific promoters, temporal promoters, spatial promoters, inducible promoters, and repressible promoters. In some embodiments, the one or more heterologous promoters are one or more of the human cytomegalovirus (HCMV) immediate early promoter, the human elongation factor-1 (EF1) promoter, the human β-actin promoter, the human UbC promoter, the human PGK promoter, the synthetic CAGG promoter, and any combinations thereof. In some embodiments, a polynucleotide of the present disclosure is operably linked to an HCMV promoter.

[0186] In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of an ichthyosis-associated gene (i.e., encoding an ichthyosis-associated polypeptide) does not comprise the coding sequence of a transglutaminase gene (i.e., does not encode a transglutaminase polypeptide), such as a TGM1 gene or a TGM5 gene. In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of a human transglutaminase gene (i.e., does not encode a human transglutaminase polypeptide), such as a human TGM1 gene or a human TGM5 gene. In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of a filaggrin gene or filaggrin 2 gene (i.e., does not encode a filaggrin or filaggrin 2 polypeptide), such as a human FLG gene or a human FLG2 gene. In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of a keratin gene (i.e., does not encode a keratin polypeptide), such as a KRT1, KRT2, KRT9, KRT10, and/or KRT17 gene. In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of a human keratin gene (i.e., does not encode a human keratin polypeptide), such as a human KRT1 gene, a human KRT2 gene, a human KRT9 gene, a human KRT10 gene, and/or a human KRT17 gene.

[0187] In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a Collagen alpha-1 (VII) chain polypeptide (COL7). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a Lysyl hydroxylase 3 polypeptide (LH3). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comparing the coding sequence of (e.g., a transgene encoding) a Keratin type I cytoskeletal 17 polypeptide (KRT17). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a transglutaminase (TGM) polypeptide (e.g., a human transglutaminase polypeptide such as a human TGM1 polypeptide and/or a human TGM5 polypeptide). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a cosmetic protein (e.g., collagen proteins, fibronectins, elastins, lumicans, vitronectins/vitronectin receptors, laminins, neuromodulators, fibrillins, additional dermal extracellular matrix proteins, etc.). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) an antibody (e.g., a full-length antibody, an antibody fragments, etc.). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a serine protease inhibitor kazal-type (SPINK) polypeptide (e.g., a human SPINK polypeptide such as a human SPINK5 polypeptide). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a laminin polypeptide (e.g., a human laminin polypeptide such as a human LAMA3, LAMB3, and/or LAMC2 polypeptide). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a cystic fibrosis transmembrane conductance regulator (CFTR) polypeptide (e.g., a human CFTR polypeptide). In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a Collagen alpha-1 (VII) chain polypeptide, a Lysyl hydroxylase 3 polypeptide, a Keratin type I cytoskeletal 17 polypeptide, and/or any chimeric polypeptides thereof. In some embodiments, a recombinant nucleic acid of the present disclosure does not comprise a polynucleotide comprising the coding sequence of (e.g., a transgene encoding) a Collagen alpha-1 (VII) chain polypeptide, a Lysyl hydroxylase 3 polypeptide, a Keratin type I cytoskeletal 17 polypeptide, a transglutaminase (TGM) polypeptide, a filaggrin polypeptide, a SPINK polypeptide, a CFTR polypeptide, a cosmetic protein, an antibody, and/or any chimeric polypeptides thereof.

Ichthyosis-Associated Polypeptides

[0188] In some embodiments, the present disclosure relates to one or more polynucleotides encoding a full-length ichthyosis-associated polypeptide, or any portions thereof (e.g., functional fragments). Ichthyosis-associated polypeptides may be encoded by any of the ichthyosis-associated genes described herein. Any suitable ichthyosis-associated polypeptide known in the art may be encoded by a polynucleotide of the present disclosure, including, for example, an ATP-binding cassette sub-family A member 12 polypeptide (such as a human ATP-binding cassette sub-family A member 12 polypeptide, e.g., as disclosed by UniProt accession number: Q86UK0), a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide (such as a human 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide, e.g., as disclosed by UniProt accession number: Q8WTS1), an Aldehyde dehydrogenase family 3 member A2 polypeptide (such as a human Aldehyde dehydrogenase family 3 member A2 polypeptide, e.g., as disclosed by UniProt accession number: P51648), an Arachidonate 12-lipoxygenase 12R-type polypeptide (such as a human Arachidonate 12-lipoxygenase 12R-type polypeptide, e.g., as disclosed by UniProt accession number: O75342), a Hydroperoxide isomerase ALOXE3 polypeptide (such as a human Hydroperoxide isomerase ALOXE3 polypeptide, e.g., as disclosed by UniProt accession number: Q9BYJ1), an AP-1 complex subunit sigma-1A polypeptide (such as a human AP-1 complex subunit sigma-1A polypeptide, e.g., as disclosed by UniProt accession number: P61966), an Arylsulfatase E polypeptide (such as a human Arylsulfatase E polypeptide, e.g., as disclosed by UniProt accession number: P51690), a Caspase-14 polypeptide (such as a human Caspase-14 polypeptide, e.g., as disclosed by UniProt accession number: P31944), a Corneodesmosin polypeptide (such as a human Corneodesmosin polypeptide, e.g., as disclosed by UniProt accession number: Q15517), a Ceramide synthase 3 polypeptide (such as a human Ceramide synthase 3 polypeptide, e.g., as disclosed by UniProt accession number: Q8IU89), a Carbohydrate sulfotransferase 8 polypeptide (such as a human Carbohydrate sulfotransferase 8 polypeptide, e.g., as disclosed by UniProt accession number: Q9H2A9), a Claudin-1 polypeptide (such as a human Claudin-1 polypeptide, e.g., as disclosed by UniProt accession number: O95832), a Cystatin-A polypeptide (such as a human Cystatin-A polypeptide, e.g., as disclosed by UniProt accession number: P01040), a Cytochrome P450 4F22 polypeptide (such as a human Cytochrome P450 4F22 polypeptide, e.g., as disclosed by

UniProt accession number: Q6NT55), a 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide (such as a human 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide, e.g., as disclosed by UniProt accession number: Q15125), an Elongation of very long chain fatty acids protein 4 polypeptide (such as a human Elongation of very long chain fatty acids protein 4 polypeptide, e.g., as disclosed by UniProt accession number: Q9GZR5), a Filaggrin polypeptide (such as a human Filaggrin polypeptide, e.g., as disclosed by UniProt accession number: P20930), a Filaggrin-2 polypeptide (such as a human Filaggrin-2 polypeptide, e.g., as disclosed by UniProt accession number: Q5D862), a Gap junction beta-2 polypeptide (such as a human Gap junction beta-2 polypeptide, e.g., as disclosed by UniProt accession number: P29033), a Gap junction beta-3 polypeptide (such as a human Gap junction beta-3 polypeptide, e.g., as disclosed by UniProt accession number: O75712), a Gap junction beta-4 polypeptide (such as a human Gap junction beta-4 polypeptide, e.g., as disclosed by UniProt accession number: Q9NTQ9), a Gap junction beta-6 polypeptide (such as a human Gap junction beta-6 polypeptide, e.g., as disclosed by UniProt accession number: O95452), a 3-ketodihydrosphingosine reductase polypeptide (such as a human 3-ketodihydrosphingosine reductase polypeptide, e.g., as disclosed by UniProt accession number: Q06136), a Keratin, type II cytoskeletal 1 polypeptide (such as a human Keratin, type II cytoskeletal 1 polypeptide, e.g., as disclosed by UniProt accession number: P04264), a Keratin, type II cytoskeletal 2 epidermal polypeptide (such as a human Keratin, type II cytoskeletal 2 epidermal polypeptide, e.g., as disclosed by UniProt accession number: P35908), a Keratin, type I cytoskeletal 9 polypeptide (such as a human Keratin, type I cytoskeletal 9 polypeptide, e.g., as disclosed by UniProt accession number: P35527), a Keratin, type I cytoskeletal 10 polypeptide (such as a human Keratin, type I cytoskeletal 10 polypeptide, e.g., as disclosed by UniProt accession number: P13645), a Lipase member N polypeptide (such as a human Lipase member N polypeptide, e.g., as disclosed by UniProt accession number: Q5VXI9), a Loricrin polypeptide (such as a human Loricrin polypeptide, e.g., as disclosed by UniProt accession number: P23490), a Membrane-bound transcription factor site-2 protease polypeptide (such as a human Membrane-bound transcription factor site-2 protease polypeptide, e.g., as disclosed by UniProt accession number: O43462), a Magnesium transporter NIPA4 polypeptide (such as a human Magnesium transporter NIPA4 polypeptide, e.g., as disclosed by UniProt accession number: Q0D2K0), a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide (such as a human Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide, e.g., as disclosed by UniProt accession number: Q15738), a Peroxisomal targeting signal 2 receptor polypeptide (such as a human Peroxisomal targeting signal 2 receptor polypeptide, e.g., as disclosed by UniProt accession number: O00628), a D-3-phosphoglycerate dehydrogenase polypeptide (such as a human Peroxisomal targeting signal 2 receptor polypeptide, e.g., as disclosed by UniProt accession number: O43175), a Phytanoyl-CoA dioxygenase, peroxisomal polypeptide (such as a human Phytanoyl-CoA dioxygenase, peroxisomal polypeptide, e.g., as disclosed by UniProt accession number: O14832), a Patatin-like phospholipase domain-containing protein 1 polypeptide (such as a human Patatin-like phospholipase domain-containing protein 1 polypeptide, e.g., as disclosed by UniProt accession number: Q8N8W4), a Proteasome maturation protein polypeptide (such as a human Proteasome maturation protein polypeptide, e.g., as disclosed by UniProt accession number: Q9Y244), a Phosphoserine aminotransferase polypeptide (such as a human Phosphoserine aminotransferase polypeptide, e.g., as disclosed by UniProt accession number: Q9Y617), a Short-chain dehydrogenase/reductase family 9C member 7 polypeptide (such as a human Short-chain dehydrogenase/reductase family 9C member 7 polypeptide, e.g., as disclosed by UniProt accession number: Q8NEX9), a Serpin B8 polypeptide (such as a human Serpin B8 polypeptide, e.g., as disclosed by UniProt accession number: P50452), a Long-chain fatty acid transport protein 4 polypeptide (such as a human Long-chain fatty acid transport protein 4 polypeptide, e.g., as disclosed by UniProt accession number: Q6P1MO), a Synaptosomal-associated protein 29 polypeptide (such as a human Synaptosomal-associated protein 29 polypeptide, e.g., as disclosed by UniProt accession number: O95721), a Suppressor of tumorigenicity 14 protein polypeptide (such as a human Suppressor of tumorigenicity 14 protein polypeptide, e.g., as disclosed by UniProt accession number: Q9Y5Y6), a Steryl-sulfatase polypeptide (such as a human Steryl-sulfatase polypeptide, e.g., as disclosed by UniProt accession number: P08842), a Sulfotransferase 2B1 polypeptide (such as a human Sulfotransferase 2B1 polypeptide, e.g., as disclosed by UniProt accession number: O00204), a Vacuolar protein sorting-associated protein 33B polypeptide (such as a human Vacuolar protein sorting-associated protein 33B polypeptide, e.g., as disclosed by UniProt accession number: Q9H267), a CAAX prenyl protease 1 homolog polypeptide (such as a human CAAX prenyl protease 1 homolog polypeptide, e.g., as disclosed by UniProt accession number: O75844), etc. Methods of identifying ichthyosis-associated polypeptide homologs/orthologs from additional species are known to one of ordinary skill in the art, including, for example, using an amino acid sequence alignment program such as the BLAST® blastp suite or OrthoDB. In some embodiments, an ichthyosis-associated polypeptide of the preset disclosure comprises a sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to the sequence of any of the ichthyosis-associated polypeptides described herein or known in the art.

[0189] In some embodiments, a polynucleotide of the present disclosure encodes a human ATP-binding cassette sub-family A member 12 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ABCA12 gene as described herein. In some embodiments, a polynucleotide encoding an ATP-binding cassette sub-family A member 12 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 102 or SEQ ID NO: 103. In some embodiments, a polynucleotide encoding an ATP-binding cassette sub-family A member 12 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 102 or SEQ ID NO: 103.

[0190] In some embodiments, a polynucleotide encoding an ATP-binding cassette sub-family A member 12 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 102. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 600, at least 700, at least 800, at least 900, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, at least 2250, at least 2500, but fewer than 2595 consecutive amino acids of SEQ ID NO: 102.

[0191] In some embodiments, a polynucleotide encoding an ATP-binding cassette sub-family A member 12 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 103. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 600, at least 700, at least 800, at least 900, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, but fewer than 2277 consecutive amino acids of SEQ ID NO: 103.

[0192] In some embodiments, a polynucleotide of the present disclosure encodes a human 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ABHD5 gene as described herein. In some embodiments, a polynucleotide encoding a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 104. In some embodiments, a polynucleotide encoding a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 104.

[0193] In some embodiments, a polynucleotide encoding a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 104. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, but fewer than 349 consecutive amino acids of SEQ ID NO: 104.

[0194] In some embodiments, a polynucleotide of the present disclosure encodes a human Aldehyde dehydrogenase family 3 member A2 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ALDH3A2 gene as described herein. In some embodiments, a polynucleotide encoding an Aldehyde dehydrogenase family 3 member A2 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 105. In some embodiments, a polynucleotide encoding an Aldehyde dehydrogenase family 3 member A2 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 105.

[0195] In some embodiments, a polynucleotide encoding an Aldehyde dehydrogenase family 3 member A2 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 105. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, but fewer than 508 consecutive amino acids of SEQ ID NO: 105.

[0196] In some embodiments, a polynucleotide of the present disclosure encodes a human Arachidonate 12-lipoxygenase 12R-type polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ALOX12B gene as described herein. In some embodiments, a polynucleotide encoding an Arachidonate 12-lipoxygenase 12R-type polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 106. In some embodiments, a polynucleotide encoding an Arachidonate 12-lipoxygenase 12R-type polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 106.

[0197] In some embodiments, a polynucleotide encoding an Arachidonate 12-lipoxygenase 12R-type polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 106. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 600, at least 700, but fewer than 701 consecutive amino acids of SEQ ID NO: 106.

[0198] In some embodiments, a polynucleotide of the present disclosure encodes a human Hydroperoxide isomerase ALOXE3 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ALOXE3 gene as described herein. In some embodiments, a polynucleotide encoding a Hydroperoxide isomerase ALOXE3 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 107 or SEQ ID NO: 108. In some embodiments, a polynucleotide encoding a Hydroperoxide isomerase ALOXE3 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 107 or SEQ ID NO: 108.

[0199] In some embodiments, a polynucleotide encoding a Hydroperoxide isomerase ALOXE3 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 107. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 600, at least 700, but fewer than 711 consecutive amino acids of SEQ ID NO: 107.

[0200] In some embodiments, a polynucleotide encoding a Hydroperoxide isomerase ALOXE3 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 108. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 600, at least 700, at least 800, but fewer than 843 consecutive amino acids of SEQ ID NO: 108.

[0201] In some embodiments, a polynucleotide of the present disclosure encodes a human AP-1 complex subunit sigma-1A polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an AP1S1 gene as described herein. In some embodiments, a polynucleotide encoding an AP-1 complex subunit sigma-1A polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 109. In some embodiments, a polynucleotide encoding an AP-1 complex subunit sigma-1A polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 109.

[0202] In some embodiments, a polynucleotide encoding an AP-1 complex subunit sigma-1A polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 109. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, but fewer than 158 consecutive amino acids of SEQ ID NO: 109.

[0203] In some embodiments, a polynucleotide of the present disclosure encodes a human Arylsulfatase E polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ARSE gene as described herein. In some embodiments, a polynucleotide encoding an Arylsulfatase E polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 110. In some embodiments, a polynucleotide encoding an Arylsulfatase E polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 110.

[0204] In some embodiments, a polynucleotide encoding an Arylsulfatase E polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 110. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, but fewer than 589 consecutive amino acids of SEQ ID NO: 110.

[0205] In some embodiments, a polynucleotide of the present disclosure encodes a human Caspase-14 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CASP14 gene as described herein. In some embodiments, a polynucleotide encoding a Caspase-14 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 111. In some embodiments, a polynucleotide encoding a Caspase-14 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 111.

[0206] In some embodiments, a polynucleotide encoding a Caspase-14 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 111. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, but fewer than 242 consecutive amino acids of SEQ ID NO: 111.

[0207] In some embodiments, a polynucleotide of the present disclosure encodes a human Corneodesmosin polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CDSN gene as described herein. In some embodiments, a polynucleotide encoding a Corneodesmosin polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 112. In some embodiments, a polynucleotide encoding a Corneodesmosin polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 112.

[0208] In some embodiments, a polynucleotide encoding a Corneodesmosin polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 112. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, but fewer than 529 consecutive amino acids of SEQ ID NO: 112.

[0209] In some embodiments, a polynucleotide of the present disclosure encodes a human Ceramide synthase 3 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CERS3 gene as described herein. In some embodiments, a polynucleotide encoding a Ceramide synthase 3 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 113. In some embodiments, a polynucleotide encoding a Ceramide synthase 3 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 113.

[0210] In some embodiments, a polynucleotide encoding a Ceramide synthase 3 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 113. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, but fewer than 383 consecutive amino acids of SEQ ID NO: 113.

[0211] In some embodiments, a polynucleotide of the present disclosure encodes a human Carbohydrate sulfotransferase 8 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CHST8 gene as described herein. In some embodiments, a polynucleotide encoding a Carbohydrate sulfotransferase 8 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 114. In some embodiments, a polynucleotide encoding a Carbohydrate sulfotransferase 8 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 114.

[0212] In some embodiments, a polynucleotide encoding a Carbohydrate sulfotransferase 8 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 114. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, but fewer than 424 consecutive amino acids of SEQ ID NO: 114.

[0213] In some embodiments, a polynucleotide of the present disclosure encodes a human Claudin-1 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CLDN1 gene as described herein. In some embodiments, a polynucleotide encoding a Claudin-1 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 115. In some embodiments, a polynucleotide encoding a Claudin-1 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 115.

[0214] In some embodiments, a polynucleotide encoding a Claudin-1 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 115. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, but fewer than 211 consecutive amino acids of SEQ ID NO: 115.

[0215] In some embodiments, a polynucleotide of the present disclosure encodes a human Cystatin-A polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CSTA gene as described herein. In some embodiments, a polynucleotide encoding a Cystatin-A polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 116. In some embodiments, a polynucleotide encoding a Cystatin-A polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 116.

[0216] In some embodiments, a polynucleotide encoding a Cystatin-A polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 116. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, but fewer than 98 consecutive amino acids of SEQ ID NO: 116.

[0217] In some embodiments, a polynucleotide of the present disclosure encodes a human Cytochrome P450 4F22 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a CYP4F22 gene as described herein. In some embodiments, a polynucleotide encoding a Cytochrome P450 4F22 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 117. In some embodiments, a polynucleotide encoding a Cytochrome P450 4F22 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 117.

[0218] In some embodiments, a polynucleotide encoding a Cytochrome P450 4F22 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 117. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, but fewer than 531 consecutive amino acids of SEQ ID NO: 117.

[0219] In some embodiments, a polynucleotide of the present disclosure encodes a human 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an EBP gene as described herein. In some embodiments, a polynucleotide encoding a 3-beta-hydroxysteroid-Delta(8), Delta(7)-isomerase polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least

91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 118. In some embodiments, a polynucleotide encoding a 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 118.

[0220] In some embodiments, a polynucleotide encoding a 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 118. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 225, but fewer than 230 consecutive amino acids of SEQ ID NO: 118.

[0221] In some embodiments, a polynucleotide of the present disclosure encodes a human Elongation of very long chain fatty acids protein 4 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an ELOVL4 gene as described herein. In some embodiments, a polynucleotide encoding an Elongation of very long chain fatty acids protein 4 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 119. In some embodiments, a polynucleotide encoding an Elongation of very long chain fatty acids protein 4 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 119.

[0222] In some embodiments, a polynucleotide encoding an Elongation of very long chain fatty acids protein 4 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 119. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, but fewer than 314 consecutive amino acids of SEQ ID NO: 119.

[0223] In some embodiments, a polynucleotide of the present disclosure encodes a human Filaggrin polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a FLG gene as described herein. In some embodiments, a polynucleotide encoding a Filaggrin polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 120. In some embodiments, a polynucleotide encoding a Filaggrin polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 120.

[0224] In some embodiments, a polynucleotide encoding a Filaggrin polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 120. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 1000, at least 1500, at least 2000, at least 2500, at least 3000, at least 3500, at least 4000, but fewer than 4061 consecutive amino acids of SEQ ID NO: 120.

[0225] In some embodiments, a polynucleotide of the present disclosure encodes a human Filaggrin 2 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a FLG2 gene as described herein. In some embodiments, a polynucleotide encoding a Filaggrin 2 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 121. In some embodiments, a polynucleotide encoding a Filaggrin 2 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 121.

[0226] In some embodiments, a polynucleotide encoding a Filaggrin 2 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 121. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 200, at least 300, at least 400, at least 500, at least 750, at least 1000, at least 1250, at least 1500, at least 1750, at least 2000, at least 2250, but fewer than 2391 consecutive amino acids of SEQ ID NO: 121.

[0227] In some embodiments, a polynucleotide of the present disclosure encodes a human Gap junction beta-2 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a GJB2 gene as described herein. In some embodiments, a polynucleotide encoding a Gap junction beta-2 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 122. In some embodiments, a polynucleotide encoding a Gap junction beta-2 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 122.

[0228] In some embodiments, a polynucleotide encoding a Gap junction beta-2 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 122. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, but fewer than 226 consecutive amino acids of SEQ ID NO: 122.

[0229] In some embodiments, a polynucleotide of the present disclosure encodes a human Gap junction beta-3 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a GJB3 gene as described herein. In some embodiments, a polynucleotide encoding a

Gap junction beta-3 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 123. In some embodiments, a polynucleotide encoding a Gap junction beta-3 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 123.

[0230] In some embodiments, a polynucleotide encoding a Gap junction beta-3 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 123. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, but fewer than 270 consecutive amino acids of SEQ ID NO: 123.

[0231] In some embodiments, a polynucleotide of the present disclosure encodes a human Gap junction beta-4 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a GJB4 gene as described herein. In some embodiments, a polynucleotide encoding a Gap junction beta-4 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 124. In some embodiments, a polynucleotide encoding a Gap junction beta-4 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 124.

[0232] In some embodiments, a polynucleotide encoding a Gap junction beta-4 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 124. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, but fewer than 266 consecutive amino acids of SEQ ID NO: 124.

[0233] In some embodiments, a polynucleotide of the present disclosure encodes a human Gap junction beta-6 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a GJB6 gene as described herein. In some embodiments, a polynucleotide encoding a Gap junction beta-6 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 125. In some embodiments, a polynucleotide encoding a Gap junction beta-6 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 125.

[0234] In some embodiments, a polynucleotide encoding a Gap junction beta-6 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 125. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, but fewer than 261 consecutive amino acids of SEQ ID NO: 125.

[0235] In some embodiments, a polynucleotide of the present disclosure encodes a human 3-ketodihydrosphingosine reductase polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a KDSR gene as described herein. In some embodiments, a polynucleotide encoding a 3-ketodihydrosphingosine reductase polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 126. In some embodiments, a polynucleotide encoding a 3-ketodihydrosphingosine reductase polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 126.

[0236] In some embodiments, a polynucleotide encoding a 3-ketodihydrosphingosine reductase polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 126. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, but fewer than 332 consecutive amino acids of SEQ ID NO: 126.

[0237] In some embodiments, a polynucleotide of the present disclosure encodes a human Keratin, type II cytoskeletal 1 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a KRT1 gene as described herein. In some embodiments, a polynucleotide encoding a Keratin, type II cytoskeletal 1 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 127. In some embodiments, a polynucleotide encoding a Keratin, type II cytoskeletal 1 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 127.

[0238] In some embodiments, a polynucleotide encoding a Keratin, type II cytoskeletal 1 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 127. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least about 300, at least about 400, at least about 500, at least about 600, but fewer than 644 consecutive amino acids of SEQ ID NO: 127.

[0239] In some embodiments, a polynucleotide of the present disclosure encodes a human Keratin, type II cytoskeletal 2 epidermal polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a KRT2 gene as described herein. In some embodiments, a polynucleotide encoding a Keratin, type II cytoskeletal 2 epidermal polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 128. In some embodiments, a polynucleotide encoding a Keratin, type II cytoskeletal 2 epidermal polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 128.

[0240] In some embodiments, a polynucleotide encoding a Keratin, type II cytoskeletal 2 epidermal polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 128. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 400, at least 500, at least 600, but fewer than 639 consecutive amino acids of SEQ ID NO: 128.

[0241] In some embodiments, a polynucleotide of the present disclosure encodes a human Keratin, type I cytoskeletal 9 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a KRT9 gene as described herein. In some embodiments, a polynucleotide encoding a Keratin, type I cytoskeletal 9 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 129. In some embodiments, a polynucleotide encoding a Keratin, type I cytoskeletal 9 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 129.

[0242] In some embodiments, a polynucleotide encoding a Keratin, type I cytoskeletal 9 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 129. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 400, at least 500, at least 600, but fewer than 623 consecutive amino acids of SEQ ID NO: 129.

[0243] In some embodiments, a polynucleotide of the present disclosure encodes a human Keratin, type I cytoskeletal 10 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a KRT10 gene as described herein. In some embodiments, a polynucleotide encoding a Keratin, type I cytoskeletal 10 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 130. In some embodiments, a polynucleotide encoding a Keratin, type I cytoskeletal 10 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 130.

[0244] In some embodiments, a polynucleotide encoding a Keratin, type I cytoskeletal 10 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 130. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 400, at least 500, but fewer than 584 consecutive amino acids of SEQ ID NO: 130.

[0245] In some embodiments, a polynucleotide of the present disclosure encodes a human Lipase member N polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a LIPN gene as described herein. In some embodiments, a polynucleotide encoding a Lipase member N polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 131. In some embodiments, a polynucleotide encoding a Lipase member N polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 131.

[0246] In some embodiments, a polynucleotide encoding a Lipase member N polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 131. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, but fewer than 398 consecutive amino acids of SEQ ID NO: 131.

[0247] In some embodiments, a polynucleotide of the present disclosure encodes a human Loricrin polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a LOR gene as described herein. In some embodiments, a polynucleotide encoding a Loricrin polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 132. In some embodiments, a polynucleotide encoding a Loricrin polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 132.

[0248] In some embodiments, a polynucleotide encoding a Loricrin polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 132. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, but fewer than 312 consecutive amino acids of SEQ ID NO: 132.

[0249] In some embodiments, a polynucleotide of the present disclosure encodes a human Membrane-bound transcription factor site-2 protease polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a MBTPS2 gene as described herein. In some embodiments, a polynucleotide encoding a Membrane-bound transcription factor site-2 protease polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 133. In some embodiments, a polynucleotide encoding a Membrane-bound transcription factor site-2 protease polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 133.

[0250] In some embodiments, a polynucleotide encoding a Membrane-bound transcription factor site-2 protease polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 133. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, but fewer than 519 consecutive amino acids of SEQ ID NO: 133.

[0251] In some embodiments, a polynucleotide of the present disclosure encodes a human Magnesium transporter NIPA4 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an NIPAL4 gene as described herein. In some embodiments, a polynucleotide encoding a Magnesium transporter NIPA4 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 134 or SEQ ID NO: 135. In some embodiments, a polynucleotide encoding a Magnesium transporter NIPA4 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 134 or SEQ ID NO: 135.

[0252] In some embodiments, a polynucleotide encoding a Magnesium transporter NIPA4 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 134. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, but fewer than 466 consecutive amino acids of SEQ ID NO: 134.

[0253] In some embodiments, a polynucleotide encoding a Magnesium transporter NIPA4 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 135. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, but fewer than 447 consecutive amino acids of SEQ ID NO: 135.

[0254] In some embodiments, a polynucleotide of the present disclosure encodes a human Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of an NSDHL gene as described herein. In some embodiments, a polynucleotide encoding a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 136. In some embodiments, a polynucleotide encoding a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 136.

[0255] In some embodiments, a polynucleotide encoding a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 136. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, but fewer than 373 consecutive amino acids of SEQ ID NO: 136.

[0256] In some embodiments, a polynucleotide of the present disclosure encodes a human Peroxisomal targeting signal 2 receptor polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a PEX7 gene as described herein. In some embodiments, a polynucleotide encoding a Peroxisomal targeting signal 2 receptor polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 137. In some embodiments, a polynucleotide encoding a Peroxisomal targeting signal 2 receptor polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 137.

[0257] In some embodiments, a polynucleotide encoding a Peroxisomal targeting signal 2 receptor polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 137. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, but fewer than 323 consecutive amino acids of SEQ ID NO: 137.

[0258] In some embodiments, a polynucleotide of the present disclosure encodes a human D-3-phosphoglycerate dehydrogenase polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a PHGDH gene as described herein. In some embodiments, a polynucleotide encoding a D-3-phosphoglycerate dehydrogenase polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 138. In some embodiments, a polynucleotide encoding a D-3-phosphoglycerate dehydrogenase polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 138.

[0259] In some embodiments, a polynucleotide encoding a D-3-phosphoglycerate dehydrogenase polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 138. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, but fewer than 533 consecutive amino acids of SEQ ID NO: 138.

[0260] In some embodiments, a polynucleotide of the present disclosure encodes a human Phytanoyl-CoA dioxygenase, peroxisomal polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a PHYH gene as described herein. In some embodiments, a polynucleotide encoding a Phytanoyl-CoA dioxygenase, peroxisomal polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 139. In some embodiments, a polynucleotide encoding a Phytanoyl-CoA dioxygenase, peroxisomal polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 139.

[0261] In some embodiments, a polynucleotide encoding a Phytanoyl-CoA dioxygenase, peroxisomal polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 139. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, but fewer than 338 consecutive amino acids of SEQ ID NO: 139.

[0262] In some embodiments, a polynucleotide of the present disclosure encodes a human Patatin-like phospholipase domain-containing protein 1 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a PNPLA1 gene as described herein. In some embodiments, a polynucleotide encoding a Patatin-like phospholipase domain-containing protein 1 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to a sequence selected from SEQ ID NOS: 140-142. In some embodiments, a polynucleotide encoding a Patatin-like phospholipase domain-containing protein 1 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence selected from SEQ ID NOS: 140-142.

[0263] In some embodiments, a polynucleotide encoding a Patatin-like phospholipase domain-containing protein 1 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 140. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, but fewer than 532 consecutive amino acids of SEQ ID NO: 140.

[0264] In some embodiments, a polynucleotide encoding a Patatin-like phospholipase domain-containing protein 1 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 141. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, but fewer than 437 consecutive amino acids of SEQ ID NO: 141.

[0265] In some embodiments, a polynucleotide encoding a Patatin-like phospholipase domain-containing protein 1 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 142. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, but fewer than 446 consecutive amino acids of SEQ ID NO: 142.

[0266] In some embodiments, a polynucleotide of the present disclosure encodes a human Proteasome maturation protein polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a POMP gene as described herein. In some embodiments, a polynucleotide encoding a Proteasome maturation protein polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 143. In some embodiments, a polynucleotide encoding a Proteasome maturation protein polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 143.

[0267] In some embodiments, a polynucleotide encoding a Proteasome maturation protein polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 143. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 125, but fewer than 141 consecutive amino acids of SEQ ID NO: 143.

[0268] In some embodiments, a polynucleotide of the present disclosure encodes a human Phosphoserine aminotransferase polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a PSAT1 gene as described herein. In some embodiments, a polynucleotide encoding a Phosphoserine aminotransferase polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 144. In some embodiments, a polynucleotide encoding a Phosphoserine aminotransferase polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 144.

[0269] In some embodiments, a polynucleotide encoding a Phosphoserine aminotransferase polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 144. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, but fewer than 370 consecutive amino acids of SEQ ID NO: 144.

[0270] In some embodiments, a polynucleotide of the present disclosure encodes a human Short-chain dehydrogenase/reductase family 9C member 7 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a SDR9C7 gene as described herein. In some embodiments, a polynucleotide encoding a Short-chain dehydrogenase/reductase family 9C member 7 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 145. In some embodiments, a polynucleotide encoding a Short-chain dehydrogenase/reductase family 9C member 7 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 145.

[0271] In some embodiments, a polynucleotide encoding a Short-chain dehydrogenase/reductase family 9C member 7 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 145. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, but fewer than 313 consecutive amino acids of SEQ ID NO: 145.

[0272] In some embodiments, a polynucleotide of the present disclosure encodes a human Serpin B8 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a SERPINB8 gene as described herein. In some embodiments, a polynucleotide encoding a Serpin B8 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 146. In some embodiments, a polynucleotide encoding a Serpin B8 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 146.

[0273] In some embodiments, a polynucleotide encoding a Serpin B8 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 146. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, but fewer than 374 consecutive amino acids of SEQ ID NO: 146.

[0274] In some embodiments, a polynucleotide of the present disclosure encodes a human Long-chain fatty acid transport protein 4 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a SLC27A4 gene as described herein. In some embodiments, a polynucleotide encoding a Long-chain fatty acid transport protein 4 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 147. In some embodiments, a polynucleotide encoding a Long-chain fatty acid transport protein 4 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 147.

[0275] In some embodiments, a polynucleotide encoding a Long-chain fatty acid transport protein 4 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 147. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, at least 550, at least 600, but fewer than 643 consecutive amino acids of SEQ ID NO: 147.

[0276] In some embodiments, a polynucleotide of the present disclosure encodes a human Synaptosomal-associated protein 29 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a SNAP29 gene as described herein. In some embodiments, a polynucleotide encoding a Synaptosomal-associated protein 29 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 148. In some embodiments, a polynucleotide encoding a Synaptosomal-associated protein 29 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 148.

[0277] In some embodiments, a polynucleotide encoding a Synaptosomal-associated protein 29 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 148. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 125, at least 150, at least 175, at least 200, at least 225, at least 250, but fewer than 258 consecutive amino acids of SEQ ID NO: 148.

[0278] In some embodiments, a polynucleotide of the present disclosure encodes a human Suppressor of tumorigenicity 14 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a ST14 gene as described herein. In some embodiments, a polynucleotide encoding a Suppressor of tumorigenicity 14 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 149. In some embodiments, a polynucleotide encoding a Suppressor of tumorigenicity 14 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 149.

[0279] In some embodiments, a polynucleotide encoding a Suppressor of tumorigenicity 14 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 149. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, at least 550, at least 600, at least 650, at least 700, at least 750, at least 800, at least 850, but fewer than 855 consecutive amino acids of SEQ ID NO: 149.

[0280] In some embodiments, a polynucleotide of the present disclosure encodes a human Steryl-sulfatase polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a STS gene as described herein. In some embodiments, a polynucleotide encoding a Steryl-sulfatase polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 150. In some embodiments, a polynucleotide encoding a Steryl-sulfatase polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 150.

[0281] In some embodiments, a polynucleotide encoding a Steryl-sulfatase polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 150. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, at least 550, but fewer than 583 consecutive amino acids of SEQ ID NO: 150.

[0282] In some embodiments, a polynucleotide of the present disclosure encodes a human Sulfotransferase 2B1 polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a SULT2B1 gene as described herein. In some embodiments, a polynucleotide encoding a Sulfotransferase 2B1 polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 155. In some embodiments, a polynucleotide encoding a Sulfotransferase 2B1 polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 155.

[0283] In some embodiments, a polynucleotide encoding a Sulfotransferase 2B1 polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 155. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, but fewer than 365 consecutive amino acids of SEQ ID NO: 155.

[0284] In some embodiments, a polynucleotide of the present disclosure encodes a human Vacuolar protein sorting-associated protein 33B polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a VPS33B gene as described herein. In some embodiments, a polynucleotide encoding a Vacuolar protein sorting-associated protein 33B polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 151. In some embodiments, a polynucleotide encoding a Vacuolar protein sorting-associated protein 33B polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 151.

[0285] In some embodiments, a polynucleotide encoding a Vacuolar protein sorting-associated protein 33B polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 151. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, at least 500, at least 550, at least 600, but fewer than 617 consecutive amino acids of SEQ ID NO: 151.

[0286] In some embodiments, a polynucleotide of the present disclosure encodes a human CAAX prenyl protease 1 homolog polypeptide. In some embodiments, the polynucleotide comprises the coding sequence of a ZMPSTE24 gene as described herein. In some embodiments, a polynucleotide encoding a CAAX prenyl protease 1 homolog polypeptide is a polynucleotide that encodes a polypeptide comprising an amino acid sequence having at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to the sequence of SEQ ID NO: 152. In some embodiments, a polynucleotide encoding a CAAX prenyl protease 1 homolog polypeptide is a polynucleotide that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 152.

[0287] In some embodiments, a polynucleotide encoding a CAAX prenyl protease 1 homolog polypeptide is a polynucleotide that encodes an N-terminal truncation, a C-terminal truncation, or a fragment of the amino acid sequence of SEQ ID NO: 152. N-terminal truncations, C-terminal truncations, or fragments may comprise at least 10, at least 12, at least 14, at least 16, at least 18, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, at least 450, but fewer than 475 consecutive amino acids of SEQ ID NO: 152.

[0288] In some embodiments, a polynucleotide of the present disclosure encoding an ichthyosis-associated polypeptide (e.g., a human ichthyosis-associated polypeptide) expresses the ichthyosis-associated polypeptide when the polynucleotide is delivered into one or more target cells of a subject (e.g., one or more cells of the epidermis of the subject). In some embodiments, expression of the ichthyosis-associated polypeptide (e.g., a human ichthyosis-associated polypeptide) enhances, increases, augments, and/or supplements the levels, function, and/or activity of the ichthyosis-associated polypeptide in one or more target cells of a subject (e.g., as compared to prior to expression of the ichthyosis-associated polypeptide). In some embodiments, expression of the ichthyosis-associated polypeptide (e.g., a human ichthyosis-associated polypeptide) provides prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of a congenital ichthyosis in the subject (e.g., as compared to prior to expression of the ichthyosis-associated polypeptide).

Recombinant Nucleic Acids

[0289] In some embodiments, the present disclosure relates to recombinant nucleic acids comprising any one or more of the polynucleotides described herein. In some embodiments, the recombinant nucleic acid is a vector (e.g., an expression vector, a display vector, etc.). In some embodiments, the vector is a DNA vector or an RNA vector. Generally, vectors suitable to maintain, propagate, and/or express polynucleotides to produce one or more polypeptides in a subject may be used. Examples of suitable vectors may include, for example, plasmids, cosmids, episomes, transposons, and viral vectors (e.g., adenoviral vectors, adeno-associated viral vectors, vaccinia viral vectors, Sindbis-viral vectors, measles vectors, herpes viral vectors, lentiviral vectors, retroviral vectors, etc.). In some embodiments, the vector is a herpes viral vector. In some embodiments, the vector is capable of autonomous replication in a host cell. In some embodiments, the vector is incapable of autonomous replication in a host cell. In some embodiments, the vector can integrate into a host DNA. In some embodiments, the vector cannot integrate into a host DNA (e.g., is episomal). Methods of making vectors containing one or more polynucleotides of interest are well known to one of ordinary skill in the art, including, for example, by chemical synthesis or by artificial manipulation of isolated segments of nucleic acids (e.g., by genetic engineering techniques).

[0290] In some embodiments, a recombinant nucleic acid of the present disclosure is a herpes simplex virus (HSV) amplicon. Herpes virus amplicons, including the structural features and methods of making the same, are generally known to one of ordinary skill in the art (see e.g., de Silva S. and Bowers W. "Herpes Virus Amplicon Vectors". Viruses 2009, 1, 594-629). In some embodiments, the herpes simplex virus amplicon is an HSV-1 amplicon. In some embodiments, the herpes simplex virus amplicon is an HSV-1 hybrid amplicon. Examples of HSV-1 hybrid amplicons may include, but are not limited to, HSV/AAV hybrid amplicons, HSV/EBV hybrid amplicons, HSV/EBV/RV hybrid amplicons, and/or HSV/Sleeping Beauty hybrid amplicons. In some embodiments, the amplicon is an HSV/AAV hybrid amplicon. In some embodiments, the amplicon is an HSV/Sleeping Beauty hybrid amplicon.

[0291] In some embodiments, a recombinant nucleic acid of the present disclosure is a recombinant herpes virus genome. The recombinant herpes virus genome may be a recombinant genome from any member of the Herpesviridae family of DNA viruses known in the art, including, for example, a recombinant herpes simplex virus genome, a recombinant varicella zoster virus genome, a recombinant human cytomegalovirus genome, a recombinant herpesvirus 6A genome, a recombinant herpesvirus 6B genome, a recombinant herpesvirus 7 genome, a recombinant Kaposi's sarcoma-associated herpesvirus genome, and any combinations or any derivatives thereof. In some embodiments, the recombinant herpes virus genome comprises one or more (e.g., one or more, two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, ten or more, etc.) inactivating mutations. In some embodiments, the one or more inactivating mutations are in one or more (e.g., one or more, two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, ten or more, etc.) herpes virus genes. In some embodiments, the recombinant herpes virus genome is attenuated (e.g., as compared to a corresponding, wild-type herpes virus genome). In some embodiments, the recombinant herpes virus genome is replication competent. In some embodiments, the recombinant herpes virus genome is replication defective.

[0292] In some embodiments, the recombinant nucleic acid is a recombinant herpes simplex virus (HSV) genome. In some embodiments, the recombinant herpes simplex virus genome is a recombinant herpes simplex virus type 1 (HSV-1) genome, a recombinant herpes simplex virus type 2 (HSV-2) genome, or any derivatives thereof. In some embodiments, the recombinant herpes simplex virus genome is a recombinant HSV-1 genome. In some embodiments, the recombinant HSV-1 genome may be from any HSV-1 strain known in the art, including, for example, strains 17, Ty25, R62, S25, Ku86, S23, R11, Ty148, Ku47, $H166_{syn}$, 1319-2005, F-13, M-12, 90237, F-17, KOS, 3083-2008, F12g, L2, CD38, H193, M-15, India 2011, 0116209, F-11I, 66-207, 2762, 369-2007, 3355, MacIntyre, McKrae, 7862, 7-hse, HF10, 1394,2005, 270-2007, OD4, SC16, M-19, 4J1037, 5J1060, J1060, KOS79, 132-1988, 160-1982, H166, 2158-2007, RE, 78326, F18g, F11, 172-2010, H129, F, E4, CJ994, F14g, E03, E22, E10, E06, E11, E25, E23, E35, E15, E07, E12, E14, E08, E19, E13, ATCC 2011, etc. (see e.g., Bowen et al. J Virol. 2019 Apr. 3; 93 (8)). In some embodiments, the recombinant HSV-1 genome is from the KOS strain. In some embodiments, the recombinant HSV-1 genome is not from the McKrae strain. In some embodiments, the recombinant herpes simplex virus genome is attenuated. In some embodiments, the recombinant herpes simplex virus genome is replication competent. In some embodiments, the recombinant herpes simplex virus genome is replication defective. In some embodiments, the recombinant herpes simplex virus genome comprises one or more (e.g., one or more, two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, ten or more, etc.) inactivating mutations. In some embodiments, the one or more inactivating mutations are in one or more (e.g., one or more, two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, ten or more, etc.) herpes simplex virus genes. As used herein, an "inactivating mutation" may refer to any mutation that results in a gene or regulon product (RNA or protein) having reduced, undetectable, or eliminated quantity and/or function (e.g., as compared to a corresponding sequence lacking the inactivating mutation). Examples of inactivating mutations may include, but are not limited to, deletions, insertions, point mutations, and rearrangements in transcriptional control sequences (promoters, enhancers, insulators, etc.) and/or coding sequences of a given gene or regulon. Any suitable method of measuring the quantity of a gene or regulon product known in the art may be used, including, for example, qPCR, Northern blots, RNAseq, western blots, ELISAs, etc.

[0293] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in at least one, at least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the Infected Cell Protein (or Infected Cell Polypeptide) (ICP) 0, ICP4, ICP22, ICP27, ICP47, thymidine kinase (tk), Long Unique Region (UL) 41 and/or UL55 herpes simplex virus genes. In some embodiments, the recombinant herpes simplex virus genome does not comprise an inactivating mutation in the ICP34.5 (one or both copies) and/or ICP47 herpes simplex virus genes (e.g., to avoid production of an immune-stimulating virus). In some embodiments, the recombinant herpes simplex virus genome does not comprise an inactivating mutation in the ICP34.5 (one or both copies) herpes simplex virus gene. In some embodiments, the recombinant herpes simplex virus genome does not comprise an inactivating mutation in the ICP47 herpes simplex virus gene. In some embodiments, the recombinant herpes simplex virus genome does not comprise an inactivating mutation in the ICP34.5 (one or both copies) and ICP47 herpes simplex virus genes. In some embodiments, the recombinant herpes simplex virus genome is not oncolytic.

[0294] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies). In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), and further comprises an inactivating mutation in the ICP4 (one or both copies), ICP22, ICP27, ICP47, UL41, and/or UL55 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), and an inactivating mutation in the ICP4 gene (one or both copies). In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), and an inactivating mutation in the ICP22 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), and an inactivating mutation in the UL41 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), an inactivating mutation in the ICP4 gene (one or both copies), and an inactivating mutation in the ICP22 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), an inactivating mutation in the ICP4 gene (one or both copies), and an inactivating mutation in the UL41 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), an inactivating mutation in the ICP22 gene, and an inactivating mutation in the UL41 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 gene (one or both copies), an inactivating mutation in the ICP4 gene (one or both copies), an inactivating mutation in the ICP22 gene, and an inactivating mutation in the UL41 gene. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, and/or UL41 genes. In some embodiments, the recombinant herpes simplex virus genome further comprises an inactivating mutation in the ICP27, ICP47, and/or UL55 genes.

[0295] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 gene (one or both copies). In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 gene (one or both copies), and further comprises an inactivating mutation in the ICP0 (one or both copies), ICP22, ICP27, ICP47, UL41, and/or UL55 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 gene (one or both copies), and an inactivating mutation in the ICP22 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 gene (one or both copies), and an inactivating mutation in the UL41 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 gene (one or both copies), an inactivating mutation in the ICP22 gene, and an inactivating mutation in the UL41 gene. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the ICP4 (one or both copies), ICP22, and/or UL41 genes. In some embodiments, the recombinant herpes simplex virus genome further comprises an inactivating mutation in the ICP0 (one or both copies), ICP27, ICP47, and/or UL55 genes.

[0296] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP22 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP22 gene, and further comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP27, ICP47, UL41, and/or UL55 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the

ICP22 gene, and an inactivating mutation UL41 gene. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the ICP22 and/or UL41 genes. In some embodiments, the recombinant herpes simplex virus genome further comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP27, ICP47, and/or UL55 genes.

[0297] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP27 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP27 gene, and further comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP47, UL41, and/or UL55 genes. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the ICP27 gene.

[0298] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP47 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP47 gene, and further comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27, UL41, and/or UL55 genes. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the ICP47 gene.

[0299] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL41 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL41 gene, and further comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27, ICP47, and/or UL55 genes. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the UL41 gene.

[0300] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL55 gene. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL55 gene, and further comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27, ICP47, and/or UL41 genes. In some embodiments, the inactivating mutation is a deletion of the coding sequence of the UL55 gene.

[0301] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in (e.g., a deletion of) the internal repeat (Joint) region comprising the internal repeat long (IR_L) and internal repeat short (IR_S) regions. In some embodiments, inactivation (e.g., deletion) of the Joint region eliminates one copy each of the ICP4 and ICP0 genes. In some embodiments, inactivation (e.g., deletion) of the Joint region further inactivates (e.g., deletes) the promoter for the ICP22 and ICP47 genes. If desired, expression of one or both of these genes can be restored by insertion of an immediate early promoter into the recombinant herpes simplex virus genome (see e.g., Hill et al. (1995). Nature 375 (6530): 411-415; Goldsmith et al. (1998). J Exp Med 187 (3): 341-348). Without wishing to be bound by theory, it is believed that inactivating (e.g., deleting) the Joint region may contribute to the stability of the recombinant herpes simplex virus genome and/or allow for the recombinant herpes simplex virus genome to accommodate more and/or larger transgenes.

[0302] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 (one or both copies), ICP22, and ICP27 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 (one or both copies), ICP27, and UL55 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP4 (one or both copies), ICP22, ICP27, ICP47, and UL55 genes. In some embodiments, the inactivating mutation in the ICP4 (one or both copies), ICP27, and/or UL55 genes is a deletion of the coding sequence of the ICP4 (one or both copies), ICP27, and/or UL55 genes. In some embodiments, the inactivating mutation in the ICP22 and ICP47 genes is a deletion in the promoter region of the ICP22 and ICP47 genes (e.g., the ICP22 and ICP47 coding sequences are intact but are not transcriptionally active). In some embodiments, the recombinant herpes simplex virus genome comprises a deletion in the coding sequence of the ICP4 (one or both copies), ICP27, and UL55 genes, and a deletion in the promoter region of the ICP22 and ICP47 genes. In some embodiments, the recombinant herpes simplex virus genome further comprises an inactivating mutation in the ICP0 (one or both copies) and/or UL41 genes.

[0303] In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 (one or both copies) and ICP4 (one or both copies) genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), and ICP22 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, and ICP27 genes. In some embodiments, the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27 and UL55 genes. In some embodiments, the inactivating mutation in the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27 and/or UL55 genes comprises a deletion of the coding sequence of the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27 and/or UL55 genes. In some embodiments, the recombinant herpes simplex virus genome further comprises an inactivating mutation in the ICP47 and/or the UL41 genes.

[0304] In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within one, two, three, four, five, six, seven or more viral gene loci. Examples of suitable viral loci may include, without limitation, the ICP0 (one or both copies), ICP4 (one or both copies), ICP22, ICP27, ICP47, tk, UL41 and/or UL55 herpes simplex viral gene loci. In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within one or both of the viral ICP4 gene loci (e.g., a recombinant virus comprising a polynucleotide encoding an ichthyosis-associated polypeptide in one or both of the ICP4 loci). In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within the viral ICP22 gene locus (e.g., a recombinant virus carrying a polynucleotide encoding an ichthyosis-associated polypeptide in the ICP22 locus). In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within the viral UL41 gene locus (e.g., a recombinant virus carrying a polynucleotide encoding an ichthyosis-associated polypeptide in the UL41 locus). In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within the viral ICP27 gene locus (e.g., a recombinant virus carrying a polynucleotide encoding an ichthyosis-associated polypeptide in the ICP27 locus). In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within the viral ICP47 gene locus (e.g., a recombinant virus carrying a polynucleotide encoding an ichthyosis-associated polypeptide in the ICP47 locus).

[0305] In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within one or both of the viral ICP4 gene loci, and one or more polynucleotides of the present disclosure within the viral ICP22 gene locus (e.g., a recombinant virus comprising a polynucleotide encoding a first ichthyosis-associated polypeptide in one or both of the ICP4 loci, and a polynucleotide encoding a second ichthyosis-associated polypeptide in the ICP22 locus; etc.). In some embodiments, the first and second ichthyosis-associated polypeptides are the same. In some embodiments, the first and second ichthyosis-associated polypeptides are different. In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within one or both of the viral ICP4 gene loci, and one or more polynucleotides of the present disclosure within the viral UL41 gene locus (e.g., a recombinant virus comprising a polynucleotide encoding a first ichthyosis-associated polypeptide in one or both of the ICP4 loci, and a polynucleotide encoding a second ichthyosis-associated polypeptide in the UL41 locus etc.). In some embodiments, the first and second ichthyosis-associated polypeptides are the same. In some embodiments, the first and second ichthyosis-associated polypeptides are different. In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within the viral UL41 gene locus, and one or more polynucleotides of the present disclosure within the viral ICP22 gene locus (e.g., a recombinant virus comprising a polynucleotide encoding a first ichthyosis-associated polypeptide in the UL41 locus, and a polynucleotide encoding second ichthyosis-associated polypeptide in the ICP22 locus; etc.). In some embodiments, the first and second ichthyosis-associated polypeptides are the same. In some embodiments, the first and second ichthyosis-associated polypeptides are different. In some embodiments, a recombinant herpes simplex virus genome comprises one or more polynucleotides of the present disclosure within one or both of the viral ICP4 gene loci, one or more polynucleotides of the present disclosure within the viral ICP22 gene locus, and one or more polynucleotides of the present disclosure within the viral UL41 gene locus (e.g., a recombinant virus comprising a polynucleotide encoding a first ichthyosis-associated polypeptide in one or both of the ICP4 loci, a polynucleotide encoding a second ichthyosis-associated polypeptide in the ICP22 locus, and a polynucleotide encoding a third ichthyosis-associated polypeptide in the UL41 locus; etc.). In some embodiments, the first, second, and/or third ichthyosis-associated polypeptides are the same. In some embodiments, the first, second, and/or third ichthyosis-associated polypeptides are different.

[0306] In some embodiments, the recombinant herpes virus genome (e.g., a recombinant herpes simplex virus genome) has been engineered to decrease or eliminate expression of one or more herpes virus genes (e.g., one or more toxic herpes virus genes), such as one or both copies of the HSV ICP0 gene, one or both copies of the HSV ICP4 gene, the HSV ICP22 gene, the HSV UL41 gene, the HSV ICP27 gene, etc. In some embodiments, the recombinant herpes virus genome (e.g., recombinant herpes simplex virus genome) has been engineered to reduce cytotoxicity of the recombinant genome (e.g., when introduced into a target cell) as compared to a corresponding wild-type herpes virus genome (e.g., a wild-type herpes simplex virus genome). In some embodiments, the target cell is a human cell. In some embodiments, the target cell is a cell of the epidermis and/or dermis (e.g., a cell of the human epidermis and/or dermis). In some embodiments, the target cell is a keratinocyte or fibroblast (e.g., a human keratinocyte or human fibroblast). In some embodiments, the target cell is a cell of the mucosa. In some embodiments, cytotoxicity (e.g., in human keratinocytes and/or fibroblast cells) of the recombinant herpes virus genome (e.g., a recombinant herpes simplex virus genome) is reduced by at least about 5%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, or at least about 99% as compared to a corresponding wild-type herpes virus genome (e.g., measuring the relative cytotoxicity of a recombinant ΔICP4 (one or both copies) herpes simplex virus genome vs. a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); measuring the relative cytotoxicity of a recombinant ΔICP4 (one or both copies)/ΔICP22 herpes simplex virus genome vs. a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); etc.). In some embodiments, cytotoxicity (e.g., in human keratinocytes and/or fibroblast cells) of the recombinant herpes virus genome (e.g., a recombinant herpes simplex virus genome) is reduced by at least about 1.5-fold, at least about 2-fold, at least about 3-fold, at least about 4-fold, at least about 5-fold, at least about 6-fold, at least about 7-fold, at least about 8-fold, at least about 9-fold, at least about 10-fold, at least about 15-fold, at least about 20-fold, at least about 25-fold, at least about 50-fold, at least about 75-fold, at least about 100-fold, at least about 250-fold, at least about 500-fold, at least about 750-fold, at least about 1000-fold, or more as compared to a corresponding wild-type herpes virus genome (e.g., measuring the relative cytotoxicity of a recombinant ΔICP4 (one or both copies) herpes simplex virus genome vs. a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); measuring the relative cytotoxicity of a recombinant ΔICP4 (one or both copies)/ΔICP22 herpes simplex virus genome vs. a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); etc.). Methods of measuring cytotoxicity are known to one of ordinary skill in the art, including, for example, through the use of vital dyes (formazan dyes), protease biomarkers, an MTT assay (or an assay using related tetrazolium salts such as XTT, MTS, water-soluble tetrazolium salts, etc.), measuring ATP content, etc.

[0307] In some embodiments, the recombinant herpes virus genome (e.g., a recombinant herpes simplex virus genome) has been engineered to reduce its impact on host cell proliferation after exposure of a target cell to the recombinant genome, as compared to a corresponding wild-type herpes virus genome (e.g., a wild-type herpes simplex virus genome). In some embodiments, the target cell is a human cell. In some embodiments, the target cell is a cell of the epidermis and/or dermis (e.g., a cell of the human epidermis and/or dermis). In some embodiments, the target cell is a keratinocyte or fibroblast (e.g., a human keratinocyte or human fibroblast. In some embodiments, host cell proliferation (e.g., of human keratinocytes and/or fibroblasts) after exposure to the recombinant genome is at least about 5%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, or at least about 99% faster as compared to host cell proliferation after exposure to a corresponding wild-type herpes virus genome (e.g., measuring the relative cellular proliferation after exposure to a recombinant ΔICP4 (one or both copies) herpes simplex virus genome vs. cellular proliferation after exposure to a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); measuring the relative cellular proliferation after exposure to a recombinant ΔICP4 (one or both copies)/ΔICP22 herpes simplex virus genome vs. cellular proliferation after exposure to a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); etc.). In some embodiments, host cell proliferation (e.g., of human keratinocytes and/or fibroblasts) after exposure to the recombinant genome is at least about 1.5-fold, at least about 2-fold, at least about 3-fold, at least about 4-fold, at least about 5-fold, at least about 6-fold, at least about 7-fold, at least about 8-fold, at least about 9-fold, at least about 10-fold, at least about 15-fold, at least about 20-fold, at least about 25-fold, at least about 50-fold, at least about 75-fold, at least about 100-fold, at least about 250-fold, at least about 500-fold, at least about 750-fold, or at least about 1000-fold faster as compared to host cell proliferation after exposure to a corresponding wild-type herpes virus genome (e.g., measuring the relative cellular proliferation after exposure to a recombinant ΔICP4 (one or both copies) herpes simplex virus genome vs. cellular proliferation after exposure to a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); measuring the relative cellular proliferation after exposure to a recombinant ΔICP4 (one or both copies)/ΔICP22 herpes simplex virus genome vs. cellular proliferation after exposure to a wild-type herpes simplex virus genome in human keratinocytes or fibroblasts (primary cells or cell lines); etc.). Methods of measuring cellular proliferation are known to one of ordinary skill in the art, including, for example, through the use of a Ki67 cell proliferation assay, a BrdU cell proliferation assay, etc.

[0308] A vector (e.g., herpes viral vector) may include one or more polynucleotides of the present disclosure in a form suitable for expression of the polynucleotide in a host cell. Vectors may include one or more regulatory sequences operatively linked to the polynucleotide to be expressed (e.g., as described above).

[0309] In some embodiments, a recombinant nucleic acid (e.g., a recombinant herpes virus genome, such as a recombinant herpes simplex virus genome) of the present disclosure comprises one or more of the polynucleotides described herein inserted in any orientation in the recombinant nucleic acid. If the recombinant nucleic acid comprises two or more polynucleotides described herein (e.g., two or more, three or more, etc.), the polynucleotides may be inserted in the same orientation or opposite orientations to one another. Without wishing to be bound be theory, incorporating two polynucleotides (e.g., two transgenes) into a recombinant nucleic acid (e.g., a vector) in an antisense orientation may help to avoid read-through and ensure proper expression of each polynucleotide.

[0310] In some embodiments, the present disclosure relates to one or more heterologous polynucleotides (e.g., a bacterial artificial chromosome (BAC)) comprising any of the recombinant nucleic acids described herein.

IV. Viruses

[0311] Certain aspects of the present disclosure relate to viruses comprising any of the polynucleotides and/or recombinant nucleic acids described herein. In some embodiments, the virus is capable of infecting one or more target cells of a subject (e.g., a human). In some embodiments, the virus is suitable for delivering the polynucleotides and/or recombinant nucleic acids into one or more target cells of a subject (e.g., a human). In some embodiments, the present disclosure relates to one or more viral particles comprising any of the polynucleotides and/or recombinant nucleic acids described herein. In some embodiments, the one or more target cells are one or more human cells. In some embodiments, the one or more target cells are one or more cells of the skin (e.g., one or more cells of the epidermis, dermis, and/or subcutis). In some embodiments, the one or more target cells are cells of the epidermis and/or dermis (e.g., cells of the human epidermis and/or dermis). In some embodiments, the one or more target cells are selected from keratinocytes, melanocytes, Langerhans cells, Merkel cells, mast cells, fibroblasts, and/or adipocytes. In some embodiments, the one or more target cells are keratinocytes. In some embodiments, the one or more target cells reside in the stratum corneum, stratum *granulosum*, stratum *spinulosum*, stratum basale, and/or basement membrane. In some embodiments, the one or more target cells are one or more epidermal cells. In some embodiments, the one or more target cells are one or more dermal cells.

[0312] Any suitable virus known in the art may be used, including, for example, adenovirus, adeno-associated virus, retrovirus, lentivirus, sendai virus, papillomavirus, herpes virus (e.g., a herpes simplex virus), vaccinia virus, and/or any hybrid or derivative viruses thereof. In some embodiments, the virus is attenuated. In some embodiments, the virus is replication defective. In some embodiments, the virus is replication competent. In some embodiments, the virus has been modified to alter its tissue tropism relative to the tissue tropism of a corresponding unmodified, wild-type virus. In some embodiments, the virus has reduced cytotoxicity as compared to a corresponding wild-type virus. Methods of producing a virus comprising recombinant nucleic acids are well known to one of ordinary skill in the art.

[0313] In some embodiments, the virus is a member of the Herpesviridae family of DNA viruses, including, for example, a herpes simplex virus, a varicella zoster virus, a human cytomegalovirus, a herpesvirus 6A, a herpesvirus 6B, a herpesvirus 7, and a Kaposi's sarcoma-associated herpesvirus, etc. In some embodiments, the herpes virus is attenuated. In some embodiments, the herpes virus is replication defective. In some embodiments, the herpes virus is replication competent. In some embodiments, the herpes virus has reduced cytotoxicity as compared to a corresponding wild-type herpes virus. In some embodiments, the herpes virus is not oncolytic.

[0314] In some embodiments, the herpes virus is a herpes simplex virus. Herpes simplex viruses comprising recombinant nucleic acids may be produced by a process disclosed, for example, in WO2015/009952 and/or WO2017/176336. In some embodiments, the herpes simplex virus is attenuated. In some embodiments, the herpes simplex virus is replication competent. In some embodiments, the herpes simplex virus is replication defective. In some embodiments, the herpes simplex virus is a herpes simplex virus type 1 (HSV-1), a herpes simplex virus type 2 (HSV-2), or any derivatives thereof. In some embodiments, the herpes simplex virus is a herpes simplex virus type 1 (HSV-1). In some embodiments, the HSV-1 is replication defective. In some embodiments, the hsv-1 is replication competent. In some embodiments, the HSV-1 is attenuated. In some embodiments, the herpes simplex virus (e.g., the HSV-1) has reduced cytotoxicity as compared to a corresponding wild-type herpes simplex virus (e.g., a wild-type HSV-1). In some embodiments, the herpes simplex virus (e.g., the HSV-1) is not oncolytic.

[0315] In some embodiments, the herpes simplex virus has been modified to alter its tissue tropism relative to the tissue tropism of an unmodified, wild-type herpes simplex virus. In some embodiments, the herpes simplex virus comprises a modified envelope. In some embodiments, the modified envelope comprises one or more (e.g., one or more, two or more, three or more, four or more, etc.) mutant herpes simplex virus glycoproteins. Examples of herpes simplex virus glycoproteins may include, but are not limited to, the glycoproteins gB, gC, gD, gH, and gL. In some embodiments, the modified envelope alters the herpes simplex virus tissue tropism relative to a wild-type herpes simplex virus.

[0316] In some embodiments, the transduction efficiency (in vitro and/or in vivo) of a virus of the present disclosure (e.g., a herpes virus such as a herpes simplex virus) for one or more target cells (e.g., one or more human keratinocytes and/or fibroblasts) is at least about 25%. For example, the transduction efficiency of the virus for one or more target cells may be at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 99%, at least about 99.5%, or more. In some embodiments, the virus is a herpes simplex virus and the transduction efficiency of the virus for one or more target cells (e.g., one or more human keratinocytes and/or fibroblasts) is about 85% to about 100%. In some embodiments, the virus is a herpes simplex virus and the transduction efficiency of the virus for one or more target cells (e.g., one or more human keratinocytes and/or fibroblasts) is at least about 85%, at least about 86%, at least about 87%, at least about 88%, at least about 89%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or 100%. Methods of measuring viral transduction efficiency in vitro or in vivo are well known to one of ordinary skill in the art, including, for example, qPCR analysis, deep sequencing, western blotting, fluorometric analysis (such as fluorescent in situ hybridization (FISH), fluorescent reporter gene expression, immunofluorescence, FACS), etc.

V. Pharmaceutical Compositions and Formulations

[0317] Certain aspects of the present disclosure relate to pharmaceutical compositions and/or formulations comprising any of the recombinant nucleic acids (e.g., recombinant herpes virus genomes) and/or viruses (e.g., herpes viruses comprising a recombinant genomes) described herein (such as a herpes simplex virus comprising a recombinant herpes simplex virus genome), and a pharmaceutically acceptable excipient or carrier.

[0318] In some embodiments, the pharmaceutical composition or formulation comprises any one or more of the viruses (e.g., herpes viruses) described herein. In some embodiments, the pharmaceutical composition or formulation comprises from about 104 to about 1012 plaque forming units (PFU)/mL of the virus. For example, the pharmaceutical composition or formulation may comprise from about 10^4 to about 10^{12}, about 10^5 to about 10^{12}, about 10^6 to about 10^{12}, about 10^7 to about 10^{12}, about 10^8 to about 10^{12}, about 10^9 to about 10^{12}, about 10^{10} to about 10^{12}, about 10^{11} to about 10^{12}, about 10^4 to about 10^{11}, about 10^5 to about 10^{11}, about 10^6 to about 10^{11}, about 10^7 to about 10^{11}, about 10^8 to about 10^{11}, about 10^9 to about 10^{11}, about 10^{10} to about 10^{11}, about 10^4 to about 10^{10} , about 10^5 to about 10^{10} , about 10^6 to about 10^{10} , about 10^7 to about 10^{10} , about 10^8 to about 10^{10} , about 10^9 to about 10^{10} , about 10^4 to about 10^9, about 10^5 to about 10^9, about 10^6 to about 10^9, about 10^7 to about 10^9, about 10^8 to about 10^9, about 10^4 to about 10^8, about 10^5 to about 10^8, about 10^6 to about 10^8, about 10^7 to about 10^8, about 10^4 to about 10^7, about 10^5 to about 10^7, about 10^6 to about 10^7, about 10^4 to about 10^6, about 10^5 to about 10^6, or about 10^4 to about 10^5 PFU/mL of the virus. In some embodiments, the pharmaceutical composition or formulation comprises about 10^4, about 10^5, about 10^6, about 10^7, about 10^8, about 10^9, about 10^{10} , about 10^{11}, or about 10^{12} PFU/mL of the virus.

[0319] Pharmaceutical compositions and formulations can be prepared by mixing the active ingredient(s) (such as a recombinant nucleic acid and/or a virus) having the desired degree of purity with one or more pharmaceutically acceptable carriers or excipients. Pharmaceutically acceptable carriers or excipients are generally nontoxic to recipients at the dosages and concentrations employed, and may include, but are not limited to: buffers (such as phosphate, citrate, acetate, and other organic acids); antioxidants (such as ascorbic acid and methionine); preservatives (such as octadecyldimethylbenzyl ammonium chloride, benzalkonium chloride, benzethonium chloride, phenol, butyl or benzyl alcohol, alkyl parabens, catechol, resorcinol, cyclohexanol, 3-pentanol, and m-cresol); amino acids (such as glycine, glutamine, asparagine, histidine, arginine, or lysine); low molecular weight (less than about 10 residues) polypeptides; proteins (such as serum albumin, gelatin, or immunoglobulins); polyols (such as glycerol, e.g., formulations including 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, etc. glycerol); hydrophilic polymers (such as polyvinylpyrrolidone); monosaccharides, disaccharides, and other carbohydrates (including glucose, mannose, or dextrins); chelating agents (such as EDTA); sugars (such as sucrose, mannitol, trehalose, or sorbitol); salt-forming counter-ions (such as sodium); metal complexes (such as Zn-protein complexes); and/or non-ionic surfactants (such as polyethylene glycol (PEG)). A thorough discussion of pharmaceutically acceptable carriers is available in REMINGTON'S PHARMACEUTICAL SCIENCES (Mack Pub. Co., N.J. 1991).

[0320] In some embodiments, the pharmaceutical composition or formulation comprises one or more lipid (e.g., cationic lipid) carriers. In some embodiments, the pharmaceutical composition or formulation comprises one or more nanoparticle carriers. Nanoparticles are submicron (less than about 1000 nm) sized drug delivery vehicles that can carry encapsulated drugs (such as synthetic small molecules, proteins, peptides, cells, viruses, and nucleic acid-based biotherapeutics) for rapid or controlled release. A variety of molecules (e.g., proteins, peptides, recombinant nucleic acids, etc.) can be efficiently encapsulated in nanoparticles using processes well known in the art. In some embodiments, a molecule "encapsulated" in a nanoparticle may refer to a molecule (such as a virus) that is contained within the nanoparticle or attached to and/or associated with the surface of the nanoparticle, or any combination thereof. Nanoparticles for use in the compositions or formulations described herein may be any type of biocompatible nanoparticle known in the art, including, for example, nanoparticles comprising poly(lactic acid), poly(glycolic acid), PLGA, PLA, PGA, and any combinations thereof (see e.g., Vauthier et al. Adv Drug Del Rev. (2003) 55:519-48; US2007/0148074; US2007/0092575; US2006/0246139; U.S. Pat. Nos. 5,753,234; 7,081,483; and WO2006/052285).

[0321] In some embodiments, the pharmaceutically acceptable carrier or excipient may be adapted for or suitable for any administration route known in the art, including, for example, intravenous, intramuscular, subcutaneous, cutaneous, oral, nasal, intratracheal, sublingual, buccal, topical, transdermal, intradermal, intraperitoneal, intraorbital, subretinal, intravitreal, transmucosal, intraarticular, by implantation, by inhalation, intrathecal, intraventricular, and/or intranasal administration. In some embodiments, the pharmaceutical composition or formulation is adapted for or suitable for any administration route known in the art, including, for example, intravenous, intramuscular, subcutaneous, cutaneous, oral, nasal, intratracheal, sublingual, buccal, topical, transdermal, intradermal, intraperitoneal, intraorbital, intravitreal, subretinal, transmucosal, intraarticular, by implantation, by inhalation, intrathecal, intraventricular, and/or intranasal administration. In some embodiments, the pharmaceutically acceptable carrier or excipient is adapted for or suitable for topical, transdermal, subcutaneous, intradermal, and/or transmucosal administration. In some embodiments, the pharmaceutical composition or formulation is adapted for or suitable for topical, transdermal, subcutaneous, intradermal, and/or transmucosal administration. In some embodiments, the pharmaceutically acceptable carrier or excipient is adapted for or suitable for topical, transdermal, subcutaneous, and/or intradermal administration. In some embodiments, the pharmaceutical composition or formulation is adapted for or suitable for topical, transdermal, subcutaneous, and/or intradermal administration. In some embodiments, the pharmaceutically acceptable carrier or excipient is adapted for or suitable for topical, transdermal, and/or intradermal administration. In some embodiments, the pharmaceutical composition or formulation is adapted for or suitable for topical, transdermal, and/or intradermal administration. In some embodiments, the pharmaceutically acceptable carrier or excipient is adapted for or suitable for topical administration. In some embodiments, the pharmaceutical composition or formulation is adapted for or suitable for topical administration.

[0322] Examples of carriers or excipients adapted for or suitable for use in pharmaceutical compositions or formulations of the present disclosure may include, but are not limited to, ointments, oils, pastes, creams, aerosols, suspensions, emulsions, fatty ointments, gels (e.g., methylcellulose gels, such as carboxy methylcellulose, hydroxypropyl methylcellulose, etc.), powders, liquids, lotions, solutions, sprays, patches (e.g., transdermal patches or microneedle patches), adhesive strips, a microneedle or microneedle arrays, and inhalants. In some embodiments, the carrier or excipient (e.g., the pharmaceutically acceptable carrier or excipient) comprises one or more (e.g., one or more, two or more, three or more, four or more, five or more, etc.) of an ointment, oil, paste, cream, aerosol, suspension, emulsion, fatty ointment, gel, powder, liquid, lotion, solution, spray, patch, adhesive strip and an inhalant. In some embodiments, the carrier comprises a patch (e.g. a patch that adheres to the skin), such as a transdermal patch or a microneedle patch. In some embodiments, the carrier comprises a microneedle or microneedle array. Methods for making and using microneedle arrays suitable for composition delivery are generally known in the art (see e.g., Kim Y. et al. "Microneedles for drug and vaccine delivery". *Advanced Drug Delivery Reviews* 2012, 64 (14): 1547-68).

[0323] In some embodiments, the pharmaceutical composition or formulation further comprises one or more additional components. Examples of additional components may include, but are not limited to, binding agents (e.g., pregelatinized maize starch, polyvinylpyrrolidone or hydroxypropyl methylcellulose, etc.); fillers (e.g., lactose and other sugars, microcrystalline cellulose, pectin, gelatin, calcium sulfate, ethyl cellulose, polyacrylates or calcium hydrogen phosphate, etc.); lubricants (e.g., magnesium stearate, talc, silica, colloidal silicon dioxide, stearic acid, metallic stearates, hydrogenated vegetable oils, corn starch, polyethylene glycols, sodium benzoate, sodium acetate, etc.); disintegrants (e.g., starch, sodium starch glycolate, etc.); wetting agents (e.g., sodium lauryl sulphate, etc.); salt solutions; alcohols; polyethylene glycols; gelatin; lactose; amylase; magnesium stearate; talc; silicic acid; viscous paraffin; methylcellulose (e.g., carboxy methylcellulose, hydroxypropyl methylcellulose, etc.); polyvinylpyrrolidone; sweetenings; flavorings; perfuming agents; colorants; moisturizers; sunscreens; antibacterial agents; agents able to stabilize polynucleotides or prevent their degradation, and the like. In some embodiments, the pharmaceutical composition or formulation comprises a methylcellulose gel, such as hydroxypropyl methylcellulose, carboxy methylcellulose, etc., (e.g., at about 0.5%, at about 1%, at about 1.5%, at about 2%, at about 2.5%, at about 3%, at about 3.5%, at about 4%, at about 4.5%, at about 5%, at about 5.5%, at about 6%, at about 6.5%, at about 7%, at about 7.5%, at about 8%, at about 8.5%, at about 9%, at about 9.5%, at about 10%, at about 10.5%, at about 11%, at about 11.5%, at about 12%, etc.). In some embodiments, the pharmaceutical composition or formulation comprises a phosphate buffer. In some embodiments, the pharmaceutical composition or formulation comprises glycerol (e.g., at about 1%, about 2%, about 3%, about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15%, etc.). In some embodiments, the pharmaceutical composition or formulation comprises a methylcellulose gel (e.g., hydroxypropyl methylcellulose, carboxy methylcellulose, etc.), a phosphate buffer, and/or glycerol.

[0324] Compositions and formulations (e.g., pharmaceutical compositions and formulations) to be used for in vivo administration are generally sterile. Sterility may be readily accomplished, e.g., by filtration through sterile filtration membranes.

[0325] In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used to deliver one or more polynucleotides encoding an ichthyosis-associated polypeptide (e.g., a human Steryl-sulfatase polypeptide) into one or more cells of a subject (e.g., one or more Steryl-sulfatase deficient cells, one or more cells harboring an STS gene mutation, etc.). In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in a therapy. In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the treatment of a disease, disorder, defect, or condition that would benefit from the expression of an ichthyosis-associated polypeptide (e.g., one or more forms of congenital ichthyosis; a disease, disorder, defect, or condition associated with an ichthyosis-associated polypeptide deficiency (such as X-linked ichthyosis); a disease, disorder, defect, or condition associated a ichthyosis-associated gene mutation, etc.). In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the treatment of one or more forms of congenital ichthyosis (e.g., used in the treatment of one or more of harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and/or restrictive dermopathy). In some embodiments, the congenital ichthyosis is not ARCI. In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the treatment of X-linked ichthyosis.

[0326] In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the preparation or manufacture of a medicament. In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the preparation or manufacture of a medicament useful for delivering one or more polynucleotides encoding an ichthyosis-associated polypeptide (e.g., a human Steryl-sulfatase polypeptide) into one or more cells of a subject (e.g., one or more Steryl-sulfatase deficient cells, one or more cells harboring an STS gene mutation, etc.). In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the preparation or manufacture of a medicament useful for the treatment of a disease, disorder, defect, or condition that would benefit from the expression of an ichthyosis-associated polypeptide (e.g., one or more forms of congenital ichthyosis; a disease, disorder, defect, or condition associated with an ichthyosis-associated polypeptide deficiency (such as X-linked ichthyosis); a disease, disorder, defect, or condition associated with a ichthyosis-associated gene mutation, etc.). In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the preparation or manufacture of a medicament useful for the treatment of one or more forms of congenital ichthyosis (e.g., useful for the treatment of one or more of harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and/or restrictive dermopathy). In some embodiments, the congenital ichthyosis is not ARCI. In some embodiments, any of the recombinant nucleic acids, viruses, and/or pharmaceutical compositions or formulations described herein may be used in the preparation or manufacture of a medicament useful for the treatment of X-linked ichthyosis.

VI. Methods

[0327] Certain aspects of the present disclosure relate to enhancing, increasing, augmenting, and/or supplementing the levels of an ichthyosis-associated polypeptide (e.g., a human Steryl-sulfatase polypeptide) in one or more cells of a subject comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject suffers from one or more forms of congenital ichthyosis (e.g., one or more of harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and/or restrictive dermopathy). In some embodiments, the congenital ichthyosis is not ARCI. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in an endogenous ichthyosis-associated gene (one or both copies), such as a loss-of-function mutation in the STS gene.

[0328] In some embodiments, administration of the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation to the subject increases ichthyosis-associated polypeptide levels (transcript or protein levels) by at least about 25% in one or more contacted or treated cells of the subject, as compared to the endogenous levels of the ichthyosis-associated polypeptide in one or more corresponding untreated cells of the subject. In some embodiments, administration of the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation to the subject increases ichthyosis-associated polypeptide levels (transcript or protein levels) by at least about 25%, at least about 30%, at least about 40%, at least about 50%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 99%, or more in one or more contacted or treated cells of the subject, as compared to the endogenous levels of the ichthyosis-associated polypeptide in one or more corresponding untreated cells of the subject. In some embodiments, administration of the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation to the subject increases ichthyosis-associated polypeptide levels (transcript or protein levels) by at least about 2-fold in one or more contacted or treated cells of the subject, as compared to the endogenous levels of the ichthyosis-associated polypeptide in one or more corresponding untreated cells in the subject. For example, administration of the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation may increase ichthyosis-associated polypeptide levels (transcript or protein levels) by at least about 2-fold, at least about 3-fold, at least about 4-fold, at least about 5-fold, at least about 6-fold, at least about 7-fold, at least about 8-fold, at least about 9-fold, at least about 10-fold, at least about 15-fold, at least about 20-fold, at least about 25-fold, at least about 50-fold, at least about 75-fold, at least about 100-fold, at least about 250-fold, at least about 500-fold, at least about 750-fold, at least about 1000-fold, or more in one or more contacted or treated cells of the subject, as compared to the endogenous levels of the ichthyosis-associated polypeptide in one or more corresponding untreated cells in the subject. In some embodiments, the one or more contacted or treated cells are one or more cells of the epidermis, dermis, and/or mucosa. In some embodiments, the one or more contacted or treated cells are one or more cells of the epidermis and/or dermis (e.g., a keratinocyte or fibroblast). Methods of measuring transcript or protein levels from a sample are well known to one of ordinary skill in the art, including, for example, by qPCR, western blot, mass spectrometry, etc.

[0329] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of congenital ichthyosis in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject suffers from one or more of harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and restrictive dermopathy. In some embodiments, the congenital ichthyosis is not ARCI. In some embodiments, the congenital ichthyosis is X-linked ichthyosis.

[0330] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of harlequin ichthyosis in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the ABCA12 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding an ATP-binding sub-family A member 12 polypeptide (ABCA12), e.g., a human ABCA12 polypeptide. Signs and/or symptoms of harlequin ichthyosis may include, but are not limited to, thick plate-like scales of the skin, ectropion, eclabium, severe restriction of the chest and abdomen due to tightness of the skin, difficulty breathing and/or eating, low body temperature, swelling of the mouth, dehydration, lack of hydration to the corneas, hypernatremia, etc.

[0331] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Chanarin-Dorfman syndrome (CDS) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the ABHD5 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide (ABHD5), e.g., a human ABHD5 polypeptide. Signs and/or symptoms of CDS may include, but are not limited to, redness, fine scaling, dark pigmentation, and severe itching of the skin, liver disease with lipid storage, progressive weakness of the proximal arms and legs, CK elevation in the blood, early fatigability, cataracts, ectropion, progressive hearing loss, cognitive impairment, short stature, growth retardation, steatorrhea, an enlarged spleen, orthopedic problems, kidney dysfunction, etc.

[0332] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Sjogren-Larsson syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the ALDH3A2 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Aldehyde dehydrogenase family 3 member A2 polypeptide (ALDH3A2), e.g., a human ALDH3A2 polypeptide. Signs and/or symptoms of Sjogren-Larsson syndrome may include, but are not limited to, erythema, dry/rough/scaly skin with a brownish or yellowish tone, mild to severe itchiness, leukoencephalopathy, mild to profound intellectual disabilities, delayed speech and speech difficulties, seizures, delayed development of motor skills, abnormal muscle stiffness, tiny crystals/white dots formed in the light-sensitive tissues of the eye, etc.

[0333] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of autosomal recessive congenital ichthyosis (ARCI) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in one or more of the ALOX12B, ALOXE3, CASP14, CERS3, CYP4F22, LIPN, NIPAL4, PNPLA1, SDR9C7, SLC27A4, ST14, and/or SULT2B1 genes (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or more (e.g., one or more, two or more, three or more, four our more, five or more, six or more, seven or more, eight or more, nine or more, 10 or more, 11 or more, or all 12) of a Arachidonate 12-lipoxygenase 12R-type polypeptide (ALOX12B), Hydroperoxide isomerase ALOXE3 polypeptide (ALOXE3), Caspase-14 polypeptide (CASP14), Ceramide synthase 3 polypeptide (CERS3), Cytochrome P450 4F22 polypeptide (CYP4F22), Lipase member N polypeptide (LIPN), Magnesium transporter NIPA4 polypeptide (NIPAL4), Patatin-like phospholipase domain-containing protein 1 polypeptide (PNPLA1), Short-chain dehydrogenase/reductase family 9C member 7 polypeptide (SDR9C7), Long-chain fatty acid transport protein 4 polypeptide (SLC27A4), Suppressor of tumorigenicity 14 protein polypeptide (ST14), and/or Sulfotransferase 2B1 polypeptide (SULT2B1), e.g., one or more of a human ALOX12B, ALOXE3, CASP14, CERS3, CYP4F22, LIPN, NIPAL4, PNPLA1, SDR9C7, SLC27A4, ST14, and/or SULT2B1 polypeptide. In some embodiments, the ARCI is not TGM1-deficient ARCI and/or TGM5-deficient ARCI. Signs and/or symptoms of ARCI may include, but are not limited to, an abnormal stratum corneum, incomplete thickening of the cornified cell envelope, defects in the intercellular lipid layers in the stratum corneum, generalized scaling with variable redness of the skin, formation of large plate-like scales, accelerated epidermal turnover, palmoplantar hyperkeratosis, defective barrier function, recurrent skin infections, exposure keratitis, hypohidrosis, heat intolerance, corneal perforation, rickets, nail abnormalities, dehydration, respiratory problems, ectropion, eclabium, hypoplasia of joint and nasal cartilage, scarring alopecia, etc.

[0334] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of erythrokeratodermia *variabilis*/mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MED-

NIK) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in one or more of the AP1S1, GJB3, and/or GJB4 genes (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or more (e.g., one or more, two or more, or all three) of an AP-1 complex subunit sigma-1A polypeptide (AP1S1), Gap junction beta-3 polypeptide (GJB3), and/or Gap junction beta-4 polypeptide (GJB4), e.g., one or more of a human AP1S1, GJB3, and/or GJB4 polypeptide. Signs and/or symptoms of erythrokeratodermia *variabilis*/MEDNIK syndrome may include, but are not limited to, hyperkeratosis and red patches of variable size, shape, and duration on the skin, sensorineural deafness, peripheral neuropathy, psychomotor retardation, elevations of very long chain fatty acids, upslanting palpebral fissures, hypotonia, ichthyosiform erythroderma, gastrointestinal problems, hepatic fibrosis, cirrhosis, cholestasis, cataracts, etc.

[0335] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of chondrodysplasia *punctata* 1 (CDPX1) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the ARSE gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding an Arylsulfatase E polypeptide (ARSE), e.g., a human ARSE polypeptide. Signs and/or symptoms of CDPX1 may include, but are not limited to, ichthyosis, an abnormal spine, anosmia, a depressed nasal bridge, epiphyseal stippling, hearing impairment, hypogonadism, microcephaly, shortened fingers, breathing abnormalities, etc.

[0336] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of chondrodysplasia *punctata* 2 (CDPX2) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the EBP gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide (EBP), e.g., a human EBP polypeptide. Signs and/or symptoms of CDPX2 may include, but are not limited to, congenital ichthyosiform erythroderma, erythema, hyperkeratotic scaling, follicular atrophoderma (particularly in the trunk, forearms, and dorsal aspect of the hands), cicatricial alopecia, asymmetric shortening of the limbs, facial dysmorphism (low nasal bridge, frontal bossing, hypertelorism, high arched palate), joint contractures, moderate to severe sclerosis of the vertebral column, cataracts, etc.

[0337] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of peeling skin syndrome (PSS) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in one or more of the CDSN, CHST8, CSTA, FLG2, and/or SERPINB8 genes (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or more (e.g., one or more, two or more, three or more, four our more, or all five) of a Corneodesmosin polypeptide (CDSN), Carbohydrate sulfotransferase 8 polypeptide (CHST8), Cystatin-A polypeptide (CSTA), Filaggrin 2 polypeptide (FLG2), and/or SerpinB8 polypeptide (SERPINB8), e.g., one or more of a human CDSN, CHST8, CSTA, FLG2, and/or SERPINB8 polypeptide. Signs and/or symptoms of PSS may include, but are not limited to, spontaneous, painless shedding or peeling of the outermost layer of the skin, itching, short stature, blisters or erosions on the hands and feet, etc.

[0338] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the CLDN1 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Claudin-1 polypeptide (CLDN1), e.g., a human CLDN1 polypeptide. Signs and/or symptoms of NISCH may include, but are not limited to, ichthyosis with diffuse white scales, scalp hypotrichosis, cicatricial alopecia, sparse eyelashes/eyebrows, oligodontia, hypodontia, enamel dysplasia, neonatal sclerosing cholangitis with jaundice and pruritus, hepatomegaly, cholestasis, portal hypertension, patent extrahepatic bile duct obstruction, splenomegaly, leukocyte vacuolization, etc.

[0339] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis vulgaris in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the FLG gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Filaggrin polypeptide (FLG), e.g., a human FLG polypeptide. Signs and/or symptoms of ichthyosis vulgaris may include, but are not limited to, flaky scalp, itchy skin, polygon-shaped white, brown, or gray scales on the skin, severely dry skin, thickened skin, etc.

[0340] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of keratitis-ichthyosis-deafness (KID) syndrome, Clouston syndrome, and/or palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the GJB2 and/or GJB6 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or both of a Gap junction beta-2 polypeptide (GJB2) and/or Gap junction beta-6 polypeptide (GJB6), e.g., a human GJB2 and/or GJB6 polypeptide. Signs and/or symptoms of keratitis-ichthyosis-deafness (KID) syndrome, Clouston syndrome, and/or palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL) may include, but are not limited to, palmoplantar keratoderma, erythrokeratoderma, ichthyosis, keratitis, sensitivity to light, extra blood vessel growth in the eye, scarring of the eye, visual loss or blindness, nail abnormalities, progressive hair loss, hyperpigmentation of the skin, clubbing of the fingers, etc.

[0341] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of progressive symmetric erythrokeratodermia in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the KDSR gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a 3-ketodihydrosphingosine reductase polypeptide (KDSR), e.g., a human KDSR polypeptide. Signs and/or symptoms of progressive symmetric erythrokeratodermia may include, but are not limited to, reddened plaques of thickened, rough, and/or scaly skin (especially on the face, buttocks, arms, and legs) with an almost perfectly symmetrical distribution, palmoplantar keratoderma, etc.

[0342] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), and/or epidermolytic palmoplantar keratoderma in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in one or more of the KRT1, KRT2, KRT9, and/or KRT10 genes (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or more (e.g., one or more, two or more, three or more, or all four) of a Keratin, type II cytoskeletal 1 polypeptide (KRT1), Keratin, type II cytoskeletal 2 epidermal polypeptide (KRT2), Keratin, type I cytoskeletal 9 polypeptide (KRT9), and/or Keratin, type I cytoskeletal 10 polypeptide (KRT10), e.g., one or more of a human KRT1, KRT2, KRT9, and/or KRT10 polypeptide. Signs and/or symptoms of epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), and/or epidermolytic palmoplantar keratoderma may include, but are not limited to, thick, blistering, warty hardening of the skin (particularly in the skin creases over joints), scales forming in parallel rows of spines or ridges, skin fragility that may blister easily following injury, generalized erythroderma, recurrent skin infections (often *Staphylococcus* or *Streptococcus*), severe scalp involvement and hair loss, heat intolerance, even, widespread thickened skin (keratosis) over the palms and soles, a red band at the edges of the keratosis, keratotic lesions appearing on the tops of the hands, feet, knees, and elbows, excessive perspiration, nail thickening, etc.

[0343] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of loricrin keratoderma in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the LOR gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Loricrin polypeptide (LOR), e.g., a human LOR polypeptide. Signs and/or symptoms of loricrin keratoderma may include, but are not limited to, diffuse palmoplantar keratoderma, honeycomb palmoplantar hyperkeratosis (associated with pseudoainhum of the fifth digit of the hand), deafness, ichthyosis, etc.

[0344] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the MBTPS2 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Membrane-bound transcription factor site-2 protease polypeptide (MBTPS2), e.g., a human MBTPS2 polypeptide. Signs and/or symptoms of IFAP syndrome may include, but are not limited to, dry, scaly skin, absence of hair, excessive sensitivity to light, short stature, mental retardation, seizures, a tendency for respiratory infections, etc.

[0345] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the NSDHL gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide (NSDHL), e.g., a human NSDHL polypeptide. Signs and/or symptoms of CHILD syndrome may include, but are not limited to, dry, itchy, red and scaly skin on one side of the body, absence of hair on one side of the head, limb defects (e.g., underdevelopment of fingers and toes, complete absence of limbs) that often occur on the same side of the body as the major skin symptoms, skeletal defects (e.g., abnormal ribs, anomalies of the shoulder blades) webbing of the skin between joints, absence of muscles of the breast, defects in the walls between auricles and/or ventricles, abnormalities of the central nervous system, blood vessels, kidneys, thyroid, lungs, adrenal glands, reproductive system, and urinary system (often underdevelopment on the affected side of the body), etc.

[0346] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Refsum disease in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in one or both of the PEX7 and/or PHYH genes (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or both of a Peroxisomal targeting signal 2 receptor polypeptide (PEX7) and/Phytanoyl-CoA dioxygenase, peroxisomal polypeptide (PHYH), e.g., one or both of a human PEX7 and/or PHYH polypeptide. Signs and/or symptoms of Refsum disease may include, but are not limited to, dry scaly skin, retinitis pigmentosa, anosmia, bone abnormalities of the hands and feet, progressive muscle weakness and wasting, ataxia, hearing loss, abnormal heart rhythm, etc.

[0347] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Neu-Laxova syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in one or both of the PHGDH and/or PSAT1 genes (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding one or both of a D-3-phosphoglycerate dehydrogenase polypeptide (PHGDH) and/Phosphoserine aminotransferase polypeptide (PSAT1), e.g., one or both of a human PHGDH and/or PSAT1 polypeptide. Signs and/or symptoms of Neu-Laxova syndrome may include, but are not limited to, proptosis with eyelid malformations, nose malformations, round and gaping mouth, micrognathia, low set or malformed ears, cleft lip, cleft palate, syndactyly, edema and flexion deformities, ichthyosis and hyperkeratosis, microcephaly, lissencephaly, microgyria, hypoplasia of the cerebellum, agenesis of the corpus callosum, neural tube defects, etc.

[0348] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the POMP gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Proteasome maturation protein polypeptide (POMP), e.g., a human POMP polypeptide. Signs and/or symptoms of KLICK syndrome may include, but are not limited to, linear hyperkeratosis (without evidence of Koebner phenomenon), moderate, non-blistering ichthyosis, palmoplantar keratoderma, sclerosing flexion deformities of the fingers, noninflamed keratotic striae, etc.

[0349] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis prematurity syndrome (IPS) in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the SLC27A4 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Long-chain fatty acid transport protein 4 polypeptide (SLC27A4), e.g., a human SLC27A4 polypeptide. Signs and/or symptoms of IPS may include, but are not limited to, a thick caseous layer of skin, red endemic skin, spongy and desquamating skin, respiratory problems, eosinophilia, etc.

[0350] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the SNAP29 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Synaptosomal-associated protein 29 polypeptide (SNAP29), e.g., a human SNAP29 polypeptide. Signs and/or symptoms of CEDNIK syndrome may include, but are not limited to, failure to thrive, roving eye movements, poor head and trunk control, progressive microcephaly and facial dysmorphism consisting of elongated facies, downward-slanting palpebral fissures, mild hypertelorism, flat, broad nasal root, palmoplantar keratosis, ichthyosis, psychomotor retardation, hypoplastic optic discs, sensorineural deafness, etc.

[0351] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of X-linked ichthyosis in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the STS gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Steryl-sulfatase polypeptide (STS), e.g., a human STS polypeptide. Signs and/or symptoms of X-linked ichthyosis may include, but are not limited to, brownish scales that adhere to the skin (often affecting the back and legs), comma-shaped corneal opacities, etc.

[0352] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the VPS33B gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a Vacuolar protein sorting-associated protein 33B polypeptide (VPS33B), e.g., a human VPS33B polypeptide. Signs and/or symptoms of ARC syndrome may include, but are not limited to, congenital joint contractures, renal tubular dysfunction, cholestasis, ichthyosis, central nervous system malformation, platelet anomalies, agenesis of the corpus callosum, deafness, recurrent sepsis, hypothyroidism, nephrogenic diabetes insipidus, etc.

[0353] Other aspects of the present disclosure relate to a method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of restrictive dermopathy in a subject in need thereof comprising administering to the subject an effective amount of any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the subject is a human. In some embodiments, the subject's genome comprises a mutation (e.g., a loss-of-function mutation, a pathogenic variant) in the ZMPSTE24 gene (one or both copies). In some embodiments, the recombinant nucleic acid (e.g., a recombinant herpes virus genome) comprises one or more polynucleotides encoding a CAAX prenyl protease 1 homolog polypeptide (ZMPSTE24), e.g., a human ZMPSTE24 polypeptide. Signs and/or symptoms of restrictive dermopathy may include, but are not limited to, very tight and thin skin with erosions, scaling, typical facial dysmorphism, arthrogryposis multiplex, fetal akinesia or hypokinesia deformation sequence (FADS), pulmonary hypoplasia, etc.

[0354] The recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein may be administered by any suitable method or route known in the art, including, without limitation, by oral administration, sublingual administration, buccal administration, topical administration, rectal administration, via inhalation, transdermal administration, subcutaneous injection, intradermal injection, intravenous injection, intra-arterial injection, intramuscular injection, intracardiac injection, intraosseous injection, intraperitoneal injection, transmucosal administration, vaginal administration, intravitreal administration, intraorbital administration, subretinal administration, subconjunctival administration (e.g., the use of subconjunctival depots), suprachoroidal administration, intraarticular administration, peri-articular administration, local administration, epicutaneous administration, or any combinations thereof. In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered cutaneously, topically, transdermally, subcutaneously, or intradermally to the subject. In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered topically, transdermally, subcutaneously, or intradermally to the subject. In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered topically, transdermally, or intradermally to the subject. In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered topically to the subject. The present disclosure thus encompasses methods of delivering any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein to an individual (e.g., an individual having, or at risk of developing, one or more signs or symptoms of congenital ichthyosis).

[0355] In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered once to the subject. In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered at least twice (e.g., at least 2 times, at least 3 times, at least 4 times, at least 5 times, at least 10 times, etc.) to the subject. In some embodiments, at least about 1 hour (e.g., at least about 1 hour, at least about 6 hours, at least about 12 hours, at least about 18 hours, at least about 1 day, at least about 2 days, at least about 3 days, at least about 4 days, at least about 5 days, at least about 6 days, at least about 7 days, at least about 15 days, at least about 20 days, at least about 30 days, at least about 40 days, at least about 50 days, at least about 60 days, at least about 70 days, at least about 80 days, at least about 90 days, at least about 100 days, at least about 120 days, etc.) pass between administrations (e.g., between the first and second administrations, between the second and third administrations, etc.). In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered one, two, three, four, five or more times per day to the subject. In some embodiments, the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation is administered to one or more affected and/or unaffected areas of the subject.

[0356] In some embodiments, one or more portions of the skin of the subject is abraded or made more permeable prior to treatment with a recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation described herein. Any suitable method of abrading the skin or increasing skin permeability known in the art may be used, including, for example, use of a dermal roller, repeated use of adhesive strips to remove layers of skin cells (tape stripping), scraping with a scalpel or blade, use of sandpaper, use of chemical permeation enhancers or electrical energy, use of sonic or ultrasonic energy, use of light (e.g., laser) energy, use of micron-sized needles or blades with a length suitable to pierce but not completely pass through the epidermis, etc.

VII. Host Cells

[0357] Certain aspects of the present disclosure relate to one or more host cells comprising any of the recombinant nucleic acids described herein. Any suitable host cell (prokaryotic or eukaryotic) known in the art may be used, including, for example: prokaryotic cells including eubacteria, such as Gram-negative or Gram-positive organisms, for example Enterobacteriaceae such as *Escherichia* (e.g., *E. coli*), *Enterobacter, Erminia, Klebsiella, Proteus, Salmonella* (e.g., *S. typhimurium*), *Serratia* (e.g., *S. marcescans*), and *Shigella*, as well as Bacilli such as *B. subtilis* and *B. licheniformis*; fungal cells (e.g., *S. cerevisiae*); insect cells (e.g., S2 cells, etc.); and mammalian cells, including monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651), human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture), baby hamster kidney cells (BHK, ATCC CCL 10), mouse Sertoli cells (TM4), monkey kidney cells (CV1 ATCC CCL 70), African green monkey kidney cells (VERO-76, ATCC CRL-1587), human cervical carcinoma cells (HELA, ATCC CCL 2), canine kidney cells (MDCK, ATCC CCL 34), buffalo rat liver cells (BRL 3A, ATCC CRL 1442), human lung cells (W138, ATCC CCL 75), human liver cells (Hep G2, HB 8065), mouse mammary tumor (MMT O60562, ATCC CCL51), TRI cells, MRC 5 cells, FS4 cells, human hepatoma line (Hep G2), Chinese hamster ovary (CHO) cells, including DHFR" CHO cells, and myeloma cell lines such as NS0 and Sp2/0. In some embodiments, the host cell is a human or non-human primate cell. In some embodiments, the host cells are cells from a cell line. Examples of suitable host cells or cell lines may include, but are not limited to, 293, HeLa, SH-Sy5y, Hep G2, CACO-2, A549, L929, 3T3, K562, CHO-K1, MDCK, HUVEC, Vero, N20, COS-7, PSN1, VCaP, CHO cells, and the like.

[0358] In some embodiments, the recombinant nucleic acid is a herpes simplex viral vector. In some embodiments, the recombinant nucleic acid is a herpes simplex virus amplicon. In some embodiments, the recombinant nucleic acid is an HSV-1 amplicon or HSV-1 hybrid amplicon. In some embodiments, a host cell comprising a helper virus is contacted with an HSV-1 amplicon or HSV-1 hybrid amplicon described herein, resulting in the production of a virus comprising one or more recombinant nucleic acids described herein. In some embodiments, the virus is collected from the supernatant of the contacted host cell. Methods of generating virus by contacting host cells comprising a helper virus with an HSV-1 amplicon or HSV-1 hybrid amplicon are known in the art.

[0359] In some embodiments, the host cell is a complementing host cell. In some embodiments, the complementing host cell expresses one or more genes that are inactivated in any of the viral vectors described herein. In some embodiments, the complementing host cell is contacted with a recombinant herpes virus genome (e.g., a recombinant herpes simplex virus genome) described herein. In some embodiments, contacting a complementing host cell with a recombinant herpes virus genome results in the production of a herpes virus comprising one or more recombinant nucleic acids described herein. In some embodiments, the virus is collected from the supernatant of the contacted host cell. Methods of generating virus by contacting complementing host cells with a recombinant herpes simplex virus are generally described in WO2015/009952 and/or WO2017/176336.

VIII. Articles of Manufacture or Kits

[0360] Certain aspects of the present disclosure relate to an article of manufacture or a kit comprising any of the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations described herein. In some embodiments, the article of manufacture or kit comprises a package insert comprising instructions for administering the recombinant nucleic acid, virus, medicament, and/or pharmaceutical composition or formulation to treat an ichthyosis-associated polypeptide deficiency (e.g., in a subject harboring an STS loss-of-function mutation) and/or to provide prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of a congenital ichthyosis.

[0361] Suitable containers for the recombinant nucleic acids, viruses, medicaments, and/or pharmaceutical compositions or formulations may include, for example, bottles, vials, bags, tubes, and syringes. The container may be formed from a variety of materials such as glass, plastic (such as polyvinyl chloride or polyolefin), or metal alloy (such as stainless steel or hastelloy). In some embodiments, the container comprises a label on, or associated with the container, wherein the label indicates directions for use. The article of manufacture or kit may further include other materials desirable from a commercial and user standpoint, including other buffers, diluents, filters, needles, syringes, package inserts, and the like.

EXAMPLES

[0362] The present disclosure will be more fully understood by reference to the following examples. It should not, however, be construed as limiting the scope of the present disclosure. It is understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art, and are to be included within the spirit and purview of this application and scope of the appended claims.

Example 1: Modified Herpes Simplex Virus Vectors Encoding One or More Ichthyosis-Associated Polypeptides

[0363] To make modified herpes simplex virus genome vectors capable of expressing functional polypeptides of ichthyosis-associated genes in a target mammalian cell (such as a cell of the skin), a herpes simplex virus genome (FIG. **1**A) is first modified to inactivate one or more herpes simplex virus genes. Such modifications may decrease the toxicity of the genome in mammalian cells. Next, variants of these modified/attenuated recombinant viral constructs are generated such that they carry one or more polynucleotides encoding an ichthyosis-associated polypeptide. These variants include: 1) a recombinant ΔICP4-modified HSV-1 genome comprising expression cassettes containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at each ICP4 locus (FIG. **1**B); 2) a recombinant ΔICP4/AUL41-modified HSV-1 genome comprising expression cassettes containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at each ICP4 locus (FIG. **1**C); 3) a recombinant ΔICP4/AUL41-modified HSV-1 genome comprising an expression cassette containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at the UL41 locus (FIG. **1**D); 4) a recombinant ΔICP4/ΔICP22-modified HSV-1 genome comprising expression cassettes containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at each ICP4 locus (FIG. **1**E); 5) a recombinant ΔICP4/ΔICP22-modified HSV-1 genome comprising an expression cassette containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at the ICP22 locus (FIG. **1**F); 6) a recombinant ΔICP4/AUL41/ΔICP22-modified HSV-1 genome comprising expression cassettes containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at each ICP4 locus (FIG. **1**G); 7) a recombinant ΔICP4/AUL41/ΔICP22-modified HSV-1 genome comprising an expression cassette containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at the UL41 locus (FIG. **1**H); and 8) a recombinant ΔICP4/AUL41/ΔICP22-modified HSV-1 genome comprising an expression cassette containing the coding sequence of an ichthyosis-associated polypeptide under the control of a heterologous promoter integrated at the ICP22 locus (FIG. **1**I)

[0364] These modified herpes simplex virus genome vectors are transfected or transduced into engineered cells that are modified to express one or more herpes virus genes. These engineered cells secrete into the supernatant of the cell culture a replication defective herpes simplex virus with the modified genomes packaged therein. The supernatant is then collected, concentrated, and sterile filtered through a 5 μm filter.

Example 2: KB105 Pharmacology In Vitro

[0365] Congenital ichthyoses are a diverse group of cornification diseases associated with often severe clinical complications and a decreased quality of life. Germline mutations in the TGM1 gene, which encodes the enzyme transglutaminase 1 (TGM1), are the predominant cause of autosomal recessive congenital ichthyosis (ARCI). These TGM1 mutations trigger the abnormal epidermal differentiation and impaired cutaneous barrier function observed in ARCI patients. Unfortunately, like many forms of congenital ichthyoses, current ARCI therapies focus solely on symptomatic relief. Here, the ability of KB105, a gene therapy vector encoding full-length human TGM1, to deliver functional human TGM1 to keratinocytes was investigated.

[0366] Purified KB105 was first evaluated for transduction efficiency and effector expression in two-dimensional cell-based assays. These assays employed immortalized human keratinocytes harvested from a TGM1-deficient ARCI patient homozygous for a c.877-2A>G splice-site mutation, the most commonly reported TGM1 mutation in humans (Herman et al, 2009). Cells were infected with KB105 at multiplicities of infection (MOIs) ranging from 0.3 to 3.0 for 48 hours, and vector transduction and effector expression were analyzed by qPCR, qRT-PCR, western blot, and immunofluorescence. Negative controls included uninfected cells (mock) and cells infected with an mCherry-expressing vector (mCherry).

[0367] KB105 vector genomes and TGM1 transcript expression were detected in TGM1-deficient ARCI patient-derived keratinocytes at an MOI as low as 0.3, and showed a dose-dependent increase in TGM1 DNA (FIG. **2**A) and RNA (FIG. **2**B) levels. Increased TGM1 protein expression was observed by western blot and immunofluorescent analysis relative to mock-infected controls (FIGS. **2**C-**2**D). No detectable endogenous TGM1 was observed in the uninfected immortalized keratinocytes, confirming that these cells were isolated from a patient harboring a natural TGM1 deficiency.

[0368] Functionality of the KB105-expressed human TGM1 was next examined by determining whether the exogenous protein catalyzed covalent cross-linking between glutamine and lysine residues, a function essential for TGM1-mediated assembly of the cornified envelope. Protein functionality was assessed using an in situ TGM1-specific peptide cross-linking activity assay employing a biotinylated peptide that mimics a natural TGM1 substrate. TGM1-mediated conjugation of biotinylated peptides was visualized by incubating the treated cells with fluorescently labelled streptavidin. A dose-dependent increase in TGM1 enzymatic activity was observed in KB105-infected cells by immunofluorescence, with TGM1-mediated peptide cross-linking in infected cells surpassing the levels of endogenous TGM1 activity in normal primary keratinocytes (NPKs) (FIG. **2**E). Uninfected (mock) cells showed no detectable TGM1 activity. A similar trend in TGM1 protein expression and subsequent restoration of functional activity were observed in immortalized ARCI keratinocytes grown in high calcium medium to stimulate cell differentiation (FIGS. **3**A-**3**B).

[0369] The ability of KB105 to transduce a more clinically relevant cell type, i.e., primary TGM1-deficient patient keratinocytes, was next examined. Restoration of TGM1 protein expression was observed by western blot analysis in the KB105-infected primary patient cells (FIG. **2**F). As expected, no endogenous TGM1 was observed in the negative control primary ARCI keratinocytes. Supporting the western blot, immunofluorescence data revealed a dose-dependent increase in TGM1 protein between an MOI of 0.3 and 1.0 (FIG. **2**G). Rescue of TGM1 protein expression in primary patient keratinocytes was also observed by IF after growth in high calcium cell culture medium (FIG. **4**). No significant impact on cell morphology was observed upon KB105 transduction in low or high calcium environments (FIG. **5**A). Mild cytotoxic effects were observed at high dosages of the vector in primary cells, which may account for the decreased TGM1 protein levels observed at an MOI of 3 in the western blot and analyses (FIG. **5**B).

[0370] In vitro, KB105 efficiently infected TGM1-deficient human keratinocytes, produced TGM1 protein, and rescued transglutaminase enzyme function.

Example 3: In Vivo Evaluation of KB105

[0371] Because homozygous deletion of TGM1 is neonatal-lethal in mice (Matsuki et al., 1998), a mouse model harboring a genetic lesion in endogenous TGM1 was not practicable for preclinical evaluations. As such, in vivo pharmacology of KB105 was conducted in immunocompetent BALB/c mice. This approach allowed determination of the vector's ability to deliver properly localized TGM1 after single or repeated topical administration while concurrently monitoring the vector's toxicity within a fully intact immune system.

[0372] First, an evaluation of KB105 was conducted in mechanically or chemically disrupted dorsal skin of treated animals. Sequential tape stripping or wiping the skin surface with acetone are commonly used techniques for skin barrier disruption (Rissmann et al, 2009). A single low or high KB105 dose formulated in a methylcellulose gel carrier was topically administered to two regions of the prepared skin on each mouse. Skin biopsies were harvested 48 hours after topical administration and processed for analysis.

[0373] A histological examination of skin samples harvested from each treatment group was conducted to evaluate KB105-induced physiological changes, which may indicate potential safety concerns in vivo. No obvious signs of fibrosis, necrosis, or acute inflammation were detected in any KB105-treated samples as compared to vehicle control (FIG. **6**A). Post-sacrifice quantitative PCR analysis of the topically treated skin indicated that KB105 effectively transduced both the acetone treated- and tape strip-permeabilized skin (FIG. **6**B), and high levels of human TGM1 transcripts were expressed after infection (FIG. **6**C). While acetone treatment or tape stripping of the skin was found to induce endogenous mouse TGM1 transcription, no significant differences in endogenous TGM1 expression were observed between low or high dose KB105 and vehicle control (FIG. **7**).

[0374] Exogenous TGM1 protein expression and tissue localization were assessed by immunofluorescence. Human TGM1 was detected in mouse epidermis upon topical application of KB105 (FIG. **6**D). Paralleling the qPCR and qRT-PCR results, a qualitative increase in TGM1 protein was observed in the high vs. low dose samples. Samples were also co-stained for mouse loricrin (a natural substrate for TGM1) and mouse integrin alpha-6 (a marker of the basal layer of the epidermis) to determine whether the exogenously expressed TGM1, originating from KB105, was correctly localized to the stratum *granulosum*, the tissue layer where endogenous TGM1 is expressed and functionally active. Mouse loricrin colocalized with human TGM1 in all KB105-treated samples, while TGM1 was detected in a more superficial layer than mouse integrin alpha-6. Together, this data demonstrates that KB105 successfully transduced the targeted epidermal layer.

[0375] A short-term pharmacokinetics study was conducted in the tape stripped skin of BALB/c mice after topical administration of KB105 (FIGS. **8**A-**8**C). Vector genomes in the transduced cells remained relatively stable over the course of the 48-hour study, indicated by similar numbers of human TGM1 DNA copies between the 8- and 48-hour timepoints. Human TGM1 transcripts were detected as early as two hours after topical application, and steadily increased over time, peaking 24 hours after treatment before declining (while remaining detectable) at 48 hours. Similar transgene kinetics were observed at the protein level, as assessed by immunofluorescence.

[0376] The safety and feasibility of repeated in vivo vector applications using two different dosing intervals (Days 1 and 3 vs. Days 1 and 12) was next investigated. Briefly, mice were tape stripped and treated topically with KB105 (or vehicle control) on Day 1. Skin tissues from a first cohort of animals were harvested on Day 3 to act as positive (KB105) and negative (vehicle) controls. Additional animal cohorts were re-tape stripped and re-treated topically with KB105 on Days 3 or 12, with tissues subsequently harvested on Days 5 and 14, respectively.

[0377] Histological examination of skin samples found no obvious signs of toxicity or tissue reorganization, including fibrosis, necrosis, or acute inflammation, in any of the single or repeat KB105-treated samples (FIG. **9**A). High vector genome copy numbers were detected 48 hours after a single administration of KB105; comparable genome copy numbers were also detected 48 hours after a repeated KB105 dose administered at either Day 3 or Day 12 (FIG. **9**B). Similar human TGM1 transcript levels were detected after 48 hours in animals receiving a single KB105 dose on Day 1 or a second dose on Days 3 or 12 (FIG. **9**C).

[0378] Human TGM1 protein levels were qualitatively measured. Epidermal localization was assessed by colocalization with mouse loricrin. Significant levels of TGM1 protein, as well as proper epidermal localization, were detected in skin tissue biopsies harvested from mice treated either once or twice with KB105 (FIG. **9**D). While some variability was observed in TGM1 transcript numbers in these mouse cohorts, no gross differences in TGM1 protein expression were observed by immunofluorescence after single vs. repeat administration of KB105.

[0379] Toxicity and biodistribution of KB105 were evaluated in a Good Laboratory Practice (GLP) repeat-dose study in male and female BALB/c mice (Table 1). Animals were dosed once a week for five weeks with topical KB105 (group 2) or vehicle control gel (group 1) after skin permeabilization via tape stripping. Six animals/sex/group were administered one dose on Day 1 and necropsied on Day 3. Six animals/sex/group were dosed on Days 1, 8, 15, 22, and 29 and necropsied on Day 30. All remaining surviving animals were dosed on Days 1, 8, 15, 22, and 29, and were then subjected to a 33-day recovery phase before necropsy.

TABLE 1

Design of the GLP repeat-dose biodistribution and toxicity study

Group No.	No. of Animals[a] Male	Female	Dose Level (PFU/application)
1 (Vehicle Control)[b]	18	18	0
2 (KB105)	18	18	1.07×10^9

PFU = plaque forming unit

[a]Animals designated for interim sacrifice (six animals/sex/group) were euthanized on Day 3 of the dosing phase. Animals designated for terminal necropsy (six animals/sex/group) were euthanized on Day 30 of the dosing phase. All remaining surviving animals underwent a 33-day recovery phase following completion of dose administration and were euthanized on Day 34 of the recovery phase (Day 63 of the dosing phase).

[b]Group 1 was administered vehicle control formulated in gel excipient only.

[0380] Assessment of toxicity was based on mortality, clinical observations, body weights, food consumption, dermal observations, and clinical and anatomic pathology. No KB105-related mortality, clinical observations, body weight or food consumption changes, macroscopic findings, or effects on organ weight parameters were noted. All animals survived until their scheduled necropsy. Microscopic examination was limited to select tissues, including application/dose site, sternum with bone, bone marrow, brain, epididymis, heart, kidneys, liver, lungs, axillary lymph node, inguinal lymph node, ovaries, oviducts, prostate, spleen, testes, thymus, and uterus with cervix. Microscopic findings localized to the treated skin site, including hyperkeratosis, epithelial hyperplasia, inflammation, edema, and fibrosis, were evaluated at the interim and terminal sacrifices. These findings were consistent with repair following abrasion and were considered related to the dosing procedure. At the recovery sacrifice, most animals exhibited complete reversibility of the dosing procedure-related microscopic findings. As such, the no observed adverse effect level (NOAEL) for topical application of KB105 was found to be 1.07×10^9 PFU/day.

[0381] Blood and tissue samples were analyzed by qPCR for the determination of KB105 biodistribution. Nearly all blood and tissue samples collected from the vehicle control animals (Group 1) over the three intervals were negative for KB105, except for six dose site samples. A root cause analysis indicated that a contamination occurred during the preparation of vehicle specifically used for these 6 animals. Detection of high levels of KB105 in Group 2 animals was generally limited to the dose site, with no pronounced accumulation of the vector in other analyzed tissues. Vector persistence was minimal, as indicated by low-to-negative detection of vector copies obtained in samples analyzed from recovery phase animals.

[0382] In vivo studies demonstrated that both single and repeated topical KB105 administration induced TGM1 protein expression in the target epidermal layer without triggering fibrosis, necrosis, or acute inflammation. Toxicity and biodistribution assessments upon repeat-dosing indicated that KB105 was well tolerated and restricted to the dose site. Without wishing to be bound by theory, these results provide a proof-of-concept for the use of a recombinant herpes virus as a gene therapy platform for safely and non-invasively treating ichthyosis.

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source                   1..477
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                         organism = synthetic construct
SEQUENCE: 16
atgatgcggt tcatgctgct gttcagcaga cagggcaagc tgcggctgca gaagtggtat  60
ctggccacca gcgacaaaga acggaagaag atggtccgag agctgatgca ggtcgtgctg  120
gccagaaagc ccaagatgtg cagcttcctg gaatggcggg acctgaaggt ggtgtacaag  180
agatacgcca gcctgtactt ctgctgcgcc atcgaaggcc aggacaacga gctgatcacc  240
ctggaactga tccacagata cgtggaactg ctggacaagt acttcggcag cgtgtgcgag  300
ctggacatca tcttcaactt cgagaaggcc tactttatcc tggacgagtt cctgatgggc  360
ggcgacgtgc aggacaccag caagaaatct gtgctgaagg ccatcgagca ggccgacctg  420
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ctgcaagaag aggacgagag ccctagaagc gtgctggaag aaatgggcct cgcctga     477

SEQ ID NO: 17           moltype = DNA  length = 1770
FEATURE                 Location/Qualifiers
source                  1..1770
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 17
atgttacatc tgcaccattc ttgtttgtgt ttcaggagct ggctgccagc gatgctcgct  60
gtactgctaa gtttggcacc atcagcttcc agcgacattt ccgcctcccg accgaacatc  120
cttcttctga tggcggacga ccttggcatt ggggacattg gctgctatgg caacaacacc  180
atgaggactc cgaatattga ccgccttgca gaggacggcg tgaagctgac ccaacacatc  240
tctgccgcat ctttgtgcac cccaagcaga gccgccttcc tcacgggcag ataccctgtg  300
cgatcaggga tggtttccag cattggttac cgtgttcttc agtggaccgg agcatctgga  360
ggtcttccaa caaatgagac aacttttgca aaaatactga aagagaaagg ctatgccact  420
ggactcattg gaaaatggca tctgggtctc aactgtgagt cagccagtga tcattgccac  480
caccctctcc atcatggctt tgaccatttc tacggaatgc ctttctcctt gatgggtgat  540
tgcgcccgct gggaactctc agagaagcgt gtcaacctgg aacaaaaact caacttcctc  600
ttccaagtcc tggccttggt tgccctcaca ctggtagcag ggaagctcac acacctgata  660
cccgtctcgt ggatgccggt catctggtca gccctttcgg ccgtcctcct cctcgcaagc  720
tcctattttg tgggtgctct gattgtccat gccgattgct ttctgatgag aaaccacacc  780
atcacggagc agcccatgtg cttccaaaga acgacacccc ttattctgca ggaggttgcg  840
tcctttctca aaaggaataa gcatgggcct ttcctcctct ttgtttcctt tctacacgtt  900
cacatccctc ttatcactat ggagaacttc ctcgggaaga gtctccacgg gctgtatggg  960
gacaacgtag aggagatgga ctggatggta ggacggatcc ttgacacttt ggacgtggag  1020
ggtttgagca acagcaccct catttatttt acgtcggatc acggcggttc cctagagaat  1080
caacttggaa acacccagta tggtggctgg aatggaattt ataaaggtgg gaagggcatg  1140
ggaggatggg aaggtgggat ccgcgtgccc gggatcttcc gctggcccgg ggtgctcccg  1200
gccggccgag tgattggcga gcccacgagt ctgatggacg tgttccccac cgtggtccgg  1260
ctggcgggcg gcgaggtgcc ccaggacaga gtgattgacg gccaagacct tctgcccttg  1320
ctcctgggga cagcccaaca ctcagaccac gagttcctga tgcattattg tgagaggttt  1380
ctgcacgcag ccaggtggca tcaacgggac agaggaacaa tgtggaaagt ccactttgtg  1440
acgcctgtgt tccagccaga gggagccggt gcctgctatg gaagaaaggt ctgcccgtgc  1500
tttggggaaa aagtagtcca ccacgatcca cctttgctct ttgacctctc aagagaccct  1560
tctgagaccc acatcctcac accagcctca gagcccgtgt tctatcaggt gatggaacga  1620
gtccagcagg cggtgtggga acaccagcgg acactcagcc cagttcctct gcagctggac  1680
aggctgggca acatctggag accgtggctg cagccctgct gtggcccgtt ccccctctgc  1740
tggtgcctta gggaagatga cccacaataa                                   1770

SEQ ID NO: 18           moltype = DNA  length = 1770
FEATURE                 Location/Qualifiers
misc_feature            1..1770
                        note = Synthetic
source                  1..1770
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 18
atgctgcatc tgcaccacag ctgcctgtgc ttcagatctt ggctgcctgc catgctggcc  60
gtgctgttgt ctcttgcccc tagcgccagc agcgatatca gcgcctccag acctaacatc  120
ctgctgctga tggccgacga tctcggcatc ggcgatatcg gctgctacgg caacaacacc  180
atgcggaccc ctaacatcga cagactggcc gaggatggcg tgaagctgac acagcatatt  240
agcgccgcca gcctgtgcac cccttctaga gctgcatttc tgaccggcag ataccctgtg  300
cggagcggca tggtgtctag catcggctat agagtgctgc agtggacagg cgcttctggc  360
ggccttccta ccaacgagac aaccttcgcc aaaatcctga aagagaaggg ctacgccacc  420
ggcctgatcg gaaaatggca cctgggactg aattgcgaga gcgccagcga ccactgtcac  480
catcctctgc accatggctt cgaccacttc tacggcatgc ccttcagcct gatgggcgat  540
tgtgccagat gggagctgag cgagaagaga gtgaacctgg aacagaagct gaacttcctg  600
ttccaagtgc tggccctggt ggccctgaca ctggttgctg gcaaactgac ccatctgatc  660
cccgtgtcct ggatgcccgt gatttggagc gctctgtctg ctgtgctgct gctggctagc  720
agctattttg tgggcgccct gatcgtgcac gccgactgct ttctgatgcg gaaccacacc  780
atcaccgagc agcccatgtg cttccagaga acaacccctc tgatcctgca agaggtggcc  840
agcttcctga agcggaacaa gcacggccca tttctgctgt tcgtgtcctt cctccacgtg  900
cacatccctc tgattacgat ggaaaacttc ctgggcaaga gcctgcacgg cctgtacggc  960
gataacgtgg aagagatgga ctggatggtc ggaagaatcc tggacaccct ggacgtggaa  1020
ggcctgagca atagcaccct gatctacttc acctccgacc acggcggcag cctggaaaat  1080
cagctgggca atacccagta cggcggctgg aacggcatct acaaaggcgg caaaggcatg  1140
ggcggatggg aaggcggaat tagagtgccc ggcatcttca gatggcctgg cgttttgcct  1200
gccggaagag tgatcggcga gcccacctct ctgatggacg tgttccctac agttgtgcgg  1260
ctggctggcg gagaagtgcc tcaggataga gtgattgacg gccaggacct gctgcctctg  1320
ttgctgggaa cagcccagca cagcgatcac gagttcctca tgcactactg cgagcggttt  1380
ctgcacgccg ccagatggca ccagcgagat agaggcacaa tgtggaaggt gcacttcgtg  1440
acccctgtgt tccagcctga aggtgccggc gcttgttacg gcagaaaagt gtgcccttgc  1500
ttcggcgaga aggtggtgca tcacgatcct ccactgctgt ttgacctgag cagggaccct  1560
agcgagacac acattctgac accagccagc gagcccgtgt tctaccaagt gatggaaagg  1620
gtgcagcagg ccgtgtggga gcaccagaga actctgtctc cagtgcctct gcagctggat  1680
cggctgggca acatttggag gccttggctc cagccttgct gcggcccttt tcctctgtgc  1740
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tggtgcctga gagaagatga ccctcagtga                               1770

SEQ ID NO: 19           moltype = DNA  length = 729
FEATURE                 Location/Qualifiers
source                  1..729
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 19
atgagcaatc cgcggtcttt ggaagaggag aaatatgata tgtcaggtgc ccgcctggcc  60
ctaatactgt gtgtcaccaa agcccgggaa ggttccgaag aagacctgga tgctctggaa  120
cacatgtttc ggcagctgag attcgaaagc accatgaaaa gagaccccac tgccgagcaa  180
ttccaggaag agctggaaaa attccagcag gccatcgatt cccgggaaga tcccgtcagt  240
tgtgccttcg tggtactcat ggctcacggg agggaaggct tcctcaaggg agaagatggg  300
gagatggtca agctggagaa tctcttcgag gccctgaaca acaagaactg ccaggccctg  360
cgagctaagc ccaaggtgta catcatacag gcctgtcgag gagaacaaag ggaccccggt  420
gaaacagtag gtggagatga gattgtgatg gtcatcaaag acagcccaca aaccatccca  480
acatacacag atgccttgca cgtttattcc acggtagagg gatacatcgc ctaccgacat  540
gatcagaaag gctcatgctt tatccagacc ctggtggatg tgttcacgaa gaggaaagga  600
catatcttgg aacttctgac agaggtgacc cggcggatgg cagaagcaga gctggttcaa  660
gaaggaaaag caaggaaaac gaaccctgaa atccaaagca ccctccggaa acggctgtat  720
ctgcagtag                                                      729

SEQ ID NO: 20           moltype = DNA  length = 729
FEATURE                 Location/Qualifiers
misc_feature            1..729
                        note = Synthetic
source                  1..729
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 20
atgagcaacc ccagaagcct ggaagaagag aagtacgaca tgagcggcgc cagactggcc  60
ctgatcctgt gtgtgacaaa ggccagagag ggcagcgaag aggatctgga tgccctggaa  120
cacatgttcc ggcagctgag attcgagagc accatgaaga gagatcccac cgccgagcag  180
ttccaagagg aactggaaaa gttccagcag gccatcgaca gcagagaaga tcccgtgtcc  240
tgcgccttcg tggtgctgat ggctcacggc agagagggat ttctgaaagg cgaggacggc  300
gagatggtca agctggaaaa tctgttcgag gccctgaaca acaagaactg tcaggccctg  360
cgggccaagc ctaaggtgta catcatccag gcctgcagag gcgagcaaag agatcctggc  420
gaaaccgtcg gcggcgacga gatcgtgatg gtcatcaagg acagccctca gacaatcccc  480
acctacacag atgccctgca cgtgtacagc accgtggaag gctatatcgc ctaccggcac  540
gatcagaagg gcagctgctt catccagaca ctggtggacg tgttcaccaa gcggaagggc  600
cacattctcg agctgctgac cgaagtgacc agaagaatgg ccgaggccga gctggtgcaa  660
gagggcaaag ccagaaagac aaaccccgag atccagagca ccctgcggaa gagactgtac  720
ctgcagtga                                                      729

SEQ ID NO: 21           moltype = DNA  length = 1590
FEATURE                 Location/Qualifiers
source                  1..1590
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 21
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ctgctggctg gtctcctcct gccagggacc ttggctaaga gcattggcac cttctcagac  120
ccttgtaagg accccacgcg tatcacctcc cctaacgacc cctgcctcac tgggaagggt  180
gactccagcg gtttcagtag ctacagtggc tccagcagtt ctggcagctc catttccagt  240
gccagaagct ctggtggtgg ctccagtggt agctccagcg gatccagcat tgcccagggt  300
ggttctgcag gatcttttaa gccaggaacg gggtattccc aggtcagcta ctcctccgga  360
tctggctcta gtctacaagg tgcatccggt tcctcccagc tggggagcag cagctctcac  420
tcgggaaaca gcggctctca ctcgggaagc agcagctctc attcgagcag cagcagcagc  480
tttcagttca gcagcagcag cttccaagta gggaatggct ctgctctgcc aaccaatgac  540
aactcttacc gcggaatact aaacccttcc cagcctggac aaagctcttc ctcttcccag  600
acctttgggg tatccagcag tggccaaagc gtcagctcca accagcgtcc ctgtagttcg  660
gacatccccg actctccctg cagtggaggg cccatcgtct cgcactccgg cccctacatc  720
cccagctccc actctgtgtc agggggtcag aggcctgtgg tggtggtggt ggaccagcac  780
ggttctggtg cccctggagt ggttcaaggt ccccccctgta gcaatggtgg ccttccaggc  840
aagccctgtc cccaatcac ctctgtagac aaatcctatg gtggctacga ggtggtgggt  900
ggctcctctg acagttatct ggttccagc atgacctaca gtaagggtaa aatctaccct  960
gtgggctact tcaccaaaga gaaccctgtg aaaggctctc caggggtccc ttcctttgca  1020
gctgggcccc ccatctctga gggcaaatac ttctccagca accccatcat ccccagccag  1080
tcggcagctt cctcggccat tgcattccag ccagtgggga ctggtggggt ccagctctgt  1140
ggaggcggct ccacgggctc caagggaccc tgctctccct ccagttctcg agtccccagc  1200
agttctagca tttccagcag ctccggttta ccctaccatc cctgcggcag tgcttcccag  1260
agcccctgct ccccaccagg caccggctcc ttcagcagca gctccagttc ccaatccagt  1320
ggcaaaatca tccttcagcc ttgtggcagc aagtccagct cttctggtca cccttgcatg  1380
tctgtctcct ccttgacact gactgggggc cccgatggct ctccccatcc tgatccctcc  1440
gctggtgcca agccctgtgg ctccagcagt gctggaaaga tcccctgccg ctccatccgg  1500
gatatcctag cccaagtgaa gcctctgggg ccccagctag ctgaccctga agtttcctta  1560
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ccccaaggag agttactcaa cagtccataa                                   1590

SEQ ID NO: 22           moltype = DNA  length = 1590
FEATURE                 Location/Qualifiers
misc_feature            1..1590
                        note = Synthetic
source                  1..1590
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 22
atgggcagtt ctagagcccc ttggatgggc agagttggcg gccatggaat gatggctctg  60
ctgctggctg gactgctgct ccctggaaca ctggccaaga gcatcggcac cttcagcgat  120
ccatgcaagg accccaccag aatcacaagc cccaacgatc cttgcctgac cggcaagggc  180
gatagcagcg gctttagcag ctacagcggc agctctagct ctggcagcag catcagcagc  240
gccagatctt ctggcggagg cagctctggt tctagctccg gatcttctat cgcccaaggc  300
ggctctgccg gctcttttaa gcctggcaca ggctacagcc aggtgtccta ctctagcggc  360
tctggaagtt ctctgcaggg cgcctctgga tcttctcagc tgggatctag cagcagccac  420
tccggcaatt ctggctctca ctctggctcc agcagctccc actctagcag ctccagctcc  480
ttccagttca gctcctccag cttccaagtc ggcaatggaa gcgccctgcc taccaacgac  540
aacagctatc ggggcattct gaacccctct cagcctggcc aaagcagcag cagctctcag  600
acatttggcg tgtccagcag cggccagtcc gtgtctagca atcagaggcc ttgcagcagc  660
gacatccctg actctccatg ttctggcggc cctatcgtgt ctcacagcgg accttacatc  720
ccttccagcc actctgtgtc tggcggacaa agacctgtgg tggtggtcgt ggatcagcat  780
ggaagcggag cacctggtgt tgtgcagggc cctccttgtt ctaatggcgg cctgcctgga  840
aagccctgtc ctcctatcac cagcgtggac aagagctacg gcggctatga ggttgtcggc  900
ggctccagcg attcttacct ggtgcctggc atgacctaca gcaagggcaa gatctacccc  960
gtgggctact tcaccaaaga aaaccccgtg aagggcagcc ctggcgtgcc atcttttgcc  1020
gctggacctc ctatctccga gggcaagtac ttcagcagca accctatcat ccccagccag  1080
agcgccgcct cttctgccat tgcttttcag cctgttggca ccggtggcgt gcaactttgt  1140
ggcggaggaa gcacaggcag caagggacct tgtagcccat ccagctctag agtgcccagc  1200
tcctctagca tctcttccag ctccggcctg ccatatcacc cttgtggatc tgccagccag  1260
tctccttgca gccctcctgg aacaggcagc ttcagttcct ctagctcctc tcagagcagc  1320
ggcaagatca tcctgcagcc ttgtggcagc aagtctagct ccagcggaca cccttgtatg  1380
agcgtgtcct ctctgacact gacaggcggc cctgatggaa gccctcatcc tgatccatct  1440
gccggcgcta agccatgtgg aagctctagc gccggaaaga tcccctgcag atccatcaga  1500
gacattctgg cccaagtgaa gcctctgggc cctcagcttg ctgaccccga agtgtttctg  1560
cctcaaggcg agctgctgaa cagcccctaa                                   1590

SEQ ID NO: 23           moltype = DNA  length = 1152
FEATURE                 Location/Qualifiers
source                  1..1152
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 23
atgttttgga cgtttaaaga atggttctgg ttggaaagat tctggcttcc tccaacaata  60
aagtggtcag atcttgagga tcacgatgga ctcgtctttg taaaaccttc tcatttatac  120
gtgacaattc catatgcttt tctcttgctg attatcagac gtgtatttga aaaatttgtt  180
gcttcacctc tagcaaaatc atttggcatt aaagagacag ttcgaaaggt tacaccaaat  240
actgtcttag agaatttttt caaacattcc acaaggcaac cattgcaaac tgatatttat  300
ggactggcaa agaagtgtaa cttgacggag cgccaggtgg aaagatggtt taggagtcgg  360
cggaatcaag agaggccttc caggctgaag aaattccagg aagcttgctg gagatttgca  420
ttttacttaa tgatcactgt tgctggaatt gcgtttcttt atgataaacc ttggctatat  480
gacttatggg aggtttggaa tggctatccc aaacagcccc tgctgccatc ccagtactgg  540
tactacattt tagaaatgag tttttattgg tctctgttat ttagacttgg ctttgatgtc  600
aagagaaagg attttctagc tcatatcatc caccacctgg ctgctattag tctgatgagc  660
ttctcttggt gtgctaatta tattcgcagt gggaccctcg tgatgattgt acacgatgtg  720
gctgacattt ggctggagtc tgctaagatg ttttcttatg ctggatggac gcagacctgt  780
aacaccctgt ttttcatctt ctccaccata tttttcatca gccgcctcat tgtttttcct  840
ttctggattt tatattgcac gctgatcttg cctatgtatc acctcgagcc tttcttttca  900
tacatcttcc tcaacctaca gctcatgatc ttgcaggtcc ttcaccttta ctggggttat  960
tacatcttga agatgctcaa cagatgtata ttcatgaaga gcatccagga tgtgaggagt  1020
gatgacgagg attatgaaga ggaagaggaa gaggaagaag aagaggctac caaaggcaaa  1080
gagatggatt gtttaaagaa cggcctcagg gctgagaggc acctcattcc caatggccag  1140
catggccatt ag                                                      1152

SEQ ID NO: 24           moltype = DNA  length = 1152
FEATURE                 Location/Qualifiers
misc_feature            1..1152
                        note = Synthetic
source                  1..1152
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 24
atgttctgga ccttcaaaga gtggttctgg ctggaaagat tctggctgcc tcctaccatc  60
aagtggagcg acctggaaga tcacgacggc ctggtgttcg tgaagcccag ccacctgtac  120
gtgacaatcc cttacgcctt cctgctgctg atcatccggc gggtgttcga gaagttcgtg  180
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gcctctcctc tggccaaatc cttcggcatc aaagaaaccg tgcggaaagt gacccctaat  240
accgtgctgg aaaacttctt caagcacagc accagacagc cctgcagacc cgatatctat  300
ggcctggcca agaagtgcaa cctgaccgag agacaggtgg aacggtggtt cagaagccgg  360
cggaatcaag agaggcccag cagactgaag aaattccaag aagcctgttg gcgcttcgcc  420
ttctacctga tgatcaccgt ggccggaatc gcctttctgt acgacaagcc ctggctgtac  480
gacctgtggg aagtgtggaa cggctacccc aaacagcctc tgctgcccag ccagtattgg  540
tactacatcc tggaaatgag cttctactgg tccctgctgt tcagactggg cttcgacgtg  600
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ttctggattc tgtactgcac cctgatcctg cctatgtacc acctggaacc tttcttcagc  900
tacatcttcc tgaacctgca gctgatgatc ctgcaggtcc tgcacctgta ctggggctac  960
tatatcctga agatgctgaa ccggtgcatc tttatgaagt ccatccagga cgtgcggagc  1020
gacgacgagg actacgagga agaagaggaa gaggaagaag aagaggccac caagggcaaa  1080
gagatggact gcctgaagaa cggcctgcgg gccgagaggc acctgattcc taatggacag  1140
cacggccact ga                                                      1152

SEQ ID NO: 25          moltype = DNA  length = 1275
FEATURE                Location/Qualifiers
source                 1..1275
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 25
atgaccctgc gacctggaac aatgcggctg gcctgcatgt tctcttccat cctgctgttc  60
ggagctgcag gcctcctcct cttcatcagc ctgcaggacc ctacggagct cgcccccccag  120
caggtgccag gaataaagtt caacatcagg ccaaggcagc cccaccacga cctcccacca  180
ggcggctccc aggatggtga cttgaaggaa cccacagaga gggtcactcg ggacttatcc  240
agtggggccc cgaggggccg caacctgcca gcgcctgacc agcctcaacc cccgctgcag  300
aggggaaccc gtctgcggct ccgccagcgc cgtcgccgtc tgctcatcaa gaaaatgcca  360
gctgcggcga ccatcccggc caacagctcg gacgcgccct tcatccggcc gggacccggg  420
acgctggatg gccgctgggt cagcctgcac cggagccagc aggagcgcaa gcgggtgatg  480
caggaggcct gcgccaagta ccgggcgagc agcagccgcc gggccgtcac gccccgccac  540
gtgtcccgta tcttcgtgga ggaccgccac cgcgtgctct actgcgaggt gcccaaggcc  600
ggctgctcca attggaagcg ggtgctcatg gtgctggccg gcctggcctc gtccactgcc  660
gacatccagc acaacaccgt ccactatggc agcgctctca agcgcctgga caccttcgac  720
cgccagggta tcttgcaccg tctcagcacc tacaccaaga tgctctttgt ccgcgagccc  780
ttcgagaggc tggtgtccgc cttccgcgac aagtttgagc accccaacag ctactatcac  840
ccggtcttcg gcaaggccat cctggcccgg taccgcgcca atgcctctcg ggaggccctg  900
cggaccggct ctggggtgcg ttttcccgag ttcgtccagt acctgctgga cgtgcaccgg  960
cccgtgggga tggacattca ctgggaccat gtcagccggc tctgcagccc ctgcctcatc  1020
gactacgatt tcgtaggcaa gttcgagagc atggaggacg atgccaactt cttcctgagc  1080
ctcatccgcg cgccgcggaa cctgaccttc cccccggttca aggaccggca ctcgcaggag  1140
gcgcggacca cagcgaggat cgcccaccag tacttcgccc aactctcggc cctgcaaagg  1200
cagcgcacct acgacttcta ctacatggat tacctgatgt tcaactattc caagcccttt  1260
gcagatctgt actga                                                   1275

SEQ ID NO: 26          moltype = DNA  length = 1275
FEATURE                Location/Qualifiers
misc_feature           1..1275
                       note = Synthetic
source                 1..1275
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 26
atgacactga ggcccggcac aatgagactg gcctgcatgt tcagcagcat cctgctgttt  60
ggagccgccg gactgctgct gttcatcagt ctgcaggacc ctacagagct ggcccctcag  120
caagtgcctg gcatcaagtt caacatcaga cccagacagc cccaccacga tcttccacct  180
ggtggatctc aggacggcga cctgaaagaa cccaccgaga gagtgaccag ggatctgtct  240
agcggagccc ctagaggcag aaacctgcct gctcctgatc agcctcagcc tccactgcag  300
agaggcacca gactgagact gcggcagaga agaagaaggc tgctgatcaa gaagatgcct  360
gccgccgcta caatccccgc caattcttct gacgcccctt tcatcagacc tggacctggc  420
acactggacg gcagatgggt ttccctgcac agatctcagc aagagcggaa gagagtgatg  480
caagaggcct gcgccaagta cagagccagc agctctagaa gggccgtgac acccagacat  540
gtgtcccgga tcttcgtgga agatagacac cgggtgctgt actgcgaggt gccaaaggcc  600
ggctgcagca attggaagag ggtgctgatg gtgctggccg gcctggcttc ttctacagcc  660
gacatccagc acaacaccgt gcactatggc agcgccctga agcggctgga cacctttgat  720
agacagggca tcctgcaccg gctgagcacc tacaccaaga tgctgttcgt gcgcgagccc  780
ttcgagagac tggtgtccgc cttcagagac aagttcgagc accccaacag ctactatcac  840
cccgtgttcg gcaaggccat cctggccaga tatagagcca acgcctctag agaggccctg  900
agaacaggct ctggcgtgcg gtttcctgag ttcgtgcagt acctgctgga cgtgcacaga  960
cccgtcggaa tggacatcca ctgggaccac gtgtccagac tgtgcagccc ttgcctgatc  1020
gactacgact tcgtgggcaa gtttgagagc atggaagatg acgccaactt ctttctgtcc  1080
ctgattcggg cccctcggaa cctgacattc cccagattca aggaccggca cagccaagag  1140
gccagaacca cagccagaat cgcccaccag tacttcgccc agctgtctgc cctgcagaga  1200
cagcggacct acgatttcta ctacatggac tacctgatgt tcaactacag caagcccttc  1260
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gccgacctgt actga                                                  1275

SEQ ID NO: 27           moltype = DNA  length = 636
FEATURE                 Location/Qualifiers
source                  1..636
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 27
atggccaacg cggggctgca gctgttgggc ttcattctcg ccttcctggg atggatcggc  60
gccatcgtca gcactgccct gccccagtgg aggatttact cctatgccgg cgacaacatc  120
gtgaccgccc aggccatgta cgaggggctg tggatgtcct gcgtgtcgca gagcaccggg  180
cagatccagt gcaaagtctt tgactccttg ctgaatctga gcagcacatt gcaagcaacc  240
cgtgccttga tggtggttgg catcctcctg ggagtgatag caatctttgt ggccaccgtt  300
ggcatgaagt gtatgaagtg cttggaagac gatgaggtgc agaagatgag gatggctgtc  360
attgggggtg cgatatttct tcttgcaggt ctggctattt tagttgccac agcatggtat  420
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tttggtcagg ctctcttcac tggctgggct gctgcttctc tctgccttct gggaggtgcc  540
ctactttgct gttcctgtcc ccgaaaaaca acctcttacc caacaccaag gccctatcca  600
aaacctgcac cttccagcgg gaaagactac gtgtga                          636

SEQ ID NO: 28           moltype = DNA  length = 636
FEATURE                 Location/Qualifiers
misc_feature            1..636
                        note = Synthetic
source                  1..636
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 28
atggccaatg ctggactgca gctgctgggc ttcatcctgg cctttctcgg atggatcggc  60
gccatcgtgt ctacagcttt gcctcagtgg cggatctaca gctacgccgg cgacaatatc  120
gtgacagccc aggctatgta cgaaggcctg tggatgagct gcgtgtccca gtctacaggc  180
cagatccagt gcaaggtgtt cgacagcctg ctgaacctga gcagcacact gcaggctaca  240
agagccctga tggtcgtggg aatcctgctg ggagtgatcg ccatctttgt ggccaccgtg  300
ggcatgaagt gtatgaagtg cctggaagat gacgaggtgc agaaaatgcg gatggccgtg  360
atcggcggag ccatctttct gctggctgga ctggctatcc tggtggccac tgcttggtac  420
ggcaaccgga tcgtgcaaga gttctacgac cccatgacac ccgtgaacgc cagatacgag  480
tttggacagg ccctgttcac aggatgggcc gctgcttctc tgtgtctgct tggcggagca  540
ctgctgtgct gcagctgtcc cagaaagacc acaagctacc ccacacctcg gccttatcct  600
aagcctgctc ctagcagcgg caaggactac gtgtaa                          636

SEQ ID NO: 29           moltype = DNA  length = 297
FEATURE                 Location/Qualifiers
source                  1..297
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 29
atgatacctg gaggcttatc tgaggccaaa cccgccactc cagaaatcca ggagattgtt  60
gataaggtta aaccacagct tgaagaaaaa acaaatgaga cttacggaaa attggaagct  120
gtgcagtata aaactcaagt tgttgctgga acaaattact acattaaggt acgagcaggt  180
gataataaat atatgcactt gaaagtattc aaaagtcttc ccggacaaaa tgaggacttg  240
gtacttactg gataccaggt tgacaaaaac aaggatgacg agctgacggg cttttag    297

SEQ ID NO: 30           moltype = DNA  length = 297
FEATURE                 Location/Qualifiers
misc_feature            1..297
                        note = Synthetic
source                  1..297
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 30
atgattcctg gcggcctgtc tgaggccaaa cctgccacac ctgagatcca agagatcgtg  60
gacaaagtga agccccagct ggaagaaaag accaacgaga catacggcaa gctggaagcc  120
gtgcagtaca agacacaggt ggtggccggc accaactact acatcaaagt gcgcgctggc  180
gacaacaagt acatgcacct gaaggtgttc aagagcctgc ctggccagaa cgaggatctg  240
gtgctgacag gctaccaggt ggacaagaac aaggacgacg agctgaccgg cttctga    297

SEQ ID NO: 31           moltype = DNA  length = 1596
FEATURE                 Location/Qualifiers
source                  1..1596
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 31
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atatacgcgg tgtccaccct tctcctcttc ctgctcttct tcctgttccg cctgctgctg  120
cggttcctga ggctctgcag gagcttctac atcacctgcc gccggctgcg ctgcttcccc  180
cagcctcccc ggcgcaactg gctgctgggc cacctgggca tgtaccttcc aaatgaggcg  240
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ggccttcaag atgagaagaa ggtactggac aacatgcacc atgtactctt ggtatggatg  300
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gcctcagctg ccatcgcccc caaggatgac ctcttctatg gcttcctaaa accttggcta  420
ggggatgggc tgctgctcag caaaggtgac aagtggagcc ggcaccgtcg cctgctgaca  480
cccgccttcc actttgacat cctgaagcct tacatgaaga tcttcaacca gagcgctgac  540
attatgcatg ctaaatggcg gcatctggca gagggctcag cggtctccct tgatatgttt  600
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aactgccaag agaagatgag tgattatatc tccgctatca ttgaactgag cgctctgtct  720
gtccggcgcc agtatcgctt gcaccactac ctcgacttca tttactaccg ctcggcggat  780
gggcggaggt tccggcaggc ctgtgacatg gtgcaccact tcaccactga agtcatccag  840
gaacggcggc gggcactgcg tcagcagggg gccgaggcct ggcttaaggc caagcagggg  900
aagaccttgg actttattga tgtgctgctc ctggccaggg atgaagatgg aaaggaactg  960
tcagacgagg atatccgagc cgaagcagac accttcatgt ttgagggtca cgacacaaca  1020
tccagtggga tctcttggat gctgttcaat ttggcaaagt atccggaata ccaggagaaa  1080
tgccgagaag agattcagga agtcatgaaa ggccgggagc tggaggagct ggagtgggac  1140
gatctgactc agctgccctt tacaactatg tgcattaagg agagcctgcg ccagtaccca  1200
cctgtcactc ttgtctctcg ccaatgcacg gaggacatca agctcccaga tgggcgcatc  1260
atccccaaag gaatcatctg cttggtcagc atctatggaa cccaccacaa ccccacagtg  1320
tggcctgact ccaaggtgta caacccctac cgctttgacc cggacaaccc acagcagcgc  1380
tctccactgg cctatgtgcc cttctctgca ggacccagga attgcatcgg acagagcttc  1440
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gaccgaacgc gcaaggtgcg gcggaagccg gagctcatac tgcgcacgga gaacgggctc  1560
tggctcaagg tggagccgct gcctccgcgg gcctga                            1596

SEQ ID NO: 32          moltype = DNA  length = 1596
FEATURE                Location/Qualifiers
misc_feature           1..1596
                       note = Synthetic
source                 1..1596
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 32
atgctgccca tcaccgacag actgctgcat ctgctgggcc tcgagaaaac cgccttcaga  60
atctacgccg tgtctaccct gctgctgttc ctgctgtttt tcctgttccg gctgctgctg  120
agattcctgc ggctgtgccg gtccttctac atcacctgtc ggcggctgag atgcttccct  180
cagcctccta gaagaaactg gctgctggga cacctgggca tgtacctgcc taatgaagcc  240
ggcctgcagg acgagaagaa agtgctggac aacatgcacc acgtgctgct cgtgtggatg  300
ggacctgttc tgcctctgct ggtgctggtg caccccgact atatcaagcc actgctggga  360
gcctctgccg ctatcgcccc taaggacgat ctgttctacg gcttcctgaa gccttggctc  420
ggagatggcc tgctgctcag caaaggcgac aagtggtcca gacacagacg gctgctgacc  480
cctgccttcc acttcgacat tctgaagccc tacatgaaga tctttaacca gagcgccgac  540
atcatgcacg ccaagtggcg acatctggcc gagggatctg ctgtgtccct ggacatgttc  600
gagcacatca gcctgatgac cctggacagc ctgcagaaat gcgtgttcag ctacaacagc  660
aactgccaag agaagatgag cgactacatc agcgccatca tcgagctgag cgctctgtct  720
gtgcggcggc agtatagact gcaccactac ctggacttca tctactacag aagcgccgac  780
ggcagacggt tcagacaggc ttgtgacatg gtgcatcact tcaccaccga agtgatccaa  840
gagcggcgga gagcactgag acagcaaggt gcagaagcct ggctgaaggc caagcagggc  900
aagaccctgg atttcatcga tgtgctgctg ctcgccagag atgaggacgg caaagagctg  960
tccgacgagg acattagagc cgaggccgac accttcatgt ttgagggcca cgataccacc  1020
agctccggca tcagctggat gctgttcaac ctggccaagt atcccgagta tcaagagaag  1080
tgccgggaag agattcaaga agtgatgaag ggccgcgagc tggaagaact ggaatgggac  1140
gatctcaccc agctgccttt caccaccatg tgcatcaaag agagcctgcg gcagtaccct  1200
cctgtgacac tggtgtccag acagtgcacc gaggacatca agctgcccga tggcagaatc  1260
atccccaagg gcatcatctg cctggtgtct atctacggca cccaccacaa tcctaccgtg  1320
tggcccgata gcaaggtgta caacccctac agattcgacc ccgacaatcc ccagcagaga  1380
agccctctgg cctacgtgcc attttccgcc ggacctagaa actgcatcgg ccagtctttt  1440
gccatggccg agctgagagt ggtggtggct ctgaccctgc tgaggttcag actgagcgtg  1500
gaccggacca gaaaagtgcg gagaaagccc gagctgatcc tgagaaccga gaacggcctg  1560
tggctgaaag tggaacccct gccacctaga gcctga                            1596

SEQ ID NO: 33          moltype = DNA  length = 693
FEATURE                Location/Qualifiers
source                 1..693
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 33
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tttgtaccta atgaccgccc cacctggcat atactggctg gcctcttctc tgtcacaggg  120
gtcttagtcg tgaccacatg gctgttgtca ggtcgtgctg cggttgtccc attggggact  180
tggcggcgac tgtccctgtg ctggtttgca gtgtgtgggt tcattcacct ggtgatcgag  240
ggctggttcg ttctctacta cgaagacctg cttggagacc aagccttctt atctcaactc  300
tggaaagagt atgccaaggg agacagccga tacatcctgg gtgacaactt cacagtgtgc  360
atggaaacca tcacagcttg cctgtgggga ccactcagcc tgtgggtggt gatcgccttt  420
ctccgccagc atcccctccg cttcattcta cagcttgtgg tctctgtggg ccagatctat  480
ggggatgtgc tctacttcct gacagagcac cgcgacggat tccagcacgg agagctgggc  540
caccctctct acttctggtt ttactttgtc ttcatgaatg ccctgtggct ggtgctgcct  600
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ggagtccttg tgcttgatgc tgtgaagcac ctcactcatg cccagagcac gctggatgcc  660
aaggccacaa aagccaagag caagaagaac tga                               693

SEQ ID NO: 34          moltype = DNA  length = 693
FEATURE                Location/Qualifiers
misc_feature           1..693
                       note = Synthetic
source                 1..693
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 34
atgaccacca atgccggacc tctgcaccct tactggcctc agcatctgcg gctggacaac  60
ttcgtgccca acgacagacc tacctggcac attctggccg gcctgttttc tgtgaccggc  120
gtgctggtgg tcaccacatg gttgctttct ggcagagccg ccgtggtgcc tcttggaact  180
tggagaaggc tgagcctgtg ttggttcgcc gtgtgcggct ttatccacct ggtcatcgaa  240
ggatggttcg tgctgtacta cgaggacctg ctgggcgatc aggcctttct gagccagctg  300
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atggaaacca tcaccgcctg tctgtggggc cctctgtctc tgtgggttgt gatcgccttc  420
ctgagacagc accctctgcg gttcattctg cagctggtgg tgtccgtggg ccagatctat  480
ggcgacgtgc tgtatttcct gaccgagcac agagatggct tccagcatgg cgaactggga  540
caccctctgt acttttggtt ctacttcgtg ttcatgaatg ccctgtggct ggtgctgcct  600
ggcgttctgg ttctggacgc cgtgaaacac ctgacacacg cccagtctac cctggacgcc  660
aaagccacaa aggccaagag caagaagaac tga                               693

SEQ ID NO: 35          moltype = DNA  length = 945
FEATURE                Location/Qualifiers
source                 1..945
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 35
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gacacggtag agttctaccg ctggacctgg tccatcgcag ataagcgtgt ggaaaattgg  120
cctctgatgc agtctccttg gcctacacta agtataagca ctctttatct cctgtttgtg  180
tggctgggtc caaaatggat gaaggaccga gaaccttttc agatgcgtct agtgctcatt  240
atctataatt ttgggatggt tttgcttaac ctctttatct tcagagagtt attcatggga  300
tcatataatg cgggatatag ctatatttgc cagagtgtgg attattctaa taatgttcat  360
gaagtcagga tagctgctgc tctgtggtgg tactttgtat ctaaaggagt tgagtatttg  420
gacacagtgt tttttattct gagaaagaaa aacaaccaag tttctttcct tcatgtgtat  480
catcactgta cgatgtttac cttgtggtgg attggaatta agtgggttgc aggaggacaa  540
gcattttttg gagcccagtt gaattccttt atccatgtga ttatgtactc atactatggg  600
ttaactgcat ttggcccatg gattcagaaa tatctttggt ggaaacgata cctgactatg  660
ttgcaactga ttcaattcca tgtgaccatt gggcacacgg cactgtctct ttacactgac  720
tgccccttcc ccaaatggat gcactgggct ctaattgcct atgcaatcag cttcatattt  780
ctctttctta acttctacat tcggacatac aaagagccta agaaaccaaa agctggaaaa  840
acagccatga atggtatttc agcaaatggt gtgagcaaat cagaaaaaca actcatgata  900
gaaaatggaa aaaagcagaa aaatggaaaa gcaaaaggag attaa                  945

SEQ ID NO: 36          moltype = DNA  length = 945
FEATURE                Location/Qualifiers
misc_feature           1..945
                       note = Synthetic
source                 1..945
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 36
atgggcctgc tggattctga gcctggcagc gtgctgaacg tggtgtctac agccctgaac  60
gacaccgtcg agttctaccg gtggacctgg tctatcgccg acaagagagt ggaaaactgg  120
cccctgatgc agagcccctg gcctacactg agcatcagca cactgtacct gctgttcgtg  180
tggctgggcc ccaagtggat gaaggacaga gaacccttcc agatgcggct ggtgctgatc  240
atctacaact tcggcatggt gctgctgaac ctgttcatct tccgcgagct gttcatgggc  300
agctacaacg ccggctacag ctacatctgt cagagcgtgg actacagcaa caacgtgcac  360
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gacaccgtgt tcttcatcct gcggaagaaa aacaaccagg tgtccttcct gcacgtgtac  480
caccattgca ccatgttcac cctgtggtgg atcggcatca aatgggtggc aggcggccag  540
gcctttttg gcgcccagct gaacagcttt atccacgtga tcatgtacag ctactacggc  600
ctgaccgcct tcggaccctg gattcagaaa tacctctggt ggaagcgcta cctgaccatg  660
ctgcagctga tccagttcca cgtgaccatc ggacacactg ccctgagcct gtacaccgac  720
tgtccatttc caaagtggat gcactgggcc ctgatcgcct acgctatctc ctttatcttc  780
ctgttcctga acttctacat ccggacctac aaagagccca agaagcccaa ggccggcaag  840
accgccatga atggcattag cgcaaacggc gtgtccaaga gcgagaagca gctgatgatc  900
gagaacggca agaagcagaa aaacggcaag gccaagggcg actga                  945

SEQ ID NO: 37          moltype = DNA  length = 12186
FEATURE                Location/Qualifiers
source                 1..12186
                       mol_type = genomic DNA
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organism = Homo sapiens
SEQUENCE: 37
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tttcggcaaa tcctgaagaa tccagatgac ccagatatgg ttgatgtctt catggatcac 180
ttggatatag accacaacaa gaaaattgac ttcactgagt ttcttctgat ggtattcaag 240
ttggctcaag catattatga gtctaccaga aaagagaatt taccgatatc aggacacaag 300
cacagaaagc acagtcatca tgataaacat gaagataata aacaggaaga aaacaaagaa 360
aacagaaaaa gaccctcaag tctggaaaga agaaacaata gaaaagggaa taagggaaga 420
tccaagagcc caagagaaac aggggggaaa aggcatgaat ctagttctga aaaaaaagaa 480
agaaaaggat attcacctac tcatagagaa gaagaatatg gaaaaaacca tcataactca 540
agtaaaaaag agaaaaacaa gactgaaaat actagattag gagacaatag gaagaggcta 600
agtgaaagac ttgaagagaa agaagacaat gaagaaggag tatatgatta tgaaaataca 660
ggaagaatga ctcaaaaatg gatacaatca ggccatattg ccacatatta cacaatccag 720
gatgaagcct atgacaccac tgatagtcta ttagaagaaa acaaaatata tgaaagatca 780
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agggacagtg agggacactc agaagactct gagaggcact ctgggtcggc ttccagaaac 960
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ggagaaagac atggatccgg ccaccagcag tcagcagaca gctccagaca ctcagccact 1200
gggcgcgggc aagcttcatc tgcagtcagc gatcgtggac accgggggtc tagcggtagt 1260
caggccagtg acagtgaggg acattcagaa aactcagaca cacaatcagt gtcaggccac 1320
ggaaaggctg ggctgagaca gcagagccac caagagtcca cacgtggccg gtcaggggaa 1380
cggtctggac gttcagggtc ttccctctac caggtgagca ctcatgaaca gcctgactct 1440
gcccatggac ggaccgggac cagcactgga ggaagacaag gatcgcacca cgagcaggca 1500
cgagacagct ccaggcattc agcgtcccaa gagggtcagg acaccattcg tggacacccg 1560
gggtcaagca gaggaggaag gcagggatcc caccacgagc aatcggtaaa taggtctgga 1620
cactcaggtt cccatcacag ccacaccaca tcccagggaa ggtctgatgc ctcccatggg 1680
cagtcaggat ccagaagtgc aagcagacaa acacgaaatg aggaacaatc aggagacggc 1740
accaggcact cagggtcacg tcatcatgaa gcttcctctc aggctgacag ctctagacac 1800
tcacaggtgg gccagggaca atcatcgggg cccaggacaa gtaggaacca gggatccagt 1860
gttagccagg acagtgacag tcagggacac tcagaagact ctgagaggtg gtctgggtct 1920
gcttccagaa accatcatgg atctgctcag gagcagtcaa gagatggctc cagacacccc 1980
aggtcccatc acgaagacag agctggtcat gggcactctg cagacagctc cagaaaatca 2040
ggcactcgtc acacacagaa ttcctctagt ggacaggctg cgtcatccca tgaacaggca 2100
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agacactcag gcactgggca cggacaagct tcatctgcag tcagagacag tggacaccga 2220
gggtccagtg gtagtcaggc cactgacagt gagggacatt cagaagactc agacacacag 2280
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actcacggcc aaagtggatc tcaacatgga gagtcagaat ccataattca tgacagacac  4620
agaattactc atggacagac aggagatacc actagacatt cctactctgg tcatgaacaa  4680
accacacaga cagggtccag gacaactgga agacagagaa ctagccacag tgagtccact  4740
gacagtgaag tgcactcagg gggctcacac agaccacact cacgagaaca cacttacggc  4800
caagccggat ctcaacatga agagccagaa ttcacagttc atgagagaca cggaactact  4860
catggacaga taggagatac cactggacat tcccactctg gtcatggaca gtccacacag  4920
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gtgcactcag gggtctcaca cacacataca ggacacactc atggtcaagc tggatctcaa  5040
catggacagt cagaatccat agttcctgag agacatggaa ctactcatgg acagacagga  5100
gataccacta gacatgccca ctatcatcat ggattaacca cacagacagg gtccaggact  5160
actggaagaa ggggatctgg ccacagtgag tacagtgaca gtgaagggta ctcaggagtc  5220
tcacatacac attcaggaca cactcatggc caagccagat ctcaacatgg agagtcagaa  5280
tccatagttc atgagagaca tggaactata catggacaga caggcgatac caccagacat  5340
gcccactctg gtcatggaca gtccacacag acagggtcca ggaccactgg aagaaggtca  5400
tctggccaca gtgagtacag tgacagtgaa gggcactcag ggttctcaca aagaccacac  5460
tcacgaggac acactcacgg ccaggctgga tctcaacatg gagagtcaga atccatagtt  5520
gacgagagac atggaactac tcatggacag acaggagata ccagtggaca ttctcaatct  5580
ggtcatggac agtccacaca gtcaggatcc agtacaactg gaagaaggag atctggccac  5640
agtgagtcca gtgacagtga agtgcactca gggggctcac atacacattc aggacacaca  5700
cacagccaag ccaggtctca acatggagag tcagaatcca cagttcacaa gagacaccaa  5760
actactcatg gacagacagg agataccact gaacatggcc accctagtca tggacaaacc  5820
atacagacag ggtccaggac aactggaaga aggggatctg gccacagtga gtacagtgac  5880
agtgaagggc cctcaggggt ctcacacaca cattcaggac acactcacgg tcaagctgga  5940
tctcactatc cagagtcagg atcctcagtt catgagagac acggaactac tcatggacaa  6000
acagcagata ccactagaca tggccactct ggtcatggac agtccacaca gagagggtcc  6060
aggacaactg gaagaagggc atctggccac agtgagtaca gtgacagtga agggcactca  6120
ggggtctcac acacacattc aggacacgct catggccaag ccggatctca acatggagag  6180
tcaggatcct cagttcatga gagacacgga actactcatg gacagacagg agataccact  6240
agacatgctc actctggtca tggacagtcc acacagagag ggtcaaggac agctggaaga  6300
aggggatctg gccacagtga gtccagtgac agtgaagtgc actcaggggt ctcacacaca  6360
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cattcaggac acacttatgg ccaagccaga tctcaacatg gagagtcagg atctgccatt  6420
cacgggagac agggaactat acatggacag acaggagata ccactagaca tggccagtct  6480
ggtcatggac agtccacaca gacaggttcc aggacaactg gaagacaaag atctagtcac  6540
agtgagtcca gtgatagtga agtgcactca gaggcctcac ccacacattc aggacacact  6600
cacagccaag ccggatctcg acatggacag tcaggatcct caggtcatgg gagacaggga  6660
actactcatg gacagacagg agataccact agacatgccc actatggtta tggacaatcc  6720
acacagagag ggtccaggac aactggaaga aggggatctg gccacagtga gtccagtgac  6780
agtgaagtgc actcatgggg ctcacacaca cattcaggac acattcaggg ccaagctgga  6840
tctcaacaaa gacagccagg atccacagtt catgggagac tggaaactac tcatggacag  6900
acaggagata ccactagaca tggccattct ggttatggac aatccacaca gacaggttcc  6960
agatctagta gagcaagtca ttttcagtca catagtagtg aaaggcaaag gcatggatca  7020
agtcaggttt ggaaacatgg cagctatgga cctgcagaat atgactatgg gcacactggg  7080
tatgggcctt ctggtggcag cagaaaaagc atcagtaatt ctcacctttc atggtcaaca  7140
gacagcactg caaacaagca actgtctaga cattga                            7176

SEQ ID NO: 40          moltype = DNA  length = 681
FEATURE                Location/Qualifiers
source                 1..681
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 40
atggattggg gcacgctgca gacgatcctg gggggtgtga acaaacactc caccagcatt  60
ggaaagatct ggctcaccgt cctcttcatt tttcgcatta tgatcctcgt tgtggctgca  120
aaggaggtgt ggggagatga gcaggccgac tttgtctgca acaccctgca gccaggctgc  180
aagaacgtgt gctacgatca ctacttcccc atctcccaca tccggctatg ggccctgcag  240
ctgatcttcg tgtccacgcc agcgctccta gtggccatgc acgtggccta ccggagacat  300
gagaagaaga ggaagttcat caagggggag ataaagagtg aatttaagga catcgaggag  360
atcaaaaccc agaaggtccg catcgaaggc tccctgtggt ggacctacac aagcagcatc  420
ttcttccggg tcatcttcga agccgccttc atgtacgtct tctatgtcat gtacgacggc  480
ttctccatgc agcggctggt gaagtgcaac gcctggcctt gtcccaacac tgtggactgc  540
tttgtgtccc ggcccacgga gaagactgtc ttcacagtgt tcatgattgc agtgtctgga  600
atttgcatcc tgctgaatgt cactgaattg tgttatttgc taattagata ttgttctggg  660
aagtcaaaaa agccagttta a                                            681

SEQ ID NO: 41          moltype = DNA  length = 681
FEATURE                Location/Qualifiers
misc_feature           1..681
                       note = Synthetic
source                 1..681
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 41
atggattggg gcaccctgca gacaattctc ggcggcgtga acaagcacag caccagcatc  60
ggaaagatct ggctgaccgt gctgttcatc ttccgcatca tgatcctggt ggtggccgcc  120
aaagaagtgt ggggcgacga acaggccgac ttcgtgtgta ataccctgca gcctggatgc  180
aagaacgtgt gctacgacca ctacttcccc atcagccaca tcagactgtg ggccctgcag  240
ctgatcttcg tgtctacacc agctctgctg gtggccatgc acgtggccta tagacggcac  300
gagaagaagc ggaagttcat caagggcgag atcaagtccg agttcaagga catcgaggaa  360
atcaagaccc agaaagtgcg gatcgagggc agcctgtggt ggacctatac cagcagcatc  420
ttcttccgcg tgatcttcga ggccgccttt atgtacgtgt tctacgtgat gtacgacggc  480
ttcagcatgc agcggctggt caagtgtaac gcctggcctt gtcctaacac cgtggactgc  540
ttcgtgtccc ggcctaccga gaaaaccgtg ttcaccgtgt ttatgatcgc cgtgtccggc  600
atctgcatcc tgctgaatgt gaccgagctg tgctacctgc tgatccggta ctgtagcggc  660
aagagcaaga aacccgtgtg a                                            681

SEQ ID NO: 42          moltype = DNA  length = 813
FEATURE                Location/Qualifiers
source                 1..813
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 42
atggactgga agacactcca ggccctactg agcggtgtga acaagtactc cacagcgttc  60
gggcgcatct ggctgtccgt ggtgttcgtc ttccgggtgc tggtatacgt ggtggctgca  120
gagcgcgtgt gggggggatga gcagaaggac tttgactgca acaccaagca gcccggctgc  180
accaacgtct gctacgacaa ctacttcccc atctccaaca tccgcctctg ggccctgcag  240
ctcatcttcg tcacatgccc ctcgctgctg gtcatcctgc acgtggccta ccgtgaggag  300
cgggagcgcc ggcaccgcca gaaacacggg gaccagtgcg ccaagctgta cgacaacgca  360
ggcaagaagc acggaggcct gtggtggacc tacctgttca gcctcatctt caagctcatc  420
attgagttcc tcttcctcta cctgctgcac actctctggc atggcttcaa tatgccgcgc  480
ctggtgcagt gtgccaacgt ggccccctgc cccaacatcg tggactgcta cattgcccga  540
cctaccgaga agaaaatctt cacctacttc atggtgggcg cctccgccgt ctgcatcgta  600
ctcaccatct gtgagctctg ctacctcatc tgccacaggg tcctgcgagg cctgcacaag  660
gacaagcctc gagggggttg cagcccctcg tcctccgcca gccgagcttc cacctgccgc  720
tgccaccaca agctggtgga ggctggggag gtggatccag acccaggcaa taacaagctg  780
caggcttcag cacccaacct gacccccatc tga                               813
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SEQ ID NO: 43          moltype = DNA  length = 813
FEATURE                Location/Qualifiers
misc_feature           1..813
                       note = Synthetic
source                 1..813
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 43
atggactgga aaccctgca ggctctgctg agcggcgtga acaagtacag caccgccttc  60
ggcagaatct ggctgagcgt ggtgttcgtg ttcagagtgc tggtgtatgt ggtggccgcc  120
gaaagagttt ggggcgacga gcagaaggac ttcgactgca acacaaagca gcccggctgc  180
accaacgtgt gctacgacaa ctacttcccc atcagcaaca tccggctgtg ggccctgcag  240
ctgatcttcg tgacatgtcc tagcctgctg gtcatcctgc acgtggccta cagagaggaa  300
cgcgagagaa ggcacagaca gaaacacggc gatcagtgcg ccaagctgta cgacaatgcc  360
ggcaaaaagc acggcggcct gtggtggaca tacctgttca gcctgatctt caagctgatt  420
atcgagttcc tgttcctgta cctgctgcac accctgtggc acggcttcaa catgcctaga  480
ctggtgcagt gtgccaacgt ggccccttgt cctaacatcg tggactgcta cattgcccgg  540
cctaccgaga agaagatctt tacctacttc atggtcggag ccagcgccgt gtgtatcgtg  600
ctgaccattt gcgagctgtg ctacctgatc tgccaccggg ttctgagagg cctgcacaag  660
gataagccta gaggcggctg tagccctagc agctctgcct ctagagccag cacctgtaga  720
tgccaccaca agctggtgga agccggcgaa gtcgatcctg atcctggcaa caacaagctg  780
caggccagcg ctcctaatct gacccctatc tga                               813

SEQ ID NO: 44          moltype = DNA  length = 801
FEATURE                Location/Qualifiers
source                 1..801
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 44
atgaactggg catttctgca gggcctgctg agtggcgtga acaagtactc cacagtgctg  60
agccgcatct ggctgtctgt ggtgttcatc tttcgtgtgc tggtgtacgt ggtggcagcg  120
gaggaggtgt gggacgatga gcagaaggac tttgtctgca acaccaagca gcccggctgc  180
cccaacgtct gctatgacga gttcttcccc gtgtcccacg tgcgcctctg ggccctacag  240
ctcatcctgg tcacgtgccc ctcactgctc gtggtcatgc acgtggccta ccgcgaggaa  300
cgcgagcgca agcaccacct gaaacacggg cccaatgccc cgtccctgta cgacaacctg  360
agcaagaagc ggggcggact gtggtggacg tacttgctga gcctcatctt caaggccgcc  420
gtggatgctg gcttcctcta tatcttccac cgcctctaca aggattatga catgccccgc  480
gtggtggcct gctccgtgga gccttgcccc cacactgtgg actgttacat ctcccggccc  540
acggagaaga aggtcttcac ctacttcatg gtgaccacag ctgccatctg catcctgctc  600
aacctcagtg aagtcttcta cctggtgggc aagaggtgca tggagatctt cggccccagg  660
caccggcggc ctcggtgccg ggaatgccta cccgatacgt gcccaccata tgtcctctcc  720
cagggagggc accctgagga tgggaactct gtcctaatga aggctgggtc ggccccagtg  780
gatgcaggtg ggtatccata a                                            801

SEQ ID NO: 45          moltype = DNA  length = 801
FEATURE                Location/Qualifiers
misc_feature           1..801
                       note = Synthetic
source                 1..801
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 45
atgaactggg cctttctgca gggactgctg agcggcgtga acaagtacag caccgtgctg  60
agcagaatct ggctgagcgt ggtgttcatc ttcagagtgc tggtgtacgt ggtggccgcc  120
gaagaagttt gggacgacga gcagaaagac ttcgtgtgca acaccaagca gcctggctgc  180
cccaacgtgt gctacgacga gttcttcccc gtgtctcatg tgcggctgtg ggccctgcaa  240
ctgatcctgg tcacatgtcc tagcctgctg gtggtcatgc acgtggccta cagagaggaa  300
cgcgagagaa agcaccacct gaagcacggc cctaatgctc ccagcctgta cgacaacctg  360
agcaagaaac gcggcggact gtggtggaca tacctgctga gcctgatctt caaggccgcc  420
gtggatgccg gcttcctgta catcttccac cggctgtaca aggactacga catgcccaga  480
gtggtggcct gcagcgtgga accttgtcct cacaccgtgg actgctacat cagcagaccc  540
accgagaaga aggtgttcac ctacttcatg gtcaccaccg ccgccatctg catcctgctg  600
aatctgtccg aggtgttcta cctcgtgggc aagagatgca tggaaatctt cggccccaga  660
cacagacggc ccagatgcag agaatgcctg cctgacacct gtcctccata cgtgctgtct  720
caaggcggac accccgagga tggcaatagc gtgttgatga aggccggcag cgctcctgtt  780
gatgccggcg gatatccttg a                                            801

SEQ ID NO: 46          moltype = DNA  length = 786
FEATURE                Location/Qualifiers
source                 1..786
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 46
atggattggg ggacgctgca cactttcatc gggggtgtca acaaacactc caccagcatc  60
gggaaggtgt ggatcacagt catctttatt ttccgagtca tgatcctcgt ggtggctgcc  120
caggaagtgt ggggtgacga gcaagaggac ttcgtctgca acacactgca accgggatgc  180
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aaaaatgtgt gctatgacca ctttttcccg gtgtcccaca tccggctgtg ggccctccag  240
ctgatcttcg tctccacccc agcgctgctg gtggccatgc atgtggccta ctacaggcac  300
gaaaccactc gcaagttcag gcgaggagag aagaggaatg atttcaaaga catagaggac  360
attaaaaagc agaaggttcg gatagagggg tcgctgtggt ggacgtacac cagcagcatc  420
tttttccgaa tcatctttga agcagccttt atgtatgtgt tttacttcct ttacaatggg  480
taccacctgc cctgggtgtt gaaatgtggg attgacccct gccccaacct tgttgactgc  540
tttatttcta ggccaacaga gaagaccgtg tttaccattt ttatgatttc tgcgtctgtg  600
atttgcatgc tgcttaacgt ggcagagttg tgctacctgc tgctgaaagt gtgttttagg  660
agatcaaaga gagcacagac gcaaaaaaat caccccaatc atgccctaaa ggagagtaag  720
cagaatgaaa tgaatgagct gatttcagat agtggtcaaa atgcaatcac aggtttccca  780
agctaa                                                            786
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SEQ ID NO: 47           moltype = DNA  length = 786
FEATURE                 Location/Qualifiers
misc_feature            1..786
                        note = Synthetic
source                  1..786
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 47
atggattggg gcaccctgca caccttt atc ggcggcgtga acaagcacag caccagcatc  60
```

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ggcaaagtgt ggatcaccgt gatcttcatc ttccgcgtga tgatcctggt ggtggccgct  120
caagaagtgt ggggcgacga gcaagaggac ttcgtgtgca ataccctgca gcctggatgc  180
aagaacgtgt gctacgacca cttcttcccc gtgtctcaca tcagactgtg ggccctgcag  240
ctgatcttcg tgtctacacc agctctgctg gtggccatgc acgtggccta ctacagacac  300
gagacaaccc ggaagttcag acggggcgag aagcggaacg acttcaagga catcgaggac  360
atcaagaagc agaaagtccg gatcgagggc agcctgtggt ggacctatac cagcagcatc  420
ttcttccgga tcatcttcga ggccgccttt atgtacgtgt tctacttcct gtacaacggc  480
taccatctgc cttgggtgct gaagtgcggc atcgacccct gtcctaacct ggtggactgc  540
ttcatcagca gacccaccga gaaaaccgtg ttcaccatct tcatgatcag cgccagcgtg  600
atctgcatgc tgctgaatgt ggccgagctg tgctacctgc tgctcaaagt gtgcttccgg  660
cggtccaaga gagcccagac acagaagaat caccccaatc acgccctgaa agagtccaag  720
cagaacgaga tgaacgagct gatctccgac agcggccaga acgccatcac aggctttcca  780
agctga                                                            786
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SEQ ID NO: 48           moltype = DNA  length = 999
FEATURE                 Location/Qualifiers
source                  1..999
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 48
atgctgctgc tggctgccgc cttcctcgtg gccttcgtgc tgctgctgta catggtgtct  60
ccgctcatca gccccaagcc cctcgccctg cccggggcgc atgtggtggt tacaggaggt  120
tccagtggca tcgggaagtg cattgctatc gagtgctata aacaaggagc ttttataact  180
ctggttgcac gaaatgagga taagctgctg caggcaaaga aagaaattga aatgcactct  240
attaatgaca aacaggtggt gctttgcata tcagttgatg tatctcaaga ctataaccaa  300
gtagagaatg tcataaaaca agcacaggag aaactgggtc cagtggacat gctggtaaat  360
tgtgcaggaa tggcagtgtc aggaaaattt gaagatcttg aagttagtac ctttgaaagg  420
ttaatgagca tcaattacct gggcagcgtg taccccagcc gggccgtgat caccaccatg  480
aaggagcgcc gggtgggcag gatcgtgttt gtgtcctccc aggcaggaca gttgggatta  540
ttcggtttca cagcctactc tgcatccaag tttgccataa ggggattggc agaagctttg  600
cagatggagg tgaagccata taatgtctac atcacagttg cttacccacc agacacagac  660
acacctggct ttgccgaaga aaacagaaca aagcctttgg agactcgact tatttcagag  720
accacatctg tgtgcaaacc agaacaggtg gccaaacaaa ttgttaaaga tgccatacaa  780
ggaaatttca acagttccct tggctcagat gggtacatgc tctcggccct gacctgtggg  840
atggctccag taacttctat tactgagggg ctccagcagg tggtcaccat gggccttttc  900
cgcactattg ctttgtttta ccttggaagt tttgacagca tagttcgtcg ctgcatgatg  960
cagagagaaa aatctgaaaa tgcagacaaa actgcctaa                        999
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SEQ ID NO: 49           moltype = DNA  length = 999
FEATURE                 Location/Qualifiers
misc_feature            1..999
                        note = Synthetic
source                  1..999
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 49
atgctgcttc tggccgctgc tttcctggtg gcctttgtgc tgctgctgta catggtgtcc  60
ccactgatca gccccaagcc tcttgcactt cctggcgctc acgttgtcgt gacaggcgga  120
tcttctggca tcggcaagtg tatcgccatc gagtgctata agcagggcgc cttcatcaca  180
ctggtggcca gaaacgagga caagctgctg caggccaaga aagaaatcga gatgcacagc  240
atcaacgaca agcaggtcgt gctgtgtatc agcgtggacg tgtcccagga ctacaaccag  300
gtggaaaacg tgatcaagca ggcccaagag aagctgggcc ccgtggacat gctggttaat  360
tgtgccggaa tggccgtgtc cggcaagttc gaggatctgg aagtgtccac cttcgagcgg  420
ctgatgagca tcaactacct gggcagcgtg taccccagca gagccgtgat caccaccatg  480
aaggagcgga gagtcggccg gatcgtgttt gtgtctagtc aggctggaca gctgggcctg  540
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tttggcttta cagcctacag cgccagcaag ttcgccatca gaggactggc tgaagccctg  600
cagatggaag tgaagcccta caacgtgtac atcaccgtgg cctatcctcc agacaccgac  660
acacctggct tcgccgagga aaacaggacc aagccactgg aaacccggct gatctccgag  720
acaaccagcg tgtgcaagcc tgaacaggtg gccaagcaga tcgtgaagga tgccatccag  780
ggcaacttca acagcagcct gggctccgat ggctacatgc tgtctgccct gacatgtggc  840
atggcccctg tgacctctat caccgaggga ctccagcagg ttgtgaccat gggactgttc  900
cggacaatcg ccctgttcta tctgggcagc ttcgacagca tcgtgcggcg gtgtatgatg  960
cagagagaga agtccgagaa cgccgacaag accgcctaa                         999

SEQ ID NO: 50          moltype = DNA  length = 1935
FEATURE                Location/Qualifiers
source                 1..1935
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 50
atgagtcgac agtttagttc caggtctggg taccgaagtg gagggggctt cagctctggc  60
tctgctggga tcatcaacta ccagcgcagg accaccagca gctccacacg ccgcagtgga  120
ggaggtggtg ggagattttc aagctgtggt ggtggtggtg gtagctttgg tgctggtggt  180
ggatttggaa gtcggagtct tgttaacctt ggtggcagta aaagcatctc cataagtgtg  240
gctagaggag gtggacgtgg tagtggcttt ggtggtggtt atggtggtgg tggctttggt  300
ggtggtggct ttggtggtgg tggctttggt ggaggtggca ttgggggtgg tggctttggt  360
ggttttggca gtggtggtgg tggttttggt ggaggtggct ttgggggtgg tggatatggg  420
ggtggttatg gtcctgtctg ccctcctggt ggcatacaag aagtcactat caaccagagc  480
cttcttcagc ccctcaatgt ggagattgac cctgagatcc aaaaggtgaa gtctcgagaa  540
agggagcaaa tcaagtcact caacaaccaa tttgcctcct tcattgacaa ggtgaggttc  600
ctggagcagc agaaccaggt actgcaaaca aaatgggagc tgctgcagca ggtagatacc  660
tccactagaa cccataattt agagccctac tttgagtcat tcatcaacaa tctccgaagg  720
agagtggacc aactgaagag tgatcaatct cggttggatt cggaactgaa gaacatgcag  780
gacatggtgg aggattaccg gaacaagtat gaggatgaaa tcaacaagcg gacaaatgca  840
gagaatgaat ttgtgaccat caagaaggat gtggatggtg cttatatgac caaggtggac  900
cttcaggcca aacttgacaa cttgcagcag gaaattgatt tccttacagc actctaccaa  960
gcagagttgt ctcagatgca gactcaaatc agtgaaacta atgtcatcct ctctatggac  1020
aacaaccgca gtctcgacct ggacagcatc attgctgagg tcaaggccca gtacgaggat  1080
atagcccaga agagcaaagc tgaggccgag tccttgtacc agagcaagta tgaagagctg  1140
cagatcactg ctggcagaca tggggatagt gtgagaaatt caaagataga aatttctgag  1200
ctgaatcgtg tgatccagag acttagatct gaaatcgaca atgtcaagaa gcagatctcc  1260
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source                 1..1920
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                       organism = Homo sapiens
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FEATURE                Location/Qualifiers
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                       mol_type = genomic DNA
                       organism = Homo sapiens
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FEATURE                Location/Qualifiers
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                       organism = synthetic construct
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SEQ ID NO: 60           moltype = DNA  length = 939
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                        organism = Homo sapiens
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acccagcaga agcaggcgcc tacctggccg tccaaatag                         939

SEQ ID NO: 61           moltype = DNA  length = 939
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                        note = Synthetic
source                  1..939
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                        organism = synthetic construct
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SEQ ID NO: 62           moltype = DNA  length = 1560
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source                  1..1560
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                        organism = Homo sapiens
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SEQ ID NO: 63          moltype = DNA  length = 1560
FEATURE                Location/Qualifiers
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source                 1..1560
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SEQ ID NO: 64          moltype = DNA  length = 1401
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                       organism = Homo sapiens
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FEATURE                Location/Qualifiers
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source                 1..1401
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SEQ ID NO: 66          moltype = DNA  length = 1344
FEATURE                Location/Qualifiers
source                 1..1344
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 66
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agcaatgcca cctttcacag ctggcaggaa agaatcaggc agaactatgg cttctacatc  360
ggcctgggcc tggcattcct gtctagcttc ctcatcggca gcagcgtcat cctcaagaag  420
aaaggcctct tgcgactcgt ggccacggga gccactcgag ctgtggctgc tggagaagtt  480
gccaactttg gagcctacgc atttgcacct gcaacagtcg tcacgcctct gggagcgctg  540
agtgtcctca taagtgccat cctctcctca tatttcctga gggagagtct gaacctgctg  600
gggaagctgg gctgtgtgat ctgtgtggcc ggaagcacag tgatggtgat acatgctcct  660
gaggaagaga aggtcactac catcatggag atggcttcca agatgaaaga cacagggttc  720
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cgttacgggc aaaggaatat cctcatctac atcatcatct gctctgtgat cggggccttc  840
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gttgtccggc acccgctccc ctacatcctg tccctcatcc tggcactgtc cctcagcact  960
caggtcaact tcctcaacag agcactggac attttcaaca cttccctggt gttccccatc  1020
tactacgtgt tcttcaccac ggtggtcgtt acctcgtcca tcatcctctt caaggagtgg  1080
tacagcatgt ctgctgtgga cattgcaggc accctctcgg gctttgtcac catcatcttg  1140
ggcgtgttca tgctgcatgc tttcaaagac ctggacatca gctgcgccag cttgccccac  1200
atgcacaaaa acccaccccc ttctcccgcc ccggaaccca ctgttattag actggaagac  1260
aagaacgtcc ttgtggacaa tatagaactt gccagcacct catcaccaga agagaaaccc  1320
aaagtattta taatccattc ctga                                        1344

SEQ ID NO: 67          moltype = DNA  length = 1344
FEATURE                Location/Qualifiers
misc_feature           1..1344
                       note = Synthetic
source                 1..1344
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 67
atgcctggcg atagctctcc tggaacactg cctctgtggg acgcctcttt gtctcctcct  60
ctgggacctg atcctggcgg cttttctaga gcctctcacg ccggcgataa gagcagacct  120
cctgctcctg aactgggatc tcctggcgcc gttagaccta gagtgggatc ttgtgcccct  180
ggacctatgg aactgcgggt gtccaatacc agctgcgaga atggctctct gctgcacctg  240
tactgcagca gccaagaggt gctgtgccag atcgtgaatg atctgagccc tgaggtgccc  300
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aagggcctgc tgagactggt ggccacaggc gctacaagag ctgttgctgc tggcgaggtg  480
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agcgtgctga tctccgctat cctgtccagc tacttcctga gagagagcct gaacctgctg  600
ggcaagctgg gctgtgtgat ttgtgtggcc ggctccaccg tgatggtcat tcatgcccct  660
gaagaggaaa aagtgaccac catcatggaa atggccagca agatgaagga caccggcttc  720
atcgtgttcg ccgtgctgct gctggtgtcc tgcctgattc tgatcttcgt gatcgcccct  780
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tctgtggccg ctgtgaaagg cctgggaatc accatcaaga atttcttcca gggcctgcct  900
gtcgtgcggc accctctgcc ttatatcctg tctctgattc tggccctgag cctgtctacc  960
caagtgaact ttctgaacag agccctggac atcttcaaca caagcctggt gttccccatc  1020
tactacgtgt tcttcaccac agtggtcgtg accagctcca tcatcctgtt caaagaatgg  1080
tacagcatga gcgccgtgga cattgccggc acactgagcg gctttgtgac aatcatcctg  1140
ggcgtgttca tgctgcacgc cttcaaggac ctggatatca gctgcgccag cctgcctcac  1200
atgcacaaga accctccacc ttctccagct ccagagccta ccgtgatccg gctggaagat  1260
aagaacgtgc tggtggacaa catcgagctg gcctccacaa gcagccccga agagaagccc  1320
aaggtgttca tcatccacag ctga                                        1344

SEQ ID NO: 68           moltype = DNA  length = 1122
FEATURE                 Location/Qualifiers
source                  1..1122
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 68
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gacactccca aagtgaatgc tgacatagaa aaggttaacc agaatcaggc caagagatgc  120
acagtgatcg gtggctctgg attcctgggg cagcacatgg tggagcagtt gctggcaaga  180
ggatatgctg tcaatgtatt tgatatccag caagggtttg ataatcccca ggtgcggttc  240
tttctgggtg acctctgcag ccgacaggat ctgtacccag ctctgaaagg tgtaaacaca  300
gttttccact gtgcgtcacc cccaccatcc agtaacaaca aggagctctt ttatagagtg  360
aattacattg gcaccaagaa tgtcattgaa acttgcaaag aggctggggt tcagaaactc  420
attttaacca gcagtgccag tgtcatcttt gagggcgtcg atatcaagaa tggaactgaa  480
gaccttccct atgccatgaa acccattgac tactacacag agactaagat cttacaggag  540
agggcagttc tgggcgccaa cgatcctgag aagaatttct taaccacagc catccgccct  600
catggcattt tcggcccaag ggacccgcag ttggtaccca tcctcatcga ggcagccagg  660
aacggcaaga tgaagttcgt gattggaaat gggaagaact tggtggactt cacctttgtg  720
gagaacgtgg tccatggaca catcctggcg gcagagcagc tctcccgaga ctcgacactg  780
ggtgggaagg catttcacat caccaatgat gagcccatcc ctttctggac attcctgtct  840
cgcatcctga caggcctcaa ttatgaggcc cccaagtacc acatccccta ctgggtggcc  900
tactacctgg ccctcctgct atccctgctg gtgatggtga tcagtcctgt catccagctg  960
cagcccacct tcacacccat gcgggtcgca ctggctggca cattccacta ctacagctgc  1020
gagagagcca aaaaggccat gggctaccag ccactagtga ccatggatga tgctatggag  1080
aggaccgtgc agagctttcg ccacctgcgg agggtcaagt ga                    1122

SEQ ID NO: 69           moltype = DNA  length = 1122
FEATURE                 Location/Qualifiers
misc_feature            1..1122
                        note = Synthetic
source                  1..1122
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 69
atggaacctg ccgtgtccga gcctatgaga gatcaggtgg ccagaacaca cctgaccgag  60
gacacccccta aagtgaacgc cgacatcgag aaagtgaacc agaaccaggc caagagatgc  120
accgtgatcg gcggctctgg atttctgggc cagcacatgg ttgaacagct gctggccaga  180
ggctacgccg tgaacgtgtt cgatatccag cagggcttcg acaaccctca agtgcggttc  240
ttcctgggcg acctgtgcag cagacaggat ctgtaccctg ctctgaaggg cgtgaacacc  300
gtgttccact gtgcctctcc acctccaagc agcaacaaca aagaactgtt ctaccgcgtg  360
aactacatcg gcaccaagaa cgtgatcgag acatgcaaag aggccggcgt ccagaagctg  420
atcctgacat ctagcgccag cgtgatcttc gaaggcgtgg acatcaagaa cggcaccgag  480
gatctgccct acgccatgaa gcccatcgac tactacaccg agacaaagat cctgcaagag  540
cgggccgtgc tgggcgccaa tgatcccgag aagaatttcc tgaccaccgc catcagaccc  600
cacggcatct ttggccctag agatcctcag ctggtgccca tcctgattga ggccgccaga  660
aacggcaaga tgaagtttgt gatcggcaac gggaagaacc tggtggactt caccttcgtg  720
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ggcggcaagg ccttccacat taccaacgac gagcctattc ctttctggac cttcctgagc  840
agaatcctga ccggcctgaa ctacgaggcc cctaagtatc acatccccta ctgggtcgcc  900
tactatctgg ccctgctgct gtctctgctg gtcatggtta tcagccccgt gatccagctg  960
cagcccacct tcacaccaat gagagtggcc ctggccggca ccttccacta ctactcttgc  1020
gagagggcca agaaagccat gggctaccag cctctggtca ccatggacga cgccatggaa  1080
agaaccgtgc agagcttccg gcacctgaga agagtgaagt ga                    1122

SEQ ID NO: 70           moltype = DNA  length = 972
FEATURE                 Location/Qualifiers
source                  1..972
                        mol_type = genomic DNA
                        organism = Homo sapiens
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ggcatcgcgg gctgtggaac cctactaata ttggatccag atgaagctgg gctaaggctt  180
tttagaagct ttgactggaa tgatggtttg tttgatgtga cttggagtga gaacaacgaa  240
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gatccaactg ttggaaagtc tctgtgcacc tttagaggcc atgaaagtat tatttatagc  480
acaatctggt ctccccacat ccctggttgt tttgcttcag cctcaggtga tcagactctg  540
agaatatggg atgtgaaggc agcaggagta agaatcgtga ttcctgcaca tcaggcagaa  600
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catacctatg ctattaggag ggtgaaattt tcaccatttc atgcttctgt gctggcctct  780
tgctcgtatg attttactgt aagattctgg aacttttcaa agcctgactc tcttcttgaa  840
acagtggagc atcatacaga gtttacttgt ggtttagact tcagtcttca gagccccact  900
caggtggctg actgttcttg ggatgaaaca ataaagatct atgaccctgc ttgtcttact  960
attcctgctt ga                                                      972

SEQ ID NO: 71           moltype = DNA  length = 972
FEATURE                 Location/Qualifiers
misc_feature            1..972
                        note = Synthetic
source                  1..972
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 71
atgtctgctg tgtgtggcgg agccgccaga atgctgagaa cacctggcag acatggctac  60
gccgccgagt tctctcctta cctgcctgga aggctggcct gtgccacagc tcagcactat  120
ggaatcgccg gctgtggaac cctgctgatt ctggaccctg atgaggccgg actgcggctg  180
ttcagaagct tcgattggaa cgacggcctg ttcgatgtga cttggagcga gaacaacgaa  240
cacgtgctga tcacctgtag cggcgacgga tctctgcagc tgtgggatac agccaaagcc  300
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gaccccaccg tgggcaagag cctgtgtacc tttagaggcc acgagagcat catctacagc  480
accatctggt ccccacacat ccctggctgt tttgcctctg cctctggcga tcagaccctg  540
agaatctggg atgtgaaagc cgctggcgtg cggatcgtga ttcctgctca tcaggccgag  600
atcctgagct gcgactggtg caagtacaac gagaacctgc tcgtgacagg cgccgtggat  660
tgctctctga gaggctggga cctgagaaac gtgcggcagc ctgtgtttga gctgctgggc  720
cacacatacg ccatcagaag agtgaagttc agccccttcc acgcctccgt gctggcctct  780
tgcagctacg atttcaccgt gcggttctgg aacttcagca agcccgacag cctgctggaa  840
accgtggaac accacaccga gttcacctgt ggcctggact ttagcctgca gagccctaca  900
caggtggccg actgttcttg ggacgagaca atcaagatct acgaccccgc ctgcctgaca  960
atccctgctt aa                                                      972

SEQ ID NO: 72           moltype = DNA  length = 1602
FEATURE                 Location/Qualifiers
source                  1..1602
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 72
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aagatcttgc aagatggagg gctgcaggtg gtggaaaagc agaaccttag caaagaggag  120
ctgatagcgg agctgcagga ctgtgaaggc cttattgttc gctctgccac caaggtgacc  180
gctgatgtca tcaacgcagc tgagaaactc caggtggtgg gcagggctgg cacaggtgtg  240
gacaatgtgg atctggaggc cgcaacaagg aagggcatct tggttatgaa cacccccaat  300
gggaacagcc tcagtgccgc agaactcact tgtggaatga tcatgtgcct ggccaggcag  360
attccccagg cgacggcttc gatgaaggac ggcaaatggg agcggaagaa gttcatggga  420
acagagctga atggaaagac cctgggaatt cttggcctgg gcaggattgg gagagaggta  480
gctacccgga tgcagtcctt tgggatgaag actatagggt atgaccccat catttcccca  540
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gatttcatca ctgtgcacac tcctctcctg ccctccacga caggcttgct gaatgacaac  660
acctttgccc agtgcaagaa gggggtgcgt gtggtgaact gtgcccgtgg agggatcgtg  720
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gtgtttacgg aagagccgcc acgggaccgg gccttggtgg accatgagaa tgtcatcagc  840
tgtccccacc tgggtgccag caccaaggag gctcagagcc gctgtgggga ggaaattgct  900
gttcagttcg tggacatggt gaaggggaaa tctctcacgg gggttgtgaa tgcccaggcc  960
cttaccagtg ccttctctcc acacaccaag ccttggattg gtctggcaga agctctgggg  1020
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acatccctga agaatgctgg gaactgccta agccccgcag tcattgtcgg cctcctgaaa  1140
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agggacctgc ccctgctcct attccggact cagacctctg accctgcaat gctgcctacc  1440
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tcagatgggg agacctggca cgtcatgggc atctcctcct tgctgcccag cctggaagcg  1560
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SEQ ID NO: 73           moltype = DNA  length = 1602
FEATURE                 Location/Qualifiers
misc_feature            1..1602
                        note = Synthetic
source                  1..1602
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 73
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aagattctgc aggatggcgg cctgcaggtc gtggaaaagc agaacctgag caaagaggaa  120
ctgatcgccg agctgcagga ttgcgaggga ctgatcgtca gatccgccac caaagtgacc  180
gccgacgtga tcaatgccgc cgagaaactg caggttgtgg gcagagctgg aaccggcgtg  240
gacaatgtgg atctggaagc cgccaccaga aagggcatcc tggtcatgaa cacccctaac  300
ggcaacagcc tgtctgccgc cgaactgacc tgcggaatga tcatgtgcct ggccagacag  360
atccctcagg ccacagcctc tatgaaggac ggcaagtggg agcgcaagaa attcatgggc  420
accgagctga acggcaagac cctgggaatc ctcggcctgg gcagaatcgg aagagaagtg  480
gccacacgga tgcagagctt cggcatgaag accatcggct acgaccccat catcagccct  540
gaagtgtctg ccagctttgg cgtgcagcag ctgcctctgg aagagatctg gcccctgtgc  600
gacttcatca ccgtgcacac acctctgctg cctagcacaa ccggcctgct gaacgacaac  660
acattcgccc agtgcaagaa aggcgtgcgg gtcgtgaatt gtgccagagg cggaattgtg  720
gacgaaggcg ctctgctgag agccctgcaa tctggtcaat gtgctggcgc cgctctggac  780
gtgttcacag aggaacctcc aagagacaga gccctggtgg accacgagaa cgtgatcagc  840
tgtcctcacc tgggcgcctc tacaaaagag gcccagagcc gttgcggaga agagatcgct  900
gtgcagttcg tggacatggt caagggcaag agcctgaccg gcgtcgtgaa tgctcaggcc  960
ctgacaagcg cttttagccc tcacaccaag ccttggatcg gactggctga agccctgggc  1020
acactgatga gagcatgggc cggatctcca aagggcacca tccaagtgat cacccagggc  1080
accagcctga agaacgccgg caattgtctg agccctgccg tgattgtggg actgctgaaa  1140
gaggccagca agcaggccga tgtgaacctg gttaacgcca agctgctggt caaagaggcc  1200
ggactgaacg tcaccacctc tcactctcca gctgctccag gcgagcaagg ctttggagaa  1260
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acaacccctg ttctgcaagg actgaatggc gccgtgttca gacctgaggt gcccctgaga  1380
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atgattggcc tgttggccga ggccggcgtc agactgctga gctatcagac aagcctggtg  1500
tccgacggcg agacatggca cgtgatggga atctctagcc tgctgccaag cctggaagct  1560
tggaagcagc acgtgaccga ggccttccag ttccacttct ga                     1602

SEQ ID NO: 74           moltype = DNA  length = 1017
FEATURE                 Location/Qualifiers
source                  1..1017
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 74
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tcggccgggg ctgtcgtagc tcatcccact tcagggacta tttcctctgc cagtttccat  120
cctcaacaat tccagtatac tctggataat aacgttctaa ccctggaaca gagaaaattt  180
tatgaagaaa atgggtttct agtaatcaaa aatcttgtac ctgatgccga tattcaacgc  240
tttcggaatg agtttgaaaa aatctgcaga aaggaggtga aaccattagg attaacagta  300
atgagagatg tgaccatttc gaaatccgaa tatgctccaa gtgagaagat gatcacgaag  360
gtccaggatt tccaggaaga taaggagctc ttcagatact gcactctccc cgagattctg  420
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cggaacaacg gctgtctggt tgtgctccca ggcacacaca agggctccct gaagccccac  660
gattacccca agtgggaggg gggagttaac aaaatgttcc acgggatcca ggactacgag  720
gaaaacaagg cccgggtgca cctggtgatg gagaagggcg acactgtttt cttccatcct  780
ttgctcatcc acggatctgg tcagaataaa acccagggat tccggaaggc aatttcctgc  840
catttcgcca gtgccgattg ccactacatt gacgtgaagg gcaccagtca agaaaacatc  900
gagaaggaag ttgtaggaat agcacataaa ttctttggag ctgaaaatag cgtgaacttg  960
aaggatattt ggatgtttcg agctcgactt gtgaaaggag aaagaaccaa tctttga     1017

SEQ ID NO: 75           moltype = DNA  length = 1017
FEATURE                 Location/Qualifiers
misc_feature            1..1017
                        note = Synthetic
source                  1..1017
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 75
atggaacagc tgagagccgc tgccagactg cagattgtgc tgggacatct gggcagacca  60
tctgctggtg ctgtggtggc ccatcctaca agcggcacaa ttagcagcgc cagcttccat  120
cctcagcagt tccagtacac cctggacaac aacgtgctga ccctggaaca gcggaagttc  180
tacgaagaga acggcttcct ggtcatcaag aacctggtgc ctgacgccga catccagcgg  240
ttcagaaacg agttcgagaa gatctgccgg aaagaagtga agcccctggg cctgaccgtg  300
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atgagagatg tgaccatcag caagagcgag tacgccccta gcgagaagat gatcaccaag  360
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aaatacgtgg aatgcttcac aggccccaac attatggcca tgcacaccat gctgatcaac  480
aagcctccag acagcggcaa gaaaaccagc agacaccctc tgcaccagga cctgcactac  540
ttcccattca gacccagcga cctgatcgtg tgtgcctgga cagccatgga acacatcagc  600
cggaacaatg gctgcctggt ggtgctgcct ggaacacaca agggaagcct gaagcctcac  660
gactacccca agtgggaagg cggcgtgaac aagatgttcc acggcatcca ggactatgaa  720
gagaacaagg cccgggtgca cctggtcatg gaaaagggcg acaccgtgtt ctttcaccca  780
ctgctgatcc acggcagcgg ccagaacaag acccagggct ttagaaaggc catcagctgc  840
cacttcgcca gcgccgattg ccactacatc gacgtgaagg gcaccagcca agagaacatc  900
gagaaagaag tcgtgggaat cgcccacaag ttctttggcg ccgagaacag cgtgaacctg  960
aaggatatct ggatgttccg ggccagactg gtcaagggcg agcgcacaaa tctgtga     1017

SEQ ID NO: 76          moltype = DNA  length = 1599
FEATURE                Location/Qualifiers
source                 1..1599
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 76
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agtggattcc tctccttcta ccaggcgggg gctgtggacg ccctgcggga cctggccccc  120
cggatgctgg aaacagccca ccgctttgcg gggacatcgg caggtgctgt gatcgccgcc  180
ctggccatct gcgggattga aatggatgag tatctcagag tcctcaacgt gggtgtggcc  240
gaggtgaaga aatccttcct ggggcccttg tccccgtcct gtaagatggt gcagatgatg  300
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cacgattact actaccgagg gtacgaggat gcagttttgt acttgaggcg gctgaatgct  780
gtttatctta attcttcctc caagagagtg attttccccc gggtggaagt gtactgccag  840
atagaactcg cccttggcaa tgagtgccct gaacgcagtc aaccaagcct tcgagcacgg  900
caggccagtc tggaaggagc cacacaacct cacaaggagt gggttcccaa aggggatgga  960
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tcacctgttt cagcaccagt ctctccactt gagcagccac ctgcacagcc actggcctct  1080
tcaactccac tttctctaag tggcatgcca cctgtatcat tcccagctgt gcacaagcca  1140
cccagctcca cacctggttc atcactgccc accccaccac ctggactgtc acctctgtca  1200
cctcagcagc aggtacaacc gtctggatca ccagccagat ccctacactc tcaggcaccc  1260
acttcaccca ggccatccct ggggccttca actgtggggg cacctcaaac actgccccga  1320
agttctcttt cagccttccc tgctcagcca cctgtggagg aactaggcca agaacagccc  1380
caagctgtag ctcttcttgt ctcttcaaaa ccaaaaagcg ccgtgcctct ggttcatgtg  1440
aaggaaaccg tcagcaagcc ttatgtaacg gagagccctg ctgaagactc aaactgggtg  1500
aataaggtct tcaagaagaa caagcaaaag acaagtggca ccagaaaagg cttcccaaga  1560
cattcgggat ccaaaaaacc aagcagcaaa gtgcagtga                        1599

SEQ ID NO: 77          moltype = DNA  length = 1599
FEATURE                Location/Qualifiers
misc_feature           1..1599
                       note = Synthetic
source                 1..1599
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 77
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agcggcttcc tgagcttcta tcaggctggt gccgtggacg ccctgagaga tcttgcccct  120
agaatgctgg aaacagccca cagattcgcc ggcacatctg ctggcgctgt gattgccgct  180
ctggccatct gtggcatcga gatggacgag tacctgcggg tgctgaatgt gggcgtcgcc  240
gaagtgaaga agtcctttct gggccctctg agccccagct gcaagatggt gcagatgatg  300
cggcagttcc tgtacagagt gctgcccgag gacagctaca aagtgaccac aggcaagctg  360
cacgtgtccc tgaccagact gaccgacggc gagaatgtgg tggtgtccga gttcaccagc  420
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cccagagatt gccctgccat cttccacgac ttccggatgt tcaactgcag cttccagttc  660
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cacgactact actaccgggg ctacgaggat gccgtgctgt acctgagaag gctgaacgcc  780
gtgtacctga acagcagcag caagagagtg atcttcccca gagtggaagt gtactgccag  840
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caggcttctc tggaaggcgc cacacagccc cacaaagaat gggttcccaa aggcgacggc  960
agaggctctc atggacctcc agtttctcag cccgtgcaga ccctggaatt cacatgcgag  1020
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acaagcccca gaccaagcct gggaccttct acagttggag cccctcagac actgcccaga  1320
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caggctgtgg cactgctggt gtctagcaag cctaagtctg ccgtgcctct ggtgcacgtg  1440
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aacaaagtgt tcaagaagaa caagcagaaa accagcggca cccggaaggg cttccctaga  1560
cactctggca gcaagaaacc cagctccaag gtgcagtga                          1599

SEQ ID NO: 78          moltype = DNA  length = 1314
FEATURE                Location/Qualifiers
source                 1..1314
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 78
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tcagagttca cgtccaagga ggagctcatt gaggccctat actgcagctg cttcgtcccg  180
gtgtactgtg gcctcatccc cccgacttac cgcggtgtga ggtacatcga tgggggcttc  240
acgggcatgc agccctgtgc cttctggacc gacgccatca ccatctccac cttcagtggg  300
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gaagtgtact gccagataga actcgccctt ggcaatgagt gccctgaacg cagtcaacca  600
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cccaaagggg atggaagggg cagccatggt ccgcctgtgt cccaacctgt gcagacactt  720
gaattcacat gcgagtcacc tgtttcagca ccagtctctc cacttgagca gccacctgca  780
cagccactgg cctcttcaac tccactttct ctaagtggca tgccacctgt atcattccca  840
gctgtgcaca agccacccag ctccacacct ggttcatcac tgcccacccc accacctgga  900
ctgtcacctc tgtcacctca gcagcaggta caaccgtctg gatcaccagc cagatcccta  960
cactctcagg cacccacttc acccaggcca tccctggggc cttcaactgt gggggcacct  1020
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ggccaagaac agccccaagc tgtagctctt cttgtctctt caaaaccaaa aagcgccgtg  1140
cctctggttc atgtgaagga aaccgtcagc aagccttatg taacggagag ccctgctgaa  1200
gactcaaact gggtgaataa ggtcttcaag aagaacaagc aaaagacaag tggcaccaga  1260
aaaggcttcc caagacattc gggatccaaa aaaccaagca gcaaagtgca gtga        1314

SEQ ID NO: 79          moltype = DNA  length = 1314
FEATURE                Location/Qualifiers
misc_feature           1..1314
                       note = Synthetic
source                 1..1314
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 79
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tccgagttca ccagcaaaga ggaactgatc gaggccctgt actgcagctg cttcgtgcct  180
gtgtactgcg gactgatccc tccaacctat agaggcgtgc ggtacatcga tggcggcttc  240
actggaatgc agccctgcgc cttttggacc gacgccatca ccatcagcac cttctctggc  300
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gatctggtca tcctgcacga ctactactac cggggctacg aggatgccgt gctgtacctg  480
agaaggctga acgccgtgta cctgaacagc agcagcaaga gagtgatctt ccccagagtg  540
gaagtgtact gccagatcga gctggccctg ggcaatgagt gccctgagag atctcagcca  600
agcctgagag ccagacaggc ttctctggaa ggcgccacac agccccacaa agaatgggtt  660
cccaaaggcg acggcagagg ctctcatgga cctccagttt ctcagcccgt gcagaccctg  720
gaattcacat gcgagagccc tgtgtcagcc cctgtgtctc ctctggaaca gcctcctgct  780
cagcctctgg cctctagcac acctctgtct ctgtctggca tgcctcctgt gtcctttcca  840
gccgtgcaca agcctccaag cagcacccct ggatctagcc tgcctacacc tcctccagga  900
ctgagcccac tgtctcctca gcagcaggtt cagcctagcg gatctccagc cagaagcctg  960
cattctcagg cccctacaag ccccagacca tctctgggac cttctacagt gggagcccct  1020
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ggacaagagc agcctcaggc tgtggcactg ctggtgtcta gcaagcctaa gtctgccgtg  1140
cctctggtgc acgtgaaaga aaccgtgtcc aagccttacg tgaccgagtc tcctgccgag  1200
gactccaact gggtcaacaa ggtgttcaag aagaacaagc agaaaaccag cggcacccgg  1260
aagggcttcc ctagacactc tggcagcaag aaacccagct ccaaggtgca gtga        1314

SEQ ID NO: 80          moltype = DNA  length = 1341
FEATURE                Location/Qualifiers
source                 1..1341
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 80
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accacgggga agctccatgt gagcctcacc cgcttaacgg acggggagaa tgtggtggtt  120
tcagagttca cgtccaagga ggagctcatt gaggccctat actgcagctg cttcgtcccg  180
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gtgtactgtg gcctcatccc cccgacttac cgcggtgtgt gggcatttct cactctgccc  240
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gggtacgagg atgcagtttt gtacttgagg cggctgaatg ctgtttatct taattcttcc  540
tccaagagag tgatttccc ccgggtggaa gtgtactgcc agatagaact cgcccttggc  600
aatgagtgcc ctgaacgcag tcaaccaagc cttcgagcac ggcaggccag tctggaagga  660
gccacacaac ctcacaagga gtgggttccc aaaggggatg gaaggggcag ccatggtccg  720
cctgtgtccc aacctgtgca gacacttgaa ttcacatgcg agtcacctgt ttcagcacca  780
gtctctccac ttgagcagcc acctgcacag ccactggcct cttcaactcc actttctcta  840
agtggcatgc cacctgtatc attcccagct gtgcacaagc cacccagctc cacacctggt  900
tcatcactgc ccacccccacc acctggactg tcacctctgt cacctcagca gcaggtacaa  960
ccgtctggat caccagccag atccctacac tctcaggcac ccacttcacc caggccatcc  1020
ctggggcctt caactgtggg ggcacctcaa acactgcccc gaagttctct ttcagccttc  1080
cctgctcagc cacctgtgga ggaactaggc caagaacagc cccaagctgt agctcttctt  1140
gtctcttcaa aaccaaaaag cgccgtgcct ctggttcatg tgaaggaaac cgtcagcaag  1200
ccttatgtaa cggagagccc tgctgaagac tcaaactggg tgaataaggt cttcaagaag  1260
aacaagcaaa agacaagtgg caccagaaaa ggcttcccaa gacattcggg atccaaaaaa  1320
ccaagcagca aagtgcagtg a                                            1341

SEQ ID NO: 81          moltype = DNA  length = 1341
FEATURE                Location/Qualifiers
misc_feature           1..1341
                       note = Synthetic
source                 1..1341
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 81
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tccgagttca ccagcaaaga ggaactgatc gaggccctgt actgcagctg cttcgtgcct  180
gtgtactgcg gactgatccc tccaacctac agaggcgtgt gggcctttct gacactgcct  240
cctcagcggt atatcgacgg cggctttacc ggaatgcagc cctgtgcctt ttggaccgac  300
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gccacacagc cccacaaaga atgggtgcca aaaggcgacg gcagaggctc tcatggacct  720
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tctagcctgc ctacacctcc tccaggactg agcccactgt ctccacagca acaggtgcag  960
cctagcggat ctccagccag aagcctgcat tctcaggccc ctacaagccc cagaccatct  1020
ctgggacctt ctacagtggg agcccctcaa acactgccaa gaagcagcct gtcagctttc  1080
cccgctcagc cacctgtgga agaactggga caagagcagc ctcaggctgt ggcactgctg  1140
gtgtctagca agcctaagtc tgccgtgcct ctggtgcacg tgaaagaaac cgtgtccaag  1200
ccttacgtga ccgagtctcc tgccgaggac tccaactggg tcaacaaggt gttcaagaag  1260
aacaagcaga aaaccagcgg cacccggaag ggcttcccta gacactctgg cagcaagaaa  1320
cccagctcca aggtgcagtg a                                            1341

SEQ ID NO: 82          moltype = DNA  length = 426
FEATURE                Location/Qualifiers
source                 1..426
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 82
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gcaagtggac cttttgaaag tcatgatctt cttcggaaag gtttttcttg tgtgaaaaat  120
gaacttttgc ctagtcatcc ccttgaatta tcagaaaaaa atttccagct caaccaagat  180
aaaatgaatt tttccacact gagaaacatt caggtctat ttgctccgct aaaattacag  240
atggaattca aggcagtgca gcaggttcag cgtcttccat ttctttcaag ctcaaatctt  300
tcactggatg ttttgagggg taatgatgag actattggat ttgaggatat tcttaatgat  360
ccatcacaaa gcgaagtcat gggagagcca cacttgatgg tggaatataa acttggttta  420
ctgtaa                                                             426

SEQ ID NO: 83          moltype = DNA  length = 426
FEATURE                Location/Qualifiers
misc_feature           1..426
                       note = Synthetic
source                 1..426
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 83
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gagctgctgc cttctcaccc tctggaactg agcgagaaga acttccagct gaaccaggac  180
aagatgaact tcagcaccct gcggaacatc cagggcctgt ttgcccctct gaagctgcag  240
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agcctggatg tgctgagagg caacgacgag acaatcggct tcgaggacat cctgaacgac  360
cccagccaga gcgaagtgat gggagagcct cacctgatgg tcgagtacaa gctgggcctg  420
ctgtga                                                             426

SEQ ID NO: 84          moltype = DNA  length = 1113
FEATURE                Location/Qualifiers
source                 1..1113
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 84
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gtgttgttag agatacaaaa ggaattatta gactacaaag gagttggcat tagtgttctt  120
gaaatgagtc acaggtcatc agattttgcc aagattatta acaatacaga gaatcttgtg  180
cgggaattgc tagctgttcc agacaactat aaggtgattt ttctgcaagg aggtgggtgc  240
ggccagttca gtgctgtccc cttaaacctc attggcttga aagcaggaag gtgtgctgac  300
tatgtggtga caggagcttg gtcagctaag gccgcagaag aagccaagaa gtttgggact  360
ataaatatcg ttcaccctaa acttgggagt tatacaaaaa ttccagatcc aagcacctgg  420
aacctcaacc cagatgcctc ctacgtgtat tattgcgcaa atgagacggt gcatggtgtg  480
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aatgttggct ctgctggggt caccgtggtg attgtccgtg atgacctgct ggggtttgcc  660
ctccgagagt gcccctcggt cctggaatac aaggtgcagg ctggaaacag ctccttgtac  720
aacacgcctc catgtttcag catctacgtc atgggcttgg ttctggagtg gattaaaaac  780
aatggaggtg ccgcggccat ggagaagctt agctccatca aatctcaaac aatttatgag  840
attattgata attctcaagg attctacgtt tgtccagtgg agccccaaaa tagaagcaag  900
atgaatattc cattccgcat tggcaatgcc aaaggagatg atgctttaga aaaaagattt  960
cttgataaag ctcttgaact caatatgttg tccttgaaag ggcataggtc tgtgggaggc  1020
atccgggcct ctctgtataa tgctgtcaca attgaagacg ttcagaagct ggccgccttc  1080
atgaaaaaat ttttggagat gcatcagcta tga                               1113

SEQ ID NO: 85          moltype = DNA  length = 1113
FEATURE                Location/Qualifiers
misc_feature           1..1113
                       note = Synthetic
source                 1..1113
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 85
atggatgccc ctagacaggt ggtcaacttc ggacctggac ctgccaagct gcctcactct  60
gtgctgctgg aaatccagaa agagctgctg gactacaaag gcgtgggcat cagcgtgctc  120
gagatgagcc acagaagcag cgacttcgcc aagatcatca acaacaccga gaacctcgtg  180
cgggaactgc tggccgtgcc tgacaactac aaagtgatct tcctgcaagg cggcggatgc  240
ggccagtttt ctgctgtgcc tctgaacctg atcggcctga aggccggaag atgcgccgat  300
tatgtggtca caggcgcttg gagcgccaag gccgctgagg aagccaagaa gttcggcacc  360
atcaacatcg tgcaccccaa gctgggcagc tacacaaaga tccccgatcc tagcacctgg  420
aatctgaacc ccgatgccag ctacgtgtac tactgcgcca acgagacagt gcacggcgtg  480
gaatttgact tcatccccga tgtgaagggc gccgtgctcg tgtgtgatat gagcagcaac  540
ttcctgagca agcccgtgga cgtgtccaag ttcggcgtga tctttgccgg cgctcagaaa  600
aacgtgggat ctgccggcgt gaccgtggtc attgtcagag atgatctgct gggcttcgcc  660
ctgagagaat gcccttctgt gctcgagtac aaggtgcagg ccggcaacag cagcctgtac  720
aatacccctc catgcttcag catctacgtg atgggcctcg tgctggaatg gatcaagaac  780
aatggcggag ccgccgctat ggaaaagctg agcagcatca agagccagac catctacgag  840
atcatcgaca atagccaggg cttctacgtg tgccccgtgg aaccccagaa ccggtccaag  900
atgaacatcc ccttccggat cggcaacgcc aaaggcgacg atgccctgga aaagcggttc  960
ctggataagg ccctggaact gaatatgctg agcctgaagg gccacagatc cgtcggagga  1020
atcagagcca gcctgtataa cgccgtgacc atcgaggacg tgcagaaact ggctgccttt  1080
atgaagaaat tcctcgagat gcaccagctg tga                               1113

SEQ ID NO: 86          moltype = DNA  length = 942
FEATURE                Location/Qualifiers
source                 1..942
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 86
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ggcaacctct cagagaagta cgtcttcatc acaggctgtg actctggctt cgggaacctg  120
ctggccaaac agctggttga tcggggcatg caggtgctgg ctgcttgctt cactgaggag  180
ggatcccaga aacttcagcg ggatacctcc tatcggctgc agaccacccct actggatgtc  240
accaagagcg aaagcatcaa ggcggcggcc cagtgggtga gggacaaagt gggcgaacaa  300
ggcctctggg ccctggtgaa caatgctggt gtgggcctgc ccagtggtcc caacgaatgg  360
ctgaccaagg atgactttgt gaaggtgatt aatgtgaacc tggtgggact gatcgaagtg  420
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accettcaca tgctgcccat ggtcaagaga gcccggggca gggttgtcaa catgtccagc  480
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gccttctctg acagcataag gcgtgagctc tactactttg ggtgaaagt ctgcatcatt  600
gagccaggga actatcggac agccattctc ggcaaggaga acctggagtc acgcatgcga  660
aagctttggg agaggctgcc tcaggagacc cgggacagct acggagagga ttatttccgc  720
atctatactg acaagttaaa aaacataatg caggtggcag agcccagagt cagagatgtc  780
atcaacagca tggagcatgc tattgtttcc cggagccctc gcatccgcta caaccctggc  840
ctggatgcca aactcctcta catccctctg gctaagttgc ccacccctgt gacagatttc  900
atcctaagcc ggtaccttcc aaggccagcg gacagtgtct aa                      942

SEQ ID NO: 87          moltype = DNA  length = 942
FEATURE                Location/Qualifiers
misc_feature           1..942
                       note = Synthetic
source                 1..942
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 87
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ctggctaagc agctggtgga cagaggcatg caggttctgg ccgcctgctt tacagaggaa  180
ggctcccaga agctgcagcg ggacacaagc tacagactgc agaccacact gctggacgtg  240
accaagagcg agagcatcaa agccgccgct cagtgggtcc gagacaaagt gggagaacaa  300
ggactgtggg ccctcgtgaa caatgctggc gttggactgc ctagcggccc caatgagtgg  360
ctgaccaagg acgacttcgt gaaagtgatc aacgtgaacc ttgtgggcct gatcgaagtg  420
accctgcaca tgctgcccat ggtcaagcga gccagaggca gagtggtcaa catgagcagc  480
tctggcggaa gagtggccgt gatcggcgga ggctattgcg tgtccaagtt tggcgtggaa  540
gccttcagcg acagcatcag aagagagctg tactacttcg gagtgaaagt gtgcatcatc  600
gagcccggca actaccggac agccatcctg ggcaaagaga acctggaaag ccggatgcgg  660
aagctgtggg agagactgcc ccaagagaca agagacagct acggcgagga ctacttccgg  720
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atcaactcta tggaacacgc catcgtgtct cggagcccca gaatcagata caaccctggc  840
ctggacgcca agctgctgta tatccctctg gccaaactgc ccacacctgt gaccgacttc  900
atcctgagca gatacctgcc tagacctgcc gacagcgtgt ga                      942

SEQ ID NO: 88          moltype = DNA  length = 1125
FEATURE                Location/Qualifiers
source                 1..1125
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 88
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gaagaggaca actcaagaaa cgtattcttc tctcccatga gcatctcctc tgccctggcc  120
atggtcttca tgggggcaaa gggaagcact gcagcccaga tgtcccaggc actttgttta  180
tacaaagacg gagatattca ccgaggtttc cagtcacttc tcagtgaagt taacagaact  240
ggcactcagt acttgcttag aactgccaac agactctttg gagaaaagac gtgtgatttc  300
cttccagact ttaaagaata ctgtcagaag ttctatcagg cagagctgga ggagttgtcc  360
tttgctgaag acactgaaga gtgcaggaag catataaatg actgggtggc agagaagact  420
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gctaagttta aaatggggta tgcggatgag gtacacaccc aggtcctgga gctgccctat  660
gtggaagagg agctgagcat ggtcattctg cttcccgatg acaacacgga cctcgccgtg  720
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aaaagtaagg ttcaagtttt ccttcccaga ttaaagctgg aggagagtta tgacttggag  840
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ggaatgtcaa ctgagaagaa tgtgcctctg tccaaggttg cccacaagtg cttcgtggag  960
gtcaatgagg aaggcacaga ggctgccgca gccactgctg tggtcaggaa ttcccggtgc  1020
agcagaatgg agccaagatt ctgtgcagac caccctttc ttttcttcat caggcaccac  1080
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SEQ ID NO: 89          moltype = DNA  length = 1125
FEATURE                Location/Qualifiers
misc_feature           1..1125
                       note = Synthetic
source                 1..1125
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 89
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atggtgttta tgggcgccaa gggatctacc gccgctcaga tgtctcaggc cctgtgcctg  180
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ggcacacagt acctgctgcg gaccgccaat agactgttcg gcgagaaaac ctgcgacttc  300
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agccggatgg aaccccggtt ttgtgccgac catccttttc tgttcttcat ccggcaccac  1080
aagaccaact gcatcctgtt ctgcggcaga ttcagcagcc cttga                 1125

SEQ ID NO: 90          moltype = DNA  length = 1932
FEATURE                Location/Qualifiers
source                 1..1932
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 90
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ttcatccggg tcttcatcaa gaccatcagg cgcgatatct ttggcggcct ggtcctcctg  180
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gagaagctgt ga                                                      1932

SEQ ID NO: 91          moltype = DNA  length = 1932
FEATURE                Location/Qualifiers
misc_feature           1..1932
                       note = Synthetic
source                 1..1932
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 91
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cagaaaggca gatacgtgcc cctggatcaa gaggcctact ccagaatcca ggccggcgaa  1920
gagaagctgt ga                                                      1932

SEQ ID NO: 92          moltype = DNA  length = 777
FEATURE                Location/Qualifiers
source                 1..777
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 92
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ctcgcccgtc agcgaggagt cctggagcgc acagagaaga tggtggacaa gatggaccaa  300
gatttgaaga tcagccagaa acacatcaat agcattaaga gcgtgtttgg ggggctggtc  360
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caggccagcc acccaaacct tagaaagctg gatgatacag accctgtccc cagaggggct  540
ggttctgcca tgagtactga tgcttaccca aagaacccac accttcgagc ctatcaccag  600
aagatcgaca gcaacctaga tgagctgtcc atgggactgg gtcgtctgaa ggacatagcc  660
ctggggatgc agacagaaat tgaggagcaa gatgacattc ttgaccggct gacaaccaaa  720
gtggacaagt tagatgtcaa cataaaaagc acagaaagaa aagttcgaca actctga     777

SEQ ID NO: 93          moltype = DNA  length = 777
FEATURE                Location/Qualifiers
misc_feature           1..777
                       note = Synthetic
source                 1..777
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 93
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aagatcgaca gcaacctgga cgagctgtct atgggcctgg gcagactgaa ggatatcgcc  660
ctgggaatgc agaccgagat cgaggaacag gacgacatcc tggaccggct gaccaccaag  720
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SEQ ID NO: 94          moltype = DNA  length = 2568
FEATURE                Location/Qualifiers
source                 1..2568
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 94
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gccggtgtgg tgagctgggg agacggctgc gctcagagga acaagccagg cgtgtacaca  2520
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SEQ ID NO: 95           moltype = DNA  length = 2568
FEATURE                 Location/Qualifiers
misc_feature            1..2568
                        note = Synthetic
source                  1..2568
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 95
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KPVEKMMELF IRLKEILNQM ASGTHPLLDK MRSLKQMHLP RSVPLTQAMY RSNRMNTPQG  720
SFSTISQALC SQGITTEYLT AMLPSSQRPK GNHTKDFLTY KLTKEQIASK YGIPINSTPF  780
CFSLYKDIIN MPAGPVIWAF LKPMLLGRIL YAPYNPVTKA IMEKSNVTLR QLAELREKSQ  840
EWMDKSPLFM NSFHLLNQAI PMLQNTLRNP FVQVFVKFSV GLDAVELLKQ IDELDILRLK  900
LENNIDIIDQ LNTLSSLTVN ISSCVLYDRI QAAKTIDEME REAKRLYKSN ELFGSVIFKL  960
PSNRSWHRGY DSGNVFLPPV IKYTIRMSLK TAQTTRSLRT KIWAPGPHNS PSHNQIYGRA  1020
FIYLQDSIER AIIELQTGRN SQEIAVQVQA IPYPCFMKDN FLTSVSYSLP IVLMVAWVVF  1080
IAAFVKKLVY EKDLRLHEYM KMMGVNSCSH FFAWLIESVG FLLVTIVILI IILKFGNILP  1140
KTNGFILFLY FSDYSFSVIA MSYLISVFFN NTNIAALIGS LIYIIAFFPF IVLVTVENEL  1200
SYVLKVFMSL LSPTAFSYAS QYIARYEEQG IGLQWENMYT SPVQDDTTSF GWLCCLILAD  1260
SFIYFLIAWY VRNVFPGTYG MAAPWYFPIL PSYWKERFGC AEVKPEKSNG LMFTNIMMQN  1320
TNPSASPEYM FSSNIEPEPK DLTVGVALHG VTKIYGSKVA VDNLNLNFYE GHITSLLGPN  1380
GAGKTTTISM LTGLFGASAG TIFVYGKDIK TDLHTVRKNM GVCMQHDVLF SYLTTKEHLL  1440
LYGSIKVPHW TKKQLHEEVK RTLKDTGLYS HRHKRVGTLS GGMKRKLSIS IALIGGSRVV  1500
ILDEPSTGVD PCSRRSIWDV ISKNKTARTI ILSTHHLDEA EVLSDRIAFL EQGGLRCCGS  1560
PFYLKEAFGD GYHLTLTKKK SPNLNANAVC DTMAVTAMIQ SHLPEAYLKE DIGGELVYVL  1620
PPFSTKVSGA YLSLLRALDN GMGDLNIGCY GISDTTVEEV FLNLTKESQK NSAMSLEHLT  1680
QKKIGNSNAN GISTPDDLSV SSSNFTDRDD KILTRGERLD GFGLLLKKIM AILIKRFHHT  1740
RRNWKGLIAQ VILPIVFVTT AMGLGTLRNS SNSYPEIQIS PSLYGTSEQT AFYANYHPST  1800
EALVSAMWDF PGIDNMCLNT SDLQCLNKDS LEKWNTSGEP ITNFGVCSCS ENVQECPKFN  1860
YSPPHRRTYS SQVIYNLTGQ RVENYLISTA NEFVQKRYGG WSFGLPLTKD LRFDITGVPA  1920
NRTLAKVWYD PEGYHSLPAY LNSLNNFLLR VNMSKYDAAR HGIIMYSHPY PGVQDQEQAT  1980
ISSLIDILVA LSILMGYSVT TASFVTYVVR EHQTKAKQLQ HISGIGVTCY WVTNFIYDMV  2040
FYLVPVAFSI GIIAIFKLPA FYSENNLGAV SLLLLLFGYA TFSWMYLLAG LFHETGMAFI  2100
TYVCVNLFFG INSIVSLSVV YFLSKEKPND PTLELISETL KRIFLIFPQF CFGYGLIELS  2160
QQQSVLDFLK AYGVEYPNET FEMNKLGAMF VALVSQGTMF FSLRLLINES LIKKLRLFFR  2220
KFNSSHVRET IDEDEDVRAE RLRVESGAAE FDLVQLYCLT KTYQLIHKKI IAVNNISIGI  2280
PAGECFGLLG VNGAGKTTIF KMLTGDIIPS SGNILIRNKT GSLGHVDSHS SLVGYCPQED  2340
ALDDLVTVEE HLYFYARVHG IPEKDIKETV HKLLRRLHLM PFKDRATSMC SYGTKRKLST  2400
ALALIGKPSI LLLDEPSSGM DPKSKRHLWK IISEEVQNKC SVILTSHSME ECEALCTRLA  2460
IMVNGKFQCI GSLQHIKSRF GRGFTVKVHL KNNKVTMETL TKFMQLHFPK TYLKDQHLSM  2520
LEYHVPVTAG GVANIFDLLE TNKTALNITN FLVSQTTLEE VFINFAKDQK SYETADTSSQ  2580
GSTISVDSQD DQMES                                                 2595

SEQ ID NO: 103          moltype = AA  length = 2277
FEATURE                 Location/Qualifiers
source                  1..2277
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 103
MFTYIKIITS GDSDNITHVW NEDDGQTLSP SSLAAQLLIL ENFEDALLNI SANSPYIPYL  60
ACVRNVTDSL ARGSPENLRL LQSTIRFKKS FLRNGSYEDY FPPVPEVLKS KLSQLRNLTE  120
LLCESETFSL IEKSCQLSDM SFGSLCEESE FDLQLLEAAE LGTEIAASLL YHDNVISKKV  180
RDLLTGDPSK INLNMDQFLE QALQMNYLEN ITQLIPIIEA MLHVNNSADA SEKPGQLLEM  240
FKNVEELKED LRRTTGMSNR TIDKLLAIPI PDNRAEIISQ VFWLHSCDTN ITTPKLEDAM  300
KEFCNLSLSE RSRQSYLIGL TLLHYLNIYN FTYKVFFPRK DQKPVEKMME LFIRLKEILN  360
QMASGTHPLL DKMRSLKQMH LPRSVPLTQA MYRSNRMNTP QGSFSTISQA LCSQGITTEY  420
LTAMLPSSQR PKGNHTKDFL TYKLTKEQIA SKYGIPINST PFCFSLYKDI INMPAGPVIW  480
AFLKPMLLGR ILYAPYNPVT KAIMEKSNVT LRQLAELREK SQEWMDKSPL FMNSFHLLNQ  540
AIPMLQNTLR NPFVQVFVKF SVGLDAVELL KQIDELDILR LKLENNIDII DQLNTLSSLT  600
VNISSCVLYD RIQAAKTIDE MEREAKRLYK SNELFGSVIF KLPSNRSWHR GYDSGNVFLP  660
PVIKYTIRMS LKTAQTTRSL RTKIWAPGPH NSPSHNQIYG RAFIYLQDSI ERAIIELQTG  720
RNSQEIAVQV QAIPYPCFMK DNFLTSVSYS LPIVLMVAWV VFIAAFVKKL VYEKDLRLHE  780
YMKMMGVNSC SHFFAWLIES VGFLLVTIVI LIIILKFGNI LPKTNGFILF LYFSDYSFSV  840
IAMSYLISVF FNNTNIAALI GSLIYIIAFF PFIVLVTVEN ELSYVLKVFM SLLSPTAFSY  900
ASQYIARYEE QGIGLQWENM YTSPVQDDTT SFGWLCCLIL ADSFIYFLIA WYVRNVFPGT  960
YGMAAPWYFP ILPSYWKERF GCAEVKPEKS NGLMFTNIMM QNTNPSASPE YMFSSNIEPE  1020
PKDLTVGVAL HGVTKIYGSK VAVDNLNLNF YEGHITSLLG PNGAGKTTTI SMLTGLFGAS  1080
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AGTIFVYGKD IKTDLHTVRK NMGVCMQHDV LFSYLTTKEH LLLYGSIKVP HWTKKQLHEE  1140
VKRTLKDTGL YSHRHKRVGT LSGGMKRKLS ISIALIGGSR VVILDEPSTG VDPCSRRSIW  1200
DVISKNKTAR TIILSTHHLD EAEVLSDRIA FLEQGGLRCC GSPFYLKEAF GDGYHLTLTK  1260
KKSPNLNANA VCDTMAVTAM IQSHLPEAYL KEDIGGELVY VLPPFSTKVS GAYLSLLRAL  1320
DNGMGDLNIG CYGISDTTVE EVFLNLTKES QKNSAMSLEH LTQKKIGNSN ANGISTPDDL  1380
SVSSSNFTDR DDKILTRGER LDGFGLLLKK IMAILIKRFH HTRRNWKGLI AQVILPIVFV  1440
TTAMGLGTLR NSSNSYPEIQ ISPSLYGTSE QTAFYANYHP STEALVSAMW DFPGIDNMCL  1500
NTSDLQCLNK DSLEKWNTSG EPITNFGVCS CSENVQECPK FNYSPPHRRT YSSQVIYNLT  1560
GQRVENYLIS TANEFVQKRY GGWSFGLPLT KDLRFDITGV PANRTLAKVW YDPEGYHSLP  1620
AYLNSLNNFL LRVNMSKYDA ARHGIIMYSH PYPGVQDQEQ ATISSLIDIL VALSILMGYS  1680
VTTASFVTYV VREHQTKAKQ LQHISGIGVT CYWVTNFIYD MVFYLVPVAF SIGIIAIFKL  1740
PAFYSENNLG AVSLLLLLFG YATFSWMYLL AGLFHETGMA FITYVCVNLF FGINSIVSLS  1800
VVYFLSKEKP NDPTLELISE TLKRIFLIFP QFCFGYGLIE LSQQQSVLDF LKAYGVEYPN  1860
ETFEMNKLGA MFVALVSQGT MFFSLRLLIN ESLIKKLRLF FRKFNSSHVR ETIDEDEDVR  1920
AERLRVESGA AEFDLVQLYC LTKTYQLIHK KIIAVNNISI GIPAGECFGL LGVNGAGKTT  1980
IFKMLTGDII PSSGNILIRN KTGSLGHVDS HSSLVGYCPQ EDALDDLVTV EEHLYFYARV  2040
HGIPEKDIKE TVHKLLRRLH LMPFKDRATS MCSYGTKRKL STALALIGKP SILLLDEPSS  2100
GMDPKSKRHL WKIISEEVQN KCSVILTSHS MEECEALCTR LAIMVNGKFQ CIGSLQHIKS  2160
RFGRGFTVKV HLKNNKVTME TLTKFMQLHF PKTYLKDQHL SMLEYHVPVT AGGVANIFDL  2220
LETNKTALNI TNFLVSQTTL EEVFINFAKD QKSYETADTS SQGSTISVDS QDDQMES     2277

SEQ ID NO: 104          moltype = AA  length = 349
FEATURE                 Location/Qualifiers
source                  1..349
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 104
MAAEEEEVDS ADTGERSGWL TGWLPTWCPT SISHLKEAEE KMLKCVPCTY KKEPVRISNG  60
NKIWTLKFSH NISNKTPLVL LHGFGGGLGL WALNFGDLCT NRPVYAFDLL GFGRSSRPRF  120
DSDAEEVENQ FVESIEEWRC ALGLDKMILL GHNLGGFLAA AYSLKYPSRV NHLILVEPWG  180
FPERPDLADQ DRPIPVWIRA LGAALTPFNP LAGLRIAGPF GLSLVQRLRP DFKRKYSSMF  240
EDDTVTEYIY HCNVQTPSGE TAFKNMTIPY GWAKRPMLQR IGKMHPDIPV SVIFGARSCI  300
DGNSGTSIQS LRPHSYVKTI AILGAGHYVY ADQPEEFNQK VKEICDTVD              349

SEQ ID NO: 105          moltype = AA  length = 508
FEATURE                 Location/Qualifiers
source                  1..508
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 105
MELEVRRVRQ AFLSGRSRPL RFRLQQLEAL RRMVQEREKD ILTAIAADLC KSEFNVYSQE  60
VITVLGEIDF MLENLPEWVT AKPVKKNVLT MLDEAYIQPQ PLGVVLIIGA WNYPFVLTIQ  120
PLIGAIAAGN AVIIKPSELS ENTAKILAKL LPQYLDQDLY IVINGGVEET TELLKQRFDH  180
IFYTGNTAVG KIVMEAAAKH LTPVTLELGG KSPCYIDKDC DLDIVCRRIT WGKYMNCGQT  240
CIAPDYILCE ASLQNQIVWK IKETVKEFYG ENIKESPDYE RIINLRHFKR ILSLLEGQKI  300
AFGGETDEAT RYIAPTVLTD VDPKTKVMQE EIFGPILPIV PVKNVDEAIN FINEREKPLA  360
LYVFSHNHKL IKRMIDETSS GGVTGNDVIM HFTLNSFPFG GVGSSGMGAY HGKHSFDTFS  420
HQRPCLLKSL KREGANKLRY PPNSQSKVDW GKFFLLKRFN KEKLGLLLLT FLGIVAAVLV  480
KKYQAVLRRK ALLIFLVVHR LRWSSKQR                                     508

SEQ ID NO: 106          moltype = AA  length = 701
FEATURE                 Location/Qualifiers
source                  1..701
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 106
MATYKVRVAT GTDLLSGTRD SISLTIVGTQ GESHKQLLNH FGRDFATGAV GQYTVQCPQD  60
LGELIIIRLH KERYAFFPKD PWYCNYVQIC APNGRIYHFP AYQWMDGYET LALREATGKT  120
TADDSLPVLL EHRKEEIRAK QDFYHWRVFL PGLPSYVHIP SYRPPVRRHR NPNRPEWNGY  180
IPGFPILINF KATKFLNLNL RYSFLKTASF FVRLGPMALA FKVRGLLDCK HSWKRLKDIR  240
KIFPGKKSVV SEYVAEHWAE DTFFGYQYLN GVNPGLIRRC TRIPDKFPVT DDMVAPFLGE  300
GTCLQAELEK GNIYLADYRI MEGIPTVELS GRKQHHCAPL CLLHFGPEGK MMPIAIQLSQ  360
TPGPDCPIFL PSDSEWDWLL AKTWVRYAEF YSHEAIAHLL ETHLIAEAFC LALLRNLPMC  420
HPLYKLLIPH TRYTVQINSI GRAVLLNEGG LSAKGMSLGV EGFAGVMVRA LSELTYDSLY  480
LPNDFVERGV QDLPGYYYRD DSLAVWNALE KYVTEIITYY YPSDAAVEGD PELQSWVQEI  540
FKECLLGRES SGFPRCLRTV PELIRYVTIV IYTCSAKHAA VNTGQMEFTA WMPNFPASMR  600
NPPIQTKGLT TLETFMDTLP DVKTTCITLL VLWTLSREPD DRRPLGHFPD IHFVEEAPRR  660
SIEAFRQRLN QISHDIRQRN KCLPIPYYYL DPVLIENSIS I                      701

SEQ ID NO: 107          moltype = AA  length = 711
FEATURE                 Location/Qualifiers
source                  1..711
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 107
MAVYRLCVTT GPYLRAGTLD NISVTLVGTC GESPKQRLDR MGRDFAPGSV QKYKVRCTAE  60
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LGELLLLRVH KERYAFFRKD SWYCSRICVT EPDGSVSHFP CYQWIEGYCT VELRPGTART  120
ICQDSLPLLL DHRTRELRAR QECYRWKIYA PGFPCMVDVN SFQEMESDKK FALTKTTTCV  180
DQGDSSGNRY LPGFPMKIDI PSLMYMEPNV RYSATKTISL LFNAIPASLG MKLRGLLDRK  240
GSWKKLDDMQ NIFWCHKTFT TKYVTEHWCE DHFFGYQYLN GVNPVMLHCI SSLPSKLPVT  300
NDMVAPLLGQ DTCLQTELER GNIFLADYWI LAEAPTHCLN GRQQYVAAPL CLLWLSPQGA  360
LVPLAIQLSQ TPGPDSPIFL PTDSEWDWLL AKTWVRNSEF LVHENNTHFL CTHLLCEAFA  420
MATLRQLPLC HPIYKLLLPH TRYTLQVNTI ARATLLNPEG LVDQVTSIGR QGLIYLMSTG  480
LAHFTYTNFC LPDSLRARGV LAIPNYHYRD DGLKIWAAIE SFVSEIVGYY YPSDASVQQD  540
SELQAWTGEI FAQAFLGRES SGFPSRLCTP GEMVKFLTAI IFNCSAQHAA VNSGQHDFGA  600
WMPNAPSSMR QPPPQTKGTT TLKTYLDTLP EVNISCNNLL LFWLVSQEPK DQRPLGTYPD  660
EHFTEEAPRR SIAAFQSRLA QISRDIQERN QGLALPYTYL DPPLIENSVS I           711

SEQ ID NO: 108          moltype = AA  length = 843
FEATURE                 Location/Qualifiers
source                  1..843
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 108
MPRGAFRPCL PALYFAFLTC PTPEQRMSGT QAPDIHLGEP ARGTGCVRGK QTSIRVQDCG  60
RREEARAASR ELRREKAQEH PRESWAHPQP YPAPQPLALR PETQPCPACR SSPPGRLLLR  120
PALPGHPFLL PIMAVYRLCV TTGPYLRAGT LDNISVTLVG TCGESPKQRL DRMGRDFAPG  180
SVQKYKVRCT AELGELLLLR VHKERYAFFR KDSWYCSRIC VTEPDGSVSH FPCYQWIEGY  240
CTVELRPGTA RTICQDSLPL LLDHRTRELR ARQECYRWKI YAPGFPCMVD VNSFQEMESD  300
KKFALTKTTT CVDQGDSSGN RYLPGFPMKI DIPSLMYMEP NVRYSATKTI SLLFNAIPAS  360
LGMKLRGLLD RKGSWKKLDD MQNIFWCHKT FTTKYVTEHW CEDHFFGYQY LNGVNPVMLH  420
CISSLPSKLP VTNDMVAPLL GQDTCLQTEL ERGNIFLADY WILAEAPTHC LNGRQQYVAA  480
PLCLLWLSPQ GALVPLAIQL SQTPGPDSPI FLPTDSEWDW LLAKTWVRNS EFLVHENNTH  540
FLCTHLLCEA FAMATLRQLP LCHPIYKLLL PHTRYTLQVN TIARATLLNP EGLVDQVTSI  600
GRQGLIYLMS TGLAHFTYTN FCLPDSLRAR GVLAIPNYHY RDDGLKIWAA IESFVSEIVG  660
YYYPSDASVQ QDSELQAWTG EIFAQAFLGR ESSGFPSRLC TPGEMVKFLT AIIFNCSAQH  720
AAVNSGQHDF GAWMPNAPSS MRQPPPQTKG TTTLKTYLDT LPEVNISCNN LLLFWLVSQE  780
PKDQRPLGTY PDEHFTEEAP RRSIAAFQSR LAQISRDIQE RNQGLALPYT YLDPPLIENS  840
VSI                                                                843

SEQ ID NO: 109          moltype = AA  length = 158
FEATURE                 Location/Qualifiers
source                  1..158
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 109
MMRFMLLFSR QGKLRLQKWY LATSDKERKK MVRELMQVVL ARKPKMCSFL EWRDLKVVYK  60
RYASLYFCCA IEGQDNELIT LELIHRYVEL LDKYFGSVCE LDIIFNFEKA YFILDEFLMG  120
GDVQDTSKKS VLKAIEQADL LQEEDESPRS VLEEMGLA                          158

SEQ ID NO: 110          moltype = AA  length = 589
FEATURE                 Location/Qualifiers
source                  1..589
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 110
MLHLHHSCLC FRSWLPAMLA VLLSLAPSAS SDISASRPNI LLLMADDLGI GDIGCYGNNT  60
MRTPNIDRLA EDGVKLTQHI SAASLCTPSR AAFLTGRYPV RSGMVSSIGY RVLQWTGASG  120
GLPTNETTFA KILKEKGYAT GLIGKWHLGL NCESASDHCH HPLHHGFDHF YGMPFSLMGD  180
CARWELSEKR VNLEQKLNFL FQVLALVALT LVAGKLTHLI PVSWMPVIWS ALSAVLLLAS  240
SYFVGALIVH ADCFLMRNHT ITEQPMCFQR TTPLILQEVA SFLKRNKHGP FLLFVSFLHV  300
HIPLITMENF LGKSLHGLYG DNVEEMDWMV GRILDTLDVE GLSNSTLIYF TSDHGGSLEN  360
QLGNTQYGGW NGIYKGGKGM GGWEGGIRVP GIFRWPGVLP AGRVIGEPTS LMDVFPTVVR  420
LAGGEVPQDR VIDGQDLLPL LLGTAQHSDH EFLMHYCERF LHAARWHQRD RGTMWKVHFV  480
TPVFQPEGAG ACYGRKVCPC FGEKVVHHDP PLLFDLSRDP SETHILTPAS EPVFYQVMER  540
VQQAVWEHQR TLSPVPLQLD RLGNIWRPWL QPCCGPFPLC WCLREDDPQ              589

SEQ ID NO: 111          moltype = AA  length = 242
FEATURE                 Location/Qualifiers
source                  1..242
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 111
MSNPRSLEEE KYDMSGARLA LILCVTKARE GSEEDLDALE HMFRQLRFES TMKRDPTAEQ  60
FQEELEKFQQ AIDSREDPVS CAFVVLMAHG REGFLKGEDG EMVKLENLFE ALNNKNCQAL  120
RAKPKVYIIQ ACRGEQRDPG ETVGGDEIVM VIKDSPQTIP TYTDALHVYS TVEGYIAYRH  180
DQKGSCFIQT LVDVFTKRKG HILELLTEVT RRMAEAELVQ EGKARKTNPE IQSTLRKRLY  240
LQ                                                                 242

SEQ ID NO: 112          moltype = AA  length = 529
FEATURE                 Location/Qualifiers
source                  1..529
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                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 112
MGSSRAPWMG RVGGHGMMAL LLAGLLLPGT LAKSIGTFSD PCKDPTRITS PNDPCLTGKG  60
DSSGFSSYSG SSSSGSSISS ARSSGGGSSG SSSGSSIAQG GSAGSFKPGT GYSQVSYSSG  120
SGSSLQGASG SSQLGSSSSH SGNSGSHSGS SSSHSSSSSS FQFSSSSFQV GNGSALPTND  180
NSYRGILNPS QPGQSSSSSQ TFGVSSSGQS VSSNQRPCSS DIPDSPCSGG PIVSHSGPYI  240
PSSHSVSGGQ RPVVVVVDQH GSGAPGVVQG PPCSNGGLPG KPCPPITSVD KSYGGYEVVG  300
GSSDSYLVPG MTYSKGKIYP VGYFTKENPV KGSPGVPSFA AGPPISEGKY FSSNPIIPSQ  360
SAASSAIAFQ PVGTGGVQLC GGGSTGSKGP CSPSSSRVPS SSSISSSSGL PYHPCGSASQ  420
SPCSPPGTGS FSSSSSQSS GKIILQPCGS KSSSSGHPCM SVSSLTLTGG PDGSPHPDPS  480
AGAKPCGSSS AGKIPCRSIR DILAQVKPLG PQLADPEVFL PQGELLNSP             529

SEQ ID NO: 113          moltype = AA  length = 383
FEATURE                 Location/Qualifiers
source                  1..383
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 113
MFWTFKEWFW LERFWLPPTI KWSDLEDHDG LVFVKPSHLY VTIPYAFLLL IIRRVFEKFV  60
ASPLAKSFGI KETVRKVTPN TVLENFFKHS TRQPLQTDIY GLAKKCNLTE RQVERWFRSR  120
RNQERPSRLK KFQEACWRFA FYLMITVAGI AFLYDKPWLY DLWEVWNGYP KQPLLPSQYW  180
YYILEMSFYW SLLFRLGFDV KRKDFLAHII HHLAAISLMS FSWCANYIRS GTLVMIVHDV  240
ADIWLESAKM FSYAGWTQTC NTLFFIFSTI FFISRLIVFP FWILYCTLIL PMYHLEPFFS  300
YIFLNLQLMI LQVLHLYWGY YILKMLNRCI FMKSIQDVRS DDEDYEEEEE EEEEEATKGK  360
EMDCLKNGLR AERHLIPNGQ HGH                                         383

SEQ ID NO: 114          moltype = AA  length = 424
FEATURE                 Location/Qualifiers
source                  1..424
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 114
MTLRPGTMRL ACMFSSILLF GAAGLLLFIS LQDPTELAPQ QVPGIKFNIR PRQPHHDLPP  60
GGSQDGDLKE PTERVTRDLS SGAPRGRNLP APDQPQPPLQ RGTRLRLRQR RRRLLIKKMP  120
AAATIPANSS DAPFIRPGPG TLDGRWVSLH RSQQERKRVM QEACAKYRAS SSRRAVTPRH  180
VSRIFVEDRH RVLYCEVPKA GCSNWKRVLM VLAGLASSTA DIQHNTVHYG SALKRLDTFD  240
RQGILHRLST YTKMLFVREP FERLVSAFRD KFEHPNSYYH PVFGKAILAR YRANASREAL  300
RTGSGVRFPE FVQYLLDVHR PVGMDIHWDH VSRLCSPCLI DYDFVGKFES MEDDANFFLS  360
LIRAPRNLTF PRFKDRHSQE ARTTARIAHQ YFAQLSALQR QRTYDFYYMD YLMFNYSKPF  420
ADLY                                                              424

SEQ ID NO: 115          moltype = AA  length = 211
FEATURE                 Location/Qualifiers
source                  1..211
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 115
MANAGLQLLG FILAFLGWIG AIVSTALPQW RIYSYAGDNI VTAQAMYEGL WMSCVSQSTG  60
QIQCKVFDSL LNLSSTLQAT RALMVVGILL GVIAIFVATV GMKCMKCLED DEVQKMRMAV  120
IGGAIFLLAG LAILVATAWY GNRIVQEFYD PMTPVNARYE FGQALFTGWA AASLCLLGGA  180
LLCCSCPRKT TSYPTPRPYP KPAPSSGKDY V                                211

SEQ ID NO: 116          moltype = AA  length = 98
FEATURE                 Location/Qualifiers
source                  1..98
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 116
MIPGGLSEAK PATPEIQEIV DKVKPQLEEK TNETYGKLEA VQYKTQVVAG TNYYIKVRAG  60
DNKYMHLKVF KSLPGQNEDL VLTGYQVDKN KDDELTGF                         98

SEQ ID NO: 117          moltype = AA  length = 531
FEATURE                 Location/Qualifiers
source                  1..531
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 117
MLPITDRLLH LLGLEKTAFR IYAVSTLLLF LLFFLFRLLL RFLRLCRSFY ITCRRLRCFP  60
QPPRRNWLLG HLGMYLPNEA GLQDEKKVLD NMHHVLLVWM GPVLPLLVLV HPDYIKPLLG  120
ASAAIAPKDD LFYGFLKPWL GDGLLLSKGD KWSRHRRLLT PAFHFDILKP YMKIFNQSAD  180
IMHAKWRHLA EGSAVSLDMF EHISLMTLDS LQKCVFSYNS NCQEKMSDYI SAIIELSALS  240
VRRQYRLHHY LDFIYYRSAD GRRFRQACDM VHHFTTEVIQ ERRRALRQQG AEAWLKAKQG  300
KTLDFIDVLL LARDEDGKEL SDEDIRAEAD TFMFEGHDTT SSGISWMLFN LAKYPEYQEK  360
CREEIQEVMK GRELEELEWD DLTQLPFTTM CIKESLRQYP PVTLVSRQCT EDIKLPDGRI  420
IPKGIICLVS IYGTHHNPTV WPDSKVYNPY RFDPDNPQQR SPLAYVPFSA GPRNCIGQSF  480
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AMAELRVVVA LTLLRFRLSV DRTRKVRRKP ELILRTENGL WLKVEPLPPR A          531

SEQ ID NO: 118          moltype = AA  length = 230
FEATURE                 Location/Qualifiers
source                  1..230
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 118
MTTNAGPLHP YWPQHLRLDN FVPNDRPTWH ILAGLFSVTG VLVVTTWLLS GRAAVVPLGT  60
WRRLSLCWFA VCGFIHLVIE GWFVLYYEDL LGDQAFLSQL WKEYAKGDSR YILGDNFTVC  120
METITACLWG PLSLWVVIAF LRQHPLRFIL QLVVSVGQIY GDVLYFLTEH RDGFQHGELG  180
HPLYFWFYFV FMNALWLVLP GVLVLDAVKH LTHAQSTLDA KATKAKSKKN            230

SEQ ID NO: 119          moltype = AA  length = 314
FEATURE                 Location/Qualifiers
source                  1..314
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 119
MGLLDSEPGS VLNVVSTALN DTVEFYRWTW SIADKRVENW PLMQSPWPTL SISTLYLLFV  60
WLGPKWMKDR EPFQMRLVLI IYNFGMVLLN LFIFRELFMG SYNAGYSYIC QSVDYSNNVH  120
EVRIAAALWW YFVSKGVEYL DTVFFILRKK NNQVSFLHVY HHCTMFTLWW IGIKWVAGGQ  180
AFFGAQLNSF IHVIMYSYYG LTAFGPWIQK YLWWKRYLTM LQLIQFHVTI GHTALSLYTD  240
CPFPKWMHWA LIAYAISFIF LFLNFYIRTY KEPKKPKAGK TAMNGISANG VSKSEKQLMI  300
ENGKKQKNGK AKGD                                                   314

SEQ ID NO: 120          moltype = AA  length = 4061
FEATURE                 Location/Qualifiers
source                  1..4061
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 120
MSTLLENIFA IINLFKQYSK KDKNTDTLSK KELKELLEKE FRQILKNPDD PDMVDVFMDH  60
LDIDHNKKID FTEFLLMVFK LAQAYYESTR KENLPISGHK HRKHSHHDKH EDNKQEENKE  120
NRKRPSSLER RNNRKGNKGR SKSPRETGGK RHESSSEKKE RKGYSPTHRE EEYGKNHHNS  180
SKKEKNKTEN TRLGDNRKRL SERLEEKEDN EEGVYDYENT GRMTQKWIQS GHIATYYTIQ  240
DEAYDTTDSL LEENKIYERS RSSDGKSSSQ VNRSRHENTS QVPLQESRTR KRRGSRVSQD  300
RDSEGHSEDS ERHSGSASRN HHGSAWEQSR DGSRHPRSHD EDRASHGHSA DSSRQSGTRH  360
AETSSRGQTA SSHEQARSSP GERHGSGHQQ SADSSRHSAT GRGQASSAVS DRGHRGSSGS  420
QASDSEGHSE NSDTQSVSGH GKAGLRQQSH QESTRGRSGE RSGRSGSSLY QVSTHEQPDS  480
AHGRTGTSTG GRQGSHHEQA RDSSRHSASQ EGQDTIRGHP GSSRGGRQGS HHEQSVNRSG  540
HSGSHHSHTT SQGRSDASHG QSGSRSASRQ TRNEEQSGDG TRHSGSRHHE ASSQADSSRH  600
SQVGQGQSSG PRTSRNQGSS VSQDSDSQGH SEDSERWSGS ASRNHHGSAQ EQSRDGSRHP  660
RSHHEDRAGH GHSADSSRKS GTRHTQNSSS GQAASSHEQA RSSAGERHGS RHQLQSADSS  720
RHSGTGHGQA SSAVRDSGHR GSSGSQATDS EGHSEDSDTQ SVSGHGQAGH HQQSHQESAR  780
DRSGERSRRS GSFLYQVSTH KQSESSHGWT GPSTGVRQGS HHEQARDNSR HSASQDGQDT  840
IRGHPGSSRR GRQGSHHEQS VDRSGHSGSH HSHTTSQGRS DASRGQSGSR SASRTTRNEE  900
QSRDGSRHSG SRHHEASSHA DISRHSQAGQ GQSEGSRTSR RQGSSVSQDS DSEGHSEDSE  960
RWSGSASRNH RGSAQEQSRH GSRHPRSHHE DRAGHGHSAD SSRQSGTPHA ETSSGGQAAS  1020
SHEQARSSPG ERHGSRHQQS ADSSRHSGIP RRQASSAVRD SGHWGSSGSQ ASDSEGHSEE  1080
SDTQSVSGHG QDGPHQQSHQ ESARDWSGGR SGRSGSFIYQ VSTHEQSESA HGRTRTSTGR  1140
RQGSHHEQAR DSSRHSASQE GQDTIRAHPG SRRGGRQGSH HEQSVDRSGH SGSHHSHTTS  1200
QGRSDASHGQ SGSRSASRQT RKDKQSGDGS RHSGSRHHEA ASWADSSRHS QVGQEQSSGS  1260
RTSRHQGSSV SQDSDSERHS DDSERLSGSA SRNHHGSSRE QSRDGSRHPG FHQEDRASHG  1320
HSADSSRQSG THHTESSSHG QAVSSHEQAR SSPGERHGSR HQQSADSSRH SGIGHRQASS  1380
AVRDSGHRGS SGSQVTNSEG HSEDSDTQSV SAHGQAGPHQ QSHKESARGQ SGESSGRSRS  1440
FLYQVSSHEQ SESTHGQTAP STGGRQGSRH EQARNSSRHS ASQDGQDTIR GHPGSSRGGR  1500
QGSYHEQSVD RSGHSGYHHS HTTPQGRSDA SHGQSGPRSA SRQTRNEEQS GDGSRHSGSR  1560
HHEPSTRAGS SRHSQVGQGE SAGSKTSRRQ GSSVSQDRDS EGHSEDSERR SESASRNHYG  1620
SAREQSRHGS RNPRSHQEDR ASHGHSAESS RQSGTRHAET SSGGQAASSQ EQARSSPGER  1680
HGSRHQQSAD SSTDSGTGRR QDSSVVGDSG NRGSSGSQAS DSEGHSEESD TQSVSAHGQA  1740
GPHQQSHQES TRGQSGERSG RSGSFLYQVS THEQSESAHG RTGPSTGGRQ RSRHEQARDS  1800
SRHSASQEGQ DTIRGHPGSS RGGRQGSHYE QSVDSSGHSG SHHSHTTSQE RSDVSRGQSG  1860
SRSVSRQTRN EKQSGDGSRH SGSRHHEASS RADSSRHSQV GQGQSSGPRT SRNQGSSVSQ  1920
DSDSQGHSED SERWSGSASR NHLGSAWEQS RDGSRHPGSH HEDRAGHGHS ADSSRQSGTR  1980
HTESSSRGQA ASSHEQARSS AGERHGSHHQ LQSADSSRHS GIGHGQASSA VRDSGHRGYS  2040
GSQASDSEGH SEDSDTQSVS AQGKAGPHQQ SHKESARGQS GESSGRSGSF LYQVSTHEQS  2100
ESTHGQSAPS TGGRQGSHYD QAQDSSRHSA SQEGQDTIRG HPGPSRGGRQ GSHQEQSVDR  2160
SGHSGSHHSH TTSQGRSDAS RGQSGSRSAS RKTYDKEQSG DGSRHSGSHH HEASSWADSS  2220
RHSLVGQGQS SGPRTSRPRG SSVSQDSDSE GHSEDSERRS GSASRNHHGS AQEQSRDGSR  2280
HPRSHHEDRA GHGHSAESSR QSGTHHAENS SGGQAASSHE QARSSAGERH GSHHQQSADS  2340
SRHSGIGHGQ ASSAVRDSGH RGSSGSQASD SEGHSEDSDT QSVSAHGQAG PHQQSHQEST  2400
RGRSAGRSGR SGSFLYQVST HEQSESAHGR TGTSTGGRQG SHHKQARDSS RHSTSQEGQD  2460
TIHGHPGSSS GGRQGSHYEQ LVDRSGHSGS HHSHTTSQGR SDASHGHSGS RSASRQTRND  2520
EQSGDGSRHS GSRHHEASSR ADSSGHSQVG QGQSEGPRTS RNWGSSFSQD SDSQGHSEDS  2580
ERWSGSASRN HHGSAQEQLR DGSRHPRSHQ EDRAGHGHSA DSSRQSGTRH TQTSSGGQAA  2640
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SSHEQARSSA GERHGSHHQQ SADSSRHSGI GHGQASSAVR DSGHRGYSGS QASDNEGHSE  2700
DSDTQSVSAH GQAGSHQQSH QESARGRSGE TSGHSGSFLY QVSTHEQSES SHGWTGPSTR  2760
GRQGSRHEQA QDSSRHSASQ DGQDTIRGHP GSSRGGRQGY HHEHSVDSSG HSGSHHSHTT  2820
SQGRSDASRG QSGSRSASRT TRNEEQSGDG SRHSGSRHHE ASTHADISRH SQAVQGQSEG  2880
SRRSRRQGSS VSQDSDSEGH SEDSERWSGS ASRNHHGSAQ EQLRDGSRHP RSHQEDRAGH  2940
GHSADSSRQS GTRHTQTSSG GQAASSHEQA RSSAGERHGS HHQQSADSSR HSGIGHGQAS  3000
SAVRDSGHRG YSGSQASDNE GHSEDSDTQS VSAHGQAGSH QQSHQESARG RSGETSGHSG  3060
SFLYQVSTHE QSESSHGWTG PSTRGRQGSR HEQAQDSSRH SASQYGQDTI RGHPGSSRGG  3120
RQGYHHEHSV DSSGHSGSHH SHTTSQGRSD ASRGQSGSRS ASRTTRNEEQ SGDSSRHSVS  3180
RHHEASTHAD ISRHSQAVQG QSEGSRRSRR QGSSVSQDSD SEGHSEDSER WSGSASRNHR  3240
GSVQEQSRHG SRHPRSHHED RAGHGHSADR SRQSGTRHAE TSSGGQAASS HEQARSSPGE  3300
RHGSRHQQSA DSSRHSGIPR GQASSAVRDS RHWGSSGSQA SDSEGHSEES DTQSVSGHGQ  3360
AGPHQQSHQE SARDRSGGRS GRSGSFLYQV STHEQSESAH GRTRTSTGRR QGSHHEQARD  3420
SSRHSASQEG QDTIRGHPGS SRRGRQGSHY EQSVDRSGHS GSHHSHTTSQ GRSDASRGQS  3480
GSRSASRQTR NDEQSGDGSR HSWSHHHEAS TQADSSRHSQ SGQGQSAGPR TSRNQGSSVS  3540
QDSDSQGHSE DSERWSGSAS RNHRGSAQEQ SRDGSRHPTS HHEDRAGHGH SAESSRQSGT  3600
HHAENSSGGQ AASSHEQARS SAGERHGSHH QQSADSSRHS GIGHGQASSA VRDSGHRGSS  3660
GSQASDSEGH SEDSDTQSVS AHGQAGPHQQ SHQESTRGRS AGRSGRSGSF LYQVSTHEQS  3720
ESAHGRAGPS TGGRQGSRHE QARDSSRHSA SQEGQDTIRG HPGSRRGGRQ GSYHEQSVDR  3780
SGHSGSHHSH TTSQGRSDAS HGQSGSRSAS RETRNEEQSG DGSRHSGSRH HEASTQADSS  3840
RHSQSGQGES AGSRRSRRQG SSVSQDSDSE AYPEDSERRS ESASRNHHGS SREQSRDGSR  3900
HPGSSHRDTA SHVQSSPVQS DSSTAKEHGH FSSLSQDSAY HSGIQSRGSP HSSSSYHYQS  3960
EGTERQKGQS GLVWRHGSYG SADYDYGESG FRHSQHGSVS YNSNPVVFKE RSDICKASAF  4020
GKDHPRYYAT YINKDPGLCG HSSDISKQLG FSQSQRYYYY E                     4061

SEQ ID NO: 121         moltype = AA  length = 2391
FEATURE                Location/Qualifiers
source                 1..2391
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 121
MTDLLRSVVT VIDVFYKYTK QDGECGTLSK GELKELLEKE LHPVLKNPDD PDTVDVIMHM  60
LDRDHDRRLD FTEFLLMIFK LTMACNKVLS KEYCKASGSK KHRRGHRHQE EESETEEDEE  120
DTPGHKSGYR HSSWSEGEEH GYSSGHSRGT VKCRHGSNSR RLGRQGNLSS SGNQEGSQKR  180
YHRSSCGHSW SGGKDRHGSS SVELRERINK SHISPSRESG EEYESGSGSN SWERKGHGGL  240
SCGLETSGHE SNSTQSRIRE QKLGSSCSGS GDSGRRSHAC GYSNSSGCGR PQNASSSCQS  300
HRFGGQGNQF SYIQSGCQSG IKGGQGHGCV SGGQPSGCGQ PESNPCSQSY SQRGYGAREN  360
GQPQNCGGQW RTGSSQSSCC GQYGSGGSQS CSNGQHEYGS CGRFSNSSSS NEFSKCDQYG  420
SGSSQSTSFE QHGTGLSQSS GFEQHVCGSG QTCGQHESTS SQSLGYDQHG SSSGKTSGFG  480
QHGSGSGQSS GFGQCGSGSG QSSGFGQHGS VSGQSSGFGQ HGSVSGQSSG FGQHESRSRQ  540
SSYGQHGSGS SQSSGYGQYG SRETSGFGQH GLGSGQSTGF GQYGSGSGQS SGFGQHGSGS  600
GQSSGFGQHE SRSGQSSYGQ HSSGSSQSSG YGQHGSRQTS GFGQHGSGSS QSTGFGQYGS  660
GSGQSSGFGQ HVSGSGQSSG FGQHESRSGH SSYGQHGFGS SQSSGYGQHG SSSGQTSGFG  720
QHELSSGQSS SFGQHGSGSG QSSGFGQHGS GSGQSSGFGQ HESRSGQSSY GQHSSGSSQS  780
SGYGQHGSRQ TSGFGQHGSG SSQSTGFGQY GSGSGQSAGF GQHGSGSGQS SGFGQHESRS  840
HQSSYGQHGS GSSQSSGYGQ HGSSSGQTSG FGQHRSSSGQ YSGFGQHGSG SGQSSGFGQH  900
GTGSGQYSGF GQHESRSHQS SYGQHGSGSS QSSGYGQHGS SSGQTFGFGQ HRSGSGQSSG  960
FGQHGSGSGQ SSGFGQHESG SGKSSGFGQH ESRSSQSNYG QHGSGSSQSS GYGQHGSSSG  1020
QTTGFGQHRS SSGQYSGFGQ HGSGSDQSSG FGQHGTGSGQ SSGFGQYESR SRQSSYGQHG  1080
SGSSQSSGYG QHGSNSGQTS GFGQHRPGSG QSSGFGQYGS GSGQSSGFGQ HGSGTGKSSG  1140
FAQHEYRSGQ SSYGQHGTGS SQSSGCGQHE SGSGPTTSFG QHVSGSDNFS SSGQHISDSG  1200
QSTGFGQYGS GSGQSTGLGQ GESQQVESGS TVHGRQETTH GQTINTTRHS QSGQGQSTQT  1260
GSRVTRRRRS SQSENSDSEV HSKVSHRHSE HIHTQAGSHY PKSGSTVRRR QGTTHGQRGD  1320
TTRHGHSGHG QSTQTGSRTS GRQRFSHSDA TDSEVHSGVS HRPHSQEQTH SQAGSQHGES  1380
ESTVHERHET TYGQTGEATG HGHSGHGQST QRGSRTTGRR GSGHSESSDS EVHSGGSHRP  1440
QSQEQTHGQA GSQHGESGST VHGRHGTTHG QTGDTTRHAH YHHGKSTQRG SSTTGRRGSG  1500
HSESSDSEVH SGGSHTHSGH THGQSGSQHG ESESIIHDRH RITHGQTGDT TRHSYSGHEQ  1560
TTQTGSRTTG RQRTSHSEST DSEVHSGGSH RPHSREHTYG QAGSQHEEPE FTVHERHGTT  1620
HGQIGDTTGH SHSGHGQSTQ RGSRTTGRQR SSHSESSDSE VHSGVSHTHT GHTHGQAGSQ  1680
HGQSESIVPE RHGTTHGQTG DTTRHAHYHH GLTTQTGSRT TGRRGSGHSE YSDSEGYSGV  1740
SHTHSGHTHG QARSQHGESE SIVHERHGTI HGQTGDTTRH AHSGHGQSTQ TGSRTTGRRS  1800
SGHSEYSDSE GHSGFSQRPH SRGHTHGQAG SQHGESESIV DERHGTTHGQ TGDTSGHSQS  1860
GHGQSTQSGS STTGRRRSGH SESSDSEVHS GGSHTHSGHT HSQARSQHGE SESTVHKRHQ  1920
TTHGQTGDTT EHGHPSHGQT IQTGSRTTGR RGSGHSEYSD SEGPSGVSHT HSGHTHGQAG  1980
SHYPESGSSV HERHGTTHGQ TADTTRHGHS GHGQSTQRGS RTTGRRASGH SEYSDSEGHS  2040
GVSHTHSGHA HGQAGSQHGE SGSSVHERHG TTHGQTGDTT RHAHSGHGQS TQRGSRTAGR  2100
RGSGHSESSD SEVHSGVSHT HSGHTYGQAR SQHGESGSAI HGRQGTIHGQ TGDTTRHGQS  2160
GHGQSTQTGS RTTGRQRSSH SESSDSEVHS EASPTHSGHT HSQAGSRHGQ SGSSGHGRQG  2220
TTHGQTGDTT RHAHYGYGQS TQRGSRTTGR RGSGHSESSD SEVHSWGSHT HSGHIQGQAG  2280
SQQRQPGSTV HGRLETTHGQ TGDTTRHGHS GYGQSTQTGS RSSRASHFQS HSSERQRHGS  2340
SQVWKHGSYG PAEYDYGHTG YGPSGGSRKS ISNSHLSWST DSTANKQLSR H           2391

SEQ ID NO: 122         moltype = AA  length = 226
FEATURE                Location/Qualifiers
source                 1..226
                       mol_type = protein
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                          organism = Homo sapiens
SEQUENCE: 122
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KNVCYDHYFP ISHIRLWALQ LIFVSTPALL VAMHVAYRRH EKKRKFIKGE IKSEFKDIEE  120
IKTQKVRIEG SLWWTYTSSI FFRVIFEAAF MYVFYVMYDG FSMQRLVKCN AWPCPNTVDC  180
FVSRPTEKTV FTVFMIAVSG ICILLNVTEL CYLLIRYCSG KSKKPV                226

SEQ ID NO: 123            moltype = AA  length = 270
FEATURE                   Location/Qualifiers
source                    1..270
                          mol_type = protein
                          organism = Homo sapiens
SEQUENCE: 123
MDWKTLQALL SGVNKYSTAF GRIWLSVVFV FRVLVYVVAA ERVWGDEQKD FDCNTKQPGC  60
TNVCYDNYFP ISNIRLWALQ LIFVTCPSLL VILHVAYREE RERRHRQKHG DQCAKLYDNA  120
GKKHGGLWWT YLFSLIFKLI IEFLFLYLLH TLWHGFNMPR LVQCANVAPC PNIVDCYIAR  180
PTEKKIFTYF MVGASAVCIV LTICELCYLI CHRVLRGLHK DKPRGGCSPS SSASRASTCR  240
CHHKLVEAGE VDPDPGNNKL QASAPNLTPI                                  270

SEQ ID NO: 124            moltype = AA  length = 266
FEATURE                   Location/Qualifiers
source                    1..266
                          mol_type = protein
                          organism = Homo sapiens
SEQUENCE: 124
MNWAFLQGLL SGVNKYSTVL SRIWLSVVFI FRVLVYVVAA EEVWDDEQKD FVCNTKQPGC  60
PNVCYDEFFP VSHVRLWALQ LILVTCPSLL VVMHVAYREE RERKHHLKHG PNAPSLYDNL  120
SKKRGGLWWT YLLSLIFKAA VDAGFLYIFH RLYKDYDMPR VVACSVEPCP HTVDCYISRP  180
TEKKVFTYFM VTTAAICILL NLSEVFYLVG KRCMEIFGPR HRRPRCRECL PDTCPPYVLS  240
QGGHPEDGNS VLMKAGSAPV DAGGYP                                      266

SEQ ID NO: 125            moltype = AA  length = 261
FEATURE                   Location/Qualifiers
source                    1..261
                          mol_type = protein
                          organism = Homo sapiens
SEQUENCE: 125
MDWGTLHTFI GGVNKHSTSI GKVWITVIFI FRVMILVVAA QEVWGDEQED FVCNTLQPGC  60
KNVCYDHFFP VSHIRLWALQ LIFVSTPALL VAMHVAYYRH ETTRKFRRGE KRNDFKDIED  120
IKKQKVRIEG SLWWTYTSSI FFRIIFEAAF MYVFYFLYNG YHLPWVLKCG IDPCPNLVDC  180
FISRPTEKTV FTIFMISASV ICMLLNVAEL CYLLLKVCFR RSKRAQTQKN HPNHALKESK  240
QNEMNELISD SGQNAITGFP S                                           261

SEQ ID NO: 126            moltype = AA  length = 332
FEATURE                   Location/Qualifiers
source                    1..332
                          mol_type = protein
                          organism = Homo sapiens
SEQUENCE: 126
MLLLAAAFLV AFVLLLYMVS PLISPKPLAL PGAHVVVTGG SSGIGKCIAI ECYKQGAFIT  60
LVARNEDKLL QAKKEIEMHS INDKQVVLCI SVDVSQDYNQ VENVIKQAQE KLGPVDMLVN  120
CAGMAVSGKF EDLEVSTFER LMSINYLGSV YPSRAVITTM KERRVGRIVF VSSQAGQLGL  180
FGFTAYSASK FAIRGLAEAL QMEVKPYNVY ITVAYPPDTD TPGFAEENRT KPLETRLISE  240
TTSVCKPEQV AKQIVKDAIQ GNFNSSLGSD GYMLSALTCG MAPVTSITEG LQQVVTMGLF  300
RTIALFYLGS FDSIVRRCMM QREKSENADK TA                               332

SEQ ID NO: 127            moltype = AA  length = 644
FEATURE                   Location/Qualifiers
source                    1..644
                          mol_type = protein
                          organism = Homo sapiens
SEQUENCE: 127
MSRQFSSRSG YRSGGGFSSG SAGIINYQRR TTSSSTRRSG GGGGRFSSCG GGGGSFGAGG  60
GFGSRSLVNL GGSKSISISV ARGGGRGSGF GGGYGGGGFG GGGFGGGGFG GGGIGGGGFG  120
GFGSGGGGFG GGGFGGGGYG GGYGPVCPPG GIQEVTINQS LLQPLNVEID PEIQKVKSRE  180
REQIKSLNNQ FASFIDKVRF LEQQNQVLQT KWELLQQVDT STRTHNLEPY FESFINNLRR  240
RVDQLKSDQS RLDSELKNMQ DMVEDYRNKY EDEINKRTNA ENEFVTIKKD VDGAYMTKVD  300
LQAKLDNLQQ EIDFLTALYQ AELSQMQTQI SETNVILSMD NNRSLDLDSI IAEVKAQYED  360
IAQKSKAEAE SLYQSKYEEL QITAGRHGDS VRNSKIEISE LNRVIQRLRS EIDNVKKQIS  420
NLQQSISDAE QRGENALKDA KNKLNDLEDA LQQAKEDLAR LLRDYQELMN TKLALDLEIA  480
TYRTLLEGEE SRMSGECAPN VSVSVSTSHT TISGGGSRGG GGGGYGSGGS SYGSGGGSYG  540
SGGGGGGGRG SYGSGGSSYG SGGGSYGSGG GGGGHGSYGS GSSSGGYRGG SGGGGGGSSG  600
GRGSGGGSSG GSIGGRGSSS GGVKSSGGSS SVKFVSTTYS GVTR                  644

SEQ ID NO: 128            moltype = AA  length = 639
FEATURE                   Location/Qualifiers
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source                  1..639
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 128
MSCQISCKSR GRGGGGGGFR GFSSGSAVVS GGSRRSTSSF SCLSRHGGGG GGFGGGGFGS  60
RSLVGLGGTK SISISVAGGG GGFGAAGGFG GRGGGFGGGS SFGGGSGFSG GGFGGGGFGG  120
GRFGGFGGPG GVGGLGGPGG FGPGGYPGGI HEVSVNQSLL QPLNVKVDPE IQNVKAQERE  180
QIKTLNNKFA SFIDKVRFLE QQNQVLQTKW ELLQQMNVGT RPINLEPIFQ GYIDSLKRYL  240
DGLTAERTSQ NSELNNMQDL VEDYKKKYED EINKRTAAEN DFVTLKKDVD NAYMIKVELQ  300
SKVDLLNQEI EFLKVLYDAE ISQIHQSVTD TNVILSMDNS RNLDLDSIIA EVKAQYEEIA  360
QRSKEEAEAL YHSKYEELQV TVGRHGDSLK EIKIEISELN RVIQRLQGEI AHVKKQCKNV  420
QDAIADAEQR GEHALKDARN KLNDLEEALQ QAKEDLARLL RDYQELMNVK LALDVEIATY  480
RKLLEGEECR MSGDLSSNVT VSVTSSTISS NVASKAAFGG SGGRGSSSGG GYSSGSSSYG  540
SGGRQSGSRG GSGGGGSISG GGYGSGGGSG GRYGSGGGSK GGSISGGGYG SGGGKHSSGG  600
GSRGGSSSGG GYGSGGGGSS SVKGSSGEAF GSSVTFSFR                         639

SEQ ID NO: 129          moltype = AA  length = 623
FEATURE                 Location/Qualifiers
source                  1..623
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 129
MSCRQFSSSY LSRSGGGGGG GLGSGGSIRS SYSRFSSSGG GGGGGRFSSS SGYGGGSSRV  60
CGRGGGGSFG YSYGGGSGGG FSASSLGGGF GGGSRGFGGA SGGGYSSSGG FGGGFGGGSG  120
GGFGGGYGSG FGGFGGFGGG AGGGDGGILT ANEKSTMQEL NSRLASYLDK VQALEEANND  180
LENKIQDWYD KKGPAAIQKN YSPYYNTIDD LKDQIVDLTV GNNKTLLDID NTRMTLDDFR  240
IKFEMEQNLR QGVDADINGL RQVLDNLTME KSDLEMQYET LQEELMALKK NHKEEMSQLT  300
GQNSGDVNVE INVAPGKDLT KTLNDMRQEY EQLIAKNRKD IENQYETQIT QIEHEVSSSG  360
QEVQSSAKEV TQLRHGVQEL EIELQSQLSK KAALEKSLED TKNRYCGQLQ MIQEQISNLE  420
AQITDVRQEI ECQNQEYSLL LSIKMRLEKE IETYHNLLEG GQEDFESSGA GKIGLGGRGG  480
SGGSYGRGSR GGSGGSYGGG GSGGGYGGGS GSRGGSGGSY GGGSGSGGGS GGGYGGGSGG  540
GHSGGSGGGH SGGSGGNYGG GSGSGGGSGG GYGGGSGSRG GSGGSHGGGS GFGGESGGSY  600
GGGEEASGSG GGYGGGSGKS SHS                                          623

SEQ ID NO: 130          moltype = AA  length = 584
FEATURE                 Location/Qualifiers
source                  1..584
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 130
MSVRYSSSKH YSSSRSGGGG GGGGCGGGGG VSSLRISSSK GSLGGGFSSG GFSGGSFSRG  60
SSGGGCFGGS SGGYGGLGGF GGGSFRGSYG SSSFGGSYGG SFGGGSFGGG SFGGGSFGGG  120
GFGGGGFGGG FGGGFGGDGG LLSGNEKVTM QNLNDRLASY LDKVRALEES NYELEGKIKE  180
WYEKHGNSHQ GEPRDYSKYY KTIDDLKNQI LNLTTDNANI LLQIDNARLA ADDFRLKYEN  240
EVALRQSVEA DINGLRRVLD ELTLTKADLE MQIESLTEEL AYLKKNHEEE MKDLRNVSTG  300
DVNVEMNAAP GVDLTQLLNN MRSQYEQLAE QNRKDAEAWF NEKSKELTTE IDNNIEQISS  360
YKSEITELRR NVQALEIELQ SQLALKQSLE ASLAETEGRY CVQLSQIQAQ ISALEEQLQQ  420
IRAETECQNT EYQQLLDIKI RLENEIQTYR SLLEGEGSSG GGGRGGGSFG GGYGGGSSGG  480
GSSGGGHGGG HGGSSGGGYG GGSSGGGSSG GGYGGGSSSG GHGGSSSGGY GGGSSGGGGG  540
GYGGGSSGGG SSSGGGYGGG SSSGGHKSSS SGSVGESSSK GPRY                   584

SEQ ID NO: 131          moltype = AA  length = 398
FEATURE                 Location/Qualifiers
source                  1..398
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 131
MMWLLLTTTC LICGTLNAGG FLDLENEVNP EVWMNTSEII IYNGYPSEEY EVTTEDGYIL  60
LVNRIPYGRT HARSTGPRPV VYMQHALFAD NAYWLENYAN GSLGFLLADA GYDVWMGNSR  120
GNTWSRRHKT LSETDEKFWA FSFDEMAKYD LPGVIDFIVN KTGQEKLYFI GHSLGTTIGF  180
VAFSTMPELA QRIKMNFALG PTISFKYPTG IFTRFFLLPN SIIKAVFGTK GFFLEDKKTK  240
IASTKICNNK ILWLICSEFM SLWAGSNKKN MNQSRMDVYM SHAPTGSSVH NILHIKQLYH  300
SDEFRAYDWG NDADNMKHYN QSHPPIYDLT AMKVPTAIWA GGHDVLVTPQ DVARILPQIK  360
SLHYFKLLPD WNHFDFVWGL DAPQRMYSEI IALMKAYS                          398

SEQ ID NO: 132          moltype = AA  length = 312
FEATURE                 Location/Qualifiers
source                  1..312
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 132
MSYQKKQPTP QPPVDCVKTS GGGGGGGGSG GGGCGFFGGG GSGGGSSGSG CGYSGGGGYS  60
GGGCGGGSSG GGGGGGIGGC GGGSGGSVKY SGGGGSSGGG SGCFSSGGGG SGCFSSGGGG  120
SSGGGSGCFS SGGGGSSGGG SGCFSSGGGG FSGQAVQCQS YGGVSSGGSS GGGSGCFSSG  180
GGGGSVCGYS GGGSGCGGGS SGGSGSGYVS SQQVTQTSCA PQPSYGGGSS GGGGSGGSGC  240
FSSGGGGGSS GCGGGSSGIG SGCIISGGGS VCGGGSSGGG GGGSSVGGSG SGKGVPICHQ  300
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TQQKQAPTWP SK                                                 312

SEQ ID NO: 133          moltype = AA  length = 519
FEATURE                 Location/Qualifiers
source                  1..519
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 133
MIPVSLVVVV VGGWTVVYLT DLVLKSSVYF KHSYEDWLEN NGLSISPFHI RWQTAVFNRA  60
FYSWGRRKAR MLYQWFNFGM VFGVIAMFSS FFLLGKTLMQ TLAQMMADSP SSYSSSSSSS  120
SSSSSSSSSS SSSSSSLHNE QVLQVVVPGI NLPVNQLTYF FTAVLISGVV HEIGHGIAAI  180
REQVRFNGFG IFLFIIYPGA FVDLFTTHLQ LISPVQQLRI FCAGIWHNFV LALLGILALV  240
LLPVILLPFY YTGVGVLITE VAEDSPAIGP RGLFVGDLVT HLQDCPVTNV QDWNECLDTI  300
AYEPQIGYCI SASTLQQLSF PVRAYKRLDG STECCNNHSL TDVCFSYRNN FNKRLHTCLP  360
ARKAVEATQV CRTNKDCKKS SSSSFCIIPS LETHTRLIKV KHPPQIDMLY VGHPLHLHYT  420
VSITSFIPRF NFLSIDLPVV VETFVKYLIS LSGALAIVNA VPCFALDGQW ILNSFLDATL  480
TSVIGDNDVK DLIGFFILLG GSVLLAANVT LGLWMVTAR                        519

SEQ ID NO: 134          moltype = AA  length = 466
FEATURE                 Location/Qualifiers
source                  1..466
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 134
MPGDSSPGTL PLWDASLSPP LGPDPGGFSR ASHAGDKSRP PAPELGSPGA VRPRVGSCAP  60
GPMELRVSNT SCENGSLLHL YCSSQEVLCQ IVNDLSPEVP SNATFHSWQE RIRQNYGFYI  120
GLGLAFLSSF LIGSSVILKK KGLLRLVATG ATRAVDGGFG YLKDAMWWAG FLTMAAGEVA  180
NFGAYAFAPA TVVTPLGALS VLISAILSSY FLRESLNLLG KLGCVICVAG STVMVIHAPE  240
EEKVTTIMEM ASKMKDTGFI VFAVLLLVSC LILIFVIAPR YGQRNILIYI IICSVIGAFS  300
VAAVKGLGIT IKNFFQGLPV VRHPLPYILS LILALSLSTQ VNFLNRALDI FNTSLVFPIY  360
YVFFTTVVVT SSIILFKEWY SMSAVDIAGT LSGFVTIILG VFMLHAFKDL DISCASLPHM  420
HKNPPPSPAP EPTVIRLEDK NVLVDNIELA STSSPEEKPK VFIIHS                466

SEQ ID NO: 135          moltype = AA  length = 447
FEATURE                 Location/Qualifiers
source                  1..447
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 135
MPGDSSPGTL PLWDASLSPP LGPDPGGFSR ASHAGDKSRP PAPELGSPGA VRPRVGSCAP  60
GPMELRVSNT SCENGSLLHL YCSSQEVLCQ IVNDLSPEVP SNATFHSWQE RIRQNYGFYI  120
GLGLAFLSSF LIGSSVILKK KGLLRLVATG ATRAVAAGEV ANFGAYAFAP ATVVTPLGAL  180
SVLISAILSS YFLRESLNLL GKLGCVICVA GSTVMVIHAP EEEKVTTIME MASKMKDTGF  240
IVFAVLLLVS CLILIFVIAP RYGQRNILIY IIICSVIGAF SVAAVKGLGI TIKNFFQGLP  300
VVRHPLPYIL SLILALSLST QVNFLNRALD IFNTSLVFPI YYVFFTTVVV TSSIILFKEW  360
YSMSAVDIAG TLSGFVTIIL GVFMLHAFKD LDISCASLPH MHKNPPPSPA PEPTVIRLED  420
KNVLVDNIEL ASTSSPEEKP KVFIIHS                                     447

SEQ ID NO: 136          moltype = AA  length = 373
FEATURE                 Location/Qualifiers
source                  1..373
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 136
MEPAVSEPMR DQVARTHLTE DTPKVNADIE KVNQNQAKRC TVIGGSGFLG QHMVEQLLAR  60
GYAVNVFDIQ QGFDNPQVRF FLGDLCSRQD LYPALKGVNT VFHCASPPPS SNNKELFYRV  120
NYIGTKNVIE TCKEAGVQKL ILTSSASVIF EGVDIKNGTE DLPYAMKPID YYTETKILQE  180
RAVLGANDPE KNFLTTAIRP HGIFGPRDPQ LVPILIEAAR NGKMKFVIGN GKNLVDFTFV  240
ENVVHGHILA AEQLSRDSTL GGKAFHITND EPIPFWTFLS RILTGLNYEA PKYHIPYWVA  300
YYLALLLSLL VMVISPVIQL QPTFTPMRVA LAGTFHYYSC ERAKKAMGYQ PLVTMDDAME  360
RTVQSFRHLR RVK                                                    373

SEQ ID NO: 137          moltype = AA  length = 323
FEATURE                 Location/Qualifiers
source                  1..323
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 137
MSAVCGGAAR MLRTPGRHGY AAEFSPYLPG RLACATAQHY GIAGCGTLLI LDPDEAGLRL  60
FRSFDWNDGL FDVTWSENNE HVLITCSGDG SLQLWDTAKA AGPLQVYKEH AQEVYSVDWS  120
QTRGEQLVVS GSWDQTVKLW DPTVGKSLCT FRGHESIIYS TIWSPHIPGC FASASGDQTL  180
RIWDVKAAGV RIVIPAHQAE ILSCDWCKYN ENLLVTGAVD CSLRGWDLRN VRQPVFELLG  240
HTYAIRRVKF SPFHASVLAS CSYDFTVRFW NFSKPDSLLE TVEHHTEFTC GLDFSLQSPT  300
QVADCSWDET IKIYDPACLT IPA                                         323

SEQ ID NO: 138          moltype = AA  length = 533
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FEATURE                Location/Qualifiers
source                 1..533
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 138
MAFANLRKVL ISDSLDPCCR KILQDGGLQV VEKQNLSKEE LIAELQDCEG LIVRSATKVT  60
ADVINAAEKL QVVGRAGTGV DNVDLEAATR KGILVMNTPN GNSLSAAELT CGMIMCLARQ  120
IPQATASMKD GKWERKKFMG TELNGKTLGI LGLGRIGREV ATRMQSFGMK TIGYDPIISP  180
EVSASFGVQQ LPLEEIWPLC DFITVHTPLL PSTTGLLNDN TFAQCKKGVR VVNCARGGIV  240
DEGALLRALQ SGQCAGAALD VFTEEPPRDR ALVDHENVIS CPHLGASTKE AQSRCGEEIA  300
VQFVDMVKGK SLTGVVNAQA LTSAFSPHTK PWIGLAEALG TLMRAWAGSP KGTIQVITQG  360
TSLKNAGNCL SPAVIVGLLK EASKQADVNL VNAKLLVKEA GLNVTTSHSP AAPGEQGFGE  420
CLLAVALAGA PYQAVGLVQG TTPVLQGLNG AVFRPEVPLR RDLPLLLFRT QTSDPAMLPT  480
MIGLLAEAGV RLLSYQTSLV SDGETWHVMG ISSLLPSLEA WKQHVTEAFQ FHF         533

SEQ ID NO: 139         moltype = AA  length = 338
FEATURE                Location/Qualifiers
source                 1..338
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 139
MEQLRAAARL QIVLGHLGRP SAGAVVAHPT SGTISSASFH PQQFQYTLDN NVLTLEQRKF  60
YEENGFLVIK NLVPDADIQR FRNEFEKICR KEVKPLGLTV MRDVTISKSE YAPSEKMITK  120
VQDFQEDKEL FRYCTLPEIL KYVECFTGPN IMAMHTMLIN KPPDSGKKTS RHPLHQDLHY  180
FPFRPSDLIV CAWTAMEHIS RNNGCLVVLP GTHKGSLKPH DYPKWEGGVN KMFHGIQDYE  240
ENKARVHLVM EKGDTVFFHP LLIHGSGQNK TQGFRKAISC HFASADCHYI DVKGTSQENI  300
EKEVVGIAHK FFGAENSVNL KDIWMFRARL VKGERTNL                         338

SEQ ID NO: 140         moltype = AA  length = 532
FEATURE                Location/Qualifiers
source                 1..532
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 140
MEEQVFKGDP DTPHSISFSG SGFLSFYQAG AVDALRDLAP RMLETAHRFA GTSAGAVIAA  60
LAICGIEMDE YLRVLNVGVA EVKKSFLGPL SPSCKMVQMM RQFLYRVLPE DSYKVTTGKL  120
HVSLTRLTDG ENVVVSEFTS KEELIEALYC SCFVPVYCGL IPPTYRGVRY IDGGFTGMQP  180
CAFWTDAITI STFSGQQDIC PRDCPAIFHD FRMFNCSFQF SLENIARMTH ALFPPDLVIL  240
HDYYYRGYED AVLYLRRLNA VYLNSSSKRV IFPRVEVYCQ IELALGNECP ERSQPSLRAR  300
QASLEGATQP HKEWVPKGDG RGSHGPPVSQ PVQTLEFTCE SPVSAPVSPL EQPPAQPLAS  360
STPLSLSGMP PVSFPAVHKP PSSTPGSSLP TPPPGLSPLS PQQQVQPSGS PARSLHSQAP  420
TSPRPSLGPS TVGAPQTLPR SSLSAFPAQP PVEELGQEQP QAVALLVSSK PKSAVPLVHV  480
KETVSKPYVT ESPAEDSNWV NKVFKKNKQK TSGTRKGFPR HSGSKKPSSK VQ          532

SEQ ID NO: 141         moltype = AA  length = 437
FEATURE                Location/Qualifiers
source                 1..437
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 141
MVQMMRQFLY RVLPEDSYKV TTGKLHVSLT RLTDGENVVV SEFTSKEELI EALYCSCFVP  60
VYCGLIPPTY RGVRYIDGGF TGMQPCAFWT DAITISTFSG QQDICPRDCP AIFHDFRMFN  120
CSFQFSLENI ARMTHALFPP DLVILHDYYY RGYEDAVLYL RRLNAVYLNS SSKRVIFPRV  180
EVYCQIELAL GNECPERSQP SLRARQASLE GATQPHKEWV PKGDGRGSHG PPVSQPVQTL  240
EFTCESPVSA PVSPLEQPPA QPLASSTPLS LSGMPPVSFP AVHKPPSSTP GSSLPTPPPG  300
LSPLSPQQQV QPSGSPARSL HSQAPTSPRP SLGPSTVGAP QTLPRSSLSA FPAQPPVEEL  360
GQEQPQAVAL LVSSKPKSAV PLVHVKETVS KPYVTESPAE DSNWVNKVFK KNKQKTSGTR  420
KGFPRHSGSK KPSSKVQ                                                437

SEQ ID NO: 142         moltype = AA  length = 446
FEATURE                Location/Qualifiers
source                 1..446
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 142
MVQMMRQFLY RVLPEDSYKV TTGKLHVSLT RLTDGENVVV SEFTSKEELI EALYCSCFVP  60
VYCGLIPPTY RGVWAFLTLP PQRYIDGGFT GMQPCAFWTD AITISTFSGQ QDICPRDCPA  120
IFHDFRMFNC SFQFSLENIA RMTHALFPPD LVILHDYYYR GYEDAVLYLR RLNAVYLNSS  180
SKRVIFPRVE VYCQIELALG NECPERSQPS LRARQASLEG ATQPHKEWVP KGDGRGSHGP  240
PVSQPVQTLE FTCESPVSAP VSPLEQPPAQ PLASSTPLSL SGMPPVSFPA VHKPPSSTPG  300
SSLPTPPPGL SPLSPQQQVQ PSGSPARSLH SQAPTSPRPS LGPSTVGAPQ TLPRSSLSAF  360
PAQPPVEELG QEQPQAVALL VSSKPKSAVP LVHVKETVSK PYVTESPAED SNWVNKVFKK  420
NKQKTSGTRK GFPRHSGSKK PSSKVQ                                      446

SEQ ID NO: 143         moltype = AA  length = 141
FEATURE                Location/Qualifiers
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source                  1..141
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 143
MNARGLGSEL KDSIPVTELS ASGPFESHDL LRKGFSCVKN ELLPSHPLEL SEKNFQLNQD  60
KMNFSTLRNI QGLFAPLKLQ MEFKAVQQVQ RLPFLSSSNL SLDVLRGNDE TIGFEDILND  120
PSQSEVMGEP HLMVEYKLGL L                                           141

SEQ ID NO: 144          moltype = AA  length = 370
FEATURE                 Location/Qualifiers
source                  1..370
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 144
MDAPRQVVNF GPGPAKLPHS VLLEIQKELL DYKGVGISVL EMSHRSSDFA KIINNTENLV  60
RELLAVPDNY KVIFLQGGGC GQFSAVPLNL IGLKAGRCAD YVVTGAWSAK AAEEAKKFGT  120
INIVHPKLGS YTKIPDPSTW NLNPDASYVY YCANETVHGV EFDFIPDVKG AVLVCDMSSN  180
FLSKPVDVSK FGVIFAGAQK NVGSAGVTVV IVRDDLLGFA LRECPSVLEY KVQAGNSSLY  240
NTPPCFSIYV MGLVLEWIKN NGGAAAMEKL SSIKSQTIYE IIDNSQGFYV CPVEPQNRSK  300
MNIPFRIGNA KGDDALEKRF LDKALELNML SLKGHRSVGG IRASLYNAVT IEDVQKLAAF  360
MKKFLEMHQL                                                        370

SEQ ID NO: 145          moltype = AA  length = 313
FEATURE                 Location/Qualifiers
source                  1..313
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 145
MAALTDLSFM YRWFKNCNLV GNLSEKYVFI TGCDSGFGNL LAKQLVDRGM QVLAACFTEE  60
GSQKLQRDTS YRLQTTLLDV TKSESIKAAA QWVRDKVGEQ GLWALVNNAG VGLPSGPNEW  120
LTKDDFVKVI NVNLVGLIEV TLHMLPMVKR ARGRVVNMSS SGGRVAVIGG GYCVSKFGVE  180
AFSDSIRREL YYFGVKVCII EPGNYRTAIL GKENLESRMR KLWERLPQET RDSYGEDYFR  240
IYTDKLKNIM QVAEPRVRDV INSMEHAIVS RSPRIRYNPG LDAKLLYIPL AKLPTPVTDF  300
ILSRYLPRPA DSV                                                    313

SEQ ID NO: 146          moltype = AA  length = 374
FEATURE                 Location/Qualifiers
source                  1..374
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 146
MDDLCEANGT FAISLFKILG EEDNSRNVFF SPMSISSALA MVFMGAKGST AAQMSQALCL  60
YKDGDIHRGF QSLLSEVNRT GTQYLLRTAN RLFGEKTCDF LPDFKEYCQK FYQAELEELS  120
FAEDTEECRK HINDWVAEKT EGKISEVLDA GTVDPLTKLV LVNAIYFKGK WNEQFDRKYT  180
RGMLFKTNEE KKTVQMMFKE AKFKMGYADE VHTQVLELPY VEEELSMVIL LPDDNTDLAV  240
VEKALTYEKF KAWTNSEKLT KSKVQVFLPR LKLEESYDLE PFLRRLGMID AFDEAKADFS  300
GMSTEKNVPL SKVAHKCFVE VNEEGTEAAA ATAVVRNSRC SRMEPRFCAD HPFLFFIRHH  360
KTNCILFCGR FSSP                                                   374

SEQ ID NO: 147          moltype = AA  length = 643
FEATURE                 Location/Qualifiers
source                  1..643
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 147
MLLGASLVGV LLFSKLVLKL PWTQVGFSLL FLYLGSGGWR FIRVFIKTIR RDIFGGLVLL  60
KVKAKVRQCL QERRTVPILF ASTVRRHPDK TALIFEGTDT HWTFRQLDEY SSSVANFLQA  120
RGLASGDVAA IFMENRNEFV GLWLGMAKLG VEAALINTNL RRDALLHCLT TSRARALVFG  180
SEMASAICEV HASLDPSLSL FCSGSWEPGA VPPSTEHLDP LLKDAPKHLP SCPDKGFTDK  240
LFYIYTSGTT GLPKAAIVVH SRYYRMAALV YYGFRMRPND IVYDCLPLYH SAGNIVGIGQ  300
CLLHGMTVVI RKKFSASRFW DDCIKYNCTI VQYIGELCRY LLNQPPREAE NQHQVRMALG  360
NGLRQSIWTN FSSRFHIPQV AEFYGATECN CSLGNFDSQV GACGFNSRIL SFVYPIRLVR  420
VNEDTMELIR GPDGVCIPCQ PGEPGQLVGR IIQKDPLRRF DGYLNQGANN KKIAKDVFKK  480
GDQAYLTGDV LVMDELGYLY FRDRTGDTFR WKGENVSTTE VEGTLSRLLD MADVAVYGVE  540
VPGTEGRAGM AAVASPTGNC DLERFAQVLE KELPLYARPI FLRLLPELHK TGTYKFQKTE  600
LRKEGFDPAI VKDPLFYLDA QKGRYVPLDQ EAYSRIQAGE EKL                   643

SEQ ID NO: 148          moltype = AA  length = 258
FEATURE                 Location/Qualifiers
source                  1..258
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 148
MSAYPKSYNP FDDDGEDEGA RPAPWRDARD LPDGPDAPAD RQQYLRQEVL RRAEATAAST  60
SRSLALMYES EKVGVASSEE LARQRGVLER TEKMVDKMDQ DLKISQKHIN SIKSVFGGLV  120
NYFKSKPVET PPEQNGTLTS QPNNRLKEAI STSKEQEAKY QASHPNLRKL DDTDPVPRGA  180
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GSAMSTDAYP KNPHLRAYHQ KIDSNLDELS MGLGRLKDIA LGMQTEIEEQ DDILDRLTTK  240
VDKLDVNIKS TERKVRQL                                                258

SEQ ID NO: 149         moltype = AA  length = 855
FEATURE                Location/Qualifiers
source                 1..855
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 149
MGSDRARKGG GGPKDFGAGL KYNSRHEKVN GLEEGVEFLP VNNVKKVEKH GPGRWVVLAA  60
VLIGLLLVLL GIGFLVWHLQ YRDVRVQKVF NGYMRITNEN FVDAYENSNS TEFVSLASKV  120
KDALKLLYSG VPFLGPYHKE SAVTAFSEGS VIAYYWSEFS IPQHLVEEAE RVMAEERVVM  180
LPPRARSLKS FVVTSVVAFP TDSKTVQRTQ DNSCSFGLHA RGVELMRFTT PGFPDSPYPA  240
HARCQWALRG DADSVLSLTF RSFDLASCDE RGSDLVTVYN TLSPMEPHAL VQLCGTYPPS  300
YNLTFHSSQN VLLITLITNT ERRHPGFEAT FFQLPRMSSC GGRLRKAQGT FNSPYYPGHY  360
PPNIDCTWNI EVPNNQHVKV RFKFFYLLEP GVPAGTCPKD YVEINGEKYC GERSQFVVTS  420
NSNKITVRFH SDQSYTDTGF LAEYLSYDSS DPCPGQFTCR TGRCIRKELR CDGWADCTDH  480
SDELNCSCDA GHQFTCKNKF CKPLFWVCDS VNDCGDNSDE QGCSCPAQTF RCSNGKCLSK  540
SQQCNGKDDC GDGSDEASCP KVNVVTCTKH TYRCLNGLCL SKGNPECDGK EDCSDGSDEK  600
DCDCGLRSFT RQARVVGGTD ADEGEWPWQV SLHALGQGHI CGASLISPNW LVSAAHCYID  660
DRGFRYSDPT QWTAFLGLHD QSQRSAPGVQ ERRLKRIISH PFFNDFTFDY DIALLELEKP  720
AEYSSMVRPI CLPDASHVFP AGKAIWVTGW GHTQYGGTGA LILQKGEIRV INQTTCENLL  780
PQQITPRMMC VGFLSGGVDS CQGDSGGPLS SVEADGRIFQ AGVVSWGDGC AQRNKPGVYT  840
RLPLFRDWIK ENTGV                                                   855

SEQ ID NO: 150         moltype = AA  length = 583
FEATURE                Location/Qualifiers
source                 1..583
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 150
MPLRKMKIPF LLLFFLWEAE SHAASRPNII LVMADDLGIG DPGCYGNKTI RTPNIDRLAS  60
GGVKLTQHLA ASPLCTPSRA AFMTGRYPVR SGMASWSRTG VFLFTASSGG LPTDEITFAK  120
LLKDQGYSTA LIGKWHLGMS CHSKTDFCHH PLHHGFNYFY GISLTNLRDC KPGEGSVFTT  180
GFKRLVFLPL QIVGVTLLTL AALNCLGLLH VPLGVFFSLL FLAALILTLF LGFLHYFRPL  240
NCFMMRNYEI IQQPMSYDNL TQRLTVEAAQ FIQRNTETPF LLVLSYLHVH TALFSSKDFA  300
GKSQHGVYGD AVEEMDWSVG QILNLLDELR LANDTLIYFT SDQGAHVEEV SSKGEIHGGS  360
NGIYKGGKAN NWEGGIRVPG ILRWPRVIQA GQKIDEPTSN MDIFPTVAKL AGAPLPEDRI  420
IDGRDLMPLL EGKSQRSDHE FLFHYCNAYL NAVRWHPQNS TSIWKAFFFT PNFNPVGSNG  480
CFATHVCFCF GSYVTHHDPP LLFDISKDPR ERNPLTPASE PRFYEILKVM QEAADRHTQT  540
LPEVPDQFSW NNFLWKPWLQ LCCPSTGLSC QCDREKQDKR LSR                    583

SEQ ID NO: 151         moltype = AA  length = 617
FEATURE                Location/Qualifiers
source                 1..617
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 151
MAFPHRPDAP ELPDFSMLKR LARDQLIYLL EQLPGKKDLF IEADLMSPLD RIANVSILKQ  60
HEVDKLYKVE NKPALSSNEQ LCFLVRPRIK NMRYIASLVN ADKLAGRTRK YKVIFSPQKF  120
YACEMVLEEE GIYGDVSCDE WAFSLLPLDV DLLSMELPEF FRDYFLEGDQ RWINTVAQAL  180
HLLSTLYGPF PNCYGIGRCA KMAYELWRNL EEEEDGETKG RRPEIGHIFL LDRDVDFVTA  240
LCSQVVYEGL VDDTFRIKCG SVDFGPEVTS SDKSLKVLLN AEDKVFNEIR NEHFSNVFGF  300
LSQKARNLQA QYDRRRGMDI KQMKNFVSQE LKGLKQEHRL LSLHIGACES IMKKKTKQDF  360
QELIKTEHAL LEGFNIREST SYIEEHIDRQ VSPIESLRLM CLLSITENGL IPKDYRSLKT  420
QYLQSYGPEH LLTFSNLRRA GLLTEQAPGD TLTAVESKVS KLVTDKAAGK ITDAFSSLAK  480
RSNFRAISKK LNLIPRVDGE YDLKVPRDMA YVFGGAYVPL SCRIIEQVLE RRSWQGLDEV  540
VRLLNCSDFA FTDMTKEDKA SSESLRLILV VFLGGCTFSE ISALRFLGRE KGYRFIFLTT  600
AVTNSARLME AMSEVKA                                                 617

SEQ ID NO: 152         moltype = AA  length = 475
FEATURE                Location/Qualifiers
source                 1..475
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 152
MGMWASLDAL WEMPAEKRIF GAVLLFSWTV YLWETFLAQR QRRIYKTTTH VPPELGQIMD  60
SETFEKSRLY QLDKSTFSFW SGLYSETEGT LILLFGGIPY LWRLSGRFCG YAGFGPEYEI  120
TQSLVFLLLA TLFSALTGLP WSLYNTFVIE EKHGFNQQTL GFFMKDAIKK FVVTQCILLP  180
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VSSLLLYIIK IGGDYFFIYA WLFTLVVSLV LVTIYADYIA PLFDKFTPLP EGKLKEEIEV  240
MAKSIDFPLT KVYVVEGSKR SSHSNAYFYG FFKNKRIVLF DTLLEEYSVL NKDIQEDSGM  300
EPRNEEEGNS EEIKAKVKNK KQGCKNEEVL AVLGHELGHW KLGHTVKNII ISQMNSFLCF  360
FLFAVLIGRK ELFAAFGFYD SQPTLIGLLI IFQFIFSPYN EVLSFCLTVL SRRFEFQADA  420
FAKKLGKAKD LYSALIKLNK DNLGFPVSDW LFSMWHYSHP PLLERLQALK TMKQH       475

SEQ ID NO: 153          moltype = DNA  length = 1098
FEATURE                 Location/Qualifiers
source                  1..1098
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 153
atggacgggc ccgccgagcc ccagatcccg ggcttgtggg acacctatga agatgacatc  60
tcggaaatca gccagaagtt gccaggtgaa tacttccggt acaagggcgt ccccttcccc  120
gtcggcctgt actcgctcga gagcatcagc ttggcggaga acacccaaga tgtgcgggac  180
gacgacatct ttatcatcac ctaccccaag tcaggcacga cctggatgat cgagatcatc  240
tgcttaatcc tgaaggaagg ggatccatcc tggatccgct ccgtgcccat ctgggagcgg  300
gcaccctggt gtgagaccat tgtgggtgcc ttcagcctcc cggaccagta cagcccccgc  360
ctcatgagct cccatcttcc catccagatc ttcaccaagg ccttcttcag ctccaaggcc  420
aaggtgatct acatgggccg caacccccgg gacgttgtgg tctccctcta tcattactcc  480
aagatcgccg ggcagttaaa ggacccgggc acacccgacc agttcctgag ggacttcctc  540
aaaggcgaag tgcagtttgg ctcctggttc gaccacatta agggctggct tcggatgaag  600
ggcaaagaca acttcctatt tatcacctac gaggagctgc agcaggactt acagggctcc  660
gtggagcgca tctgtgggtt cctgggccgt ccgctgggca aggaggcact gggctccgtc  720
gtggcacact caaccttcag cgccatgaag gccaacacca tgtccaacta cacgctgctg  780
cctcccagcc tgctggacca ccgtcgcggg gccttcctcc ggaaaggggt ctgcggcgac  840
tggaagaacc acttcacggt ggcccagagc gaagccttcg atcgtgccta ccgcaagcag  900
atgcgggggа tgccgacctt ccccctgggat gaagacccgg aggaggacgg cagcccagat  960
cctgagccca gccctgagcc tgagcccaag cccagccttg agcccaacac cagcctggag  1020
cgtgagccca gacccaactc cagccccagc cccagccccg gccaggcctc tgagaccccg  1080
cacccacgac cctcataa                                                1098

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atgcggggca tgcctacctt tccatgggac gaagatcccg aagaggacgg ctctcctgat  960
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MRGMPTFPWD EDPEEDGSPD PEPSPEPEPK PSLEPNTSLE REPRPNSSPS PSPGQASETP  360
HPRPS                                                              365
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What is claimed is:

1. A recombinant herpes virus genome comprising one or more polynucleotides encoding an ichthyosis-associated polypeptide.

2. The recombinant herpes virus genome of claim 1, wherein the recombinant herpes virus genome is replication competent.

3. The recombinant herpes virus genome of claim 1, wherein the recombinant herpes virus genome is replication defective.

4. The recombinant herpes virus genome of any one of claims 1-3, wherein the recombinant herpes virus genome comprises the one or more polynucleotides encoding an ichthyosis-associated polypeptide within one or more viral gene loci.

5. The recombinant herpes virus genome of any one of claims 1-4, wherein the recombinant herpes virus genome is selected from the group consisting of a recombinant herpes simplex virus genome, a recombinant varicella zoster virus genome, a recombinant human cytomegalovirus genome, a recombinant herpesvirus 6A genome, a recombinant herpesvirus 6B genome, a recombinant herpesvirus 7 genome, a recombinant Kaposi's sarcoma-associated herpesvirus genome, and any derivatives thereof.

6. The recombinant herpes virus genome of any one of claims 1-5, wherein the recombinant herpes virus genome is a recombinant herpes simplex virus genome.

7. The recombinant herpes virus genome of claim 6, wherein the recombinant herpes simplex virus genome is a recombinant type 1 herpes simplex virus (HSV-1) genome, a recombinant type 2 herpes simplex virus (HSV-2) genome, or any derivatives thereof.

8. The recombinant herpes virus genome of claim 6 or claim 7, wherein the recombinant herpes simplex virus genome is a recombinant type 1 herpes simplex virus (HSV-1) genome.

9. The recombinant herpes virus genome of any one of claims 5-8, wherein the recombinant herpes simplex virus genome has been engineered to reduce or eliminate expression of one or more toxic herpes simplex virus genes.

10. The recombinant herpes virus genome of any one of claims 5-9, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation.

11. The recombinant herpes virus genome of claim 10, wherein the inactivating mutation is in a herpes simplex virus gene.

12. The recombinant herpes virus genome of claim 11, wherein the inactivating mutation is a deletion of the coding sequence of the herpes simplex virus gene.

13. The recombinant herpes virus genome of claim 11 or claim 12, wherein the herpes simplex virus gene is selected from the group consisting of Infected Cell Protein (ICP) 0, ICP4, ICP22, ICP27, ICP47, thymidine kinase (tk), Long Unique Region (UL) 41, and UL55.

14. The recombinant herpes virus genome of claim 13, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in one or both copies of the ICP4 gene.

15. The recombinant herpes virus genome of claim 13 or claim 14, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP22 gene.

16. The recombinant herpes virus genome of any one of claims 13-15, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL41 gene.

17. The recombinant herpes virus genome of any one of claims 13-16, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in one or both copies of the ICP0 gene.

18. The recombinant herpes virus genome of any one of claims 13-17, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP27 gene.

19. The recombinant herpes virus genome of any one of claims 13-18, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in the ICP47 gene.

20. The recombinant herpes virus genome of any one of claims 13-19, wherein the recombinant herpes simplex virus genome comprises an inactivating mutation in the UL55 gene.

21. The recombinant herpes virus genome of any one of claims 6-20, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within one or both of the ICP4 viral gene loci.

22. The recombinant herpes virus genome of any one of claims 6-21, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the ICP22 viral gene locus.

23. The recombinant herpes virus genome of any one of claims 6-22, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the UL41 viral gene locus.

24. The recombinant herpes virus genome of any one of claims 6-23, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within one or both of the ICP0 viral gene loci.

25. The recombinant herpes virus genome of any one of claims 6-24, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the ICP27 viral gene locus.

26. The recombinant herpes virus genome of any one of claims 6-25, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the ICP47 viral gene locus.

27. The recombinant herpes virus genome of any one of claims 6-26, wherein the recombinant herpes simplex virus genome comprises the one or more polynucleotides encoding the ichthyosis-associated polypeptide within the UL55 viral gene locus.

28. The recombinant herpes virus genome of any one of claims 1-27, wherein the ichthyosis-associated polypeptide is not a transglutaminase (TGM) polypeptide.

29. The recombinant herpes virus genome of any one of claims 1-28, wherein the ichthyosis-associated polypeptide is not a transglutaminase 1 (TGM1) polypeptide or a transglutaminase 5 (TGM5) polypeptide.

30. The recombinant herpes virus genome of any one of claims 1-29, wherein the ichthyosis-associated polypeptide

is selected from the group consisting of an ATP-binding cassette sub-family A member 12 polypeptide (ABCA12), a 1-acylglycerol-3-phosphate O-acyltransferase ABHD5 polypeptide (ABHD5), an Aldehyde dehydrogenase family 3 member A2 polypeptide (ALDH3A2), an Arachidonate 12-lipoxygenase 12R-type polypeptide (ALOX12B), a Hydroperoxide isomerase ALOXE3 polypeptide (ALOXE3), an AP-1 complex subunit sigma-1A polypeptide (AP1S1), an Arylsulfatase E polypeptide (ARSE), a Caspase-14 polypeptide (CASP14), a Corneodesmosin polypeptide (CDSN), a Ceramide synthase 3 polypeptide (CERS3), a Carbohydrate sulfotransferase 8 polypeptide (CHST8), a Claudin-1 polypeptide (CLDN1), a Cystatin-A polypeptide (CSTA), a Cytochrome P450 4F22 polypeptide (CYP4F22), a 3-beta-hydroxysteroid-Delta(8),Delta(7)-isomerase polypeptide (EBP), an Elongation of very long chain fatty acids protein 4 polypeptide (ELOVL4), a Filaggrin polypeptide (FLG), a Filaggrin 2 polypeptide (FLG2), a Gap junction beta-2 polypeptide (GJB2), a Gap junction beta-3 polypeptide (GJB3), a Gap junction beta-4 polypeptide (GJB4), a Gap junction beta-6 polypeptide (GJB6), a 3-ketodihydrosphingosine reductase polypeptide (KDSR), a Keratin, type II cytoskeletal 1 polypeptide (KRT1), a Keratin, type II cytoskeletal 2 epidermal polypeptide (KRT2), a Keratin, type I cytoskeletal 9 polypeptide (KRT9), a Keratin, type I cytoskeletal 10 polypeptide (KRT10), a Lipase member N polypeptide (LIPN), a Loricrin polypeptide (LOR), a Membrane-bound transcription factor site-2 protease polypeptide (MBTPS2), a Magnesium transporter NIPA4 polypeptide (NIPAL4), a Sterol-4-alpha-carboxylate 3-dehydrogenase, decarboxylating polypeptide (NSDHL), a Peroxisomal targeting signal 2 receptor polypeptide (PEX7), a D-3-phosphoglycerate dehydrogenase polypeptide (PHGDH), a Phytanoyl-CoA dioxygenase, peroxisomal polypeptide (PHYH), Patatin-like phospholipase domain-containing protein 1 polypeptide (PNPLA1), a Proteasome maturation protein polypeptide (POMP), a Phosphoserine aminotransferase polypeptide (PSAT1), a Short-chain dehydrogenase/reductase family 9C member 7 polypeptide (SDR9C7), a Serpin B8 polypeptide (SERPINB8), a Long-chain fatty acid transport protein 4 polypeptide (SLC27A4), a Synaptosomal-associated protein 29 polypeptide (SNAP29), a Suppressor of tumorigenicity 14 protein polypeptide (ST14), a Steryl-sulfatase polypeptide (STS), a Sulfotransferase 2B1 polypeptide (SULT2B1), a Vacuolar protein sorting-associated protein 33B polypeptide (VPS33B), and a CAAX prenyl protease 1 homolog polypeptide (ZMPSTE24).

31. The recombinant herpes virus genome of any one of claims **1-30**, wherein the ichthyosis-associated polypeptide is a human ichthyosis-associated polypeptide.

32. The recombinant herpes virus genome of any one of claims **1-31**, wherein the ichthyosis-associated polypeptide comprises a sequence having at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to an amino acid sequence selected from the group consisting of SEQ ID NOS: 102-152 or 155.

33. The recombinant herpes virus genome of any one of claims **1-32**, wherein the ichthyosis-associated polypeptide is selected from the group consisting of ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, ARSE, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, ELOVL4, KDSR, LIPN, MBTPS2, NIPAL4, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4, SNAP29, ST14, STS, SULT2B1, VPS33B, and ZMPSTE24.

34. The recombinant herpes virus genome of any one of claims **1-33**, wherein the ichthyosis-associated polypeptide is selected from the group consisting of ABCA12, ABHD5, ALDH3A2, ALOX12B, ALOXE3, AP1S1, CASP14, CDSN, CERS3, CHST8, CLDN1, CSTA, CYP4F22, ELOVL4, KDSR, LIPN, NIPAL4, PEX7, PHGDH, PHYH, PNPLA1, POMP, PSAT1, SDR9C7, SERPINB8, SLC27A4, SNAP29, ST14, SULT2B1, VPS33B, and ZMPSTE24.

35. The recombinant herpes virus genome of any one of claims **1-33**, wherein the ichthyosis-associated polypeptide is selected from the group consisting of ARSE, MBTPS2, and STS.

36. The recombinant herpes virus genome of any one of claims **1-35**, wherein the recombinant herpes virus genome has reduced cytotoxicity when introduced into a target cell as compared to a corresponding wild-type herpes virus genome.

37. The recombinant herpes virus genome of claim **36**, wherein the target cell is a cell of the epidermis and/or dermis.

38. The recombinant herpes virus genome of claim **36** or claim **37**, wherein the target cell is a human cell.

39. A herpes virus comprising the recombinant herpes virus genome of any one of claims **1-38**.

40. The herpes virus of claim **39**, wherein the herpes virus is replication competent.

41. The herpes virus of claim **39**, wherein the herpes virus is replication defective.

42. The herpes virus of any one of claims **39-41**, wherein the herpes virus has reduced cytotoxicity as compared to a corresponding wild-type herpes virus.

43. The herpes virus of any one of claims **39-42**, wherein the herpes virus is selected from the group consisting of a herpes simplex virus, a varicella zoster virus, a human cytomegalovirus, a herpesvirus 6A, a herpesvirus 6B, a herpesvirus 7, and a Kaposi's sarcoma-associated herpesvirus.

44. The herpes virus of any one of claims **39-43**, wherein the herpes virus is a herpes simplex virus.

45. The herpes virus of claim **43** or claim **44**, wherein the herpes simplex virus is a type 1 herpes simplex virus (HSV-1), a type 2 herpes simplex virus (HSV-2), or any derivatives thereof.

46. The herpes virus of any one of claims **43-45**, wherein the herpes simplex virus is a type 1 herpes simplex virus (HSV-1).

47. A pharmaceutical composition comprising the recombinant herpes virus genome of any one of claims **1-38** or the herpes virus of any one of claims **39-46** and a pharmaceutically acceptable excipient.

48. The pharmaceutical composition of claim **47**, wherein the pharmaceutical composition is suitable for topical, transdermal, subcutaneous, intradermal, oral, intranasal, intratracheal, sublingual, buccal, rectal, vaginal, inhaled, intravenous, intraarterial, intramuscular, intracardiac, intraosseous, intraperitoneal, transmucosal, intravitreal, subretinal, intraarticular, peri-articular, local, or epicutaneous administration.

49. The pharmaceutical composition of claim **47** or claim **48**, wherein the pharmaceutical composition is suitable for topical, transdermal, subcutaneous, intradermal, or transmucosal administration.

50. The pharmaceutical composition of any one of claims **47-49**, wherein the pharmaceutical composition is suitable for topical, transdermal, or intradermal administration.

51. The pharmaceutical composition of any one of claims **47-50**, wherein the pharmaceutical composition is suitable for topical administration.

52. The herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51** for use as a medicament.

53. The herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51** for use in a therapy.

54. Use of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51** in the manufacture of a medicament for treating one or more forms of congenital ichthyosis.

55. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of congenital ichthyosis in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

56. The method of claim **55**, wherein the congenital ichthyosis is selected from the group consisting of harlequin ichthyosis (HI), autosomal recessive congenital ichthyosis (ARCI), lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE), Chanarin-Dorfman syndrome (CDS), Sjogren-Larsson syndrome (SLS), mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome, chondrodysplasia *punctata* 1 (CDPX1), chondrodysplasia *punctata* 2 (CDPX2), peeling skin syndrome (PSS), neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome, ichthyosis vulgaris, keratitis-ichthyosis-deafness syndrome (KID), palmoplantar keratoderma (PPK), palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL), epidermolytic palmoplantar keratoderma (EPPK), erythrokeratodermia *variabilis* (EKV), Clouston syndrome, progressive symmetric erythrokeratodermia, epidermolytic ichthyosis (EI), superficial epidermolytic ichthyosis (SEI), loricrin keratoderma, ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome, Olmsted syndrome, congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome, Refsum disease, Neu-Laxova syndrome, keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome, ichthyosis prematurity syndrome (IPS), cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome, X-linked ichthyosis, arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome, and restrictive dermopathy.

57. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of harlequin ichthyosis (HI) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

58. The method of claim **57**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an ABCA12 polypeptide.

59. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Chanarin-Dorfman syndrome (CDS) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

60. The method of claim **59**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an ABHD5 polypeptide.

61. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Sjogren-Larsson syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

62. The method of claim **61**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an ALDH3A2 polypeptide.

63. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of autosomal recessive congenital ichthyosis (ARCI) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

64. The method of claim **63**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from the group consisting of ALOX12B, ALOXE3, CASP14, CERS3, CYP4F22, LIPN, NIPAL4, PNPLA1, SDR9C7, SLC27A4, ST14, and SULT2B1.

65. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of mental retardation, enteropathy, deafness, peripheral neuropathy, ichthyosis, and keratoderma (MEDNIK) syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

66. The method of claim **65**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an AP1S1 polypeptide.

67. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of chondrodysplasia *punctata* 1 (CDPX1) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

68. The method of claim **67**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an ARSE polypeptide.

69. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of chondrodysplasia *punctata* 2 (CDPX2) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

70. The method of claim **69**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an EBP polypeptide.

71. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of peeling skin syndrome (PSS) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

72. The method of claim **71**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from the group consisting of CDSN, CHST8, CSTA, FLG2, and SERPINB8.

73. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of neonatal ichthyosis-sclerosing cholangitis (NISCH) syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

74. The method of claim **73**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a CLDN1 polypeptide.

75. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis vulgaris in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

76. The method of claim **75**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a FLG polypeptide.

77. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of keratitis-ichthyosis-deafness (KID) syndrome, Clouston syndrome, and/or palmoplantar keratoderma with sensorineural hearing loss (PPK/SNHL) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

78. The method of claim **77**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a GJB2 or GJB6 polypeptide.

79. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of erythrokeratodermia *variabilis* (EKV) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

80. The method of claim **79**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a GJB3 or GJB4 polypeptide.

81. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of progressive symmetric erythrokeratodermia in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

82. The method of claim **81**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a KDSR polypeptide.

83. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of epidermolytic ichthyosis (EI) and/or superficial epidermolytic ichthyosis (SEI) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

84. The method of claim **83**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a polypeptide selected from the group consisting of KRT1, KRT2, and KRT10.

85. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of epidermolytic palmoplantar keratoderma (EPPK) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

86. The method of claim **85**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a KRT9 polypeptide.

87. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of loricrin keratoderma in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

88. The method of claim **87**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a LOR polypeptide.

89. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis follicularis, alopecia, and photophobia (IFAP) syndrome and/or Olmsted syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

90. The method of claim **89**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an MBTPS2 polypeptide.

91. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of congenital hemidysplasia with ichthyosiform nevus and limb defects (CHILD) syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

92. The method of claim **91**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a NSDHL polypeptide.

93. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Refsum disease in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

94. The method of claim **93**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a PEX7 or PHYH polypeptide.

95. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of Neu-Laxova syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

96. The method of claim **95**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a PHGDH or PSAT1 polypeptide.

97. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of keratosis *linearis* with ichthyosis congenita and sclerosing keratoderma (KLICK) syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

98. The method of claim **97**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a POMP polypeptide.

99. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of ichthyosis prematurity syndrome (IPS) in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

100. The method of claim **99**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a SLC27A4 polypeptide.

101. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of cerebral dysgenesis, neuropathy, ichthyosis, and palmoplantar keratoderma (CEDNIK) syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

102. The method of claim **101**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a SNAP29 polypeptide.

103. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of X-linked ichthyosis in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

104. The method of claim **103**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding an STS polypeptide.

105. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of arthrogryposis-renal dysfunction-cholestasis (ARC) syndrome in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

106. The method of claim **105**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a VPS33B polypeptide.

107. A method of providing prophylactic, palliative, or therapeutic relief to one or more signs or symptoms of restrictive dermopathy in a subject in need thereof, the method comprising administering to the subject an effective amount of the herpes virus of any one of claims **39-46** or the pharmaceutical composition of any one of claims **47-51**.

108. The method of claim **107**, wherein the recombinant herpes virus genome comprises one or more polynucleotides encoding a ZMPSTE24 polypeptide.

109. The method of any one of claims **55-108**, wherein the subject is a human.

110. The method of any one of claims **55-109**, wherein the subject's genome comprises a pathogenic variant of an ichthyosis-associated gene.

111. The method of any one of claims **55-110**, wherein the subject's genome comprises a loss-of-function mutation in an ichthyosis-associated gene.

112. The method of any one of claims **55-111**, wherein the herpes virus or pharmaceutical composition is administered topically, transdermally, subcutaneously, epicutaneously, intradermally, orally, sublingually, buccally, rectally, vaginally, intravenously, intraarterially, intramuscularly, intraosseously, intracardially, intraperitoneally, transmucosally, intravitreally, subretinally, intraarticularly, peri-articularly, locally, or via inhalation to the subject.

113. The method of any one of claims **55-112**, wherein the herpes virus or pharmaceutical composition is administered topically, transdermally, subcutaneously, intradermally, or transmucosally to the subject.

114. The method of any one of claims **55-113**, wherein the herpes virus or pharmaceutical composition is administered topically, transdermally, or intradermally to the subject.

115. The method of any one of claims **55-114**, wherein the herpes virus or pharmaceutical composition is administered topically to the subject.

116. The method of any one of claims **55-115**, wherein the skin of the subject is abraded prior to administration.

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